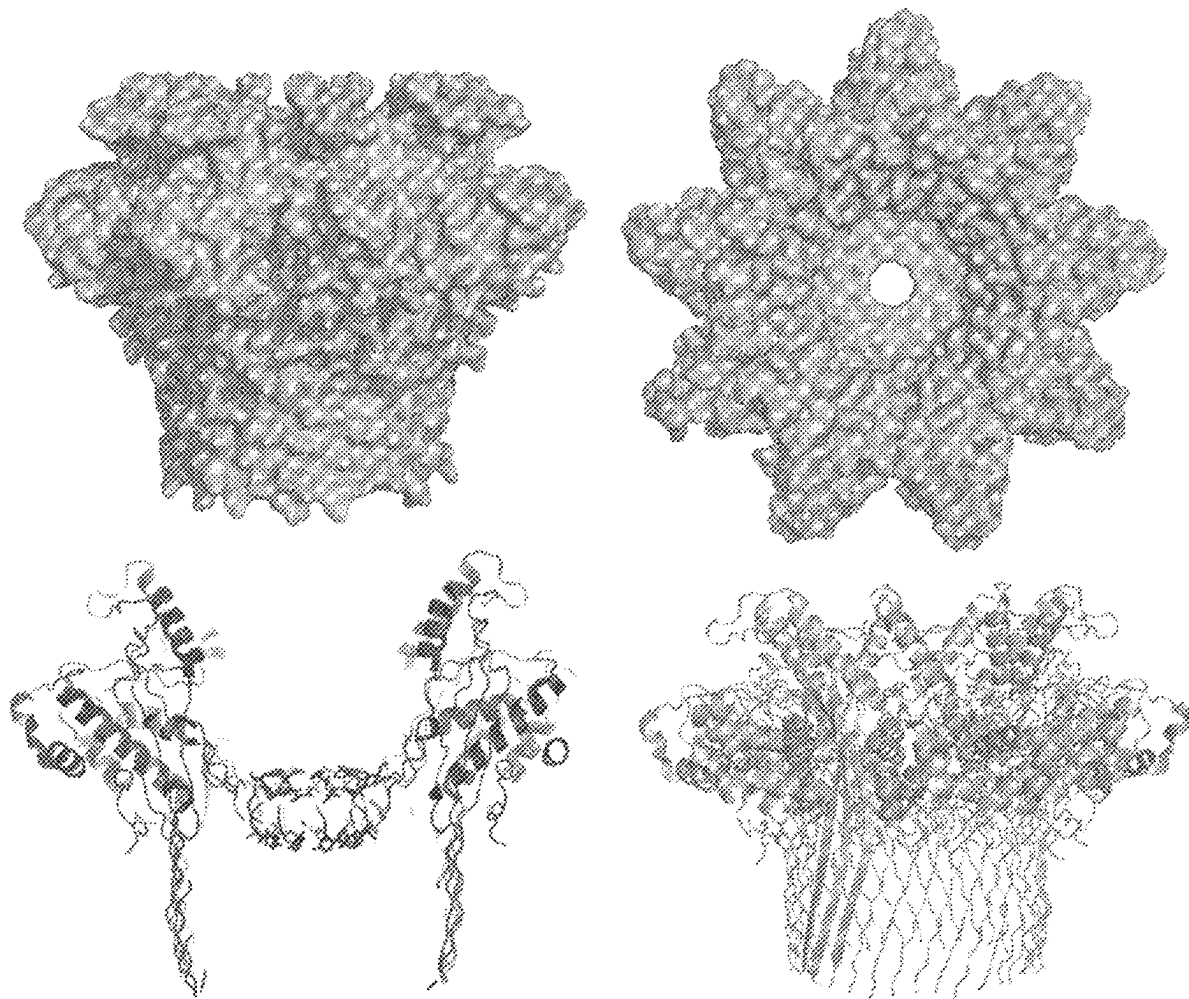




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Jayasinghe et al.(10) **Pub. No.: US 2025/0283165 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **MUTANT PORES**(71) Applicant: **Oxford Nanopore Technologies PLC**,
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(GB); **Elizabeth Jayne Wallace**,
Oxford (GB)(73) Assignee: **Oxford Nanopore Technologies PLC**,
Oxford (GB)(21) Appl. No.: **19/013,448**(22) Filed: **Jan. 8, 2025****Related U.S. Application Data**(60) Continuation of application No. 18/159,718, filed on
Jan. 26, 2023, now Pat. No. 12,227,800, which is a
continuation of application No. 17/215,506, filed on
Mar. 29, 2021, now Pat. No. 11,597,970, which is a
division of application No. 16/081,891, filed on Aug.
31, 2018, filed as application No. PCT/GB2017/
050571 on Mar. 2, 2017, now Pat. No. 10,995,372.(30) **Foreign Application Priority Data**Mar. 2, 2016 (GB) 1603656.8
Mar. 2, 2016 (GB) 1603657.6
Mar. 2, 2016 (GB) 1603658.4**Publication Classification**(51) **Int. Cl.**
C12Q 1/6869 (2018.01)
C07K 14/245 (2006.01)
G01N 33/487 (2006.01)
(52) **U.S. Cl.**
CPC **C12Q 1/6869** (2013.01); **C07K 14/245**
(2013.01); **G01N 33/48721** (2013.01); **C12Q**
2565/631 (2013.01)(57) **ABSTRACT**The invention relates to mutant forms of CsgG. The inven-
tion also relates to analyte detection and characterisation
using CsgG.**Specification includes a Sequence Listing.**

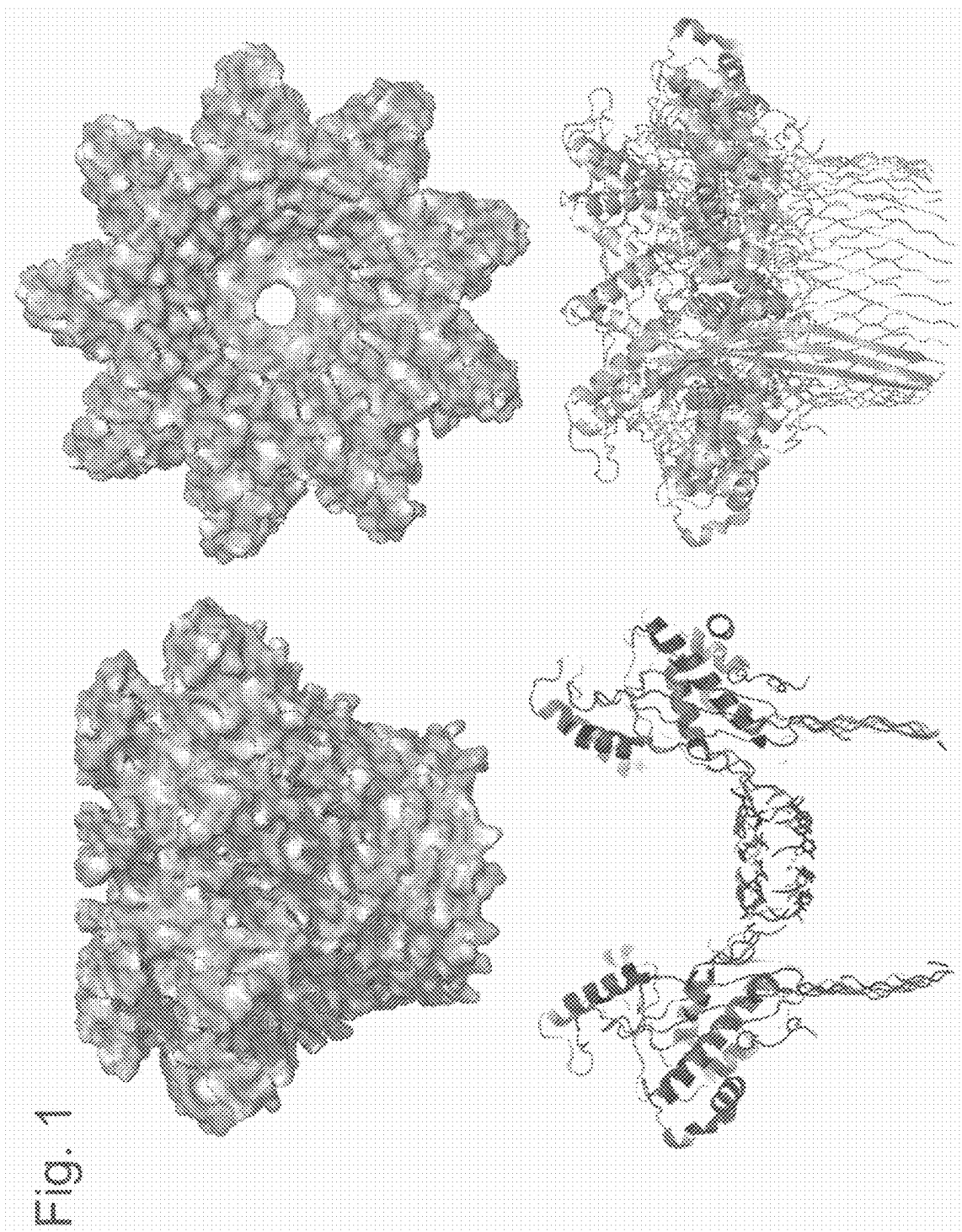


Fig. 2

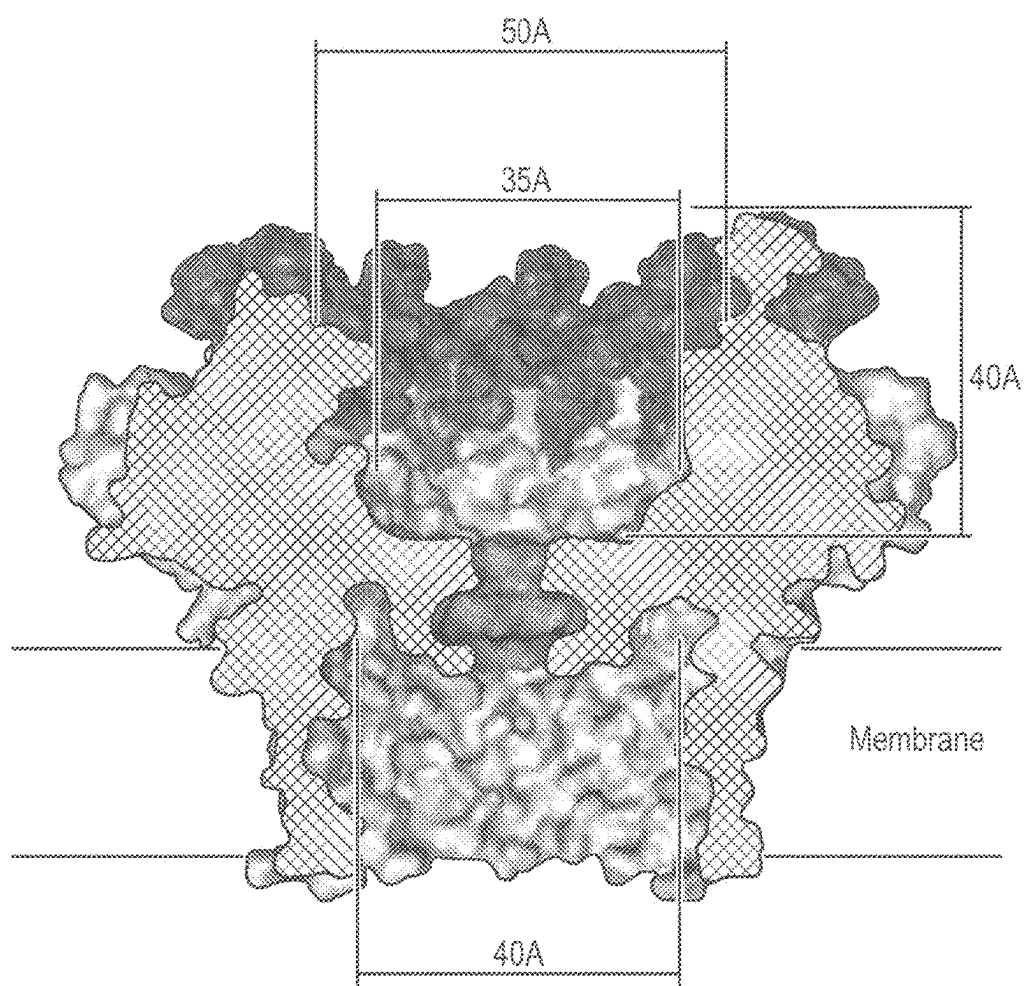
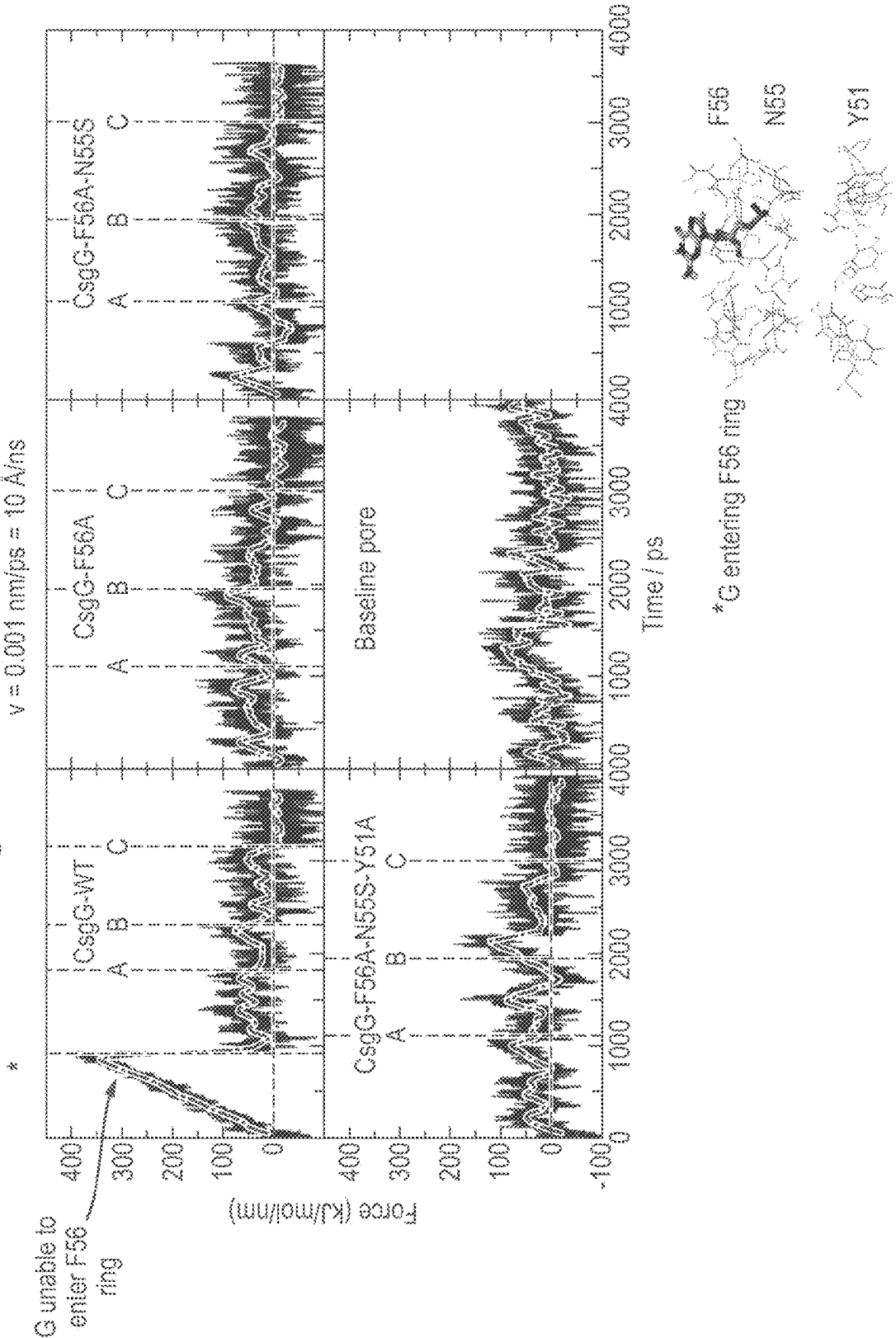
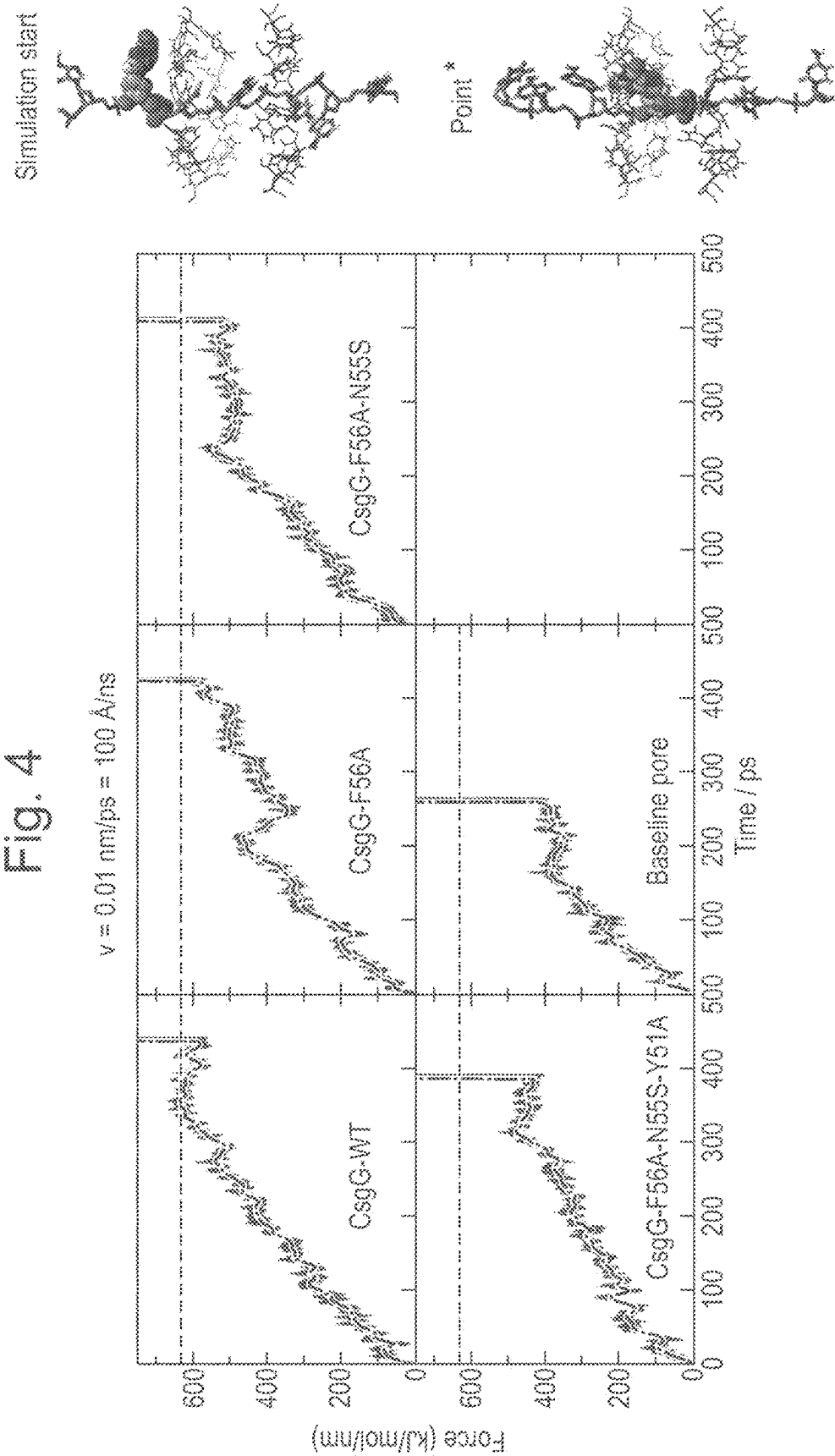


Fig. 3





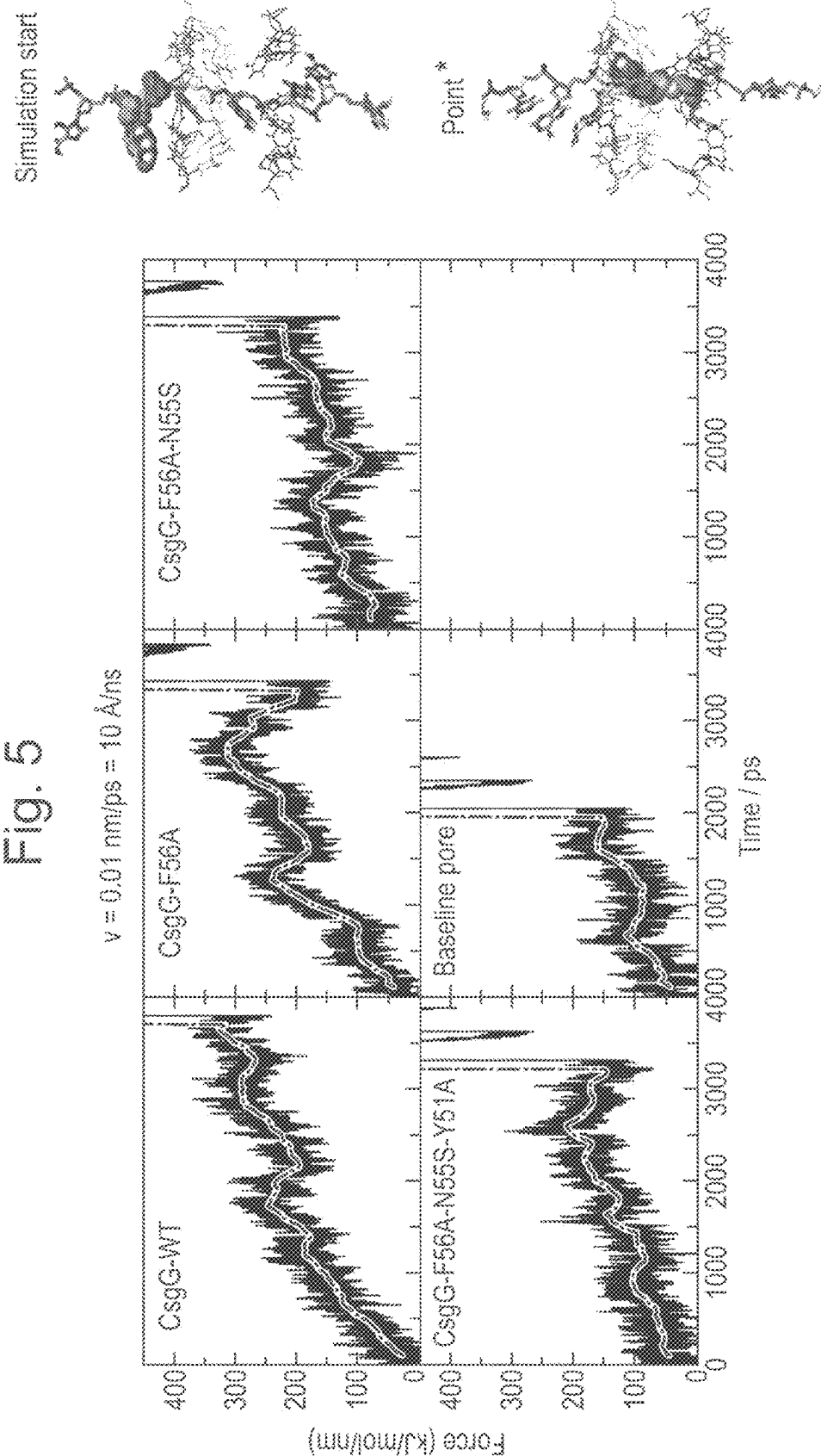


Fig. 6A

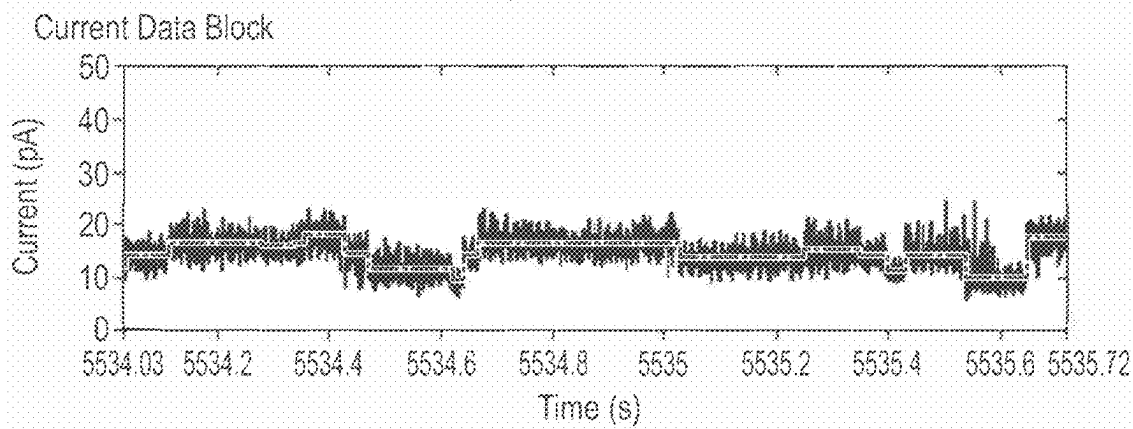


Fig. 6B

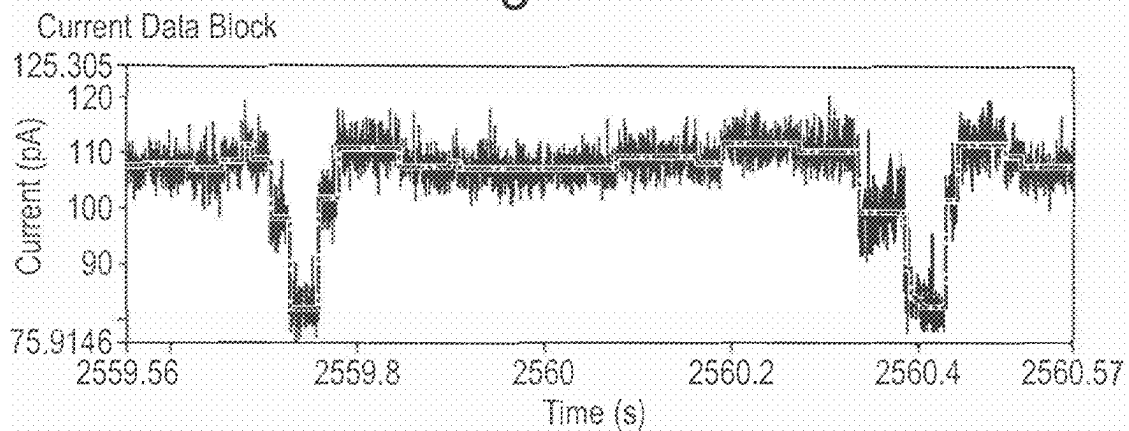


Fig. 6C

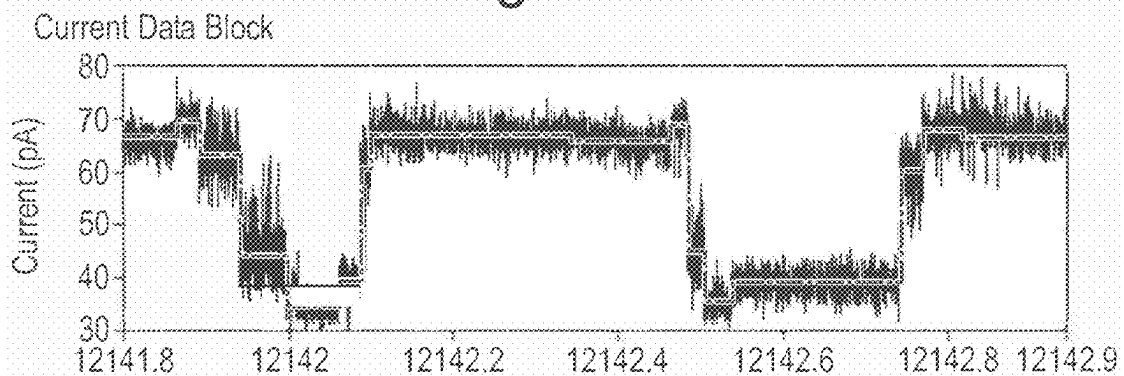


Fig. 7A

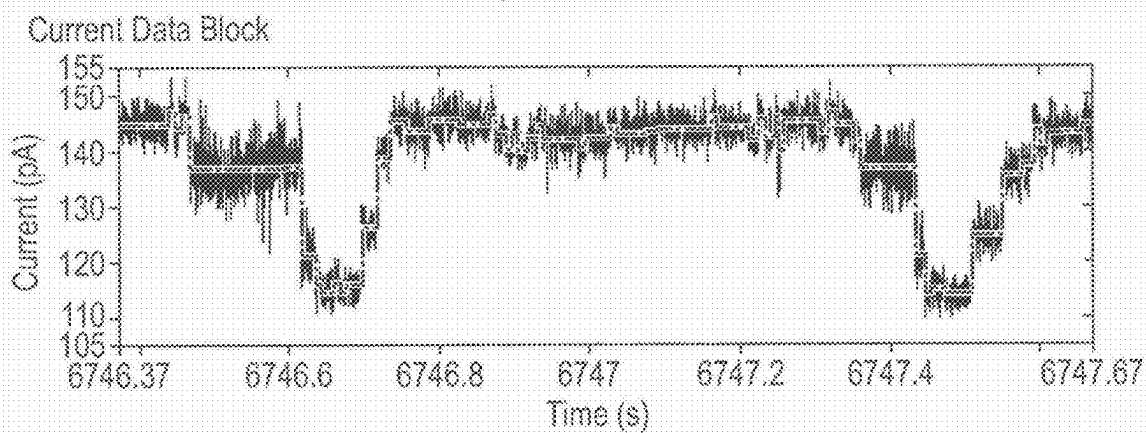


Fig. 7B

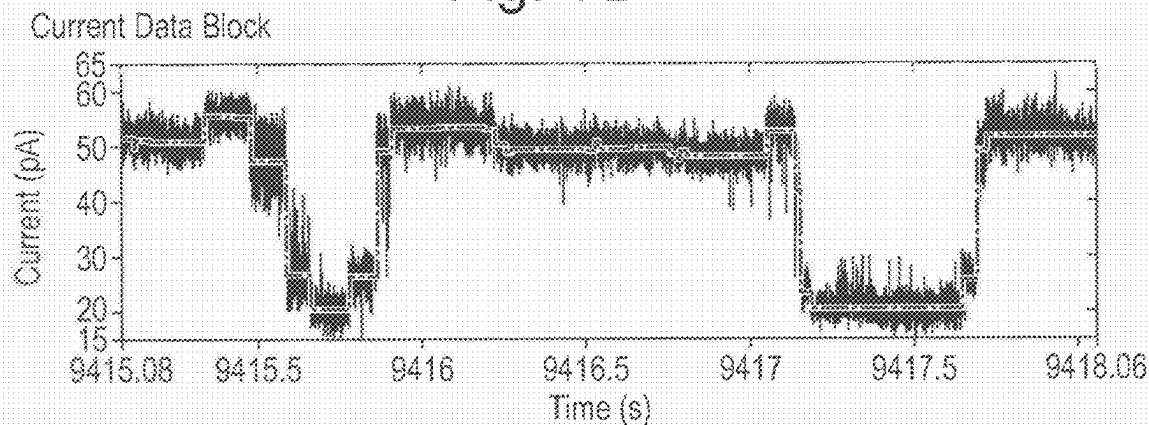


Fig. 7C

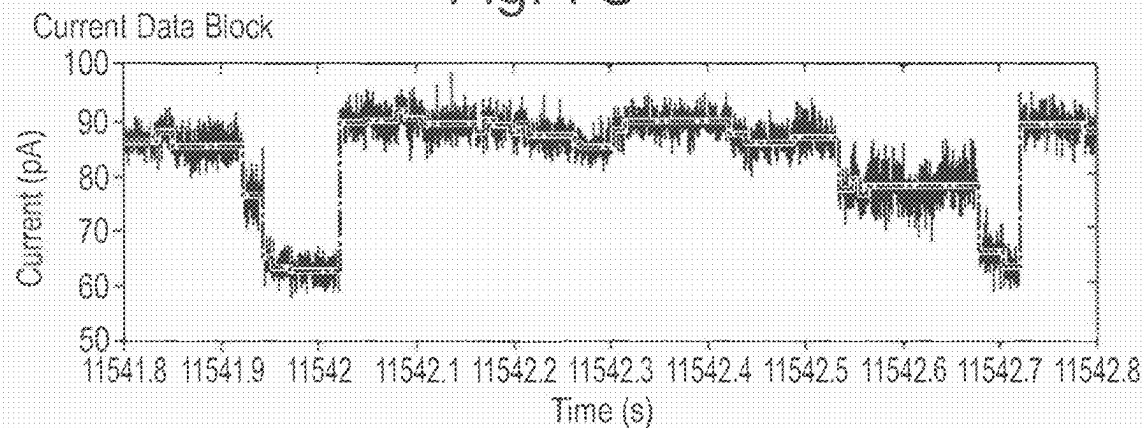


Fig. 8A

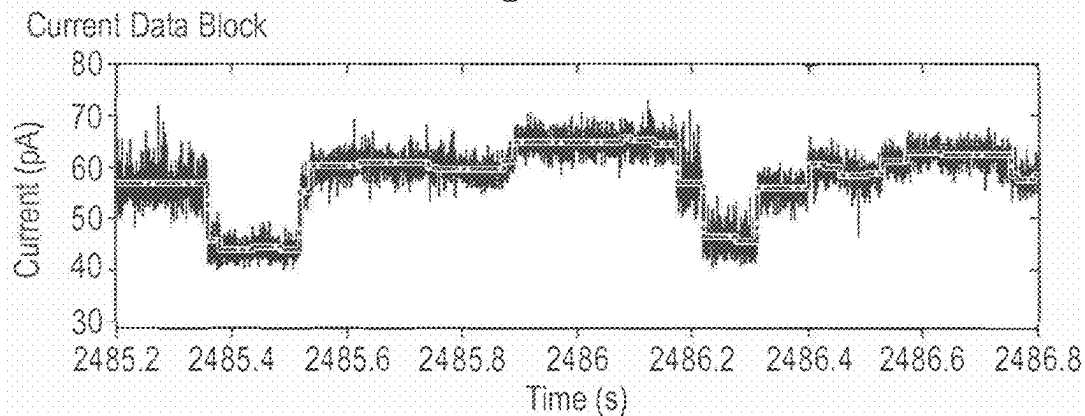


Fig. 8B

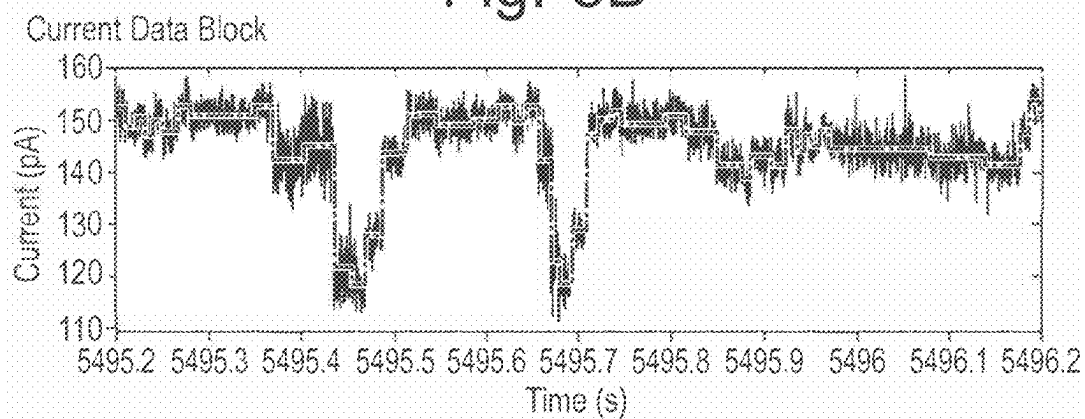


Fig. 8C

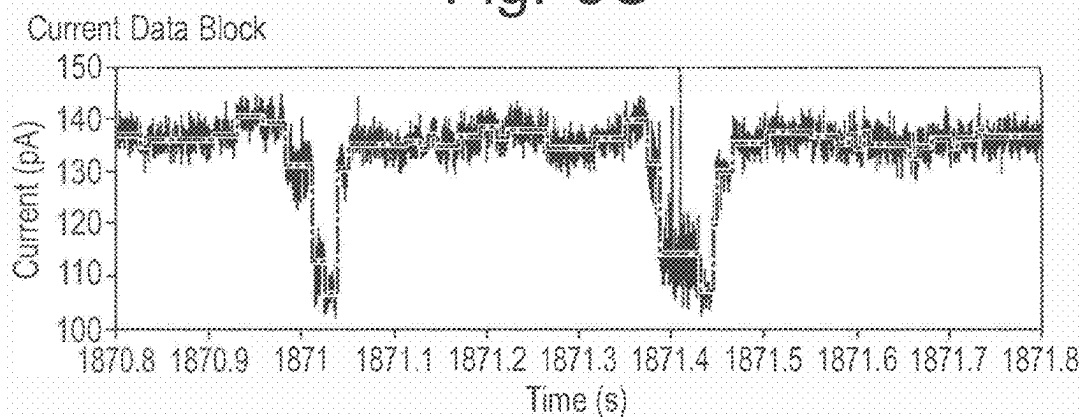


Fig. 9A

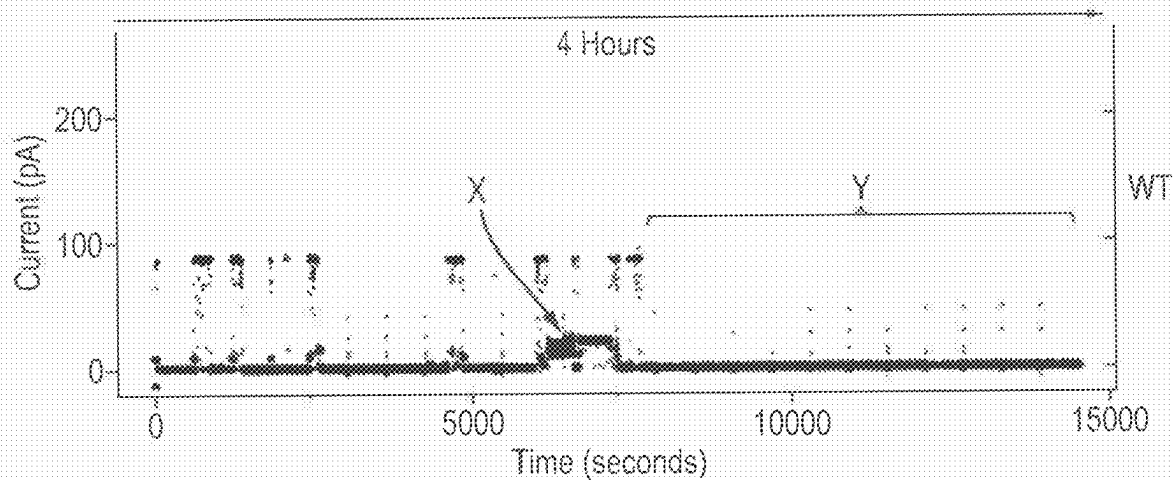


Fig. 9B

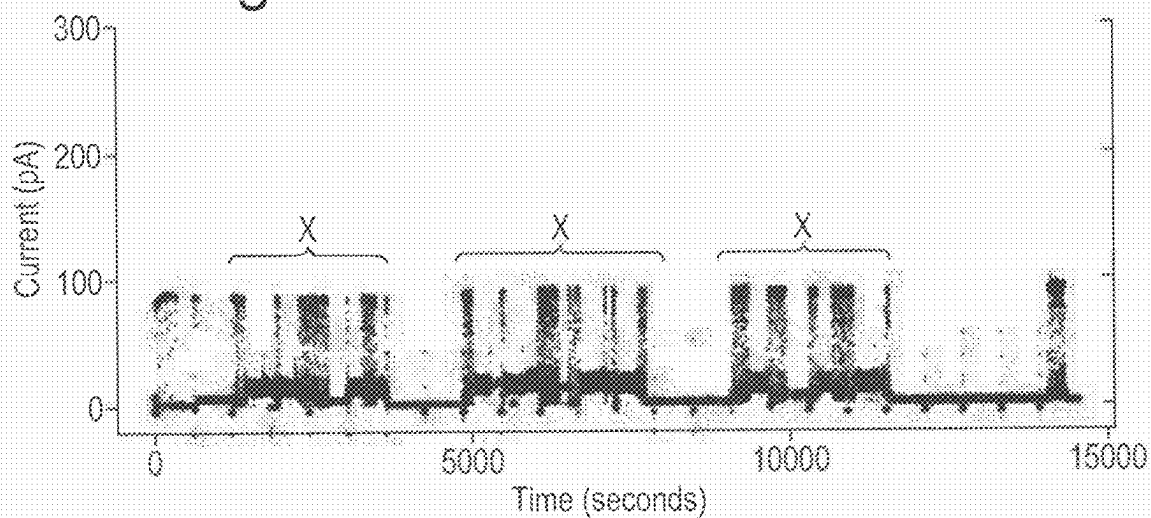


Fig. 9C

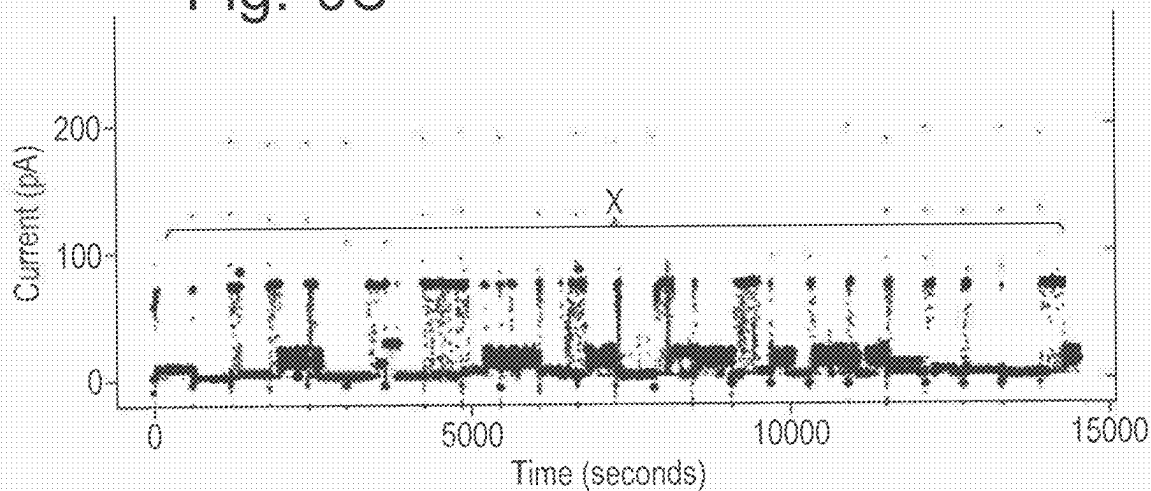


Fig. 10A

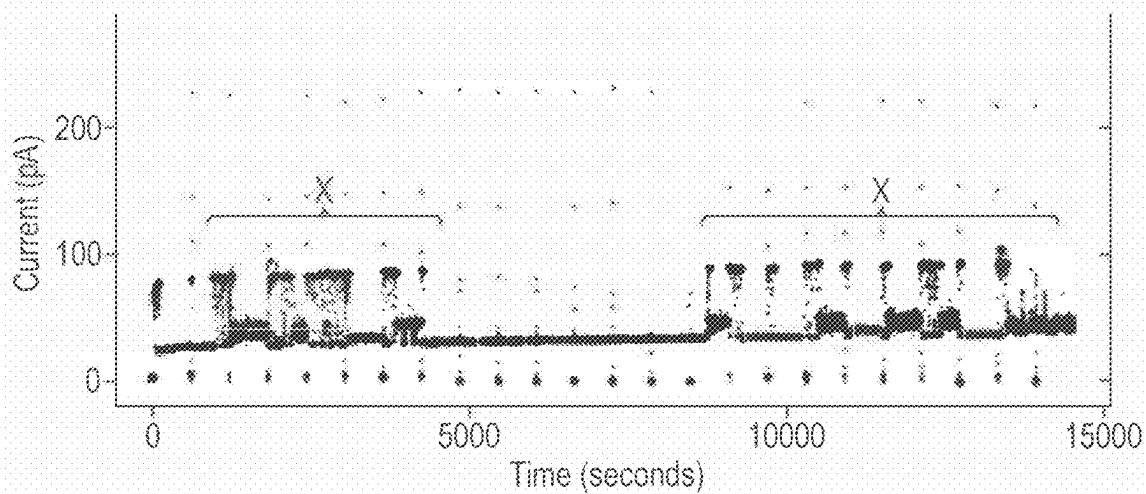


Fig. 10B

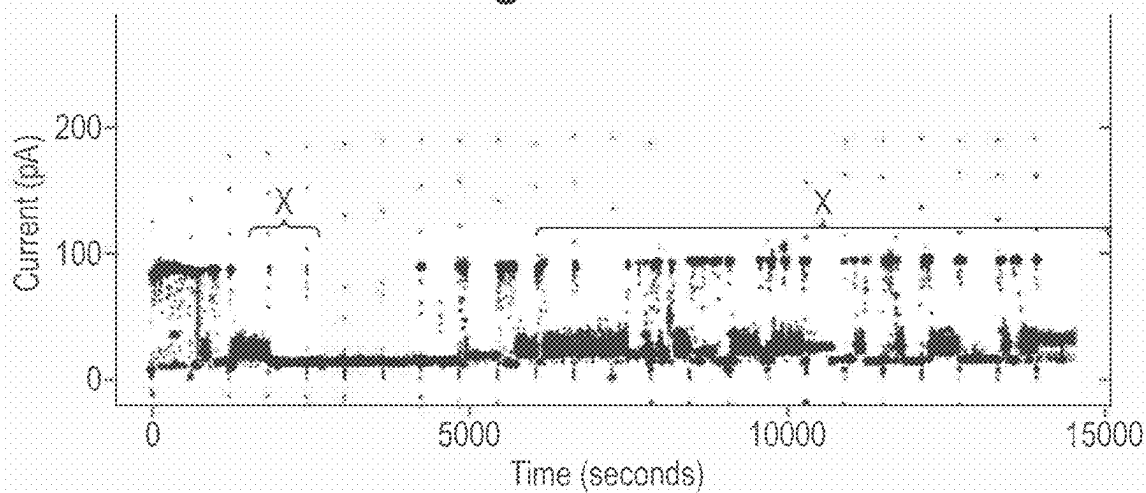


Fig. 11

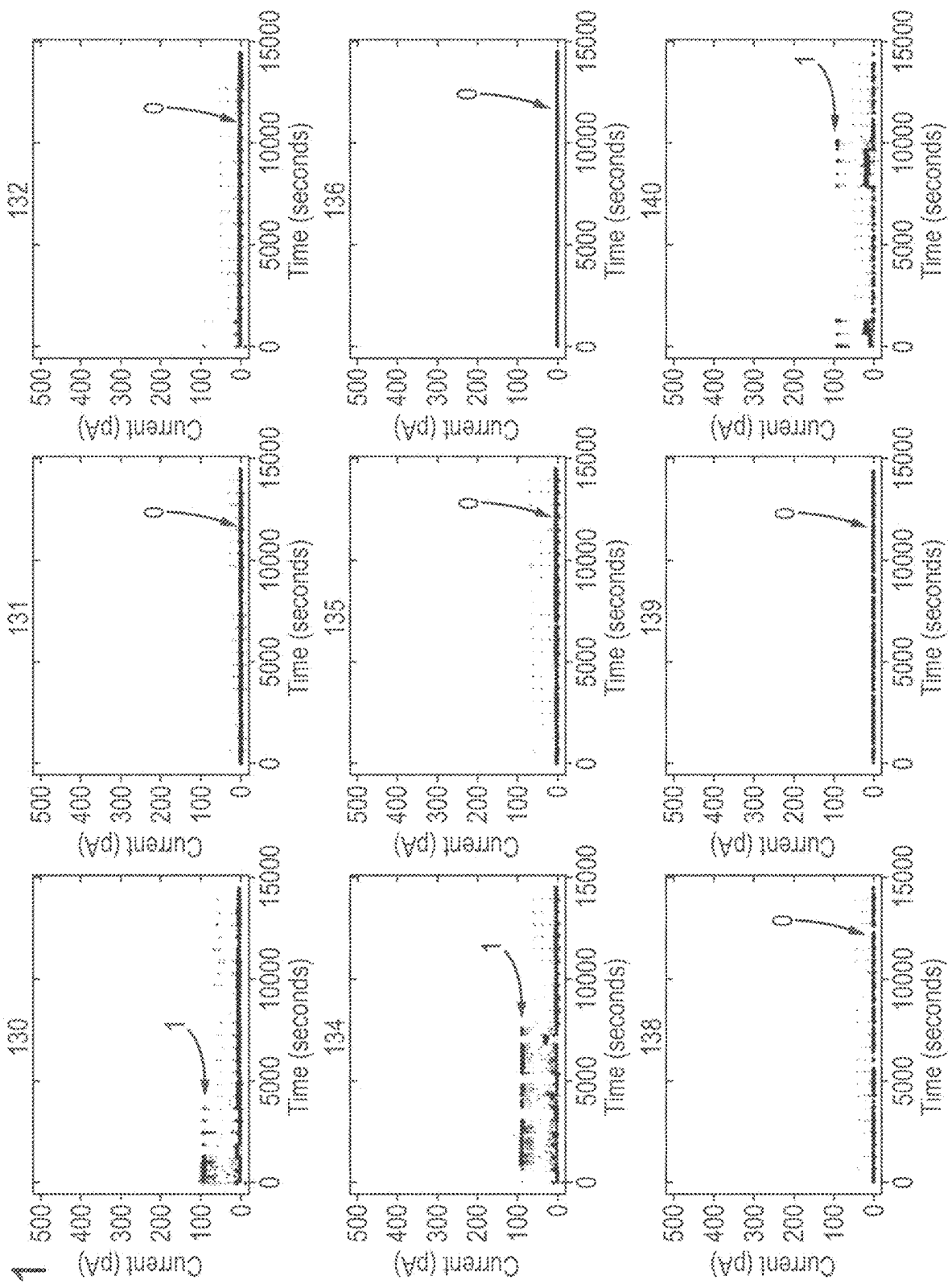


Fig. 12

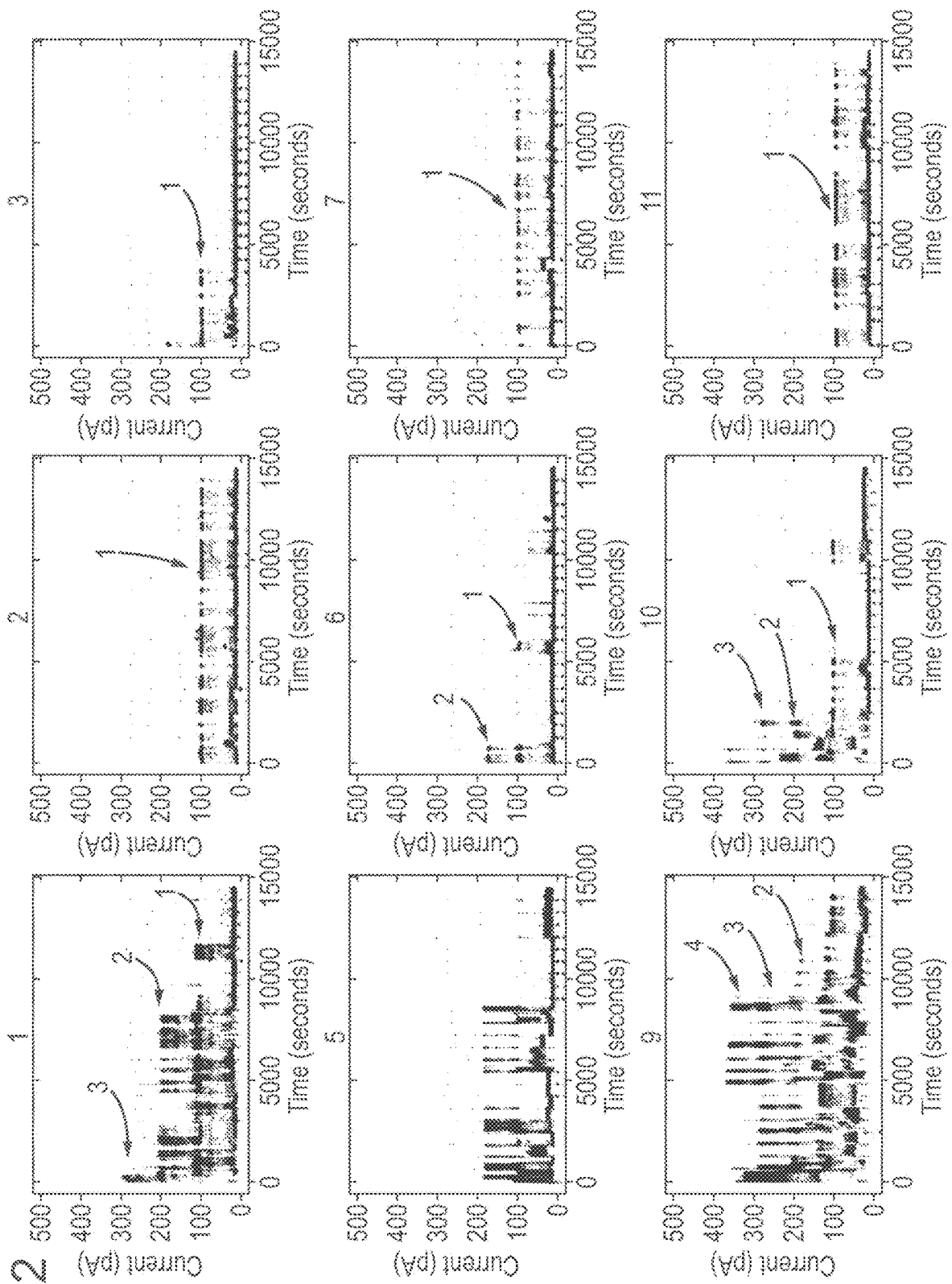
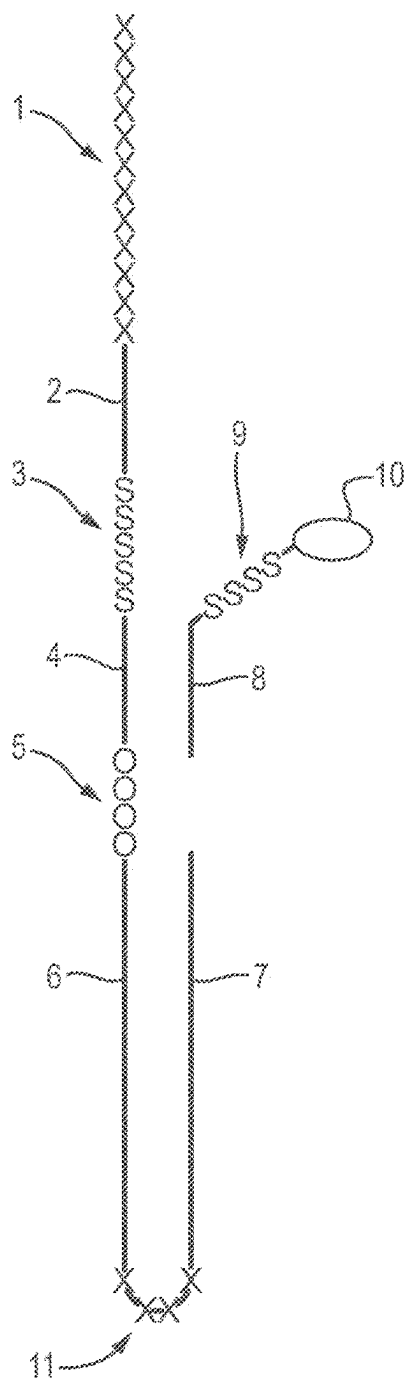


Fig. 13



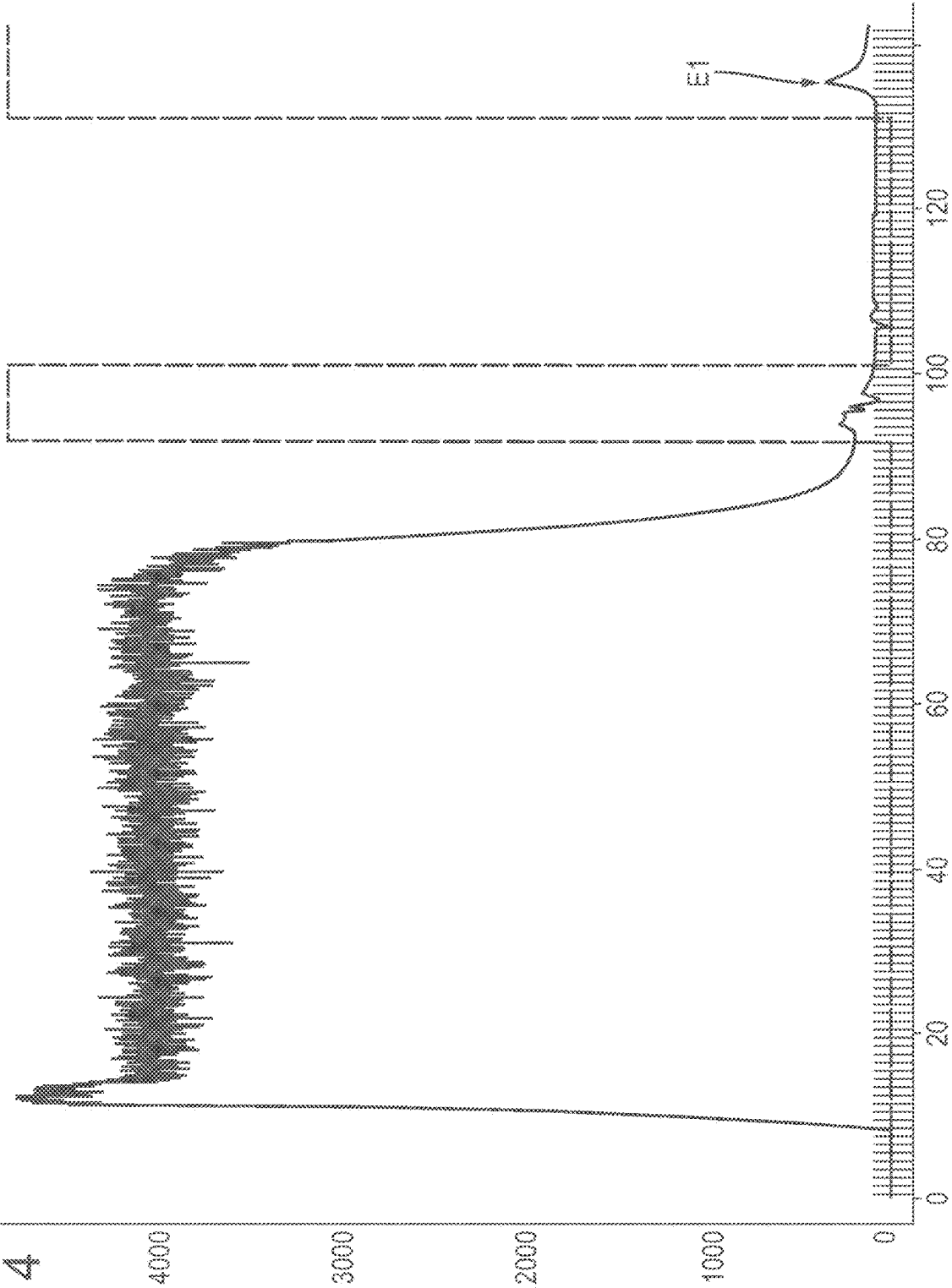


Fig. 14

Fig. 15

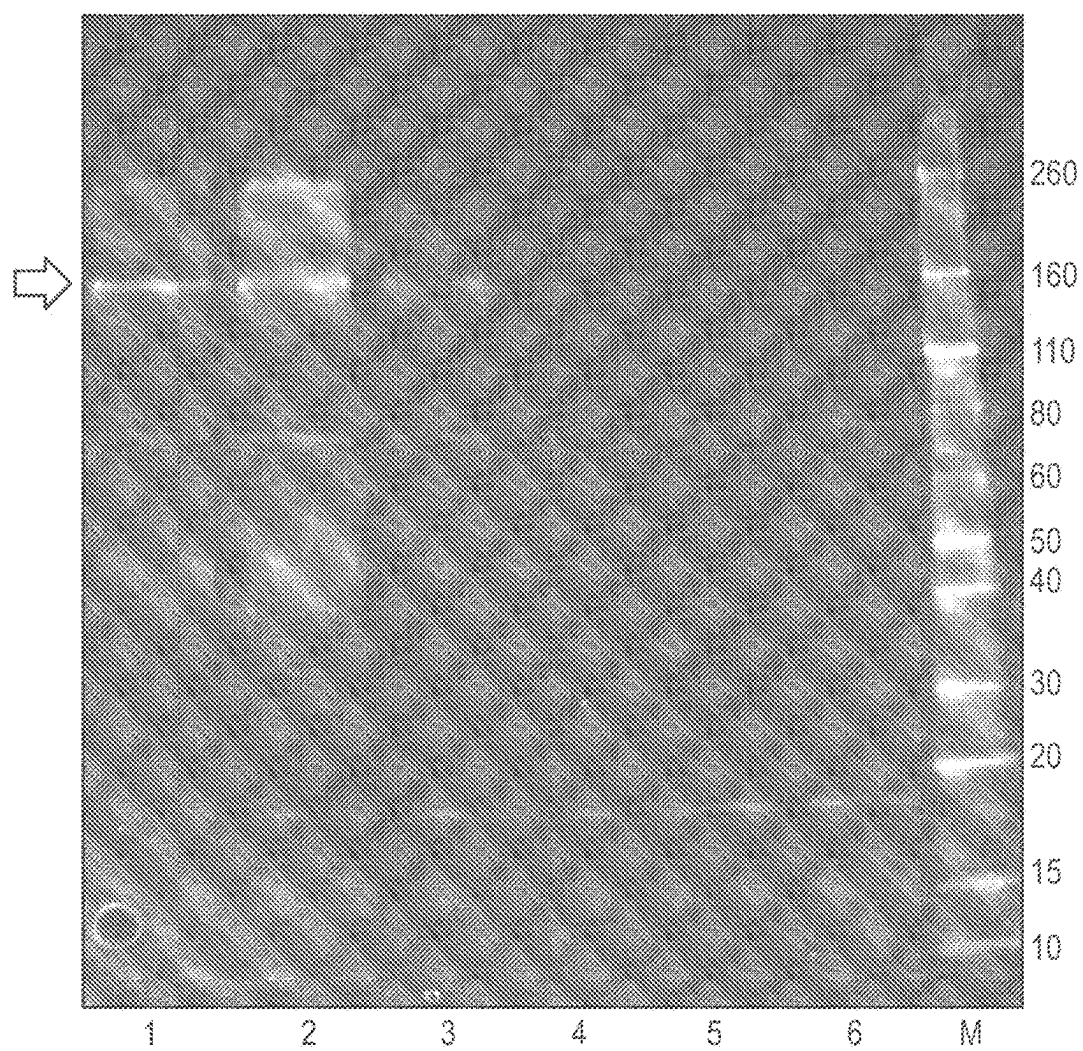


Fig. 16

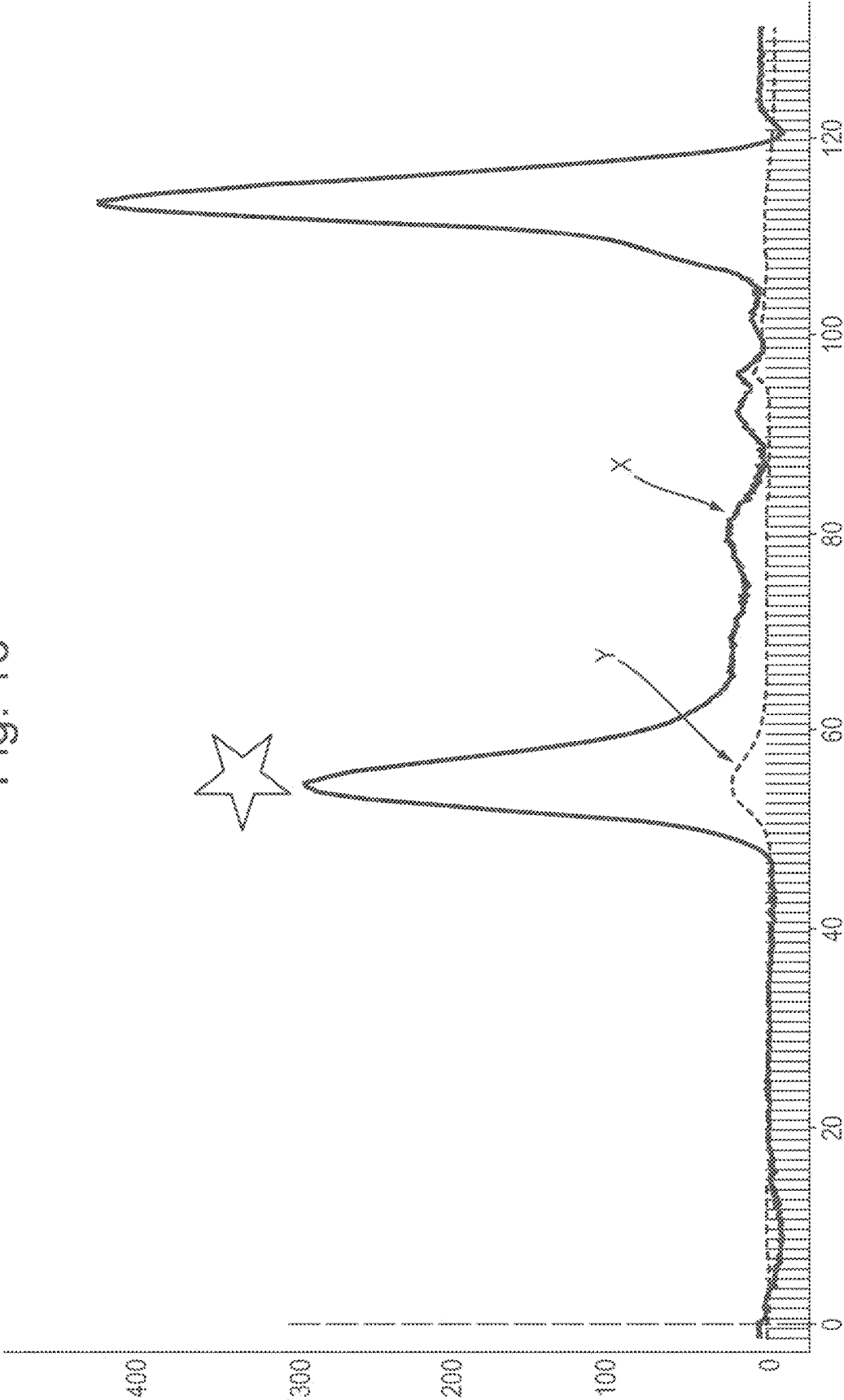


Fig. 17

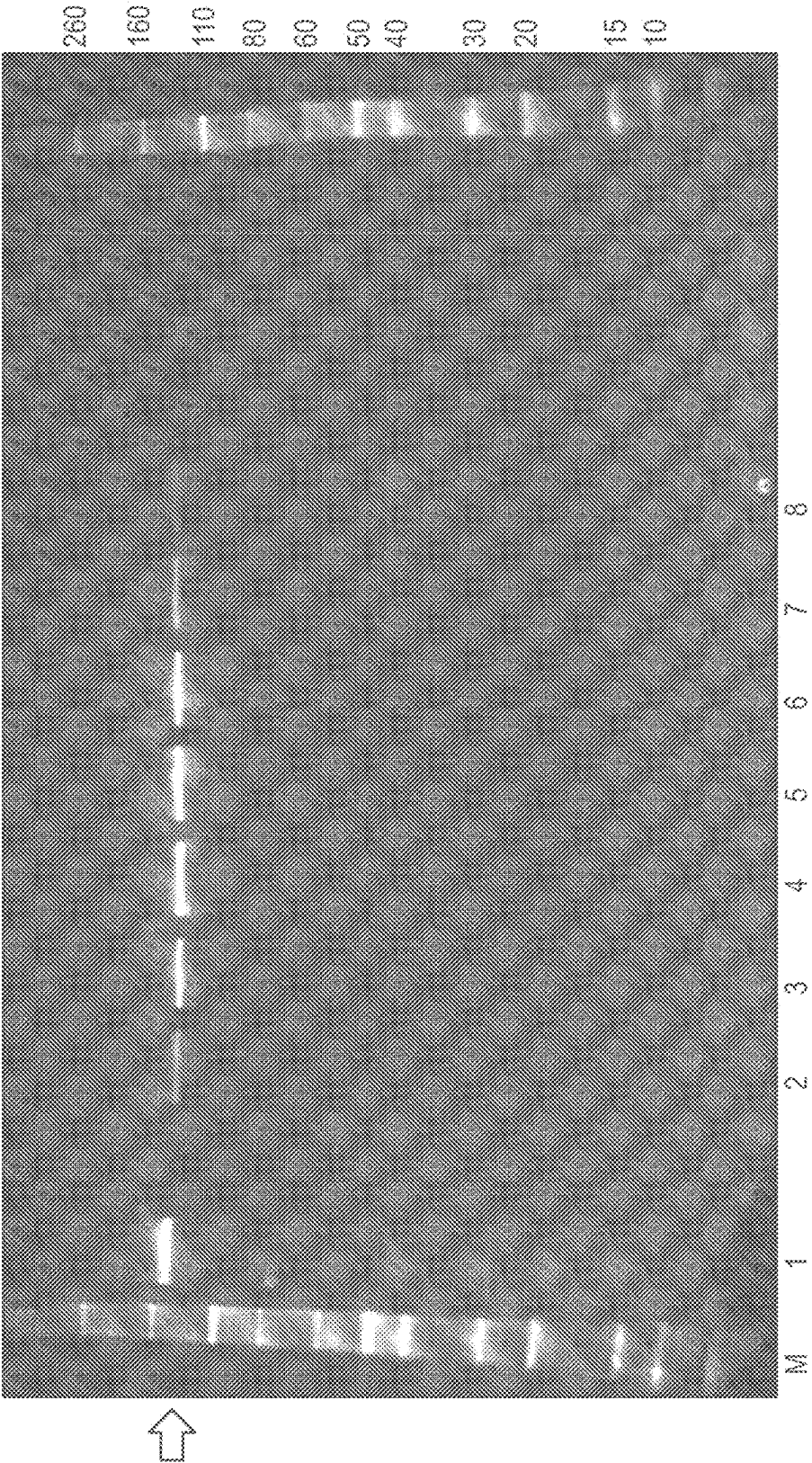


Fig. 18

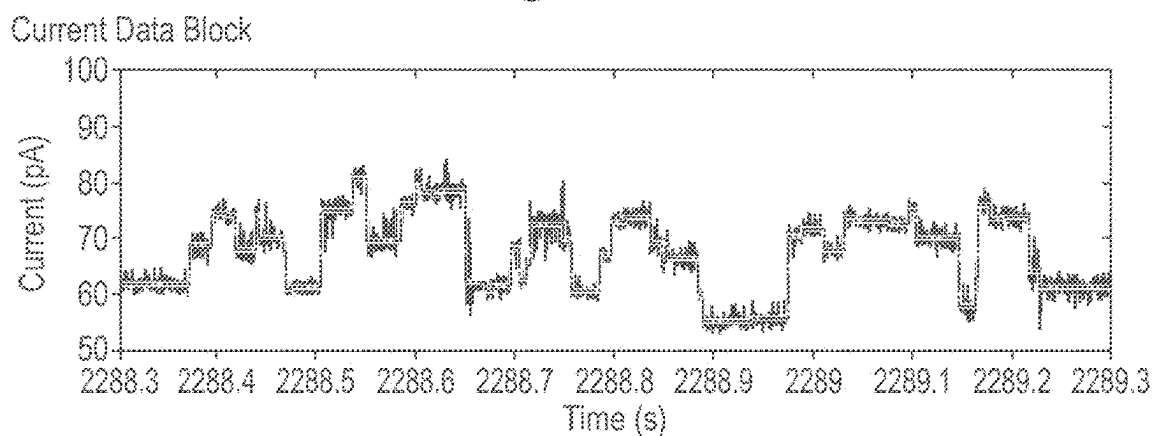


Fig. 19

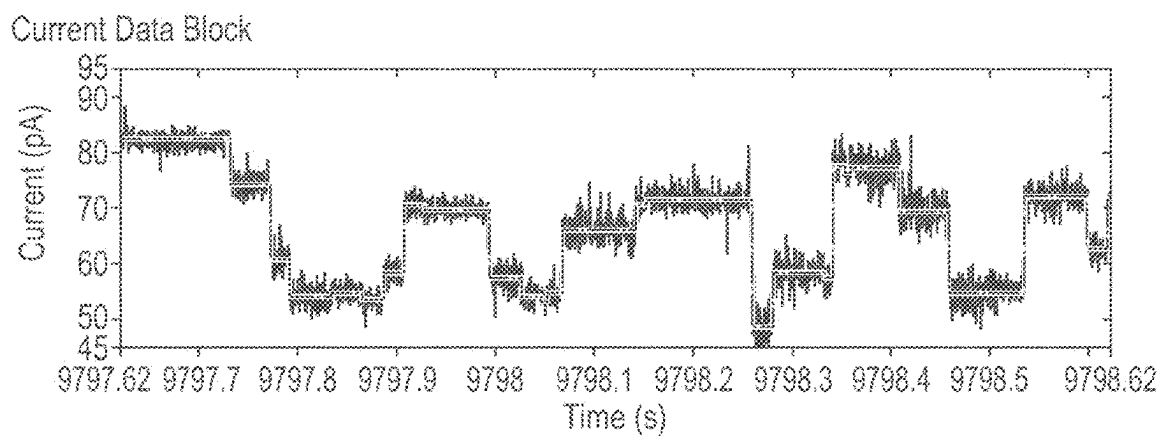


Fig. 20

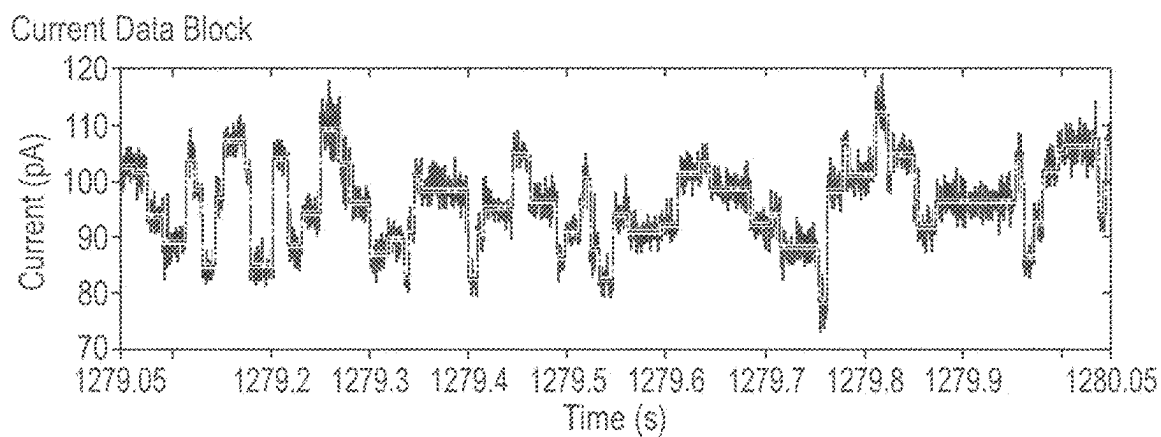


Fig. 21

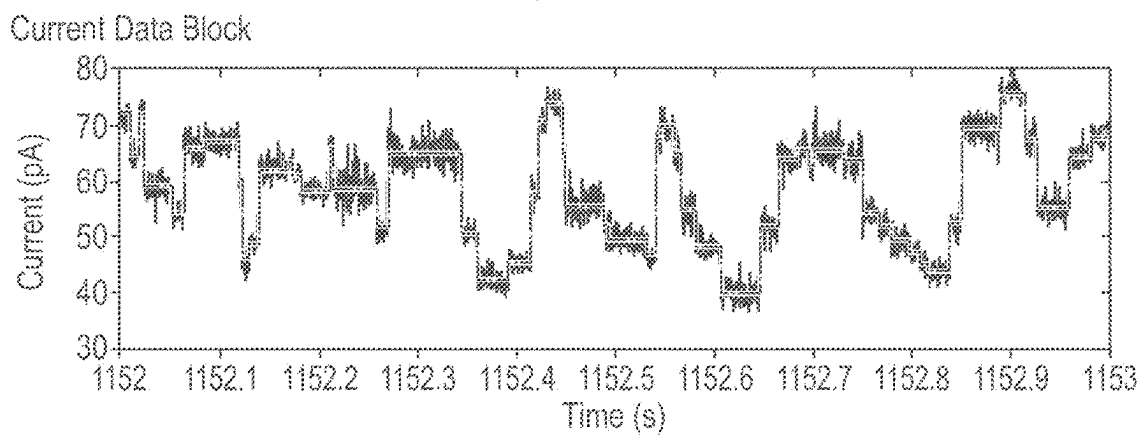


Fig. 22

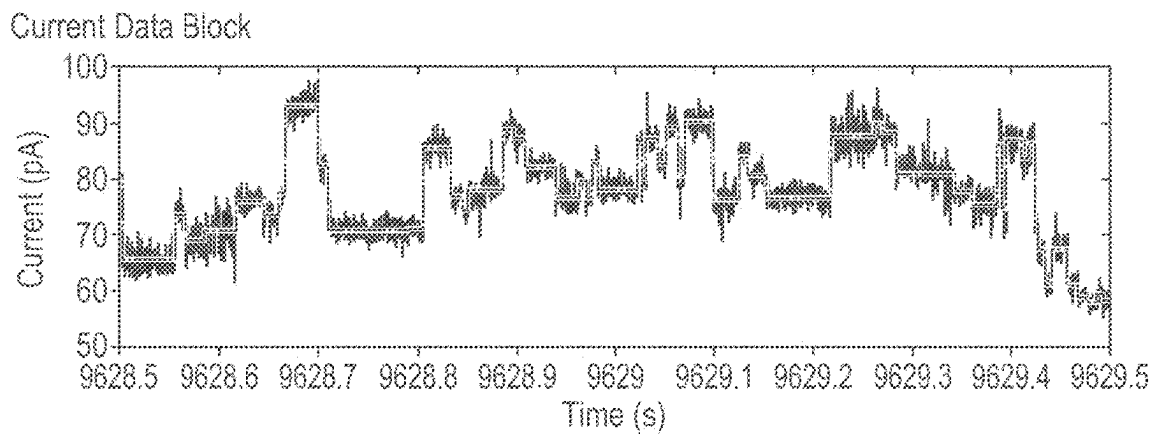


Fig. 23

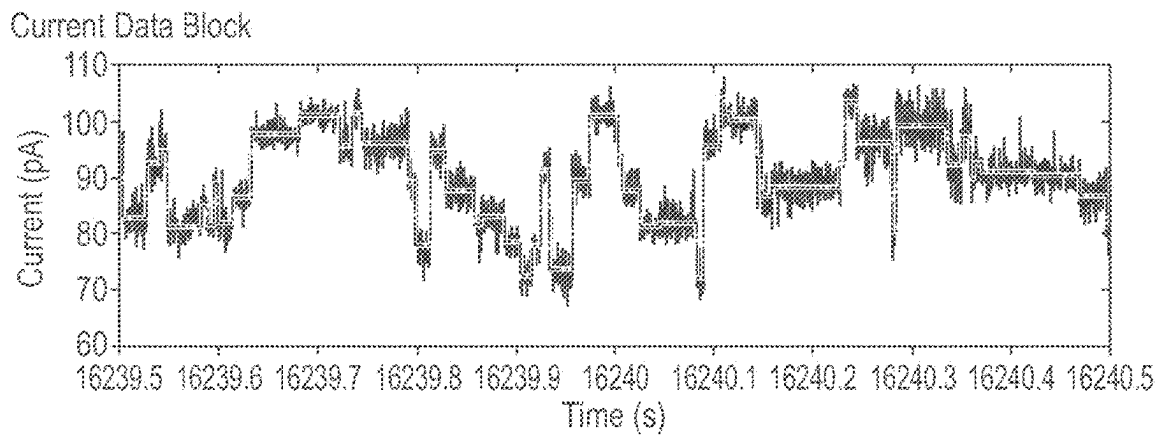


Fig. 24

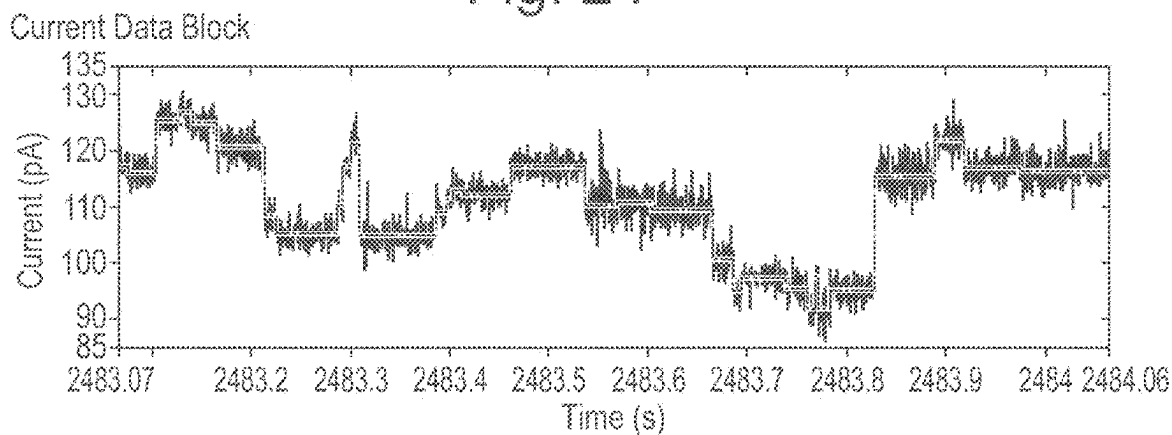


Fig. 25

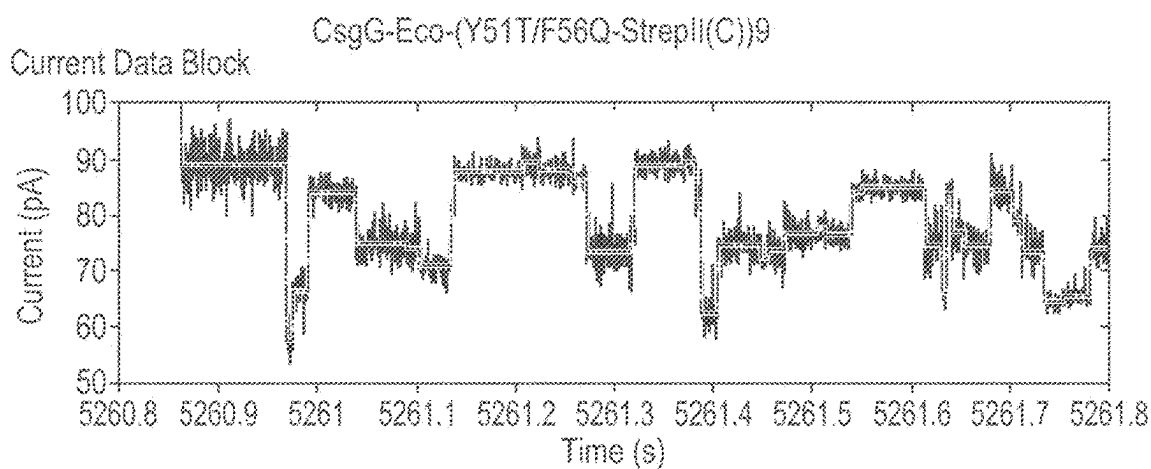


Fig. 26

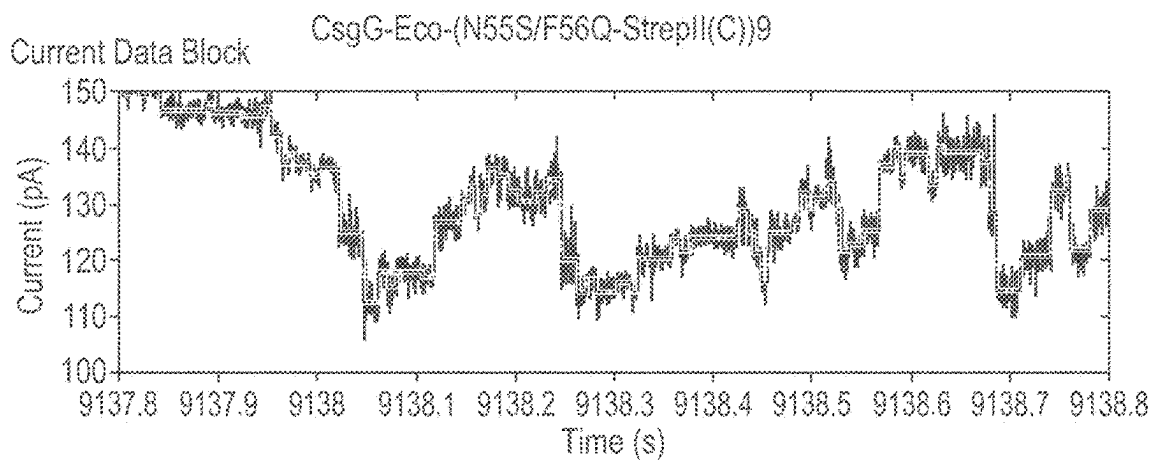


Fig. 27

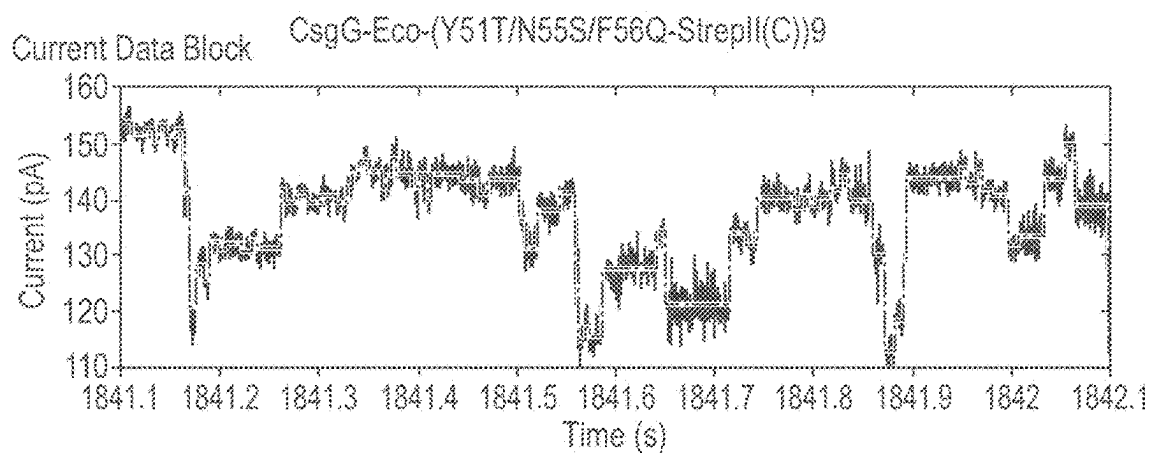


Fig. 28

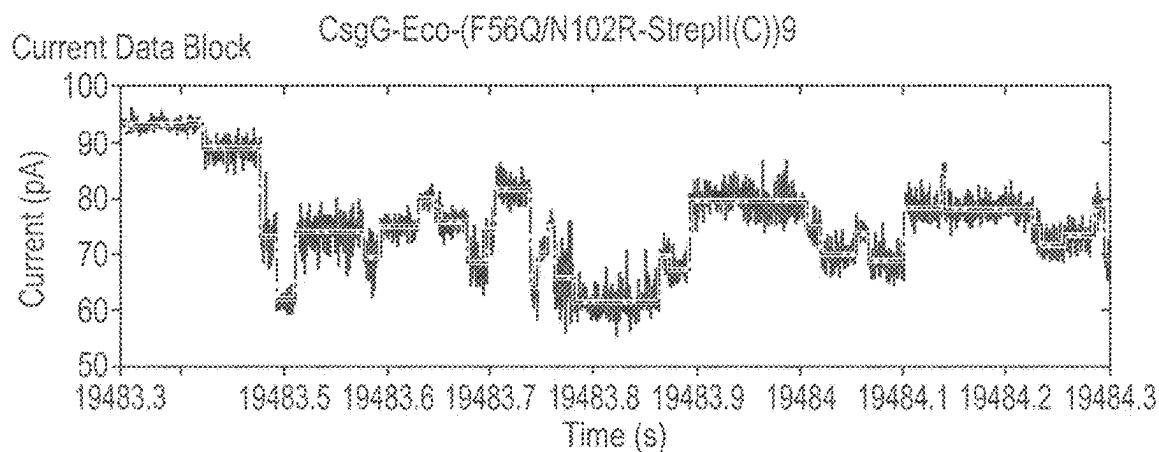


Fig. 29

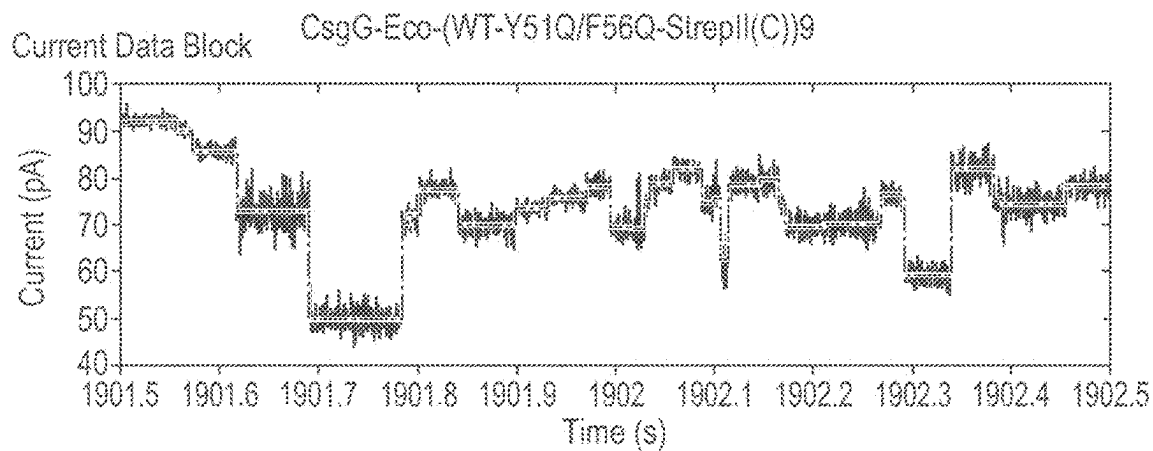


Fig. 30

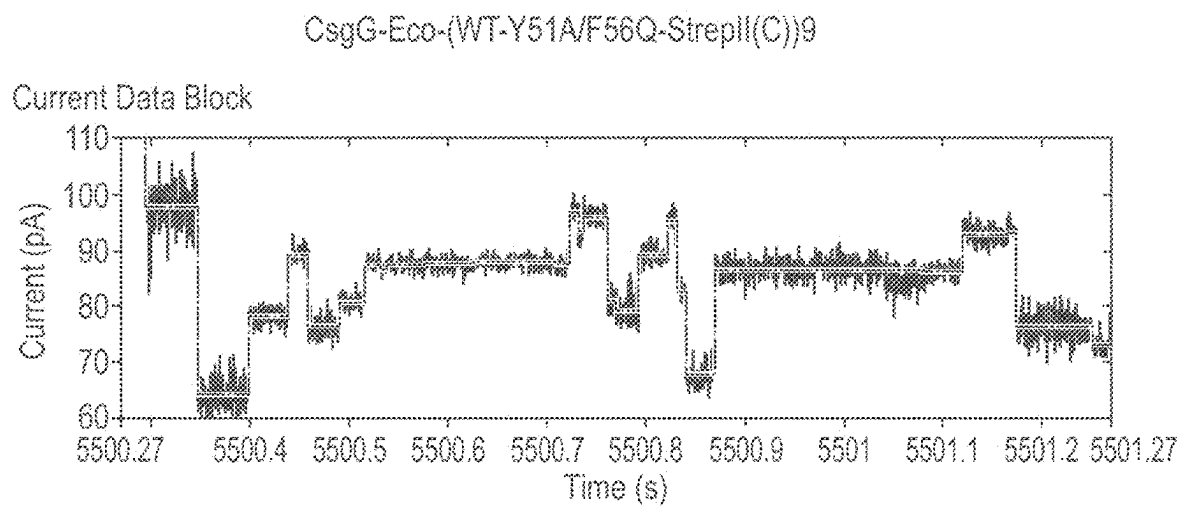


Fig. 31

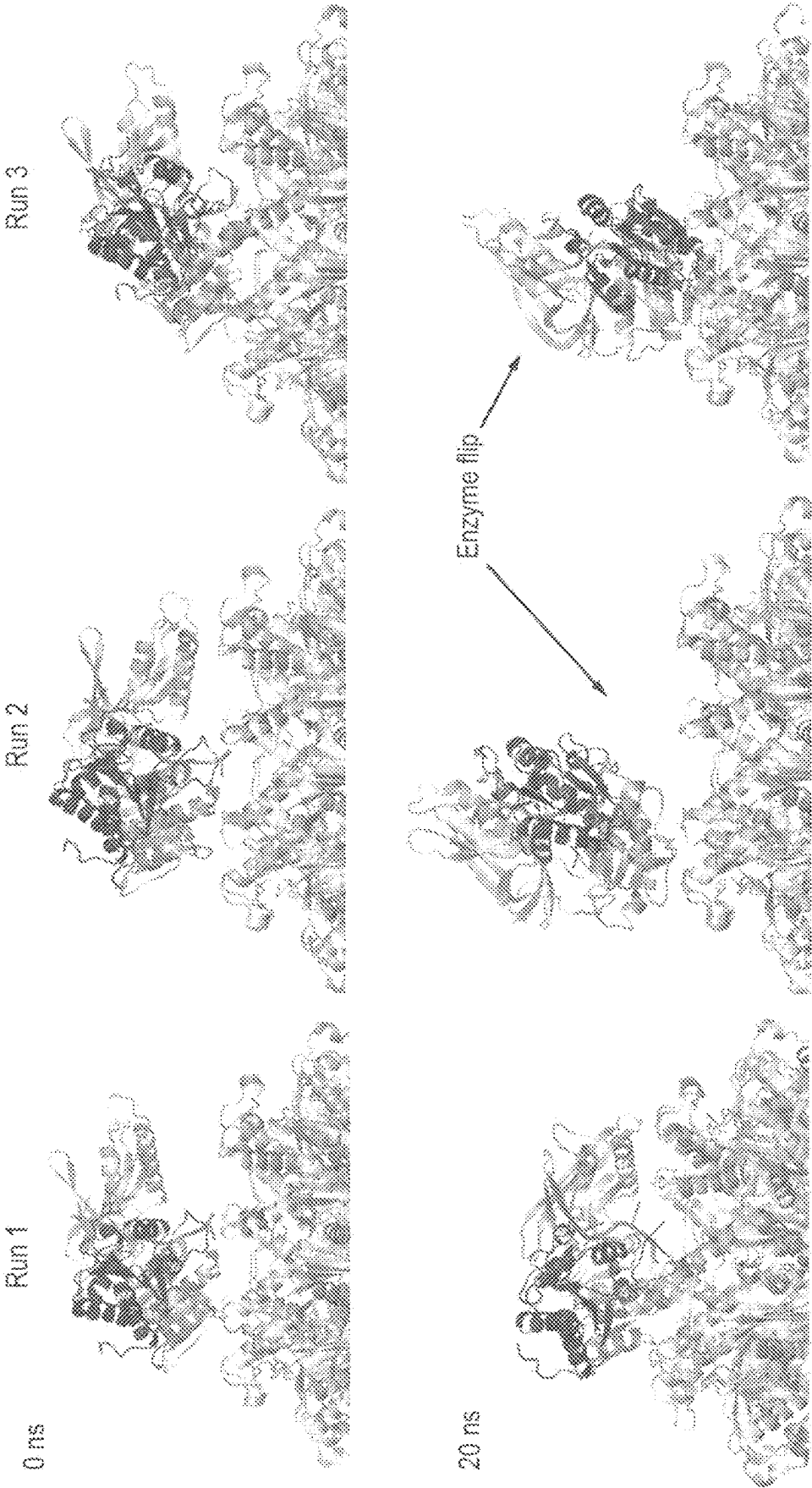


Fig. 32

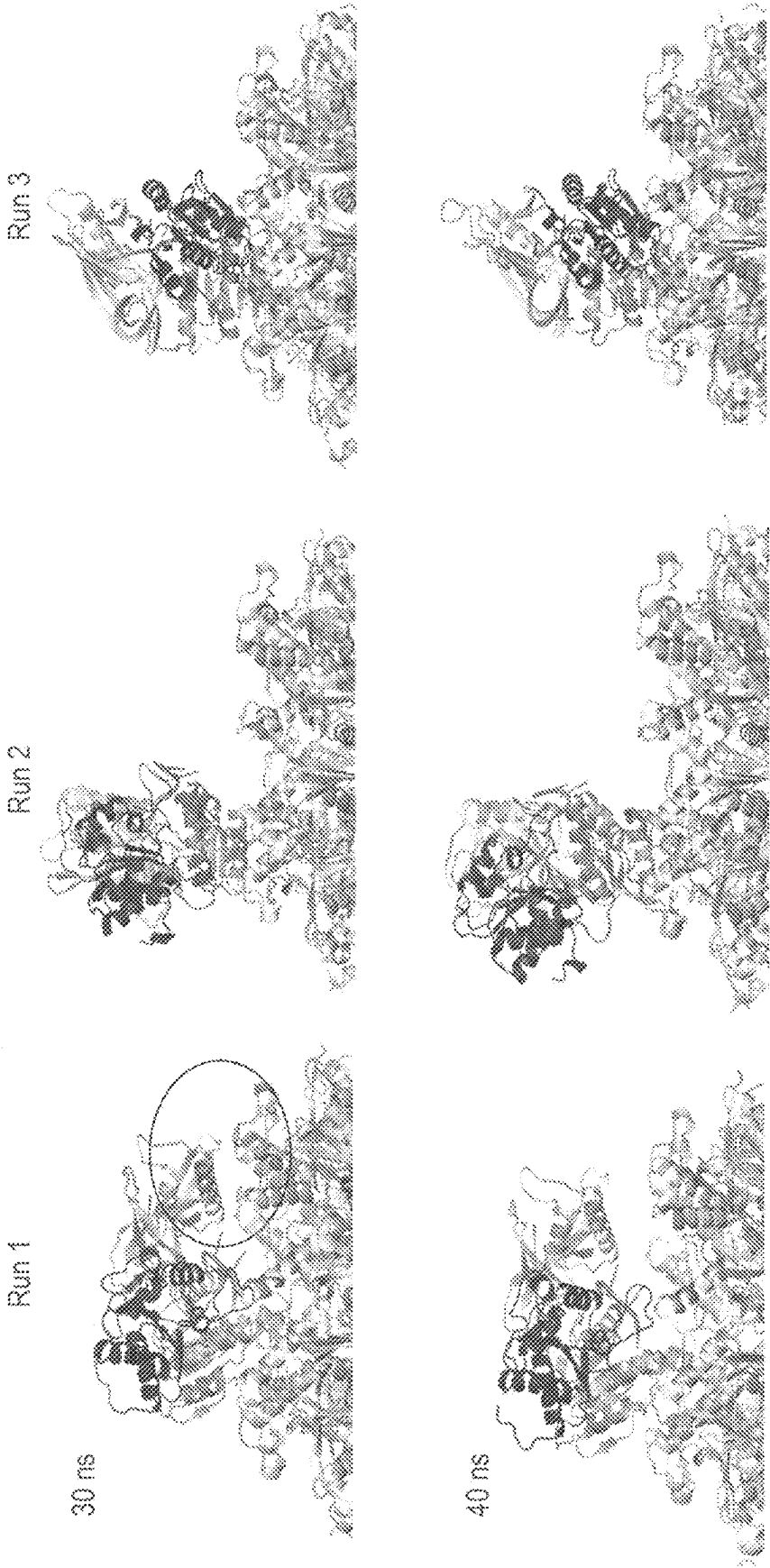


Fig. 33

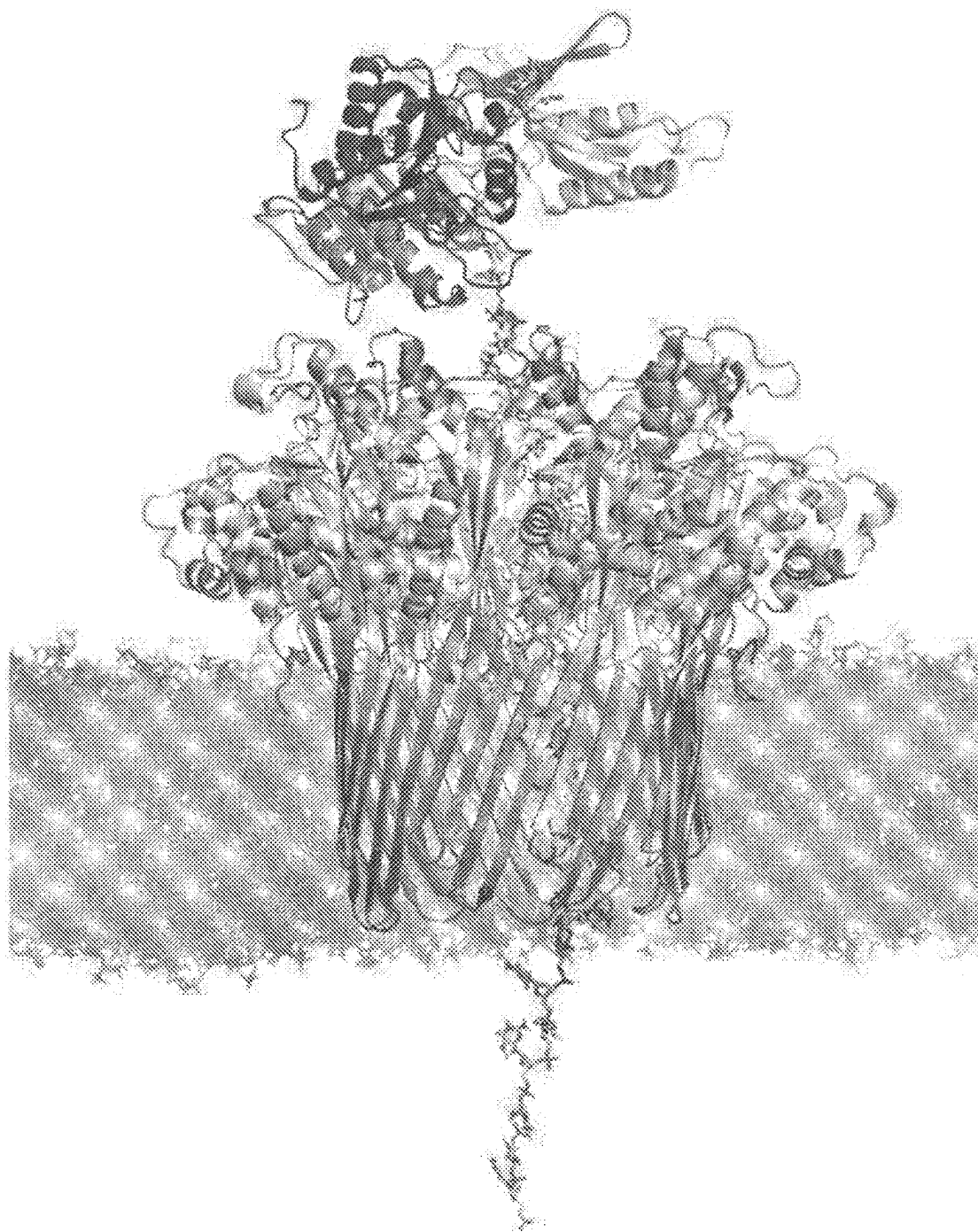


Fig. 34

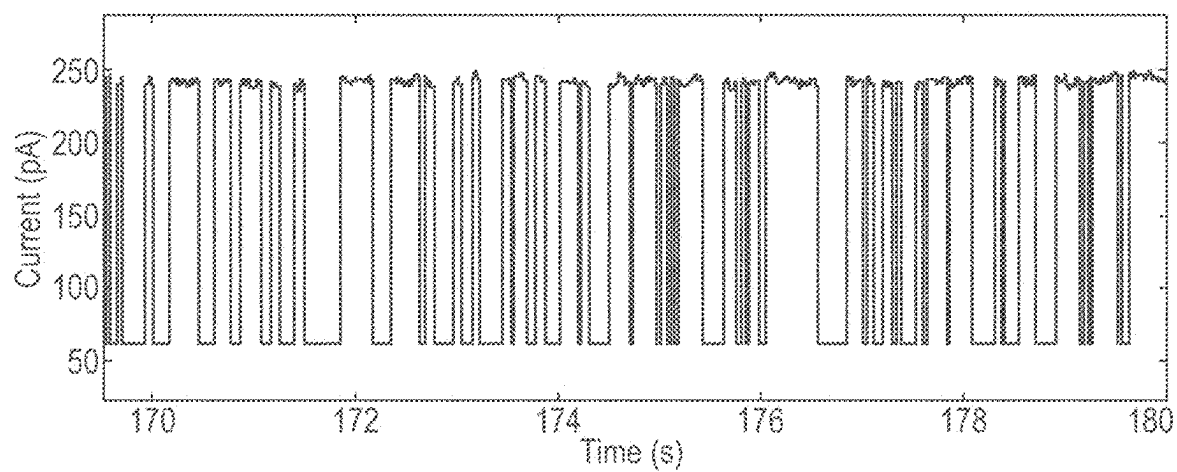
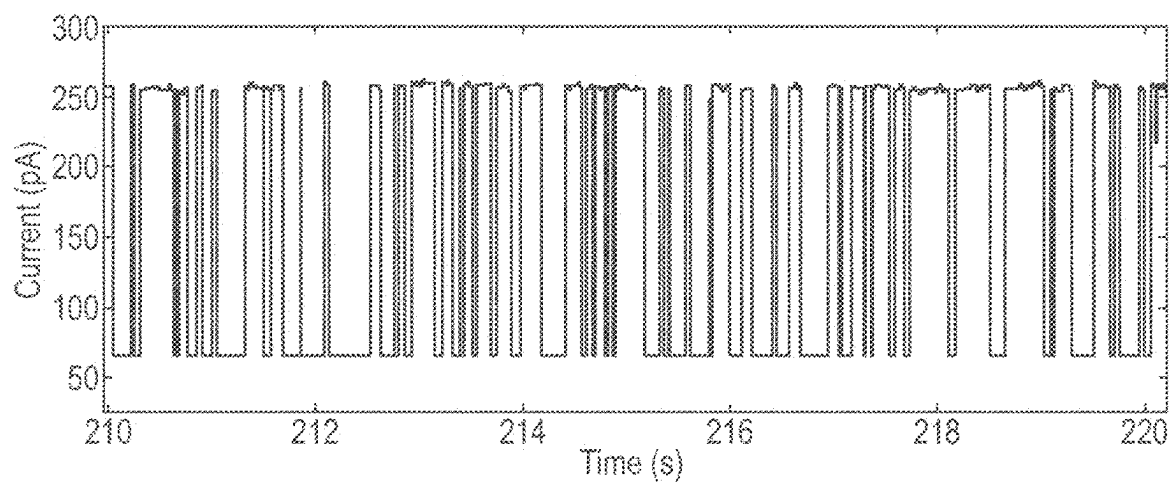


Fig. 35

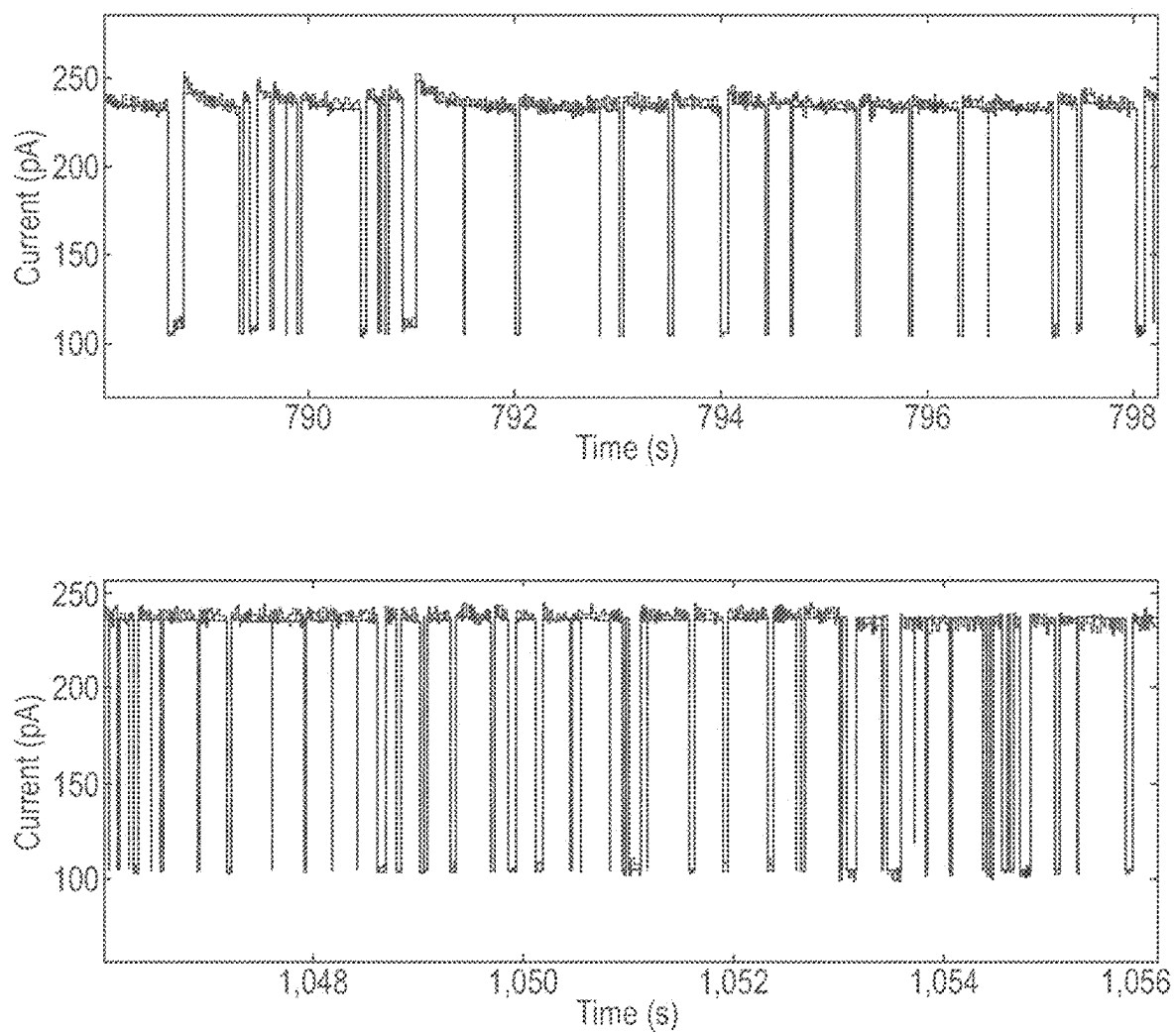


Fig. 36

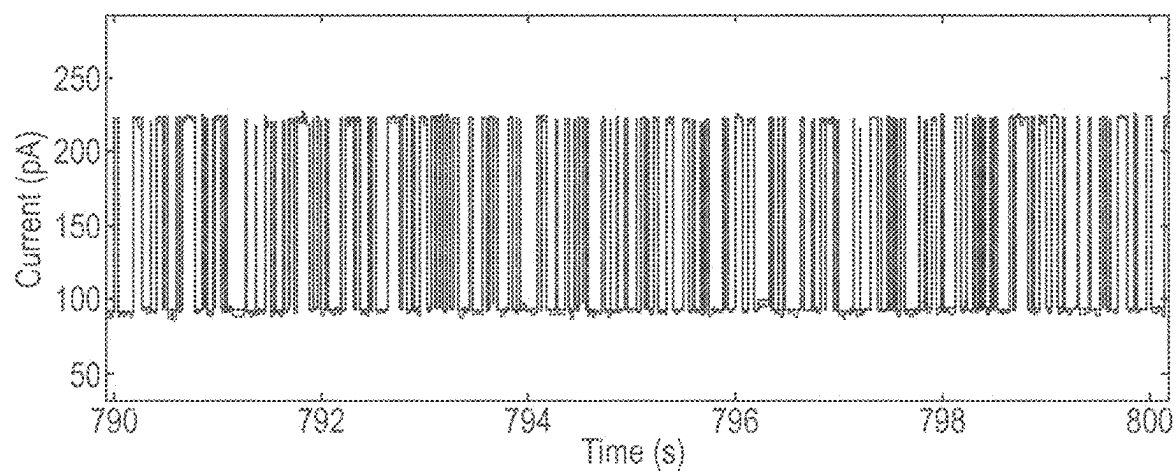
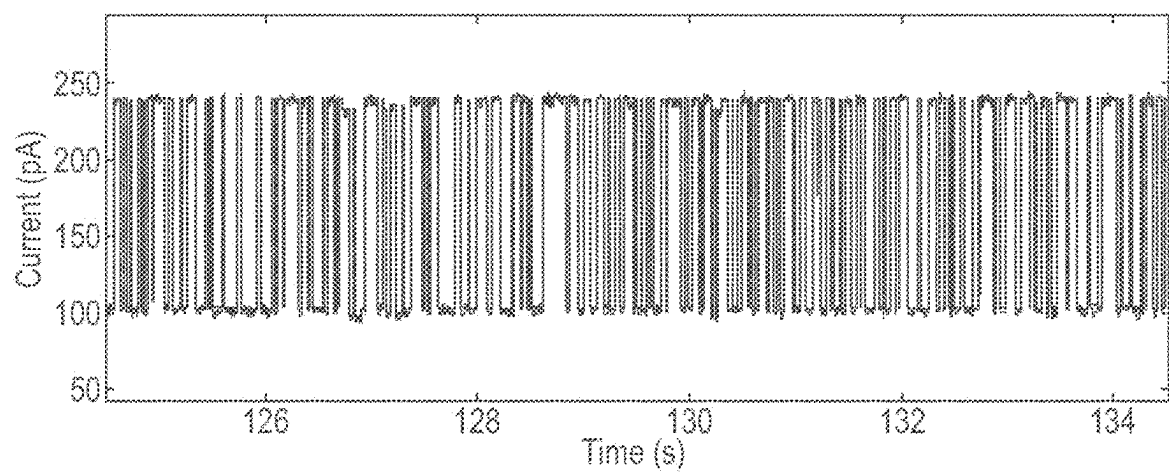


Fig. 37

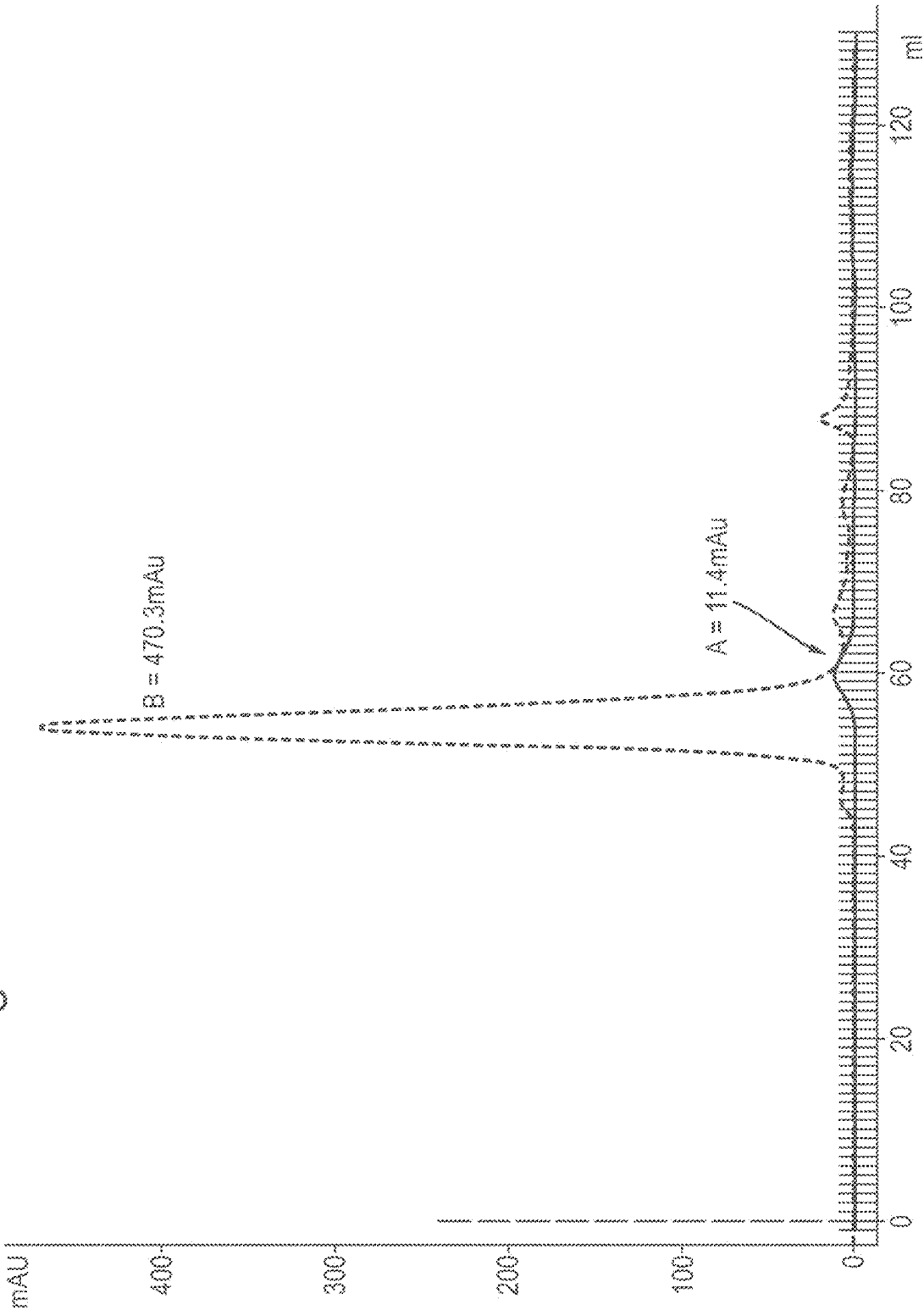


Fig. 38

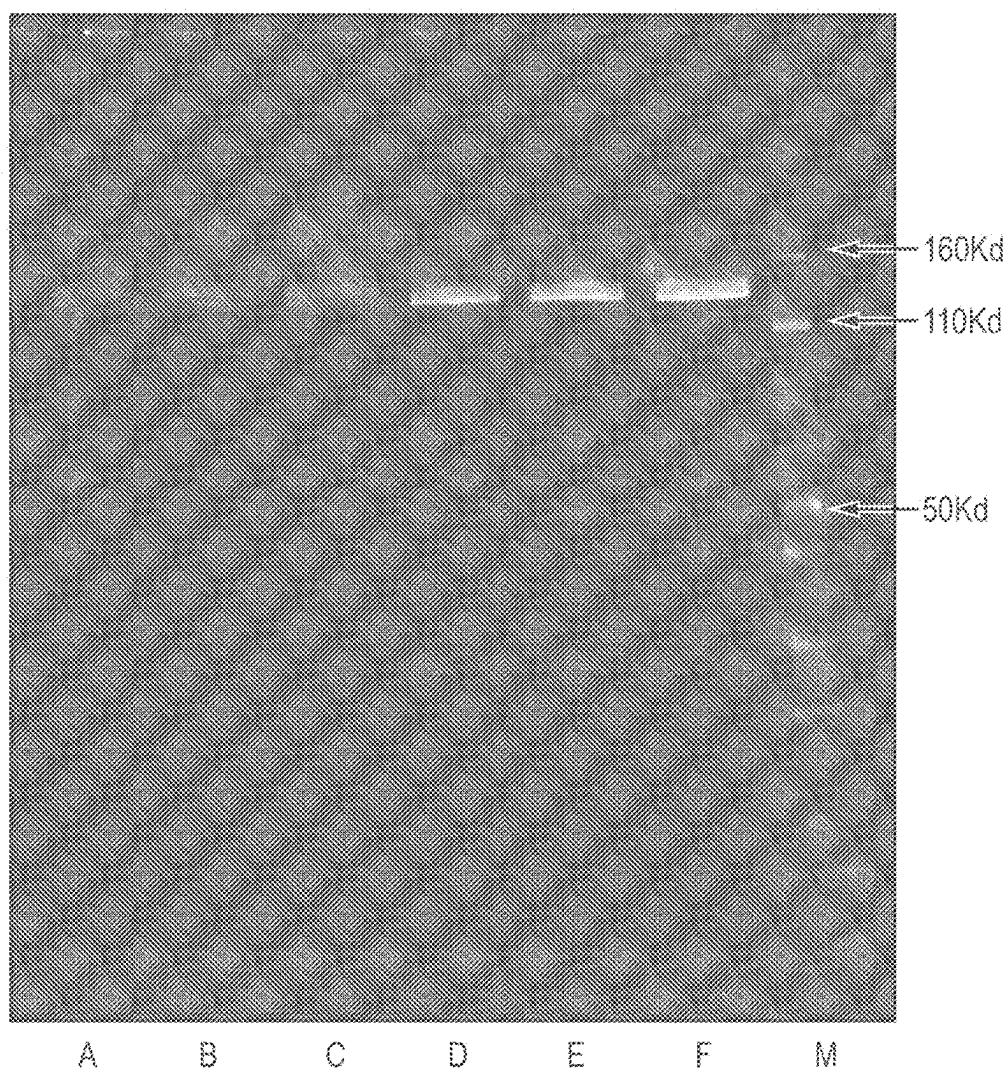


Fig. 39

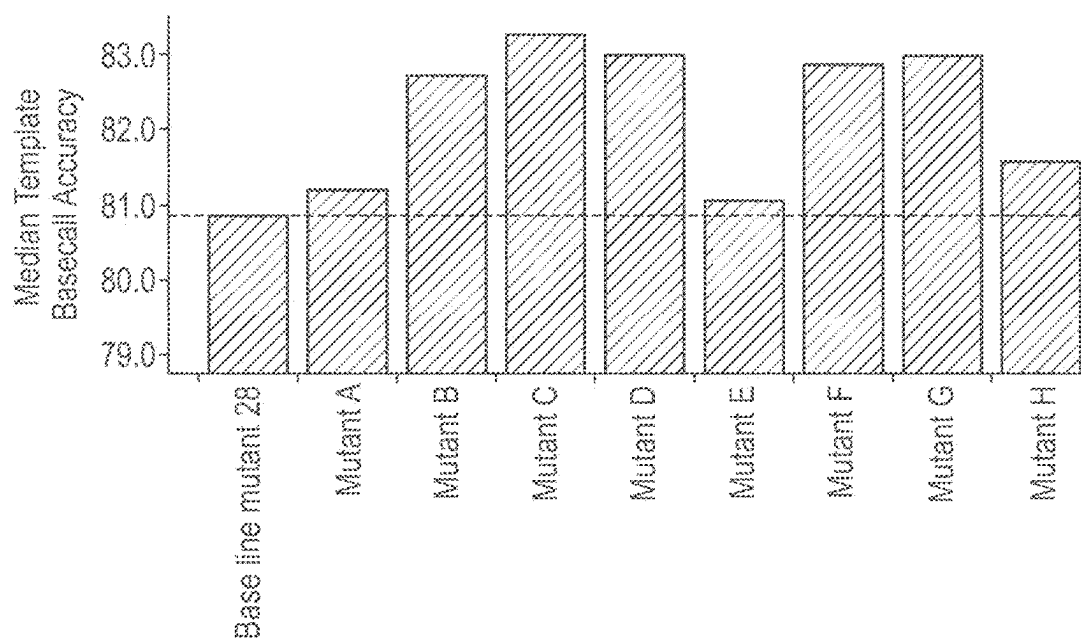


Fig. 40A

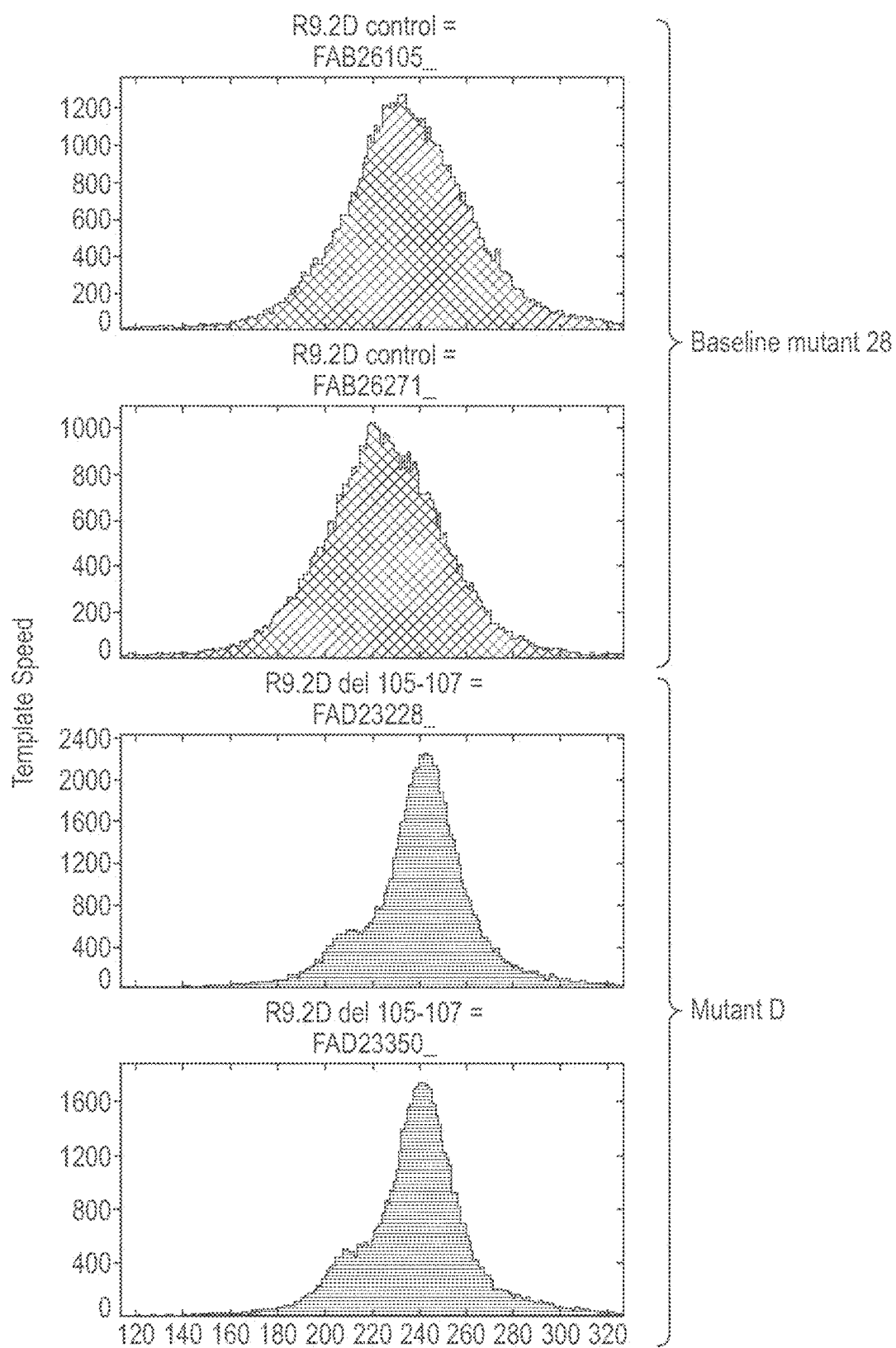


Fig. 40B

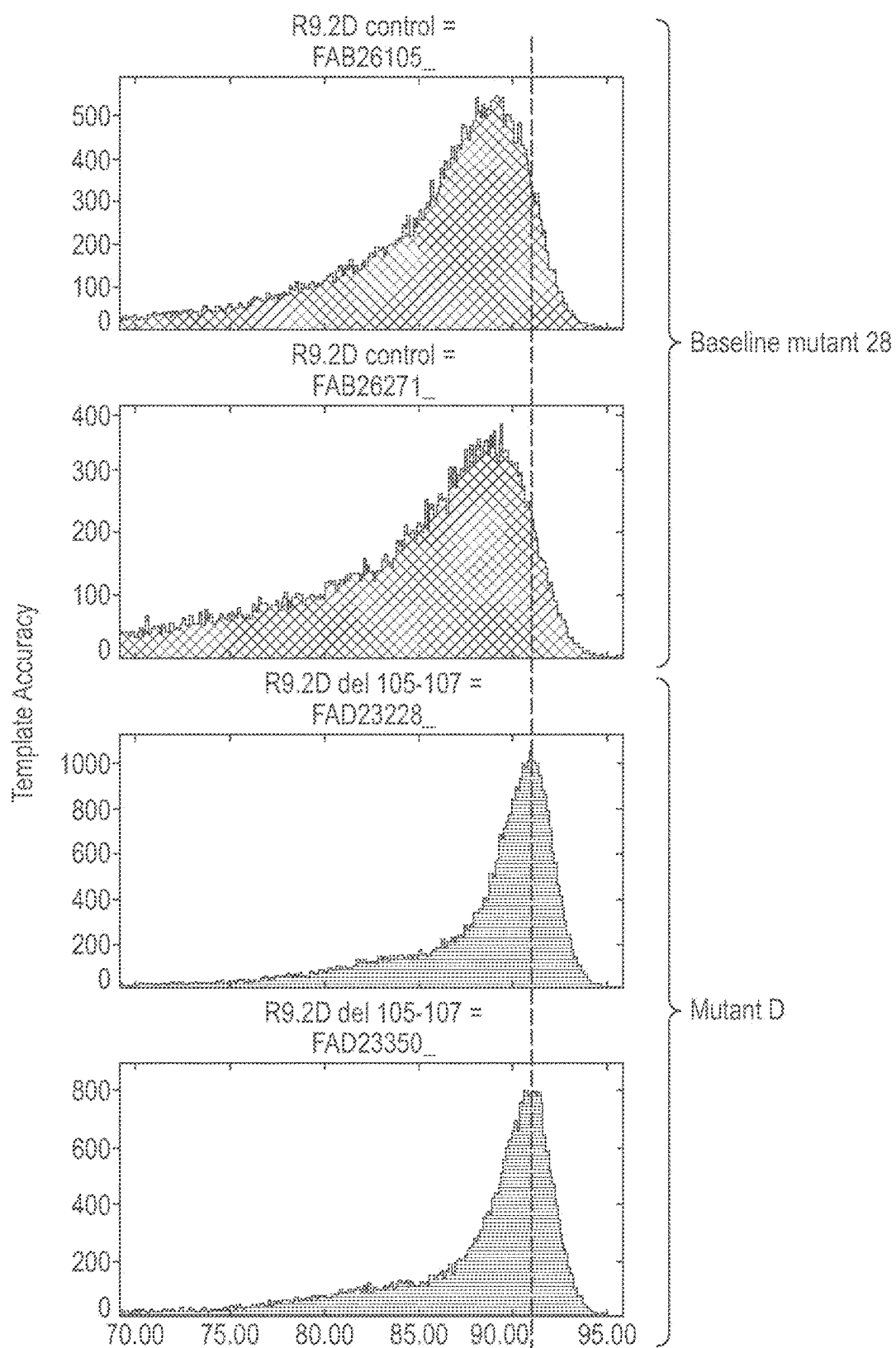


Fig. 41

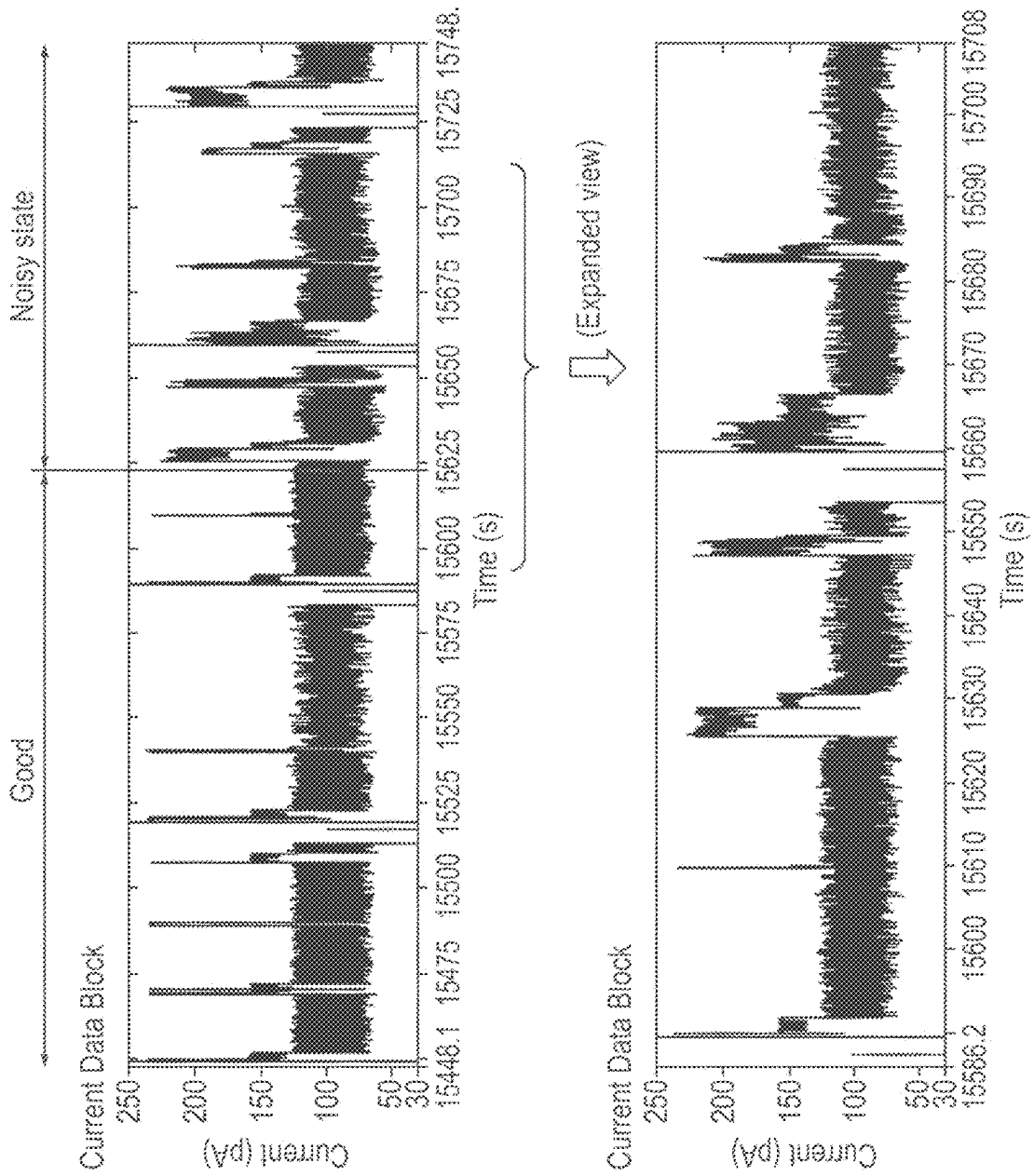


Fig. 42

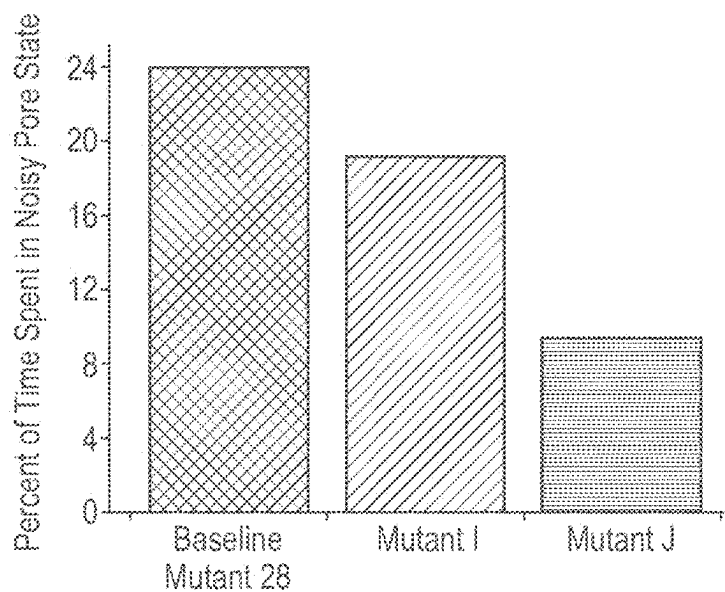


Fig. 43

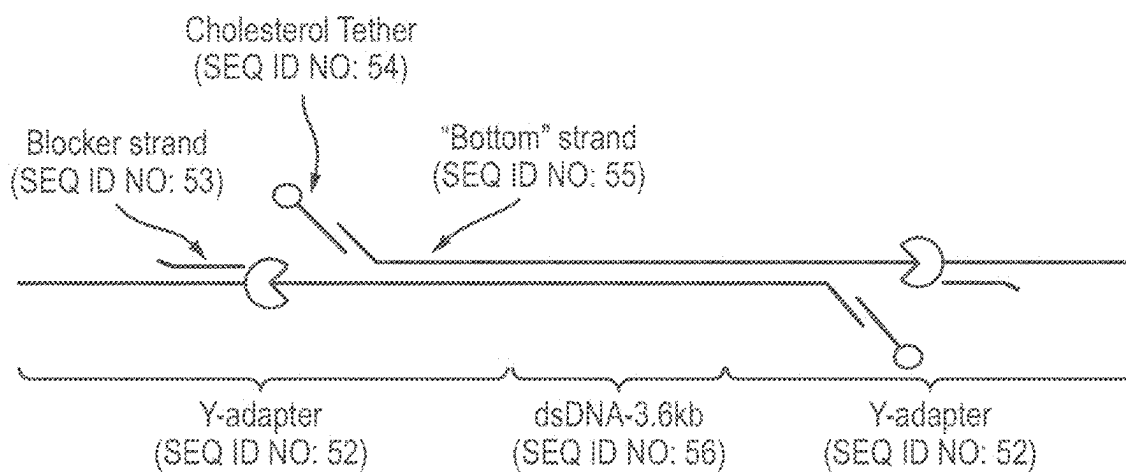


Fig. 44

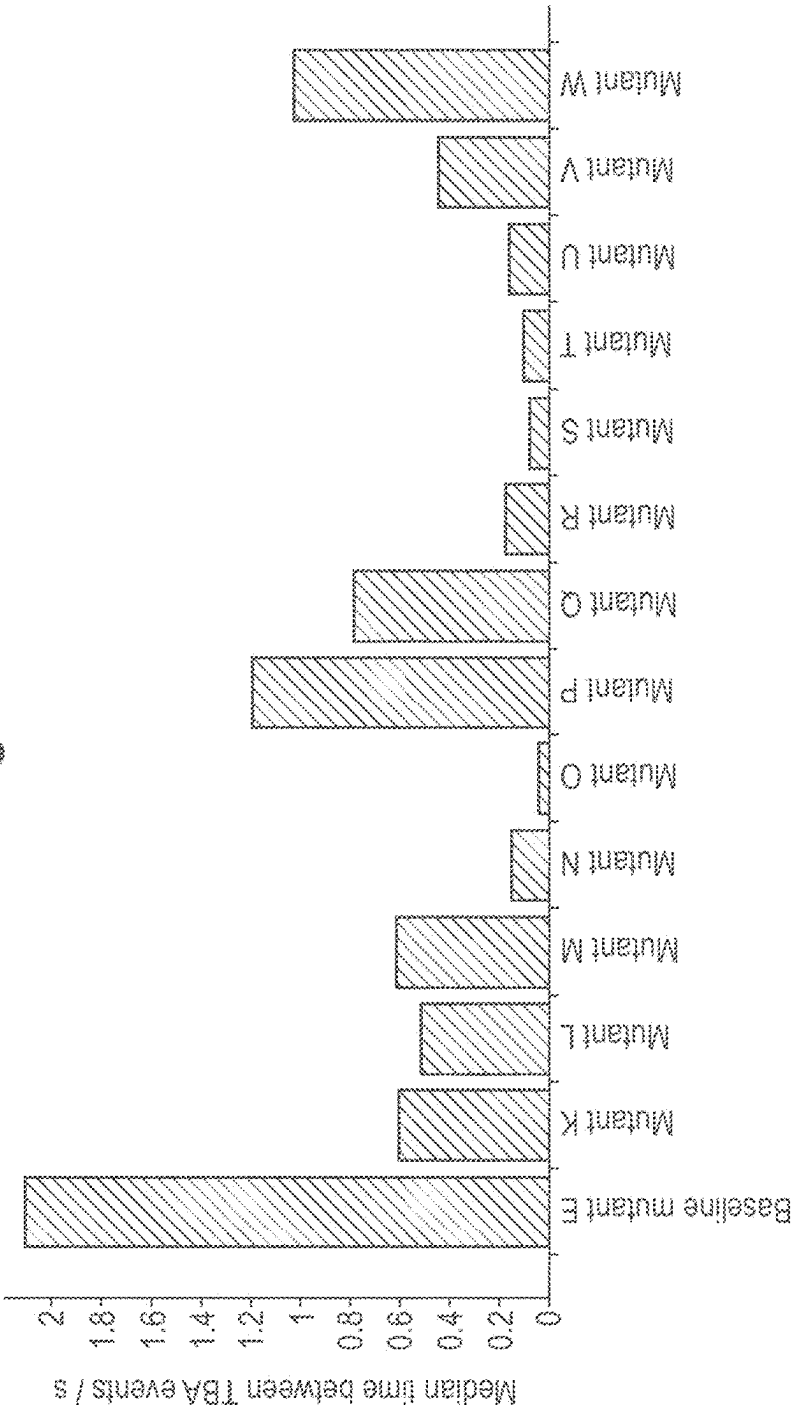


Fig. 45A

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SEQ_ID_NO_3	1 CLT--APPKEAAKPTLMPRAQSYKDLTHLP IPTGKIFVS
SEQ_ID_NO_6	1 CLT--TPPKEAAKPTLMPRAQSYKDLTHLPVPTGKIFVS
SEQ_ID_NO_7	1 CLT--APPKEAAKPTLMPRAQSYRDLTHLPAPTGKIFVS
SEQ_ID_NO_27	1 CLT--APPKEAAKPTLMPRAQSYRDLTNLPDPKGRKFVS
SEQ_ID_NO_28	1 CLT--AAPKEAARPTLLPRAPSYTDLTHLPSPQGRIFVS
SEQ_ID_NO_29	1 CIT--SPFKQAAKPTLLPRSQSYQDLTHLPEPQGRKFVS
SEQ_ID_NO_30	1 CLT--APPKQAAKPTLMPRAQSYQDLTHLPEPAGKLFVS
SEQ_ID_NO_32	1 -IT--EVPKEAAKPTLMPRASTYKDLVALPKPNGKIVS
SEQ_ID_NO_36	1 CATHIGSPVADEKATLMPRSVSYKELISLPKPKGKIVAA
SEQ_ID_NO_3	1-----MPRAQSYKDLTHLPMPTGKIFVS
SEQ_ID_NO_35	1-----PETSKEPTLMARGTAYQDLVSLPLPKGKVVVS
SEQ_ID_NO_31	1 CIT--TPPQEAAPKPTLLPRQATYKDLVSLPOPRGKIYVA
SEQ_ID_NO_40	1-----LTRRMSTYQDLIDMPAPRGKIVTA
SEQ_ID_NO_33	1-----PETSESPTLMORGANYIDLISLPKPOGKIFVS
SEQ_ID_NO_34	1-----PDASESPTLMORGATYLDLISLPKPOGKIYVS
SEQ_ID_NO_37	1-----ASSSLMPKGESYYDLINLPAPQGVMLAA
SEQ_ID_NO_39	1-----MPKSDTYDDLIGLPHPOGGMMLAA
SEQ_ID_NO_38	1-----QDSETPTLTPRASTYYDLINMPRPKGRLMAV
SEQ_ID_NO_41	1-----PSDPERSTMGELTPSTAELRNLPNEKIVIG
SEQ_ID_NO_4	1 CLT--APPKQAAKPTLMPRAQSYKDLTHLPAPTGKIFVS

Conservation



SEQ_ID_NO_2	130 YESNVKSGGVGARYFGIGADTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_3	130 YESNVKSGGVGARYFGIGADTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_6	130 YESNVKSGGAGARYFGIGADTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_7	130 YESNVKSGGVGARYFGIGADTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_27	130 YESNVKSGGVGARYFGIGDDTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_28	130 YESNVKSGGVGARYFGIGASTOYQLDDIAVNLRVVDVNT
SEQ_ID_NO_29	130 YESNVKSGGAGARYFGIGASTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_30	130 YESNVKSGGAGARFFFGIGASTOYQLDDIAVNLRVVDVNT
SEQ_ID_NO_32	129 YDSDIKTGGAGARYFGIGADGKYRVDDVAVNLRVVDVRT
SEQ_ID_NO_36	132 YDSNVRTGGAGAKYFGIGASGEYRVDDVTVNLRVVDIRT
SEQ_ID_NO_3	116 YESNVKSGGVGARYFGIGADTOYQLDDIAVNLRVVNVST
SEQ_ID_NO_35	125 YDTNIQTGGAGARYLGVGASGOYRTDDVTVNIRVVDVRT
SEQ_ID_NO_31	130 YASNVTGGFGARYFGIGGSTOYQLDDVAVNLRIVNVHT
SEQ_ID_NO_40	117 YDSNIHTGGAGARYFGIGASEKYRVDEVTVNLRVIDIRT
SEQ_ID_NO_33	125 YDSNIKTGGAGARYLGIGGSGOYRADDDVTVNIRVVDVRS
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SEQ_ID_NO_37	115 YDTNVRTGGAGARYLGIGAAATQFRVDTVTVNLRVVDIRT
SEQ_ID_NO_39	110 YDTNIKTGGAGARYLGIGVNSKFRVDTVTVNLRVVDIRT
SEQ_ID_NO_38	124 YDTNVRSGGEGARYLGIDISREYRVDDVTVNLRVVDVRT
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Conservation

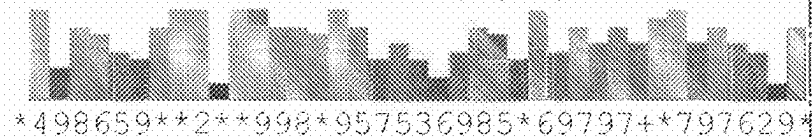


Fig. 45B

VYNIQDETGOFKPYPA-SNFSTAVPQSATAMLV TALKDSRWF I PLERQGLQNL
VYNIQDETGOFKPYPA-SNFSTAVPQSATAMLV TALKDSRWFVPLERQGLQNL
VYNIQDETGOFKPYPA-SNFSTAVPQSATAMLV TALKDSRWF I PLERQGLQNL
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VYKFRDQTGOYKPPSENGNNWSTAVPOGTTTILIKALEDSRWF I PIERENIANL
VYNIQDETGOFKPYPA-SNFSTAVPQSATAMLV TALKDSRWF I PLERQGLQNL



*376*89**957457-799**8*8*87678*536*15*2**679**87*6*+

R192

*

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GEVLSSVNTSKTILSYEQAGVFRFIDYQRLLEGEAGFTTNEFVMTCLMSAIE
GRILHSVLTSSKILSREIRSDVYRFIEFKHLLMEAGITTNDAOLCVLSAIE
GKILT SVTTSKTILSYEVSAGAFRFVDYKELLELEVELGYTTNEFVNIALMSAID
GKILT SVTTSKTILSYEL SAGAFRFVDYKELLELEVELGYTTNEFVNIALMSAID
GRLLSSVT TTKSILSKEITAGVFKFIDAQELLESELGYTSNEFVSLCVAASAIE
GRLLSSVT TTKSILSKEVSAGVFKFIDAQD LLESELGYTSNEFVSLCVAQAIE
GOVLANVMTSKTIYSVGRSAGVFKFIEFKKLLAEVGYTTNEPAOLCVLSAIE
GEILKTVYTSKTILSTSVNGSFFRYIDTERLLEAEVGLTONEFVOLAVTEAIE
GEILSSVNTSKTILSYEQAGVFRFIDYQRLLEGEIGYTSNEFVMLCLMSAIE

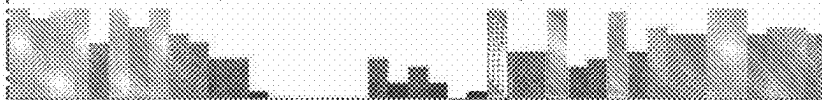


69*29*5*9*597*243675699999565***5*7*795*9*8669856*+9

Fig. 45C

R97

✱

[illegible]

*+**7*9*87551000005242-1*66*45*6988**8998

```

TGVIFLINDGIDRGLWDLONKAE-RONDILVKYRHMSPVPE 261
TGVIFLINDGIDRGLWDLONKAD-RONDILVKYRHMSPVPE 261
TGVIFLINDGIDRGLWDLONKAD-RONDILVKYRQMSVPPE 261
TGVIFLINDGIDRGLWDLONKAD-AQNPVLVKYRDMSPVPE 261
TGVIHLLINDGINRGLWELKNKGD-AKNTILAKYRSMAPVPE 261
SGVIYLVNDGIERNLWOLONPSE-INSPILORYKNNIVPAE 261
TGVYLVNDGISPNLWOLKNAED-INSPVLEKYKSIIVP-- 259
TGVYLVNDGINRNLWTLKNPQD-AKSSVLERYKSTIVP-- 259
AGVAHLIVEGIRQNLWSLONPSD-INNPILORYMKEDVP-- 258
SGVIRLIANGVYRDNLWOLADQD-IDNPILQEYLDQNPV-- 261
TGVIFLINDGIDRGLWDLONKAE-RONDILVKYRHMSPVPE 247
AAVVHVIVDGIKTGLWEPLAGED-LQHPILQEYMNRSKP-- 254
EGVIHLINDGINKKLWALSNAAD-INSEVLTRYRK----- 255
SAVAHLIVDGVIKKSWSLADPNE-LNSPVIQAYQDQRI--- 245
SAVIHLIVKGVQOGLWRPANLDT-RNNPIFEKY----- 248
SAVIHLIVKGIIEGLWRPPENQNG-KENPIFRKY----- 246
SAVVHMIADGIWKGAWNLAQDASGLRSPVLOKY----- 239
SAVVHMIADGIWKRAWNLADTASGLNNPVLOKY----- 234
SAVGHLLAQGIEQRLWQV----- 238
KAVRSLIIESTRDKIW----- 234
TG----- 232

```



B+3001201300100300001000-011021003-----

Fig. 46A

SEQ_ID_NO_2	1 CLT--APPKEAARPTLMPRAGSYDGLNLPAPTGKIFVS
SEQ_ID_NO_5	1 CLT--APPKEAAKPTLMPRAGSYDGLNLPAPTGKIFVS
SEQ_ID_NO_6	1 CLT--TPPKAAKPTLMPRAGSYDGLNLPVPTGKIFVS
SEQ_ID_NO_7	1 CLT--APPKEAAKPTLMPRAGSYDGLNLPAPTGKIFVS
SEQ_ID_NO_27	1 CLT--APPKEAAKPTLMPRAGSYDGLNLPDPKPKGLFVS
SEQ_ID_NO_28	1 CLT--AAPKEAARPTLLPRAGSYDGLNLPSPQGRIFVS
SEQ_ID_NO_29	1 CIT--SPPKQAAKPTLLPRAGSYDGLNLPFPGGRIFVS
SEQ_ID_NO_30	1 CLT--APPKQAAKPTLMPRAGSYDGLNLPFPAAGKLFVS
SEQ_ID_NO_32	1 -IT--EVPKEAAKPTLMPRAGSYDGLNLPKPKNGKIIVS
SEQ_ID_NO_36	1 CATHIGSPVADEKATLMPRAGSYDGLNLPKPKGKIIVAA
SEQ_ID_NO_3	1 -----MPRAGSYDGLNLPMPPTGKIFVS
SEQ_ID_NO_35	1 -----PETSKEPTLMARSTAGSYDGLNLPKPKGVVVS
SEQ_ID_NO_31	1 CIT--TPPKAAKPTLLPRAGSYDGLNLPQPRGKIIVVA
SEQ_ID_NO_40	1 -----LTRMSTAGSYDGLNLPMPAPRGKIIVTA
SEQ_ID_NO_33	1 -----PETSESPTLMORGANSTAGSYDGLNLPKPKGKIIVS
SEQ_ID_NO_34	1 -----PDASESPTLMORGANSTAGSYDGLNLPKPKGKIIVS
SEQ_ID_NO_37	1 -----ASSSLMPKESSTAGSYDGLNLPAPQGVMLAA
SEQ_ID_NO_39	1 -----MPKESSTAGSYDGLNLPFPGGSMMLAA
SEQ_ID_NO_38	1 -----QDSETPTLTPTSTAGSYDGLNMPRPKGRIMAV
SEQ_ID_NO_41	1 -----PSDPERSTMGELTSTAGSYDGLNLPKPKGKIIVS
SEQ_ID_NO_4	1 CLT--APPKQAAKPTLMPRAGSYDGLNLPAPTGKIFVS

Conservation

-----200101121525427729*429*1*4438656*

SEQ_ID_NO_2	190 YESNVKSGGVGARYFGIGADTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_5	190 YESNVKSGGVGARYFGIGADTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_6	190 YESNVKSGGAGARYFGIGADTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_7	190 YESNVKSGGVGARYFGIGADTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_27	190 YESNVKSGGVGARYFGIGGDTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_28	190 YESNVKSGGVGARYFGIGASTQYQLDQIAVNLRVVDVNT
SEQ_ID_NO_29	190 YESNVKSGGAGARYFGIGASTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_30	190 YESNVKSGGAGARFFFGIGASTQYQLDQIAVNLRVVDVNT
SEQ_ID_NO_32	129 YDSDIKTGGAGARYFGIGADGKYRVDOQVAVNLRVVDVRT
SEQ_ID_NO_36	192 YDGNVRTGGAGAKYFGIGASGEYRVDOQVTYNLRAVDIRS
SEQ_ID_NO_3	116 YESNVKSGGVGARYFGIGADTQYQLDQIAVNLRVVNVST
SEQ_ID_NO_35	125 YDTNIOGTGGAGARYLGVGASGQYRTDOQVTYNIRAVDVRT
SEQ_ID_NO_31	190 YASNVKTGGFGARYFGIGGSTQYQLDQVAVNLRIVNVHT
SEQ_ID_NO_40	117 YDSNIHTGGAGARYFGIGASEKYRVDOQVTYNLRAVDIRT
SEQ_ID_NO_33	125 YDSNIKTGGAGARYLGIGGSGQYRADQVTYNIRAVDVRS
SEQ_ID_NO_34	125 YDSNIKTGGAGARYLGIGGSGQYRADQVTYNIRAVDVRS
SEQ_ID_NO_37	115 YDTNVRTGGAGARYLGIGAAATQFRVDTVTYNLRAVDIRT
SEQ_ID_NO_39	110 YDTNIKTGGAGARYLGIGVNSKFRVDTVTYNLRAVDIRT
SEQ_ID_NO_38	124 YDTNVRSGGEGARYLGIGIDISREYRVDOQVTYNLRAVDVRT
SEQ_ID_NO_41	127 YDSNTMTGGFGARYFGIGASTQYQRODRITIIYLRAVSTLM
SEQ_ID_NO_4	190 YESNVKSGGVGARYFGIGADTQYQLDQIAVNLRVVNVST

Conservation

*498659**2**998*957536985*69797+*797629*

Fig. 46B

VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFVPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSHWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSKWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSNWFIPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSGWFIPLE...
 VYSVODETGOFKPLPA-SNFSSTA...TALLOSGWFVPLE...
 VYDFRDQGTGOYLPAPA-SNFSSTA...TALLOSQWFVPLE...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFIPLE...
 VYDFRDQGTGOYKQPN-SNFSSTA...TALLOSRWFMPLE...
 VYNIODETGOFOPIPA-SNFSSTA...TALLOSRWFVPLE...
 VYSFRDOSGOYKPAPS-SNFSSTA...TALLOSGWFIPLE...
 VYDFRDQGTGOYKQPN-SNFSSTA...TALLOSEWFVPLE...
 VYDFRDQGTGOYKQPN-SNFSSTA...TALLOSEWFVPLE...
 VYDFRDQGTGOYKPIPS-SNFSSTA...TALLOSSWFIPVE...
 VYDFRDQGTGOYKAIPS-SNFSSTA...TALLOSSWFVPLE...
 VYDFRDQGTGOYKPTPA-SNFSSTA...TALLOSGWFVPLE...
 VYKFRDQGTGOYKPSPENGNWSTAVPOCTTTL...TALLOSRWFIFERENT...
 VYNIODETGOFKPYPA-SNFSSTA...TALLOSRWFIPLE...

*376*89**957457-799**8*8*87678*536*15*2**679**87+6*+

R192

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GEILSSVNTSKTILSYEVQAGVFRFIDYORLLEGEVGYTSNEVMECLNSAIF
GEILSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTSNEVMECLNSAIF
GEILSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTSNEVMECLNSAIF
GEVLSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTSNEVMECLNSAIF
GEVLSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTANEVMECLNSAIF
GEVLSSVNTSKTILSYEVQAGVFRFIDYORLLEGEELGYTTNEVMECLNSAIF
GEVLSSVNTSKTILSYEFQAGVFRYIDYORLLEGEVGYTTVNEVMECLNSAIF
GOVLSVNTSKTILSYEQAGVFRYIDYORLLEGEIGYTTNEVMECLNSAIF
GEVLLSVNTSKTILSSELQAGVFRFIEYORLLELEAGYTTNEVMECLNSAIF
GRILNSVTTSKTIVMSQQVQAGVFRFVEYERLLEAEAGFSTNEVMECLNSAIF
GEILSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTSNEVMECLNSAIF
GRILLSVTTSKTILSKELOTQVFKFVQYKDLLEAELGYTTNEVMECLNSAIF
GEVLSSVNTSKTILSYEQAGVFRFIDYORLLEGEAGFTTNEVMECLNSAIF
GRILHGVLTSEKKILSRIRSDVYRFIEFKHLLMEAGITTNDAECLNSAIF
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GKILTSVTTSKTILSYELSAGAFRFVQYKELLELEVELGYTTNNEVMECLNSAIF
GRLLSSVTTTTSILSKEITAGVFRFIDAEELLESELGYTSNEVMECLNSAIF
GRLLSSVTTTTSILSKEVSAGVFKFIDAEELLESELGYTSNEVMECLNSAIF
GOVLANVNTSKTIYSVGRSAGVFRFIEFKHLLAEVGYTTTNEVMECLNSAIF
GEILKTVYTSKTILSTSVNSFFRYIDTERLLEAEVGLTQNEVMECLNSAIF
GEILSSVNTSKTILSYEVQAGVFRFIDYORLLEGEIGYTSNEVMECLNSAIF

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69*29*5*9*597*243675699999565***3*7*795*9*8669856*+9

Fig. 46C

```

NEEKYIIRAAQNGTVAINNRIPLQSLTAANIMVEGSIIG 129
NEEKYIIRAAQNGTVAINNRIPLQSLTAANIMVEGSIIG 129
NEEKYIIRAAQNGTVAINNRIPLQSLTAANIMVEGSIIG 129
NEEKYIIRAAQNGTVAANNRIPLQSLAAANVMIEGSIIG 123
NEEKYIIRAAQNGTVAENNRIPLQSLVAANVMIEGSIIG 129
NEEKYIIRAAQNGGVAANNRIPLQSLVAANILIEGSIIG 129
NEEKYIIRAAQNGTVAVNNRIPLQSLVAANILIEGSIIG 129
NEEKYIIRAAQNGTAAVNNRIPLQSLVAANVLVEGSIIG 129
NEEKYIIRAAQNGTVAANNQQLPQSLLSANVIEGSIIG 120
TEEKYIIRAAQNKPNVPGNNANPLQSLVAANILIEGSIIG 191
NEEKYIIRAAQNGTVAINNRIPLQSLTAANIMVEGSIIG 115
TEEKYIIRAAQKKDEIPTNHQVPLPSLASANIMVEGGIIV 124
NEEKYIIRAAQNGTVDGNNASPLPSLYSANVIVEGSIIG 129
TEEKYIIRAAQLKKDNVPVNNRIPLPSLLAANIMLEGGIIV 116
TEEKYIIRAAQKKQESISNHGS-TLPSELLSANVMIEGSIIV 124
TEEKYIIRAAQKKQESISNHGS-TLPSELLSANVMIEGSIIV 124
TEEKYIIRAAQGLK-----GDAN-KLPOLNSAQILMEGGIIV 114
TEEKYIIRAAQGLK-----GEAN-QLPOLNSAQILMEGGIIV 109
TEEKYIIRAAQSOKKPD-----IMG-ELPPLQAAANLMLEGGIIV 123
NEEKYIIRAAQROEYMKDADKNSQSLPFLLYASILLEGSIIV 126
NEEKYIIRAAQNGTVAANNRIPLQSLTAANIMVEGSIIG 129

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```

TBMVLELNDGGRGLWDLONKKE-RONDILVYKPM-SVPPE 261
TBMVLELNDGGRGLWDLONKKE-RONDILVYKPM-SVPPE 261
TBMVLELNDGGRGLWDLONKKE-RONDILVYKPM-SVPPE 261
TBMVLELNDGGRGLWDLONKKE-RONDILVYKPM-SVPPE 261
TBMVLELNDGGRGLWELKNKGG-RANTIAKPM-SVPPE 261
TBMVLELNDGGRNLWQLONPSE-INSPILQYKPM-SVPPE 261
TBMVLELNDGGRNLWOLKNKE-INSPVLEKPM-SVPPE 259
TBMVLELNDGGRNLWTLKNE-INSVSEKPM-SVPPE 259
TBMVLELNDGGRNLWOLQNPSE-INSVSEKPM-SVPPE 258
TBMVLELNDGGRNLWOLADORD-IDNVLQYKPM-SVPPE 261
TBMVLELNDGGRGLWDLONKKE-RONDILVYKPM-SVPPE 247
TBMVLELNDGGRGLWEPLRGED-LOPPILQYKPM-SVPPE 254
TBMVLELNDGGRKLWALSNAAD-INSVSEKPM-SVPPE 255
TBMVLELNDGGRKSWSLADPNE-LNSVSEKPM-SVPPE 245
TBMVLELNDGGRGLWRFPANLDT-RNMPLQYKPM-SVPPE 248
TBMVLELNDGGRGLWRFPENONG-RNMPLQYKPM-SVPPE 248
TBMVLELNDGGRGAWNLAQQASGLRSPVLOKY- 239
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TBMVLELNDGGRRLWQV- 239
TBMVLELNDGGRKIW- 234

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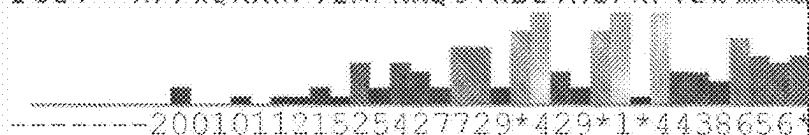
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Fig. 47A

SEQ_ID_NO_2	1 CLT--APPKEAAKPTLMPRAQSYKDLTHLPAPTGK
SEQ_ID_NO_5	1 CLT--APPKEAAKPTLMPRAQSYKDLTHLP IPTGK
SEQ_ID_NO_6	1 CLT--TPPKKEAAKPTLMPRAQSYKDLTHLPVPTGK
SEQ_ID_NO_7	1 CLT--APPKEAAKPTLMPRAQSYRDLTHLPAPTGK
SEQ_ID_NO_27	1 CLT--APPKEAAKPTLMPRAQSYRDLTNLPDPKPKGL
SEQ_ID_NO_28	1 CLT--AA PKBAARPTLLPRAPSYTDLTHLPSPQGR
SEQ_ID_NO_29	1 CIT--SPPKCAAKPTLLPRSQSYODLTHLPEPQGR
SEQ_ID_NO_30	1 CLT--APPKQAAKPTLMPRAQSYQDLTHLPEPAGKL
SEQ_ID_NO_32	1 -IT--EVPKEAAKPTLMPRASTYKDLVALPKPNSK
SEQ_ID_NO_36	1 CATHIGSPVADEKATLMPRSVSYKELISLPKPKGK
SEQ_ID_NO_3	1 -----MPRAQSYKDLTHLPAPTGK
SEQ_ID_NO_35	1 -----PETSKEPTLMARGTAYQDLVSLPLPKSKV
SEQ_ID_NO_31	1 CIT--TPPQEAARKPTLLPRDATYKDLVSLPOPGRK
SEQ_ID_NO_40	1 -----LTRRMSTYQDLIDMPAPRSGK
SEQ_ID_NO_39	1 -----PETSESPTLMORGANYIDLISLPKPOGK
SEQ_ID_NO_34	1 -----PDASESP MORGATYLDLISLPKPOGK
SEQ_ID_NO_37	1 -----ASSSLMPKGESYYDLINLPAPQSV
SEQ_ID_NO_39	1 -----MPKSDTYDYDLIGLPHPOGSM
SEQ_ID_NO_38	1 -----QDSETPTLTTPRASTYYDLINMPRPKGR
SEQ_ID_NO_41	1 -----PSDPERSTMGELTFSTAELRNLPFPNEK
SEQ_ID_NO_4	1 CLT--APPKQAAKPTLMPRAQSYKDLTHLPAPTGK

Conservation



SEQ_ID_NO_2	130	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST
SEQ_ID_NO_5	130	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST
SEQ_ID_NO_6	130	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST
SEQ_ID_NO_7	130	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST
SEQ_ID_NO_27	130	ESNVKSGGVGAG	GGDTCTGLDGLAYNLKVVST
SEQ_ID_NO_28	130	ESNVKSGGVGAG	ASTCTGLDGLAYNLKVVST
SEQ_ID_NO_29	130	ESNVKSGGVGAG	ASTCTGLDGLAYNLKVVST
SEQ_ID_NO_30	130	ESNVKSGGVGAG	ASTCTGLDGLAYNLKVVST
SEQ_ID_NO_32	129	DSNKTGGAGAR	ADGKTVDDVAVNLKVVST
SEQ_ID_NO_36	132	DSNVRTGGAGAR	GASGEVDDVAVNLKVVST
SEQ_ID_NO_3	116	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST
SEQ_ID_NO_35	125	DTN GGAGAR	GASGCATDDVAVNLKVVST
SEQ_ID_NO_31	130	ASNVKTGGFGAR	GGSTCTGLDGLAYNLKVVST
SEQ_ID_NO_40	117	DSNIHTGGAGAR	GAEECTDDVAVNLKVVST
SEQ_ID_NO_39	125	DSN GGAGAR	GGSGCTDDVAVNLKVVST
SEQ_ID_NO_34	125	YDSNKTGGAGAR	GGSGCTDDVAVNLKVVST
SEQ_ID_NO_37	115	YDTNV GGAGAR	GAATCTDDVAVNLKVVST
SEQ_ID_NO_39	110	DTN GGAGAR	GVNSCTDDVAVNLKVVST
SEQ_ID_NO_38	124	DTNVRSGGEGAR	DISKCTDDVAVNLKVVST
SEQ_ID_NO_41	127	SN GGFGAR	GASTCTDDVAVNLKVVST
SEQ_ID_NO_4	130	ESNVKSGGVGAG	ADTCTGLDGLAYNLKVVST

Conservation



Fig. 47B

```

NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSRWF-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSRWF-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSRWF-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSHWF-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPQSATSMVLVTALKDSRWF-ERQGLQNL
NIQDETGOQFKPYPA-CNFSTAVPQSATAMLVLSALKDSKWF-ERQGLQNL
NISDETGOQFKPYPA-SNFSTSVPOSATAMLVLSALKDSNWF-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPOSATAMLVLSALKDSGWF-ERQGLQNL
SDETGOQFKPLPA-SNFSTAVPQSGNAMLTALKDSGWF-ERGLQNL
DFRDQTGOQLPAPA-SNFSTAVTQGGVAMLSTALWDSQWF-EREGQLNL
NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSRWF-ERQGLQNL
DFRDQTGOYKPOPN-SNFSTAVPQSGAALLTTALLDSRWF-ERGLQNL
NIQDETGOQFQPA-SNFSTSVPOSATAMLVSSLKDSRWF-ERQGLNNL
RDQSGOYKPAPS-SSFSTAVTQGAAMLVNVLNDSGWF-EREGQLNL
DFRDQTGOYKPOPN-SNFSTAVPQGGTALLTMALLOSEWFY-ERQGLQNL
DFRDQTGOYKPOPN-SNFSTAVPQGGTALLTMALLOSEWFY-ERQGLQNL
DFRDQTGOYKPIPS-SNFSTAVPQSGTAFLAQAALNDSWF-EREGQLNL
DFRDQTGOYKAPS-SNFSTAVPQSGTAFLAQAALNDSWF-EREGQLNL
DFRDQTGOYKPTPA-SSFSTSVTQGAASMLMDALSAGGWF-EREGQLNL
YKFRDQTGOYKPSNGN-SNFSTAVPQGGTALLTMALLOSEWFY-ERQGLQNL
NIQDETGOQFKPYPA-SNFSTAVPQSATAMLVLTALKDSRWF-ERQGLQNL

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$$*376*89**957457-799**8*8*87678*536*15*2**679**87+6*+$$

R192

✱

[illegible]

69*29*5*9*597*243675699999565***5*7*795*9*8669856*+9

R97

Fig. 47C

*

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LNERKIIIRAAQENGSTVAINNRI-PLQSLTAANINDEGSIIG 129
LNERKIIIRAAQENGSTVAINNRI-PLPSLTAANINDEGSIIG 129
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LNERKIIIRAAQENGSTVAENNR-PLQSLVAANINDEGSIIG 129
LNERKIIIRAAQENGSSVAINNQR-PLSSLVAANINDEGSIIG 129
LNERKIIIRAAQENGSTVAVNNRT-QLPSLVAANINDEGSIIG 129
LNERKIIIRAAQENGTAAVNNQH-QLSSLVAANINDEGSIIG 129
LNERKIIIRAAQENGSTVAANNQD-PLPSLLSANINDEGSIIG 128
LTERKIVRAAQNKPNVPGNNAN-OLPSLVAANINDEGSIIG 131
LNERKIIIRAAQENGSTVAINNRI-PLQSLTAANINDEGSIIG 115
LTERKIIIRAAQKKDEIPTNHGV-HLPSLASANINDEGSIIG 124
LNERKIIIRAAQQNGSTVGDNNA-PLPSLYSANINDEGSIIG 129
LTERKIIIRAAALKKONVPVNNNA-GLPSLLAANINDEGSIIG 116
LTERKIIIRAAQKKQESISNHGS-TLPSSLSANINDEGSIIG 124
LTERKIIIRAAQKKQESISNHGS-TLPSSLSANINDEGSIIG 124
LTERKIVRAGLK-----GDAN-KLPQLNSAONINDEGSIIG 114
LTERKIVRAGLK-----GEAN-QLPQLSSAONINDEGSIIG 109
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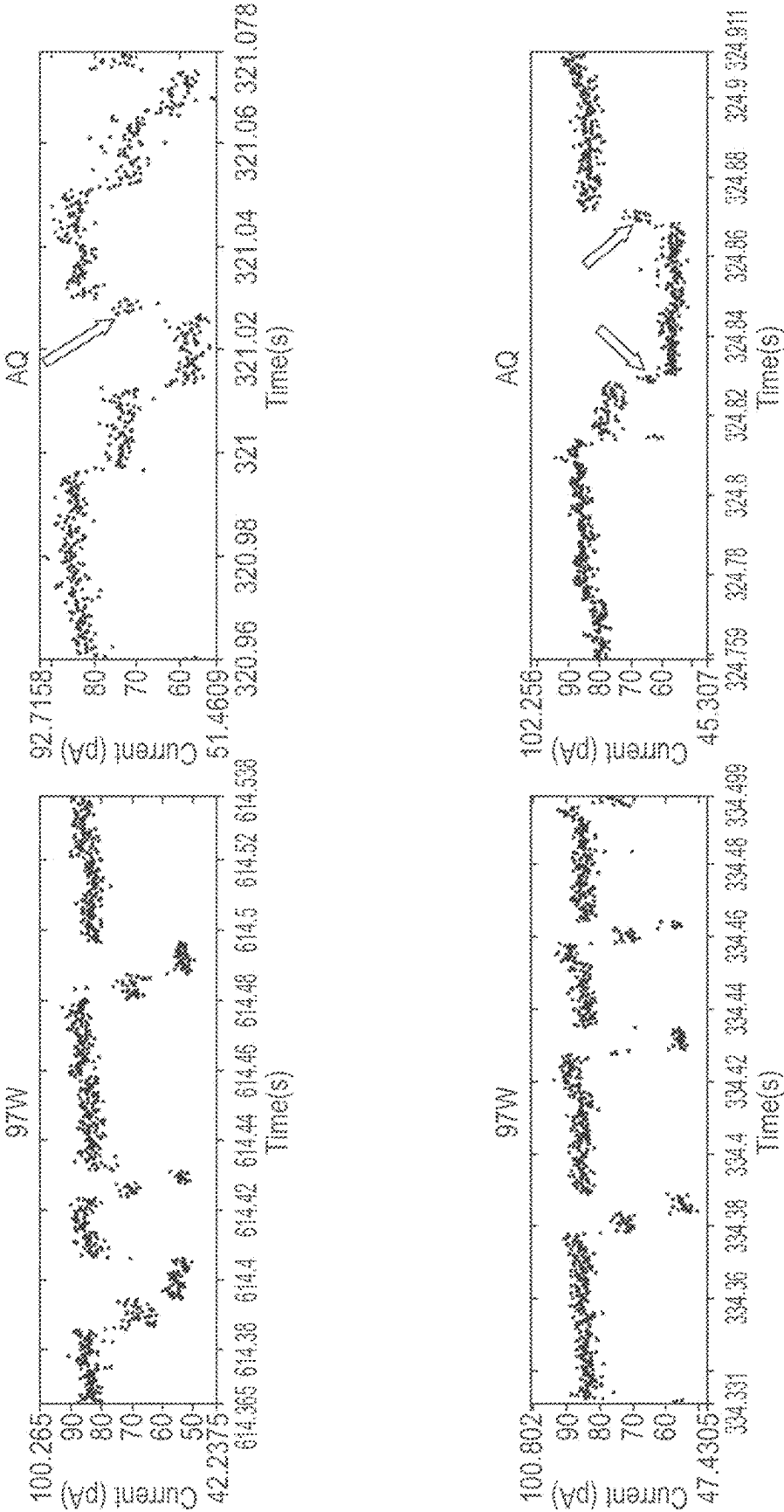
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TGVIFLINDGINRGLWELKNKGD-AKNTILAKYRSMAPPE 261
SGVIYLVNDGIERNLWOLONPSE-INSPILQRYKNNIVPAE 261
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TGVIFLINDGINRNLWTLKNPQD-AKSSVLERYKSTIVP-- 259
AGVAHLIVEGIRONLWOLONPQD-INNPILQRYMKEDVP-- 259
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EGVIFLINDGINKKLWALSNAAD-INSEVLTRYK----- 255
SAVAHLIVDGVIKKSWSLADPNE-LNSPVIQAYQQQRI--- 245
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SAVIFLIVKGVQOGLWRPENONG-KENPIFKKY----- 248
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SAVGHLLAAGIEORLWQV----- 233
KAVRSLIEGTRDKIW----- 234
TG----- 228



3+300120130010030001000-011021003-----

Fig. 48



MUTANT PORES

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/159,718, filed Jan. 26, 2023, entitled “MUTANT PORES,” which is a continuation of U.S. application Ser. No. 17/215,506, filed Mar. 29, 2021, entitled “MUTANT PORES,” which is a division of U.S. application Ser. No. 16/081,891, filed Aug. 31, 2018, entitled “MUTANT PORES,” which is a national stage filing under 35 U.S.C. 371 of International Patent Application Serial No. PCT/GB2017/050571, filed Mar. 2, 2017, entitled “MUTANT PORES,” which claims foreign priority under 35 U.S.C. § 119(a)-(d) or 35 U.S.C. § 365(b) to British application number GB 1603656.8, filed Mar. 2, 2016, British application number GB 1603657.6, filed Mar. 2, 2016, and British application number GB 1603658.4, filed Mar. 2, 2016. The contents of these applications are incorporated herein by reference in their entireties.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] The contents of the electronic sequence listing (O036670063US03-SEQ-LJG.xml; Size: 91,741 bytes; and Date of Creation: Jan. 7, 2025) are herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The invention relates to mutant forms of CsgG. The invention also relates to analyte detection and characterisation using CsgG.

BACKGROUND OF THE INVENTION

[0004] Nanopore sensing is an approach to sensing that relies on the observation of individual binding or interaction events between analyte molecules and a receptor. Nanopore sensors can be created by placing a single pore of nanometer dimensions in an insulating membrane and measuring voltage-driven ionic transport through the pore in the presence of analyte molecules. The identity of an analyte is revealed through its distinctive current signature, notably the duration and extent of current block and the variance of current levels. Such nanopore sensors are commercially available, such as the MinION™ device sold by Oxford Nanopore Technologies Ltd, comprising an array of nanopores integrated with an electronic chip.

[0005] There is currently a need for rapid and cheap nucleic acid (e.g. DNA or RNA) sequencing technologies across a wide range of applications. Existing technologies are slow and expensive mainly because they rely on amplification techniques to produce large volumes of nucleic acid and require a high quantity of specialist fluorescent chemicals for signal detection. Nanopore sensing has the potential to provide rapid and cheap nucleic acid sequencing by reducing the quantity of nucleotide and reagents required.

[0006] Two of the essential components of sequencing nucleic acids using nanopore sensing are (1) the control of nucleic acid movement through the pore and (2) the discrimination of nucleotides as the nucleic acid polymer is moved through the pore. In the past, to achieve nucleotide discrimination the nucleic acid has been passed through a mutant of hemolysin. This has provided current signatures that have been shown to be sequence dependent. It has also

been shown that a large number of nucleotides contribute to the observed current when a hemolysin pore is used, making a direct relationship between observed current and polynucleotide challenging.

[0007] While the current range for nucleotide discrimination has been improved through mutation of the hemolysin pore, a sequencing system would have higher performance if the current differences between nucleotides could be improved further. In addition, it has been observed that when the nucleic acids are moved through a pore, some current states show high variance. It has also been shown that some mutant hemolysin pores exhibit higher variance than others. While the variance of these states may contain sequence specific information, it is desirable to produce pores that have low variance to simplify the system. It is also desirable to reduce the number of nucleotides that contribute to the observed current.

SUMMARY OF THE INVENTION

[0008] The inventors have surprisingly demonstrated that CsgG and novel mutants thereof may be used to characterise analytes, such as polynucleotides. The invention concerns mutant CsgG monomers. The inventors have surprisingly demonstrated that pores comprising the novel mutant monomers have an enhanced ability to estimate the characteristics of analytes, such as the sequence of polynucleotides. The inventors have made mutant pores that surprisingly provide more consistent movement of a target polynucleotide with respect to, such as through, the pores. The inventors have made mutant pores surprisingly display improved characterisation accuracy. In particular, the mutant pores made by the inventors surprisingly display an increased current range, which makes it easier to discriminate between different nucleotides, and a reduced variance of states, which increases the signal-to-noise ratio. In addition, the mutant pores surprisingly capture nucleotides and polynucleotides more easily.

[0009] All amino-acid substitutions, deletions and/or additions disclosed herein are with reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2, unless stated to the contrary.

[0010] Reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 encompasses mutant CsgG monomers comprising variants of sequences as set out in the further SEQ ID NOS as disclosed below. Amino-acid substitutions, deletions and/or additions may be made to CsgG monomers comprising a variant of the sequence other than shown in SEQ ID NO:2 that are equivalent to those substitutions, deletions and/or additions disclosed herein with reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO:2.

[0011] A mutant monomer may be considered as an isolated monomer.

[0012] In particular, the invention concerns mutant CsgG monomers in which the arginine (R) at position 192 has been substituted with aspartic acid (D), glutamine (Q), phenylalanine (F), serine (S) or threonine (T). The inventors have surprisingly demonstrated that such monomers, and in particular a monomer comprising a R192D substitution, are much easier to express than monomers without a substitution at position 192. An increased yield is obtained when the CsgG monomer comprises this mutation.

- [0013] The mutant csgG monomer may be:
- [0014] a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 which comprises R97W;
- [0015] a mutant CsgG monomer which comprises (a) R192D; (b) R97W/Y and/or R93W/Y, preferably R97W, R93W or R93Y and R97Y; (c) K94Q/N; (d) G103K/R and/or T104KR; and/or (e) F191T, deletion of V105, A106 and I107 and/or deletion of F193, I194, D195, Y196, Q197, R198 and L199.
- [0016] The mutant CsgG monomer preferably further comprises Y51A and F56Q.
- [0017] Particular mutant CsgG monomers comprise variants of the sequence shown in SEQ ID NO: 2 that comprise the following mutations:
- [0018] (1) Y51A, F56Q and R192D;
- [0019] (2) Y51A, F56Q and R97W.
- [0020] (3) Y51A, F56Q, R192D and R97W;
- [0021] (4) Y51A, F56Q, R192D and R93W;
- [0022] (5) Y51A, F56Q, R192D, R93Y and R97Y, or
- [0023] (6) Y51A, F56Q, R192D and R93W.
- [0024] (7) the mutations of any one of (1)-(6) and:
- [0025] (a) deletion of V105, A106 and I107.
- [0026] (b) K94Q or K94N;
- [0027] (c) deletion of D195, Y196, Q197, R198 and L199 or deletion of F193, I194, D195, Y196, Q197, R198 and L199; and/or
- [0028] (d) F191T
- [0029] (8) the mutations of any one of (1)-(6) and:
- [0030] (i) K94Q and deletion of V105, A106 and I107;
- [0031] (ii) K94N and deletion of V105, A106 and I107;
- [0032] (iii) F191 T and deletion of V105, A106 and I107;
- [0033] (iv) K94Q and F191T;
- [0034] (v) K94N and F191T;
- [0035] (vi) K94Q, F191T and deletion of V105, A106 and I107; or
- [0036] (vii) K94N, F191T and deletion of V105, A106 and I107.
- [0037] (9) the mutations of any one of (1)-(8) and:
- [0038] T104K or T104R;
- [0039] L90R;
- [0040] N91R;
- [0041] I95R;
- [0042] A99R;
- [0043] E101K, E101N, E101Q, E101T or E101H;
- [0044] E44N or E44Q; and/or
- [0045] Q42K.
- [0046] The invention also provides:
- [0047] a construct comprising two or more covalently attached CsgG monomers, wherein at least one of the monomers is a mutant monomer of the invention;
- [0048] a polynucleotide which encodes a mutant monomer of the invention or a construct of the invention;
- [0049] a homo-oligomeric pore derived from CsgG comprising identical mutant monomers of the invention or identical constructs of the invention;
- [0050] a hetero-oligomeric pore derived from CsgG comprising at least one mutant monomer of the invention or at least one construct of the invention;

- [0051] a method for determining the presence, absence or one or more characteristics of a target analyte, comprising:
- [0052] (a) contacting the target analyte with a pore of the invention such that the target analyte moves with respect to the pore; and
- [0053] (b) taking one or more measurements as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte;
- [0054] a method of forming a sensor for characterising a target polynucleotide, comprising forming a complex between a pore of the invention and a polynucleotide binding protein and thereby forming a sensor for characterising the target polynucleotide;
- [0055] a sensor for characterising a target polynucleotide, comprising a complex between a pore of the invention and a polynucleotide binding protein,
- [0056] use of a pore of the invention to determine the presence, absence or one or more characteristics of a target analyte;
- [0057] a kit for characterising a target analyte comprising (a) a pore of the invention and (b) the components of a membrane;
- [0058] an apparatus for characterising target analytes in a sample, comprising (a) a plurality of a pores of the invention and (b) a plurality of membranes;
- [0059] a method of characterising a target polynucleotide, comprising:
- [0060] a) contacting the polynucleotide with a pore of the invention, a polymerase and labelled nucleotides such that phosphate labelled species are sequentially added to the target polynucleotide by the polymerase, wherein the phosphate species contain a label specific for each nucleotide, and
- [0061] b) detecting the phosphate labelled species using the pore and thereby characterising the polynucleotide; and
- [0062] a method of producing a mutant monomer of the invention or a construct of the invention, comprising expressing a polynucleotide of the invention in a suitable host cell and thereby producing a mutant monomer of the invention or a construct.

DESCRIPTION OF THE FIGURES

- [0063] FIG. 1: Illustrates CsgG from *E. coli*.
- [0064] FIG. 2: Illustrates the dimensions of CsgG.
- [0065] FIG. 3: Illustrates single G translocation at 10 Å/ns. There is a large barrier for entry of guanine into F56 ring in CsgG-Eco. * = G enters F56 ring. A G stops interacting with 56 ring. B = G stops interacting with 55 ring. C = G stops interacting with 51 ring.
- [0066] FIG. 4: Illustrates ssDNA translocation at 100 Å/ns. A larger force is required to pull the DNA through the constriction for CsgG-Eco.
- [0067] FIG. 5: Illustrates ss DNA translocation at 10 Å/ns. CsgG-F56A-N55S and CsG-F56A-N55S-Y51A mutants both have a lower barrier for ssDNA translocation.
- [0068] FIGS. 6A-6C: Mutant pores showing increased range compared with wild-type (WT).
- [0069] FIGS. 7A-7C: Mutant pores showing increased range compared with wild-type (WT).
- [0070] FIGS. 8A-8C: Mutant pores showing increased range compared with wild-type (WT).

[0071] FIGS. 9A-9C: Mutant pores showing increased throughput compared with wild-type (WT).

[0072] FIGS. 10A-10B: Mutant pores showing increased throughput compared with wild-type (WT).

[0073] FIGS. 11 and 12: Mutant pore showing increased insertion compared with wild-type (WT).

[0074] FIG. 13: shows the DNA construct X used in Example 2. The region labelled 1 corresponded to 30 SpC3 spacers. The region labelled 2 corresponded to SEQ ID NO: 42. The region labelled 3 corresponded to four iSp18 spacers. The region labelled 4 corresponded to SEQ ID NO: 43. The section labelled 5 corresponded to four 5-nitroindoles. The region labelled 6 corresponded to SEQ ID NO: 44. The region labelled 7 corresponded to SEQ ID NO: 45. The region labelled 8 corresponded to SEQ ID NO: 46 which had four iSp18 spacers (the region labelled 9) attached at the 3' end of SEQ ID NO: 46. At the opposite end of the iSp18 spacers was a 3' cholesterol tether (labelled 10). The region labelled 11 corresponded to four SpC3 spacers.

[0075] FIG. 14: shows an example chromatography trace of Strep trap (GE Healthcare) purification of CsgG protein (x-axis label=elution volume (mL), Y-axis label=Absorbance (mAu)). The sample was loaded in 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM and eluted with 10 mM desthiobiotin. The elution peak in which CsgG protein eluted is labelled as E1.

[0076] FIG. 15: shows an example of a typical SDS-PAGE visualisation of CsgG protein after the initial strep purification. A 4-20% TGX Gel (Bio Rad) was run at 300 V for 22 minutes in 1xTGS buffer. The gel was stained with Sypro Ruby stain. Lanes 1-3 show the main elution peak (labelled E1 in FIG. 14) which contained CsgG protein as indicated by the arrow. Lanes 4-6 corresponded to elution fractions of the tail of the main elution peak (labelled E1 in FIG. 14) which contained contaminants. M shows the molecular weight marker used which was a Novex Sharp Unstained (unit=kD).

[0077] FIG. 16: Shows an example of a size exclusion chromatogram (SEC) of CsgG protein (120 mL S200 GE healthcare, x-axis label=elution volume (mL), y-axis label=absorbance (mAu)). The SEC was carried out after strep purification and heating the protein sample. The running buffer for SEC was 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, pH 8.0 and the column was run at 1 mL/minute rate. The trace labelled X shows absorbance at 220 nm and the trace labelled Y shows absorbance at 280 nm. The peak labelled with a star was collected.

[0078] FIG. 17: shows an example of a typical SDS-PAGE visualisation of CsgG protein after SEC. A 4-20% TGX Gel (Bio Rad) was run at 300V for 22 minutes in 1x TGS buffer and the gel was stained with Sypro Ruby stain. Lane 1 shows CsgG protein sample after strep purification and heating but before SEC. Lanes 2-8 show fractions collected across the peak running approximately 48 mL-60 mL of FIG. 16 (mid peak=55 mL) and labelled with a star in FIG. 16. M shows the molecular weight marker used which was a Novex Sharp Unstained (unit=kD). The bar corresponding to the CsgG-Eco pore is indicated by an arrow.

[0079] FIGS. 18 to 24: Mutant pores showing increased range compared with wild-type (WT).

[0080] FIGS. 25 to 30: Mutant pores showing increased range compared with wild-type (WT).

[0081] FIG. 31 shows snap shots of the enzyme (T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (ΔM1) G1G2)) on top of the pore (CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51 T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20)) taken at 0 and 20 ns during the simulations (Runs 1 to 3).

[0082] FIG. 32 shows snap shots of the enzyme (T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (ΔM1) G1G2)) on top of the pore (CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20)) taken at 30 and 40 ns during the simulations (Runs 1 to 3).

[0083] FIG. 33 shows a snap shot of the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (ΔM1)G1G2) on top of the pore CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) taken during the simulations described in Example 5.

[0084] FIG. 34 shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through MspA mutant x=MspA-((Del-L74/G75/D118/L119)D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K)8 (SEQ ID NO: 50 with mutations D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K and deletion of the amino acids L74/G75/D118/L119) without the control of an enzyme.

[0085] FIG. 35 shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) without the control of an enzyme.

[0086] FIG. 36 shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/E101 S/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) without the control of an enzyme.

[0087] FIG. 37 shows an overlay of two gel filtration chromatograms (120 ml S200 column) of the CsgG mutants pores A) CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B) CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus). Absorbance at A280 for CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 is labelled A and for CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 is labelled B. Both constructs were grown in 500 ml cultures. Expression and purification of both proteins were carried out using exactly the same protocol and same volumes were loaded onto the column. Running Buffer was 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS pH8. The fractional delay with CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 pore was due to different connection configu-

ration used on AKTA Purifier 10. The difference in the absorbance values indicated the amount of proteins expressed with higher absorbance values indicating higher amounts of expressed protein.

[0088] FIG. 38 shows SDS-PAGE analysis of CsgG nanopores. Lanes A-C contained CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C)9, lanes D-F contained CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C)9 and lane M contained the molecular weight marker. The two pores were expressed and purified using exactly the same protocol. The pores were subjected to electrophoresis on a 4-20% TGX gel (Bio rad cat #5671093) in TGS buffer at 300 V for 22 minutes. The gel was visualised with Sypro Ruby stain (Life Technologies cat #S1200). The same volumes from each pore sample were loaded on the gel to compare the amount of proteins obtained after purification—lanes A and D contained 5 uL, lanes B and E contained 10 uL and lanes C and F contained 15 uL.

[0089] FIG. 39 shows the basecall accuracy of eight CsgC mutant pores compared to the basecall accuracy of a baseline pore, mutant 28 (CsgG-(WT-Y51A/F56Q/R97W/R192D)-StrepII)9. The deletion of D195-L199 (Mutant A), F193-L199 (Mutant B) or V105-I107 (Mutant D), or the substitution of F191 T (Mutant C) results in a further improvement in accuracy in addition to the improvement in accuracy resulting from the R97W and R 192D substitutions in mutant 28. The effect on basecall accuracy of deleting V105-I107 was also tested in a mutant pore containing an additional K94Q mutation (Mutant E) and an improvement in accuracy compared to baseline mutant 28 was still observed. Introducing a R93W mutation (Mutant F) or both R93Y and R97W mutations (Mutant H) instead of a R97W mutation (baseline mutant 28) increased the basecall accuracy. Deleting D195-L199 in addition to R93W (Mutant G) resulted in an enhancement of basecall accuracy.

[0090] FIGS. 40A-40B show the template speed distribution (FIG. 40A) and the template accuracy distribution (FIG. 40B) of the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D)-StrepII)9 and Mutant D which comprises an additional deletion of V105-I107. A template DNA was prepared and passed through the mutant pores as described in the Examples. The template speed and accuracy were determined as described in the Examples. FIG. 40A shows that the speed distribution was tighter when Mutant D was used compared to the baseline mutant. FIG. 40B shows that mutant D has a tighter distribution of template accuracy compared to the baseline mutant.

[0091] FIG. 41 displays an example “squiggle” that shows the “noisy” pore error mode exhibited by baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D)-StrepII)9. The top panel of FIG. 41 shows the difference in flow of current through the pore during the “good” and “noisy” pore states. The bottom panel of FIG. 41 shows an expanded view of the transition from “good” state to “noisy” state.

[0092] FIG. 42 shows the reduction in noisy pore state of mutant pores having the same sequence as the baseline mutant 28, which contains Y51A/F56Q/R97W/R192D mutations, with an additional K94N mutation (Mutant I) or an additional K94Q mutation (Mutant J) when compared to baseline mutant 28, averaged over at least 5 runs.

[0093] FIG. 43 illustrates the structure of the template strand having an adaptor ligated to each end thereof. The adaptor has a T4 Dda helicase enzyme prebound thereto. The

sequences of the various parts of the adaptor used in the Examples are shown in SEQ ID NOs: 52 to 55.

[0094] FIG. 44 shows the median time between Thrombin Binding Aptamer (TBA) events for mutant CsgG nanopores comprising one of the following substitutions: Q42K (Mutant K), E44N (Mutant L), E44Q (Mutant M), L90R (Mutant N), N91R (Mutant O), I95R (Mutant P), A99R (Mutant Q), E101H (Mutant R), E101K (Mutant S), E101N (Mutant T), E101Q (Mutant U), E101 T (Mutant V) and Q114K (Mutant W). The median time was significantly reduced compared to the baseline pore comprising the mutations Y51A/F56Q/K94Q/R97W/R192D-del(V105-I107) (Baseline mutant E), all of which are also included in each of the 13 mutants tested. FIG. 44 shows that each of the Q42K, E44N, E44Q, L90R, N91R, I95R, A99R, E101H, E101K, E101N, E101Q, E101T and Q114K substitutions increase template DNA capture rates.

[0095] FIGS. 45A-45C show sequence alignments of the 21 CsgG homologues corresponding to SEQ ID Nos 2, 5, 6, 7, 27, 28, 29, 30, 32, 36, 3, 35, 31, 40, 33, 34, 37, 39, 38, 41 and 4

[0096] FIGS. 46A-46C show the same relative sequence alignments as FIG. 45 with predicted alpha helical secondary structure regions additionally shaded.

[0097] FIGS. 47A-47C show the same relative sequence alignments as FIG. 45 with predicted beta sheet secondary structure regions additionally shaded.

[0098] FIG. 48 shows two examples of raw electrical data for poreAQ and pore97W.

DESCRIPTION OF THE SEQUENCE LISTING

[0099] SEQ ID NO: 1 shows the codon optimised polynucleotide sequence encoding the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. This monomer lacks the signal sequence.

[0100] SEQ ID NO: 2 shows the amino acid sequence of the mature form of the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. This monomer lacks the signal sequence. The abbreviation used for this CsgC=CsgG-Eco.

[0101] SEQ ID NO: 3 shows the amino acid sequence of YP_001453594.1: 1-248 of hypothetical protein CKO_02032 [*Citrobacter koseri* ATCC BAA-895], which is 99% identical to SEQ ID NO: 2.

[0102] SEQ ID NO: 4 shows the amino acid sequence of WP_001787128.1: 16-238 of curli production assembly/transport component CsgG, partial [*Salmonella enterica*], which is 98% to SEQ ID NO: 2.

[0103] SEQ ID NO: 5 shows the amino acid sequence of KEY44978.1: 16-277 of curli production assembly/transport protein CsgG [*Citrobacter amalonaticus*] which is 98% identical to SEQ ID NO: 2.

[0104] SEQ ID NO: 6 shows the amino acid sequence of YP_003364699.1: 16-277 of curli production assembly/transport component [*Citrobacter rodentium* ICC168], which is 97% identical to SEQ ID NO: 2.

[0105] SEQ ID NO: 7 shows the amino acid sequence of YP_004828099.1: 16-277 of curli production assembly/transport component CsgG [*Enterobacter asburiae* LF7a], which is 94% identical to SEQ ID NO: 2.

[0106] SEQ ID NO: 8 shows the polynucleotide sequence encoding the Phi29 DNA polymerase.

[0107] SEQ ID NO: 9 shows the amino acid sequence of the Phi29 DNA polymerase.

[0108] SEQ ID NO: 10 shows the codon optimised polynucleotide sequence derived from the *sbcB* gene from *E. coli*. It encodes the exonuclease I enzyme (EcoExo I) from *E. coli*.

[0109] SEQ ID NO: 11 shows the amino acid sequence of exonuclease I enzyme (EcoExo I) from *E. coli*.

[0110] SEQ ID NO: 12 shows the codon optimised polynucleotide sequence derived from the *xthA* gene from *E. coli*. It encodes the exonuclease III enzyme from *E. coli*.

[0111] SEQ ID NO: 13 shows the amino acid sequence of the exonuclease III enzyme from *E. coli*. This enzyme performs distributive digestion of 5' monophosphate nucleosides from one strand of double stranded DNA (dsDNA) in a 3'-5' direction. Enzyme initiation on a strand requires a 5' overhang of approximately 4 nucleotides.

[0112] SEQ ID NO: 14 shows the codon optimised polynucleotide sequence derived from the *recJ* gene from *T. thermophilus*. It encodes the RecJ enzyme from *T. thermophilus* (TthRecJ-cd).

[0113] SEQ ID NO: 15 shows the amino acid sequence of the RecJ enzyme from *T. thermophilus* (TthRecJ-cd). This enzyme performs processive digestion of 5' monophosphate nucleosides from ssDNA in a 5'-3' direction. Enzyme initiation on a strand requires at least 4 nucleotides.

[0114] SEQ ID NO: 16 shows the codon optimised polynucleotide sequence derived from the bacteriophage lambda *exo* (*recX*) gene. It encodes the bacteriophage lambda exonuclease.

[0115] SEQ ID NO: 17 shows the amino acid sequence of the bacteriophage lambda exonuclease. The sequence is one of three identical subunits that assemble into a trimer. The enzyme performs highly processive digestion of nucleotides from one strand of dsDNA, in a 5'-3' direction (<http://www.neb.com/nebecomm/products/productM0262.asp>). Enzyme initiation on a strand preferentially requires a 5' overhang of approximately 4 nucleotides with a 5' phosphate.

[0116] SEQ ID NO: 18 shows the amino acid sequence of Hel308 Mbu.

[0117] SEQ ID NO: 19 shows the amino acid sequence of Hel308 Csy.

[0118] SEQ ID NO: 20 shows the amino acid sequence of Hel308 Tga.

[0119] SEQ ID NO: 21 shows the amino acid sequence of Hel308 Mhu.

[0120] SEQ ID NO: 22 shows the amino acid sequence of TraI Eco.

[0121] SEQ ID NO: 23 shows the amino acid sequence of XPD Mbu.

[0122] SEQ ID NO: 24 shows the amino acid sequence of Dda 1993.

[0123] SEQ ID NO: 25 shows the amino acid sequence of TrwC Cba.

[0124] SEQ ID NO: 26 shows the amino acid sequence of WP_006819418.1: 19-280 of transporter [*Yokenella regensburgei*], which is 91% identical to SEQ ID NO: 2.

[0125] SEQ ID NO: 27 shows the amino acid sequence of WP_024556654.1: 16-277 of Curli production assembly/transport protein CsgG [*Cronobacter pulveris*], which is 89% identical to SEQ ID NO: 2.

[0126] SEQ ID NO: 28 shows the amino acid sequence of YP_005400916.1: 16-277 of curli production assembly/transport protein CsgG [*Rahnella aquatilis* HX2], which is 84% identical to SEQ ID NO 2.

[0127] SEQ ID NO: 29 shows the amino acid sequence of KFC99297.1: 20-278 of CsgG family curli production assembly/transport component [*Kluyvera ascorbata* ATCC 33433], which is 82% identical to SEQ ID NO: 2.

[0128] SEQ ID NO: 30 shows the amino acid sequence of KFC86716.1:16-274 of CsgG family curli production assembly/transport component [*Hafnia alvei* ATCC 13337], which is 81% identical to SEQ ID NO: 2.

[0129] SEQ ID NO: 31 shows the amino acid sequence of YP_007340845.1:16-270 of uncharacterised protein involved in formation of curli polymers [*Enterobacteriaceae bacterium* strain FGI 57], which is 76% identical to SEQ ID NO: 2.

[0130] SEQ ID NO: 32 shows the amino acid sequence of WP_010861740.1: 17-274 of curli production assembly/transport protein CsgG [*Plesiomonas shigelloides*], which is 70% identical to SEQ ID NO: 2.

[0131] SEQ ID NO: 33 shows the amino acid sequence of YP_205788.1: 23-270 of curli production assembly/transport outer membrane lipoprotein component CsgG [*Vibrio fischeri* ES114], which is 60% identical to SEQ ID NO: 2.

[0132] SEQ ID NO: 34 shows the amino acid sequence of WP_017023479.1: 23-270 of curli production assembly protein CsgG [*Aliivibrio lojei*], which is 59% identical to SEQ ID NO: 2.

[0133] SEQ ID NO: 35 shows the amino acid sequence of WP_007470398.1: 22-275 of Curli production assembly/transport component CsgG [*Photobacterium* sp, AK 15], which is 57% identical to SEQ ID NO: 2.

[0134] SEQ ID NO: 36 shows the amino acid sequence of WP_021231638.1: 17-277 of curli production assembly protein CsgG [*Aeromonas veronii*], which is 56% identical to SEQ ID NO: 2.

[0135] SEQ ID NO: 37 shows the amino acid sequence of WP_033538267.1: 27-265 of curli production assembly/transport protein CsgG [*Shewanella* sp. ECSMB14101], which is 56% identical to SEQ ID NO: 2.

[0136] SEQ ID NO: 38 shows the amino acid sequence of WP_003247972.1: 30-262 of curli production assembly protein CsgG [*Pseudomonas putida*], which is 54% identical to SEQ ID NO: 2.

[0137] SEQ ID NO: 39 shows the amino acid sequence of YP_003557438.1: 1-234 of curli production assembly/transport component CsgG [*Shewanella violacea* DSS12], which is 53% identical to SEQ ID NO: 2.

[0138] SEQ ID NO: 40 shows the amino acid sequence of WP_027859066.1: 36-280 of curli production assembly/transport protein CsgG [*Marinobacterium jannaschii*], which is 53% identical to SEQ ID NO: 2.

[0139] SEQ ID NO: 41 shows the amino acid sequence of CEJ702².1: 29-262 of Curli production assembly/transport component CsgG [*Chryseobacterium oranense* G311], which is 50% identical to SEQ ID NO: 2.

[0140] SEQ ID NO: 42 shows a polynucleotide sequence used in Example 2.

[0141] SEQ ID NO: 43 shows a polynucleotide sequence used in Example 2.

[0142] SEQ ID NO: 44 shows a polynucleotide sequence used in Example 2.

[0143] SEQ ID NO: 45 shows a polynucleotide sequence used in Example 2.

[0144] SEQ ID NO: 46 shows a polynucleotide sequence used in Example 2. Attached to the 3' end of SEQ ID NO:

46 is six iSp18 spacers which are attached at the opposite end to two thymines and a 3' cholesterol TEG.

[0145] SEQ ID NO: 47 shows the polynucleotide sequence of StrepII(C).

[0146] SEQ ID NO: 48 shows the polynucleotide sequence of Pro.

[0147] SEQ ID NO: 49 shows the codon optimised polynucleotide sequence encoding the wild-type MspA monomer. This mutant lacks the signal sequence.

[0148] SEQ ID NO: 50 shows the amino acid sequence of the mature form of the wild-type MspA monomer. This mutant lacks the signal sequence.

[0149] SEQ ID NO: 51 shows the polynucleotide sequence of Thrombin Binding Aptamer used in Examples 7 and 11.

[0150] SEQ ID NO: 52 shows the polynucleotide sequence of a Y-adaptor top strand.

[0151] SEQ ID NO: 53 shows the polynucleotide sequence of a Y-adaptor blocker strand.

[0152] SEQ ID NO: 54 shows the polynucleotide sequence of a Y-adaptor cholesterol tether strand.

[0153] SEQ ID NO: 55 shows the polynucleotide sequence of a Y-adaptor bottom strand. SEQ ID NO: 56 shows the polynucleotide sequence of a 3.6 kb double stranded DNA target sequence used in the Examples.

DETAILED DESCRIPTION OF THE INVENTION

[0154] It is to be understood that different applications of the disclosed products and methods may be tailored to the specific needs in the art. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments of the invention only, and is not intended to be limiting.

[0155] In addition as used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to “a polynucleotide” includes two or more polynucleotides, reference to “a polynucleotide binding protein” includes two or more such proteins, reference to “a helicase” includes two or more helicases, reference to “a monomer” refers to two or more monomers, reference to “a pore” includes two or more pores and the like.

[0156] All publications, patents and patent applications cited herein, whether supra or infra, are hereby incorporated by reference in their entirety.

Mutant CsgG Monomers

[0157] An aspect of the invention provides mutant CsgG monomers. The mutant CsgG monomers may be used to form the pores of the invention. A mutant CsgG monomer is a monomer whose sequence varies from that of a wild-type CsgG monomer and which retains the ability to form a pore. Methods for confirming the ability of mutant monomers to form pores are well-known in the art and are discussed in more detail below.

[0158] Pores constructed from the CsgG monomers of some embodiments of the invention comprising the modification R97W display an increased accuracy as compared to otherwise identical pores without the modification at 97 when characterizing (or sequencing) target polynucleotides. An increased accuracy is also seen when instead of R97W

the CsgG monomers of the invention comprise the modification R93W or the modifications R93Y and R97Y. Accordingly, pores may be constructed from one or more mutant CsgG monomers that comprise a modification at R97 or R93 of SEQ ID NO: 2 such that the modification increases the hydrophobicity of the amino acid. For example, such modification may include an amino acid substitution with any amino acid containing a hydrophobic side chain, including, e.g., but not limited to W and Y.

[0159] The CsgG monomers of the invention that comprise R192D/Q/F/S/T are easier to express than monomers which do not have a substitution at position 192 which may be due to the reduction of positive charge. Accordingly position 192 may be substituted with an amino-acid which reduces the positive charge. The monomers of the invention that comprise R192D/Q/F/S/T may also comprise additional modifications which improve the ability of mutant pores formed from the monomers to interact with and characterise analytes, such as polynucleotides.

[0160] The CsgG monomers of some embodiments of the invention that comprise a deletion of V105, A106 and I107, a deletion of F193, I194, D195, Y196, Q197, R198 and L199 or a deletion of D195, Y196, Q197, R198 and L199, and/or F191 T display an increased accuracy when characterizing (or sequencing) target polynucleotides.

[0161] Pores comprising the CsgG monomers of some embodiments of the invention that comprise K94Q or K94N show a reduction in the number of noisy pores (namely those pores that give rise to an increased signal:noise ratio) as compared to identical pores without the mutation at 94 when characterizing (or sequencing) target polynucleotides. Position 94 is found within the vestibule of the pore and was found to be a particularly sensitive position in relation to the noise of the current signal. Pores comprising the CsgG monomers of some embodiments of the invention that comprise T104K or T104R, N91R, E101K/N/Q/T/H, E44N/Q, Q114K, A99R, I95R, N91R, L90R, E44Q/N and/or Q42K all demonstrate an improved ability to capture target polynucleotides when used to characterize (or sequence) target polynucleotides as compared to identical pores without substitutions at these positions.

[0162] Characterisation, such as sequencing, of a polynucleotide using a transmembrane pore may be carried out such as disclosed in International Application No. PCT/GB2012/052343 (published as WO 2013/041878). As the target polynucleotide moves with respect to, or through the pore, the analyte may be characterised from the distinctive ion current signature produced, typically by measuring the ion current flow through the pore. The level of current measured at any particular time is typically dependent on a group of k polymer (for example nucleotide) units where k is a positive integer and the typical current signature may be represented as a series of current levels indicative of a particular k-mer. The movement of the polynucleotide with respect to, such as through, the pore can be viewed as movement from one k-mer to another or from k-mer to k-mer. Analytical techniques to characterise the polynucleotide may for example involve the use of an HMM, a neural network and for example a Forwards Backwards algorithm or Viterbi algorithm to determine the likelihood of the series of measurements corresponding to a particular sequence. Alternatively the polynucleotide may be characterised by determining a feature vector and comparing the feature vector to another feature vector, which may be known, such

as disclosed in International Application No. PCT/GB2013/050381 (published as WO 2013/121224). However, the analytical techniques used to characterise the polynucleotide are not necessarily restricted to the above examples.

[0163] When a monomer of the invention forms a transmembrane pore and is used with a polynucleotide binding protein to characterise a target polynucleotide, some of the modified positions interact with the polynucleotide binding protein. For example, when the monomer forms a transmembrane pore and is used with a polynucleotide binding protein to characterise a target polynucleotide, R97W interacts with the polynucleotide binding protein. Modifying the CsgG monomer in accordance with the invention typically provides more consistent movement of the target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. The modification(s) typically provide more consistent movement from one k-mer to another or from k-mer to k-mer as the target polynucleotide moves with respect to, such as through, the pore. The modification(s) typically allow the target polynucleotide to move with respect to, such as through, the transmembrane pore more smoothly. The modification(s) typically provide more regular or less irregular movement of the target polynucleotide with respect to, such as through, the transmembrane pore.

[0164] Modifying the CsgG monomer in accordance with the invention (e.g. R97W) typically reduces the amount of slipping forward associated with the movement of the target polynucleotide with respect to, such as through, a pore comprising the monomer. Some helicases including the Dda helicase used in the Example move along the polynucleotide in a 5' to 3' direction. When the 5' end of the polynucleotide (the end away from which the helicase moves) is captured by the pore, the helicase works with the direction of the field resulting from the applied potential and moves the threaded polynucleotide into the pore and into the trans chamber. Slipping forward involves the DNA moving forwards relative to the pore (i.e. towards its 3' end and away from its 5' end) at least 4 consecutive nucleotides and typically more than 10 consecutive nucleotides. Slipping forward may involve movement forward of 100 consecutive nucleotides or more and this may happen more than once in each strand.

[0165] Modifying the CsgG monomer in accordance with the invention (e.g. R97W) typically reduces the noise associated with the movement of the target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. Unwanted movement of the target polynucleotide in any dimension as the signal is being analysed typically results in noise in the current signature or level for the k-mer. The modification may reduce this noise by reducing unwanted movement associated with one or more k-mers, such as each k-mer, in the target polynucleotide. The modification may reduce the noise associated with the current level or signature for one or more k-mers, such as each k-mer; in the target polynucleotide.

[0166] The enzyme motors employed for moving the polynucleotide have multiple sub-steps in the full catalytic cycle where ATP is hydrolysed to move the polynucleotide forward one base (eg. binding ATP.Mg, hydrolysing to produce ADP.P.Mg, moving the polynucleotide one base forward, and releasing the ADP/P/Mg by-products). Each sub-step process has a characteristic dwell time distribution determined by the kinetics of the process. If any of these sub-steps of the catalytic cycle move the position of the polynucleotide in the reader (e.g. by moving the polynucle-

otide relative to the enzyme, or by changing the position of the enzyme on the top of the pore) then this may be observed as a change in current through the pore, as long the change lasts sufficiently long to be detected by the acquisition electronics. If the sub-step processes result in no change of conformation or shift in polynucleotide, or occur too quickly to observe, then in an ideal system the full catalytic cycle will result in only one step change in current for the polynucleotide moving one integer base forward.

[0167] For pores that do not contain R97W (eg Pro-CP1-Eco-(WT-Y51A/F56Q-StrepII(C))9), we observe long dwell time levels where predicted by the model, with an approximately exponential dwell distribution that is dependent on ATP.Mg concentration. For poreAQ we also short-lived substeps current levels in between the major levels, as marked in FIG. 48. Because the sub-step current levels are short-lived, they are most easily observed in the gap between two widely separated current levels. The sub-steps levels correspond to an intermediate approximately 0.5 base movement of the polynucleotide, and under these conditions have an ATP.Mg independent dwell time of approximately 3 milliseconds.

[0168] Pores containing R97W (e.g. Pro-CP1-Eco-(WT-Y51A/F56Q/R97W-StrepII(C))9) shows similar longer lived main levels with ATP.Mg dependent dwell times, but shows no signs of distinct intermediate sub-step current levels under these conditions or at this acquisition frequency (possible explanations being that they do not occur, occur too quickly to be observed, or that the substeps do occur and are slow enough in principle to be observed but that in practice they are not observed due to for example the way in which the enzyme interacts with the pore).

[0169] The raw data traces (FIG. 48) show the ionic current (y-axis, pA) vs. time (x-axis, seconds) trace of an enzyme controlled DNA strand translocation through a nanopore for the pores Pro-CP1-Eco-(WT-Y51A/F56Q/R97W-StrepII(C))9 (Pore 97W) and Pro-CP1-Eco-(WT-Y51A/F56Q-StrepII(C))9 (Pore AQ). Each current level is the result of the sequence held in the nanopore reader altering the flow of ions, and step-wise changes in current are observed when the polynucleotide changes position in the nanopore, for example when the enzyme moves the entire strand forward one base. In this case the DNA strand contains in part a repeating sequence (GGTT)_n. The data was acquired by loading a Dda enzyme onto synthetic DNA polynucleotides and running on a MinION recording raw data output (C is buffer: 500 mM KCl, 25 mM HEPES, pH8, 0.6 mM MgCl₂, 0.6 mM ATP, 140 mV, 37 deg C., 5 kHz acquisition frequency). Pore97W only shows the main current levels from integer step-wise movements of the polynucleotide, with no significant data density between the levels. In comparison, PoreAQ has significant intermediate sub-step levels, as marked by the arrows in FIG. 48.

[0170] The mutant monomers preferably have improved polynucleotide reading properties i.e. display improved polynucleotide capture and nucleotide discrimination. In particular, pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily than the wild type. In addition, pores constructed from the mutant monomers preferably display an increased current range, which makes it easier to discriminate between different nucleotides, and a reduced variance of states, which increases the signal-to-noise ratio.

[0171] In addition, the number of nucleotides contributing to the current as the polynucleotide moves through pores constructed from the mutants is preferably decreased. This makes it easier to identify a direct relationship between the observed current as the polynucleotide moves through the pore and the polynucleotide sequence. In addition, pores constructed from the mutant monomers may display an increased throughput, i.e. are more likely to interact with an analyte, such as a polynucleotide. This makes it easier to characterise analytes using the pores. Pores constructed from the mutant monomers may insert into a membrane more easily. A mutant monomer of the invention comprises a variant of the sequence shown in SEQ ID NO: 2. SEQ ID NO: 2 is the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. A variant of SEQ ID NO: 2 is a polypeptide that has an amino acid sequence which varies from that of SEQ ID NO: 2 and which retains its ability to form a pore. The ability of a variant to form a pore can be assayed using any method known in the art. For instance, the variant may be inserted into an amphiphilic layer along with other appropriate subunits and its ability to oligomerise to form a pore may be determined. Methods are known in the art for inserting subunits into membranes, such as amphiphilic layers. For example, subunits may be suspended in a purified form in a solution containing a triblock copolymer membrane such that it diffuses to the membrane and is inserted by binding to the membrane and assembling into a functional state.

[0172] In all of the discussion herein, the standard one letter codes for amino acids are used. These are as follows: alanine (A), arginine (R), asparagine (N), aspartic acid (D), cysteine (C), glutamic acid (E), glutamine (Q), glycine (G), histidine (H), isoleucine (I), leucine (L), lysine (K), methionine (M), phenylalanine (F), proline (P), serine (S), threonine (T), tryptophan (W), tyrosine (Y) and valine (V). Standard substitution notation is also used, i.e. Q42R means that Q at position 42 is replaced with R.

[0173] In one embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (a) one or more mutations at the following positions (i.e. mutations at one or more of the following positions) I41, R93, A98, Q100, G103, T104, A106, I107, N108, L113, S115, T117, Y130, K135, E170, S208, D233, D238 and E244 and/or (b) one or more of D43S, E44S, F48S/N/Q/Y/W/I/V/H/R/K, Q87N/R/K, N91K/R, K94R/F/Y/W/L/S/N, R97F/Y/W/V/I/K/S/Q/H, E101I/L/A/H, N102K/Q/L/T/V/S/S1, R110F/G/N, Q114R/K, R142Q/S, T150Y/A/V/L/S/Q/N, R192D/Q/F/S/T and D248S/N/Q/K/R. The variant may comprise (a); (b); or (a) and (b).

[0174] In some embodiments of the invention, the variant of SEQ ID NO: 2 comprises R97W.

[0175] In some embodiments of the invention, the variant of SEQ ID NO: 2 comprises R192D/Q/F/S/T, preferably R192D/Q, more preferably R192D. In (a), the variant may comprise modifications at any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 or 19 of the positions. In (a), the variant preferably comprises one or more of I41N, R93F/Y/W/L/I/V/N/Q/S, A98K/R, Q100K/R, G103F/W/S/N/K/R, T104R/K, A106R/K, I107R/K/W/F/Y/L/V, N108R/K, L113K/R, S115R/K, T117R/K, Y130W/F/H/O/N, K135L/V/N/Q/S, E170S/N/Q/K/R, S208V/I/F/W/Y/L/T, D233S/N/Q/K/R, D238S/N/Q/K/R and E244S/N/Q/K/R.

[0176] In (a), the variant preferably comprises one or more modifications which provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. In particular, in (a), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) R93, G103 and I107. The variant may comprise R93; G103; I107; R93 and G103; R93 and I107; G103 and I107; or R93, G103 and I107. The variant preferably comprises one or more of R93F/Y/V/L/I/V/N/Q/S, G103F/W/S/N/K/R and I107R/K/W/F/Y/L/V. These may be present in any combination shown for the positions R93, G103 and I107.

[0177] In (a), the variant preferably comprises one or more modifications which allow pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily. In particular, in (a), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) I41, T104, A106, N108, L113, S115, T117, E170, D233, D238 and E244. The variant may comprise modifications at any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or 11 of the positions. The variant preferably comprises one or more of I41N, T104R/K, A106R/K, N108R/K, L113K/R, S115R/K, T117R/K, E170S/N/Q/K/R, D233S/N/Q/K/R, D238S/N/Q/K/R and E244S/N/Q/K/R. Additionally or alternatively the variant may comprise (c) Q42K/R, E44N/Q, L90R/K, N91R/K, I95R/K, A99R/K, E101H/K/N/Q/T and/or Q114K/R.

[0178] In (a), the variant preferably comprises one or more modifications which provide more consistent movement and increase capture. In particular, in (a) the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) (i) A98, (ii) Q100, (iii) G103 and (iv) I107. The variant preferably comprises one or more of (i) A98R/K, (ii) Q100K/R, (iii) G103K/R and (iv) I107R/K. The variant may comprise {i}; {ii}; {iii}; {iv}; {i,ii}; {i,iii}; {i,iv}; {ii,iii}; {ii,iv}; {iii,iv}; {i,ii,iii}; {i,ii,iv}; {i,iii,iv}; {ii,iii,iv}; or {i,ii,iii,iv}.

[0179] Particularly preferred mutant monomers which provide for increased capture of analytes, such as a polynucleotide include a mutation at one or more of positions Q42, E44, E44, L90, N91, I95, A99, E101 and Q114, which mutation removes the negative charge and/or increases the positive charge at the mutated positions. In particular, the following mutations may be included in a mutant monomer of the invention to produce a CsgG pore that has an improved ability to capture an analyte, preferably a polynucleotide: Q42K, E44N, E44Q, L90R, N91R, I95R, A99R, E101H, E101K, E101N, E101Q, E101T and Q114K. Examples of particular mutant monomers which comprise one of these mutations in combination with other beneficial mutations are described in Example 11.

[0180] In (a), the variant preferably comprises one or more modifications which provide increased characterisation accuracy. In particular, in (a), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) Y130, K135 and S208, such as Y130; K135; S208; Y130 and K135; Y130 and S208; K135 and S208; or Y130, K135 and S208. The variant preferably comprises one or more of Y130W/F/H/Q/N, K135L/V/N/Q/S and R142Q/S. These

substitutions may be present in any number and combination as set out for Y130, K135 and S208.

[0181] In (b), the variant may comprise any number and combination of the substitutions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 of the substitutions. In (b), the variant preferably comprises one or more modifications which provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. In particular, in (b), the variant preferably comprises one or more one or more of (i) Q87N/R/K, (ii) K94R/F/Y/W/L/S/N, (iii) R97F/Y/W/V/I/K/S/Q/H, (iv) N102K/Q/L/I/V/S/H and (v) R110F/G/N. More preferably, the variant comprises K94D or K94Q and/or R97W or R97Y.

[0182] The variant may comprise {i}; {ii}; {iii}; {iv}; {v}; {i,ii}; {i,iii}; {i,iv}; {i,v}; {ii,iii}; {ii,iv}; {ii,v}; {iii,iv}; {iii,v}; {iv,v}; {i,ii,iii}; {i,ii,iv}; {i,ii,v}; {i,iii,iv}; {i,iii,v}; {i,iv,v}; {ii,iii,iv}; {ii,iii,v}; {ii,iv,v}; {iii,iv,v}; {i,ii,iii,iv}; {i,ii,iii,v}; {i,ii,iv,v}; {i,iii,iv,v}; {ii,iii,iv,v}; or {i,ii,iii,iv,v}. Other preferred variants that are modified to provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer include (vi) R93W and R93Y. A preferred variant may comprise R93W and R97W, R93Y and R97W, R93W and R97W, or more preferably R93Y and R97Y. The variant may comprise {vi}; {i,vi}; {ii,vi}; {iii,vi}; {iv,vi}; {v,vi}; {i,ii,vi}; {i,iii,vi}; {i,iv,vi}; {i,v,vi}; {ii,iii,vi}; {ii,iv,vi}; {ii,v,vi}; {iii,iv,vi}; {iii,v,vi}; {iv,v,vi}; {i,ii,iii,vi}; {i,ii,iv,vi}; {i,ii,v,vi}; {i,iii,iv,vi}; {i,iii,v,vi}; {i,iv,v,vi}; {ii,iii,iv,vi}; {ii,iii,v,vi}; {ii,iv,v,vi}; {iii,iv,v,vi}; {i,ii,iii,iv,vi}; {i,ii,iii,v,vi}; {i,ii,iv,v,vi}; {i,iii,iv,v,vi}; {ii,iii,iv,v,vi}; or {i,ii,iii,iv,v,vi}.

[0183] In (b), the variant preferably comprises one or more modifications which allow pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily. In particular, in (b), the variant preferably comprises one or more of (i) D43S, (ii) E44S, (iii) N91K/R, (iv) Q114R/K and (v) D248S/N/Q/K/R. The variant may comprise {i}; {ii}; {iii}; {iv}; {v}; {i,ii}; {i,iii}; {i,iv}; {i,v}; {ii,iii}; {ii,iv}; {ii,v}; {iii,iv}; {iii,v}; {iv,v}; {i,ii,iii}; {i,ii,iv}; {i,ii,v}; {i,iii,iv}; {i,iii,v}; {i,iv,v}; {ii,iii,iv}; {ii,iii,v}; {ii,iv,v}; {iii,iv,v}; {i,ii,iii,iv}; {i,ii,iii,v}; {i,ii,iv,v}; {i,iii,iv,v}; {ii,iii,iv,v}; or {i,ii,iii,iv,v}.

[0184] In (b), the variant preferably comprises one or more modifications which provide more consistent movement and increase capture. In particular, in (b), the variant preferably comprises one or more of Q87R/K, E101I/L/A/H and N102K, such as Q87R/K; E101I/L/A/H; N102K; Q87R/K and E101I/L/A/H; Q87R/K and N102K; E101I/L/A/H and N102K; or Q87R/K, E101I/L/A/H and N102K.

[0185] In (b), the variant preferably comprises one or more modifications which provide increased characterisation accuracy. In particular, in (a), the variant preferably comprises F48S/N/Q/Y/W/I/V.

[0186] In (b), the variant preferably comprises one or more modifications which provide increased characterisation accuracy and increased capture. In particular, in (a), the variant preferably comprises F48H/R/K.

[0187] The variant may comprise modifications in both (a) and (b) which provide more consistent movement. The variant may comprise modifications in both (a) and (b) which provide increased capture.

[0188] The invention provides variants of SEQ ID NO: 2 which provide an increased throughput of an assay for

characterising an analyte, such as a polynucleotide, using a pore comprising the variant. Such variants may comprise a mutation at K94, preferably K94Q or K94N, more preferably K94Q. Examples of particular mutant monomers which comprise a K94Q or K94N mutation in combination with other beneficial mutations are described in Examples 10 and 11.

[0189] The invention provides variants of SEQ ID NO: 2 which provide increased characterisation accuracy in an assay for characterising an analyte, such as a polynucleotide, using a pore comprising the variant. Such variants include variants that comprise: a mutation at F191, preferably F191T; deletion of V105-I107; deletion of F193-L199 or of D195-L199; and/or a mutation at R93 and/or R97, preferably R93Y, R97Y, or more preferably, R97W, R93W or both R97Y and R97Y. Examples of particular mutant monomers which comprise one or more of these mutations in combination with other beneficial mutations are described in Example 9.

[0190] In another embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (A) deletion of one or more positions R192, F193, I194, D195, Y196, Q197, R198, L199, L200 and E201 and/or (B) deletion of one or more of V139/G140/D149/T150/V186/Q187/V204/G205 (called band 1 herein), G137/G138/Q151/Y152/Y184/E185/Y206/T207 (called band 2 herein) and A141/R142/G147/A148/A188/G189/G202/E203 (called band 3 herein).

[0191] In (A), the variant may comprise deletion of any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 of the positions. In (A), the variant preferably comprises deletion of

[0192] D195, Y196, Q197, R198 and L199;

[0193] R192, F193, I194, D195, Y196, Q197, R198, L199 and L200;

[0194] Q197, R198, L199 and L200;

[0195] I194, D195, Y196, Q197, R198 and L199;

[0196] D195, Y196, Q197, R198, L199 and L200;

[0197] Y196, Q197, R198, L199, L200 and E201;

[0198] Q197, R198, L199, L200 and E201;

[0199] Q197, R198, L199; or

[0200] F193, I194, D195, Y196, Q197, R198 and L199.

More preferably, the variant comprises deletion of D195, Y196, Q197, R198 and L199 or F193, I194, D195, Y196, Q197, R198 and L199. In (B), any number and combination of bands 1 to 3 may be deleted, such as band 1; band 2; band 3; bands 1 and 2; bands 1 and 3; bands 2 and 3; or bands 1, 2 and 3.

[0201] The variant may comprise deletions according to (A); (B); or (A) and (B).

[0202] The variants comprising deletion of one or more positions according to (A) and/or (B) above may further comprise any of the modifications or substitutions discussed above and below. If the modifications or substitutions are made at one or more positions which appear after the deletion positions in SEQ ID NO: 2, the numbering of the one or more positions of the modifications or substitutions must be adjusted accordingly. For instance, if L199 is deleted, E244 becomes E243. Similarly, if band 1 is deleted, R192 becomes R186.

[0203] In another embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (C) deletion of one or more positions V105, A106 and I107. The

[illegible][illegible]

[illegible][illegible]

$x\}$ {ii,iii,iv,v,vi,vii,viii,ix,x} {ii,iii,iv,v,vi,vii,viii,x,xi} {ii,
iii,iv,v,vi,vii,ix,x,xi} {ii,iii,iv,v,vi,viii,ix,x,xi} {ii,iii,iv,v,vii,
viii,ix,x,xi} {ii,iii,iv,vi,vii,viii,ix,x,xi} {ii,iii,v,vi,vii,viii,ix,
x,xi} {ii,iv,v,vi,vii,viii,ix,x,xi} {iii,iv,v,vi,vii,viii,ix,x,xi}
{i,ii,iii,iv,v,vi,vii,viii,ix,x} {i,ii,iii,iv,v,vi,vii,viii,ix,xi} {i,ii,
iii,iv,v,vi,vii,viii,x,xi} {i,ii,iii,iv,v,vi,vii,ix,x,xi} {i,ii,iii,iv,v,
vi,viii,ix,x,xi} {i,ii,iii,iv,v,vii,viii,ix,x,xi} {i,ii,iii,iv,vi,vii,
viii,ix,x,xi} {i,ii,iii,v,vi,vii,viii,ix,x,xi} {i,ii,iv,v,vi,vii,viii,ix,
x,xi} {i,iii,iv,v,vi,vii,viii,ix,x,xi} {ii,iii,iv,v,vi,vii,viii,ix,x,
xi} or {i,ii,iii,iv,v,vi,vii,viii,ix,x,xi}.

[0209] If the variant comprises any one of (i) and (iii) to (xi), it may further comprise a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0210] In (i), the variant may comprises mutations at any number and combination of N40, D43, E44, S54, S57, Q62, R97, E101, E124, E131, R142, T150 and R192. In (i), the variant preferably comprises one or more mutations at at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, S54, S57, Q62, E101, E131 and T150. In (i), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, E101 and E131. In (i), the variant preferably comprises a mutation at S54 and/or S57. In (i), the variant more preferably comprises a mutation at (a) S54 and/or S57 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. If S54 and/or S57 are deleted in (xi), it/they cannot be mutated in (i) and vice versa. In (i), the variant preferably comprises a mutation at T150, such as T150I. Alternatively the variant preferably comprises a mutation at (a) T150 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. In (i), the variant preferably comprises a mutation at Q62, such as Q62R or Q62K. Alternatively the variant preferably comprises a mutation at (a) Q62 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. The variant may comprise a mutation at D43, E44, Q62 or any combination thereof, such as D43, E44, Q62, D43/E44, D43/Q62, E44/Q62 or D43/E44/Q62. Alternatively the variant preferably comprises a mutation at (a) D43, E44, Q62, D43/E44, D43/Q62, E44/Q62 or D43/E44/Q62 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0211] In (ii) and elsewhere in this application where different positions are separated by the/symbol, the/symbol means “and” such that Y51/N55 is Y51 and N55. In (ii), the variant preferably comprises mutations at Y51/N55. It has been proposed that the constriction in CsgG is composed of three stacked concentric rings formed by the side chains of residues Y51, N55 and F56 (Goyal et al, 2014, Nature, 516, 250-253). Mutation of these residues in (ii) may therefore decrease the number of nucleotides contributing to the current as the polynucleotide moves through the pore and thereby make it easier to identify a direct relationship between the observed current (as the polynucleotide moves through the pore) and the polynucleotide. F56 may be mutated in any of the ways discussed below with reference to variants and pores useful in the method of the invention.

[0212] In (v), the variant may comprise N102R, N102F, N102Y or N102W. The variant preferably comprises (a) N102R, N102F, N102Y or N102W and (b) a mutation at one

N55A/Y51V, F56Q/N55A/Y51A, F56Q/N55A/Y51N, F56Q/N55A/Y51Q, F56Q/N55A/Y51S, F56Q/N55A/Y51G, F56Q/N55T/Y51L, F56Q/N55T/Y51V, F56Q/N55T/Y51A, F56Q/N55T/Y51N, F56Q/N55T/Y51Q, F56Q/N55T/Y51S, F56Q/N55T/Y51G, F56R/N55Q/Y51L, F56R/N55Q/Y51V, F56R/N55Q/Y51A, F56R/N55Q/Y51N, F56R/N55Q/Y51Q, F56R/N55Q/Y51S, F56R/N55Q/Y51G, F56R/N55R/Y51L, F56R/N55R/Y51V, F56R/N55R/Y51A, F56R/N55R/Y51N, F56R/N55R/Y51Q, F56R/N55R/Y51S, F56R/N55R/Y51G, F56R/N55K/Y51L, F56R/N55K/Y51V, F56R/N55K/Y51A, F56R/N55K/Y51N, F56R/N55K/Y51Q, F56R/N55K/Y51S, F56R/N55K/Y51G, F56R/N55S/Y51L, F56R/N55S/Y51V, F56R/N55S/Y51A, F56R/N55S/Y51N, F56R/N55S/Y51Q, F56R/N55S/Y51S, F56R/N55S/Y51G, F56R/N55G/Y51L, F56R/N55G/Y51V, F56R/N55G/Y51A, F56R/N55G/Y51N, F56R/N55G/Y51Q, F56R/N55G/Y51S, F56R/N55G/Y51G, F56R/N55A/Y51L, F56R/N55A/Y51V, F56R/N55A/Y51A, F56R/N55A/Y51N, F56R/N55A/Y51Q, F56R/N55A/Y51S, F56R/N55A/Y51G, F56R/N55T/Y51L, F56R/N55T/Y51V, F56R/N55T/Y51A, F56R/N55T/Y51N, F56R/N55T/Y51Q, F56R/N55T/Y51S, F56R/N55T/Y51G, F56S/N55Q/Y51L, F56S/N55Q/Y51V, F56S/N55Q/Y51A, F56S/N55Q/Y51N, F56S/N55Q/Y51Q, F56S/N55Q/Y51S, F56S/N55Q/Y51G, F56S/N55R/Y51L, F56S/N55R/Y51V, F56S/N55R/Y51A, F56S/N55R/Y51N, F56S/N55R/Y51Q, F56S/N55R/Y51S, F56S/N55R/Y51G, F56S/N55K/Y51L, F56S/N55K/Y51V, F56S/N55K/Y51A, F56S/N55K/Y51N, F56S/N55K/Y51Q, F56S/N55K/Y51S, F56S/N55K/Y51G, F56S/N55S/Y51L, F56S/N55S/Y51V, F56S/N55S/Y51A, F56S/N55S/Y51N, F56S/N55S/Y51Q, F56S/N55S/Y51S, F56S/N55S/Y51G, F56S/N55G/Y51L, F56S/N55G/Y51V, F56S/N55G/Y51A, F56S/N55G/Y51N, F56S/N55G/Y51Q, F56S/N55G/Y51S, F56S/N55G/Y51G, F56S/N55A/Y51L, F56S/N55A/Y51V, F56S/N55A/Y51A, F56S/N55A/Y51N, F56S/N55A/Y51Q, F56S/N55A/Y51S, F56S/N55A/Y51G, F56S/N55T/Y51L, F56S/N55T/Y51V, F56S/N55T/Y51A, F56S/N55T/Y51N, F56S/N55T/Y51Q, F56S/N55T/Y51S, F56S/N55T/Y51G, F56G/N55Q/Y51L, F56G/N55Q/Y51V, F56G/N55Q/Y51A, F56G/N55Q/Y51N, F56G/N55Q/Y51Q, F56G/N55Q/Y51S, F56G/N55Q/Y51G, F56G/N55R/Y51L, F56G/N55R/Y51V, F56G/N55R/Y51A, F56G/N55R/Y51N, F56G/N55R/Y51Q, F56G/N55R/Y51S, F56G/N55R/Y51G, F56G/N55K/Y51L, F56G/N55K/Y51V, F56G/N55K/Y51A, F56G/N55K/Y51N, F56G/N55K/Y51Q, F56G/N55K/Y51S, F56G/N55K/Y51G, F56G/N55S/Y51L, F56G/N55S/Y51V, F56G/N55S/Y51A, F56G/N55S/Y51N, F56G/N55S/Y51Q, F56G/N55S/Y51S, F56G/N55S/Y51G, F56G/N55G/Y51L, F56G/N55G/Y51V, F56G/N55G/Y51A, F56G/N55G/Y51N, F56G/N55G/Y51Q, F56G/N55G/Y51S, F56G/N55G/Y51G, F56G/N55A/Y51L, F56G/N55A/Y51V, F56G/N55A/Y51A, F56G/N55A/Y51N, F56G/N55A/Y51Q, F56G/N55A/Y51S, F56G/N55A/Y51G, F56G/N55T/Y51L, F56G/N55T/Y51V, F56G/N55T/Y51A, F56G/N55T/Y51N, F56G/N55T/Y51Q, F56G/N55T/Y51S, F56G/N55T/Y51G, F56A/N55Q/Y51L, F56A/N55Q/Y51V, F56A/N55Q/Y51A, F56A/N55Q/Y51N, F56A/N55Q/Y51Q, F56A/N55Q/Y51S, F56A/N55Q/Y51G, F56A/N55R/Y51L, F56A/N55R/Y51V, F56A/N55R/Y51A, F56A/N55R/Y51N, F56A/N55R/Y51Q, F56A/N55R/Y51S, F56A/N55R/Y51G, F56A/N55K/Y51L, F56A/N55K/Y51V, F56A/N55K/Y51A, F56A/N55K/Y51N, F56A/N55K/Y51Q, F56A/N55K/Y51S, F56A/N55K/Y51G, F56A/

N55S/Y51L, F56A/N55S/Y51V, F56A/N55S/Y51A, F56A/N55S/Y51N, F56A/N55S/Y51Q, F56A/N55S/Y51S, F56A/N55S/Y51G, F56A/N55G/Y51L, F56A/N55G/Y51V, F56A/N55G/Y51A, F56A/N55G/Y51N, F56A/N55G/Y51Q, F56A/N55G/Y51S, F56A/N55G/Y51G, F56A/N55A/Y51L, F56A/N55A/Y51V, F56A/N55A/Y51A, F56A/N55A/Y51N, F56A/N55A/Y51Q, F56A/N55A/Y51S, F56A/N55A/Y51G, F56A/N55T/Y51L, F56A/N55T/Y51V, F56A/N55T/Y51A, F56A/N55T/Y51N, F56A/N55T/Y51Q, F56A/N55T/Y51S, F56A/N55T/Y51G, F56K/N55Q/Y51L, F56K/N55Q/Y51V, F56K/N55Q/Y51A, F56K/N55Q/Y51N, F56K/N55Q/Y51Q, F56K/N55Q/Y51S, F56K/N55Q/Y51G, F56K/N55R/Y51L, F56K/N55R/Y51V, F56K/N55R/Y51A, F56K/N55R/Y51N, F56K/N55R/Y51Q, F56K/N55R/Y51S, F56K/N55R/Y51G, F56K/N55K/Y51L, F56K/N55K/Y51V, F56K/N55K/Y51A, F56K/N55K/Y51N, F56K/N55K/Y51Q, F56K/N55K/Y51S, F56K/N55K/Y51G, F56K/N55S/Y51L, F56K/N55S/Y51V, F56K/N55S/Y51A, F56K/N55S/Y51N, F56K/N55S/Y51Q, F56K/N55S/Y51S, F56K/N55S/Y51G, F56K/N55G/Y51L, F56K/N55G/Y51V, F56K/N55G/Y51A, F56K/N55G/Y51N, F56K/N55G/Y51Q, F56K/N55G/Y51S, F56K/N55G/Y51G, F56K/N55A/Y51L, F56K/N55A/Y51V, F56K/N55A/Y51A, F56K/N55A/Y51N, F56K/N55A/Y51Q, F56K/N55A/Y51S, F56K/N55A/Y51G, F56K/N55T/Y51L, F56K/N55T/Y51V, F56K/N55T/Y51A, F56K/N55T/Y51N, F56K/N55T/Y51Q, F56K/N55T/Y51S, F56K/N55T/Y51G, F56E/N55R, F56E/N55K, F56D/N55R, F56D/N55K, F56R/N55E, F56R/N55D, F56K/N55E or F56K/N55D.

[0223] In (ii), the variant preferably comprises Y51R/F56Q, Y51N/F56N, Y51M/F56Q, Y51L/F56Q, Y51I/F56Q, Y51V/F56Q, Y51A/F56Q, Y51P/F56Q, Y51G/F56Q, Y51C/F56Q, Y51Q/F56Q, Y51N/F56Q, Y51S/F56Q, Y51E/F56Q, Y51D/F56Q, Y51K/F56Q or Y51H/F56Q.

[0224] In (ii), the variant preferably comprises Y51T/F56Q, Y51Q/F56Q or Y51A/F56Q.

[0225] In (ii), the variant preferably comprises Y51T/F56F, Y51T/F56M, Y51T/F56L, Y51T/F56I, Y51T/F56V, Y51T/F56A, Y51T/F56P, Y51T/F56G, Y51T/F56C, Y51T/F56Q, Y51T/F56N, Y51T/F56T, Y51T/F56S, Y51T/F56E, Y51T/F56D, Y51T/F56K, Y51T/F56H or Y51T/F56R.

[0226] In (ii), the variant preferably comprises Y51T/N55Q, Y51T/N55S or Y51T/N55A.

[0227] In (ii), the variant preferably comprises Y51A/F56F, Y51A/F56L, Y51A/F56I, Y51A/F56V, Y51A/F56A, Y51A/F56P, Y51A/F56G, Y51A/F56C, Y51A/F56Q, Y51A/F56N, Y51A/F56T, Y51A/F56S, Y51A/F56E, Y51A/F56D, Y51A/F56K, Y51A/F56H or Y51A/F56R.

[0228] In (ii), the variant preferably comprises Y51C/F56A, Y51E/F56A, Y51D/F56A, Y51K/F56A, Y51H/F56A, Y51Q/F56A, Y51N/F56A, Y51S/F56A, Y51P/F56A or Y51V/F56A.

[0229] In (xi), the variant preferably comprises deletion of Y51/P52, Y51/P52/A53, P50 to P52, P50 to A53, K49 to Y51, K49 to A53 and replacement with a single proline (P), K49 to S54 and replacement with a single P, Y51 to A53, Y51 to S54, N55/F56, N55 to S57, N55/F56 and replacement with a single P, N55/F56 and replacement with a single glycine (G), N55/F56 and replacement with a single alanine (A), N55/F56 and replacement with a single P and Y51N, N55/F56 and replacement with a single P and Y51Q, N55/F56 and replacement with a single P and Y51S, N55/F56 and replacement with a single G and Y51N, N55/F56 and

[illegible]

E203Q, D149R/E185Q/E201Q/E203N, D149R/E185Q/
E201N/E203Q, D149R/E185N/E201Q/E203Q, D149R/
E185Q/E201Q/E203Q, D149R/E185N/D195N/E201N/
E203N, D149R/E185Q/D195N/E201N/E203N, D149R/
E185N/D195Q/E201N/E203N, D149R/E185N/D195N/
E201Q/E203N, D149R/E185Q/D195N/E201N/E203Q,
D149R/E185Q/D195Q/E201N/E203N, D149R/E185Q/
D195N/E201Q/E203N, D149R/E185Q/D195N/E201N/
E203Q, D149R/E185N/D195Q/E201Q/E203N, D149R/
E185N/D195Q/E201N/E203Q, D149R/E185N/D195N/
E201Q/E203Q, D149R/E185Q/D195Q/E201Q/E203N,
D149R/E185Q/D195Q/E201N/E203Q, D149R/E185Q/
D195N/E201Q/E203Q, D149R/E185N/D195Q/E201Q/
E203Q, D149R/E185Q/D195Q/E201Q/E203Q, D149N/
E185R/D195N/E201N/E203N, D149Q/E185R/D195N/
E201N/E203N, D149N/E185R/D195Q/E201N/E203N,
D149N/E185R/D195N/E201Q/E203N, D149N/E185R/
D195N/E201N/E203Q, D149Q/E185R/D195Q/E201N/
E203N, D149Q/E185R/D195N/E201Q/E203N, D149Q/
E185R/D195N/E201N/E203Q, D149N/E185R/D195Q/
E201Q/E203N, D149N/E185R/D195Q/E201N/E203Q,
D149N/E185R/D195N/E201Q/E203Q, D149Q/E185R/
D195Q/E201Q/E203N, D149Q/E185R/D195Q/E201N/
E203Q, D149Q/E185R/D195N/E201Q/E203Q, D149N/
E185R/D195Q/E201Q/E203Q, D149Q/E185R/D195Q/
E201Q/E203Q, D149N/E185R/D195N/E201R/E203N,
D149Q/E185R/D195N/E201R/E203N, D149N/E185R/
D195Q/E201R/E203N, D149N/E185R/D195N/E201R/
E203Q, D149Q/E185R/D195Q/E201R/E203N, D149Q/
E185R/D195N/E201R/E203Q, D149N/E185R/D195Q/
E201R/E203Q, D149Q/E185R/D195Q/E201R/E203Q,
E131D/K49R, E101N/N102F, E101N/N102Y, E101N/
N102W, E101F/N102F, E101F/N102Y, E101F/N102W,
E101Y/N102F, E101Y/N102Y, E101Y/N102W, E101W/
N102F, E101W/N102Y, E101W/N102W, E101N/N102R,
E101F/N102R, E101Y/N102R or E101W/N102F.

[0231] Preferred variants of the invention which form pores in which fewer nucleotides contribute to the current as the polynucleotide moves through the pore comprise Y51A/F56A, Y51A/F56N, Y51I/F56A, Y51L/F56A, Y51T/F56A, Y51I/F56N, Y51L/F56N or Y51T/F56N or more preferably Y51I/F56A, Y51L/F56A or Y51T/F56A. As discussed above, this makes it easier to identify a direct relationship between the observed current (as the polynucleotide moves through the pore) and the polynucleotide.

[0232] Preferred variants which form pores displaying an increased range comprise mutations at the following positions:

[0233] Y51, F56, D149, E185, E201 and E203:

[0234] N55 and F56;

[0235] Y51 and F56;

[0236] Y51, N55 and F56; or

[0237] F56 and N102.

[0238] Preferred variants which form pores displaying an increased range comprise:

[0239] Y51N, F56A, D149N, E185R, E201N and E203N;

[0240] N55S and F56O:

[0241] Y51A and F56A:

[0242] Y51A and F56N;

[0243] Y511 and F56A:

[0244] Y51L and F56A;

[0245] Y51T and F56A:

[0246] Y511 and F56N;

- [0247] Y51L and F56N;
 [0248] Y51T and F56N;
 [0249] Y51T and F56Q;
 [0250] Y51A, N55S and F56A;
 [0251] Y51A, N55S and F56N,
 [0252] Y51T, N55S and F56Q; or
 [0253] F56Q and N102R.
- [0254] Preferred variants which form pores in which fewer nucleotides contribute to the current as the polynucleotide moves through the pore comprise mutations at the following positions:
- [0255] N55 and F56, such as N55X and F56Q, wherein X is any amino acid; or
 [0256] Y51 and F56, such as Y51X and F56Q, wherein X is any amino acid.
- [0257] Particularly preferred variants comprise Y51A and F56Q.
- [0258] Preferred variants which form pores displaying an increased throughput comprise mutations at the following positions:
- [0259] D149, E185 and E203;
 [0260] D149, E185, E201 and E203; or
 [0261] D149, E185, D195, E201 and E203.
- [0262] Preferred variants which form pores displaying an increased throughput comprise:
- [0263] D149N, E185N and E203N;
 [0264] D149N, E185N, E201N and E203N;
 [0265] D149N, E185R, D195N, E201N and E203N; or
 [0266] D149N, E185R, D195N, E201R and E203N.
- [0267] Preferred variants which form pores in which capture of the polynucleotide is increased comprise the following mutations:
- [0268] D43N/Y51T/F56Q;
 [0269] E44N/Y51T/F56Q;
 [0270] D43N/E44N/Y51T/F56Q;
 [0271] Y51T/F56Q/Q62R;
 [0272] D43N/Y51T/F56Q/Q62R;
 [0273] E44N/Y51T/F56Q/Q62R; or
 [0274] D43N/E44N/Y51T/F56Q/Q62R.
- [0275] Preferred variants comprise the following mutations:
- [0276] D149R/E185R/E201R/E203R or Y51T/F56Q/D149R/E185R/E201R/E203R;
 [0277] D149N/E185N/E201N/E203N or Y51T/F56Q/D149N/E185N/E201N/E203N;
 [0278] E201R/E203R or Y51T/F56Q/E201R/E203R
 [0279] E201N/E203R or Y51T/F56Q/E201N/E203R;
 [0280] E203R or Y51T/F56Q/E203R;
 [0281] E203N or Y51T/F56Q/E203N;
 [0282] E201R or Y51T/F56Q/E201R;
 [0283] E201N or Y51T/F56Q/E201N;
 [0284] E185R or Y51T/F56Q/E185R;
 [0285] E185N or Y51T/F56Q/E185N;
 [0286] D149R or Y51T/F56Q/D149R;
 [0287] D149N or Y51T/F56Q/D149N;
 [0288] R142E or Y51T/F56Q/R142E;
 [0289] R142N or Y51T/F56Q/R142N;
 [0290] R192E or Y51T/F56Q/R192E; or
 [0291] R192N or Y51T/F56Q/R192N.
- [0292] Preferred variants comprise the following mutations:
- [0293] Y51A/F56Q/E101N/N102R;
 [0294] Y51A/F56Q/R97N/N102G;
 [0295] Y51A/F56Q/R97N/N102R;

- [0296] Y51A/F56Q/R97N;
 [0297] Y51A/F56Q/R97G;
 [0298] Y51A/F56Q/R97L;
 [0299] Y51A/F56Q/N102R;
 [0300] Y51A/F56Q/N102F;
 [0301] Y51A/F56Q/N102G;
 [0302] Y51A/F56Q/E101R;
 [0303] Y51A/F56Q/E101F;
 [0304] Y51A/F56Q/E101N; or
 [0305] Y51A/F56Q/E101G

[0306] The variant preferably further comprises a mutation at T150. A preferred variant which forms a pore displaying an increased insertion comprises T150I. A mutation at T150, such as T150I, may be combined with any of the mutations or combinations of mutations discussed above.

[0307] A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) a mutation at Y51 and/or F56. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) Y51R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and/or F56R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) Y51L/V/A/N/Q/S/G and/or F56A/Q/N. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) Y51A and/or F56Q. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A and F56Q.

[0308] In the mutant monomers of the invention, the variant of SEQ ID NO: 2 preferably comprises a mutation at R192. The variant preferably comprises R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) a mutation at Y51 and/or F56 and (c) a mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and/or F56R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and (c) a mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51L/V/A/N/Q/S/G and/or F56A/Q/N and (c) a mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51A and/or F56Q and (c) a mutation at R192, such as R192 D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A, F56Q and R192D. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A, F56Q and R192Q. In the paragraphs above where different amino acids at a specific position are separated by the/symbol, the/symbol means "or". For instance, R192D/Q means R192D or R192Q.

[0309] In the mutant monomers of the invention, the variant of SEQ ID NO: 2 preferably comprises a mutation at R93. A preferred variant of SEQ ID NO: 2 comprises (a) R93W and (b) a mutation at Y51 and/or F56, preferably Y51A and F56Q. D or R192N.deletion of V105, A106 and I107.

[0310] Any of the above preferred variants of SEQ ID NO: 2 may comprise a K94N/Q mutation. Any of the above preferred variants of SEQ ID NO: 2 may comprise a F191T mutation. The invention also provides a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 comprising the combination of mutations present in a variant disclosed in the Examples.

[0311] Methods for introducing or substituting naturally-occurring amino acids are well known in the art. For instance, methionine (M) may be substituted with arginine (R) by replacing the codon for methionine (ATG) with a codon for arginine (CGT) at the relevant position in a polynucleotide encoding the mutant monomer. The polynucleotide can then be expressed as discussed below.

[0312] Methods for introducing or substituting non-naturally-occurring amino acids are also well known in the art. For instance, non-naturally-occurring amino acids may be introduced by including synthetic aminoacyl-tRNAs in the IVTT system used to express the mutant monomer. Alternatively, they may be introduced by expressing the mutant monomer in *E. coli* that are auxotrophic for specific amino acids in the presence of synthetic (i.e. non-naturally-occurring) analogues of those specific amino acids. They may also be produced by naked ligation if the mutant monomer is produced using partial peptide synthesis.

Variants

[0313] In addition to the specific mutations discussed above, the variant may include other mutations. Over the entire length of the amino acid sequence of SEQ ID NO: 2, a variant will preferably be at least 50% homologous to that sequence based on amino acid identity. More preferably, the variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95%, amino acid identity over a stretch of 100 or more, for example 125, 150, 175 or 200 or more, contiguous amino acids ("hard homology").

[0314] Standard methods in the art may be used to determine homology. For example the UWGCG Package provides the BESTFIT program which can be used to calculate homology, for example used on its default settings (Devereux et al (1984) *Nucleic Acids Research* 12, p387-395). The PILEUP and BLAST algorithms can be used to calculate homology or line up sequences (such as identifying equivalent residues or corresponding sequences (typically on their default settings)), for example as described in Altschul S. F. (1993) *J Mol Evol* 36:290-300; Altschul, S. F et al (1990) *J Mol Biol* 215:403-10. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov/>).

[0315] SEQ ID NO: 2 is the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. The variant of SEQ ID NO: 2 may comprise any of the substitutions present in another CsgG homologue. Preferred CsgG homologues are shown in SEQ ID NOs: 3 to 7 and 26 to 41. The variant may comprise combinations of one or more of the substitutions present in SEQ ID NOs: 3 to 7 and 26 to 41 compared with SEQ ID NO: 2. For example, mutations may be made at any one or more of the positions in SEQ ID NO: 2 that differ between SEQ ID NO: 2 and any one of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41. Such a mutation may be a substitution of an amino acid in SEQ ID NO: 2 with an amino acid from the corresponding position in any one of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41. Alternatively, the mutation at any one of these positions may be a substitution with any amino acid, or may be a deletion or

insertion mutation, such as deletion or insertion of 1 to 10 amino acids, such as of 2 to 8 or 3 to 6 amino acids. Other than the mutations disclosed herein, the amino acids that are conserved between SEQ ID NO: 2 and all of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41 are preferably present in a variant of the invention. However, conservative mutations may be made at any one or more of these positions that are conserved between SEQ ID NO: 2 and all of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41.

[0316] The invention provides a pore-forming CsgG mutant monomer that comprises any one or more of the amino acids described herein as being substituted into a specific position of SEQ ID NO: 2 at a position in the structure of the CsgG monomer that corresponds to the specific position in SEQ ID NO: 2. Corresponding positions may be determined by standard techniques in the art. For example, the PILEUP and BLAST algorithms mentioned above can be used to align the sequence of a CsgG monomer with SEQ ID NO: 2 and hence to identify corresponding residues.

[0317] In particular, the invention provides a pore-forming CsgG mutant monomer that comprises any one or more of the following:

- [0318]** a W at a position corresponding to R97 in SEQ ID NO:2;
- [0319]** a W at a position corresponding to R93 in SEQ ID NO:2;
- [0320]** a Y at a position corresponding to R97 in SEQ ID NO: 2;
- [0321]** a Y at a position corresponding to R93 in SEQ ID NO: 2;
- [0322]** a Y at each of the positions corresponding to R93 and R97 in SEQ ID NO: 2;
- [0323]** a D at the position corresponding to R192 in SEQ ID NO:2;
- [0324]** deletion of the residues at the positions corresponding to V105-I107 in SEQ ID NO:2;
- [0325]** deletion of the residues at one or more of the positions corresponding to F193 to L199 in SEQ ID NO: 2;
- [0326]** deletion of the residues the positions corresponding to F195 to L199 in SEQ ID NO: 2;
- [0327]** deletion of the residues the positions corresponding to F193 to L199 in SEQ ID NO: 2;
- [0328]** a T at the position corresponding to F191 in SEQ ID NO: 2;
- [0329]** a Q at the position corresponding to K49 in SEQ ID NO: 2;
- [0330]** a N at the position corresponding to K49 in SEQ ID NO: 2;
- [0331]** a Q at the position corresponding to K42 in SEQ ID NO: 2;
- [0332]** a Q at the position corresponding to E44 in SEQ ID NO: 2;
- [0333]** a N at the position corresponding to E44 in SEQ ID NO: 2;
- [0334]** a R at the position corresponding to L90 in SEQ ID NO: 2;
- [0335]** a R at the position corresponding to L91 in SEQ ID NO: 2;
- [0336]** a R at the position corresponding to 195 in SEQ ID NO: 2;
- [0337]** a R at the position corresponding to A99 in SEQ ID NO: 2;

[0338] a H at the position corresponding to E101 in SEQ ID NO: 2;

[0339] a K at the position corresponding to E101 in SEQ ID NO: 2;

[0340] a N at the position corresponding to E101 in SEQ ID NO: 2;

[0341] a Q at the position corresponding to E101 in SEQ ID NO: 2;

[0342] a T at the position corresponding to E101 in SEQ ID NO: 2;

[0343] a K at the position corresponding to Q114 in SEQ ID NO: 2.

[0344] The CsgG pore-forming monomer of the invention preferably further comprises an A at the position corresponding to Y51 in SEQ ID NO: 2 and/or a Q at the position corresponding to F56 in SEQ ID NO: 2.

[0345] The pore-forming mutant monomer typically retains the ability to form the same 3D structure as the wild-type CsgG monomer, such as the same 3D structure as a CsgG monomer having the sequence of SEQ ID NO: 2. The 3D structure of CsgG is known in the art and is disclosed, for example, in Cao et al (2014) PNAS E5439-E5444. Any number of mutations may be made in the wild-type CsgG sequence in addition to the mutations described herein provided that the CsgG mutant monomer retains the improved properties imparted on it by the mutations of the present invention.

[0346] Typically the CsgG monomer will retain the ability to form a structure comprising three alpha-helices and five beta-sheets. The present inventors have shown in particular that mutations may be made at least in the region of CsgG which is N-terminal to the first alpha helix (which starts at S63 in SEQ ID NO: 2), in the second alpha helix (from G85 to A99 of SEQ ID NO: 2), in the loop between the second alpha helix and the first beta sheet (from Q100 to N120 of SEQ ID NO: 2), in the fourth and fifth beta sheets (S173 to R192 and R198 to T107 of SEQ ID NO: 2, respectively) and in the loop between the fourth and fifth beta sheets (F193 to Q197 of SEQ ID NO: 2) without affecting the ability of the CsgG monomer to form a transmembrane pore, which transmembrane pore is capable of translocating polypeptides. Therefore, it is envisaged that further mutations may be made in any of these regions in any CsgG monomer without affecting the ability of the monomer to form a pore that can translocate polynucleotides. It is also expected that mutations may be made in other regions, such as in any of the alpha helices (S63 to R76, G85 to A99 or V211 to L236 of SEQ ID NO: 2) or in any of the beta sheets (I121 to N133, K135 to R142, I146 to R162, S173 to R192 or R198 to T107 of SEQ ID NO: 2) without affecting the ability of the monomer to form a pore that can translocate polynucleotides. It is also expected that deletions of one or more amino acids can be made in any of the loop regions linking the alpha helices and beta sheets and/or in the N-terminal and/or C-terminal regions of the CsgG monomer without affecting the ability of the monomer to form a pore that can translocate polynucleotides.

[0347] Amino acid substitutions may be made to the amino acid sequence of SEQ ID NO: 2 in addition to those discussed above, for example up to 1, 2, 3, 4, 5, 10, 20 or 30 substitutions. Conservative substitutions replace amino acids with other amino acids of similar chemical structure, similar chemical properties or similar side-chain volume. The amino acids introduced may have similar polarity,

hydrophilicity, hydrophobicity, basicity, acidity, neutrality or charge to the amino acids they replace. Alternatively, the conservative substitution may introduce another amino acid that is aromatic or aliphatic in the place of a pre-existing aromatic or aliphatic amino acid. Conservative amino acid changes are well-known in the art and may be selected in accordance with the properties of the 20 main amino acids as defined in Table 2 below. Where amino acids have similar polarity, this can also be determined by reference to the hydropathy scale for amino acid side chains in Table 3.

TABLE 2

Chemical properties of amino acids			
Ala	aliphatic, hydrophobic, neutral	Met	hydrophobic, neutral
Cys	polar, hydrophobic, neutral	Asn	polar, hydrophilic, neutral
Asp	polar, hydrophilic, charged (-)	Pro	hydrophobic, neutral
Glu	polar, hydrophilic, charged (-)	Gln	polar, hydrophilic, neutral
Phe	aromatic, hydrophobic, neutral	Arg	polar, hydrophilic, charged (+)
Gly	aliphatic, neutral	Ser	polar, hydrophilic, neutral
His	aromatic, polar, hydrophilic, charged (+)	Thr	polar, hydrophilic, neutral
Ile	aliphatic, hydrophobic, neutral	Val	aliphatic, hydrophobic, neutral
Lys	polar, hydrophilic, charged(+)	Trp	aromatic, hydrophobic, neutral
Leu	aliphatic, hydrophobic, neutral	Tyr	aromatic, polar, hydrophobic

TABLE 3

Hydropathy scale	
Side Chain	Hydropathy
Ile	4.5
Val	4.2
Leu	3.8
Phe	2.8
Cys	2.5
Met	1.9
Ala	1.8
Gly	-0.4
Thr	-0.7
Ser	-0.8
Trp	-0.9
Tyr	-1.3
Pro	-1.6
His	-3.2
Glu	-3.5
Gln	-3.5
Asp	-3.5
Asn	-3.5
Lys	-3.9
Arg	-4.5

[0348] One or more amino acid residues of the amino acid sequence of SEQ ID NO: 2 may additionally be deleted from the polypeptides described above. Up to 1, 2, 3, 4, 5, 10, 20 or 30 or more residues may be deleted.

[0349] Variants may include fragments of SEQ ID NO: 2. Such fragments retain pore forming activity. Fragments may be at least 50, at least 100, at least 150, at least 200 or at least 250 amino acids in length. Such fragments may be used to

produce the pores. A fragment preferably comprises the membrane spanning domain of SEQ ID NO: 2, namely K135-Q153 and S183-S208.

[0350] One or more amino acids may be alternatively or additionally added to the polypeptides described above. An extension may be provided at the amino terminal or carboxy terminal of the amino acid sequence of SEQ ID NO: 2 or polypeptide variant or fragment thereof. The extension may be quite short, for example from 1 to 10 amino acids in length. Alternatively, the extension may be longer, for example up to 50 or 100 amino acids. A carrier protein may be fused to an amino acid sequence according to the invention. Other fusion proteins are discussed in more detail below.

[0351] As discussed above, a variant is a polypeptide that has an amino acid sequence which varies from that of SEQ ID NO: 2 and which retains its ability to form a pore. A variant typically contains the regions of SEQ ID NO: 2 that are responsible for pore formation. The pore forming ability of CsgG, which contains a β -barrel, is provided by β -sheets in each subunit. A variant of SEQ ID NO: 2 typically comprises the regions in SEQ ID NO: 2 that form β -sheets, namely K135-Q153 and S183-S208. One or more modifications can be made to the regions of SEQ ID NO: 2 that form β -sheets as long as the resulting variant retains its ability to form a pore. A variant of SEQ ID NO: 2 preferably includes one or more modifications, such as substitutions, additions or deletions, within its-helices and/or loop regions.

[0352] The monomers derived from CsgG may be modified to assist their identification of purification, for example by the addition of a streptavidin tag or by the addition of a signal sequence to promote their secretion from a cell where the monomer does not naturally contain such a sequence. Other suitable tags are discussed in more detail below. The monomer may be labelled with a revealing label. The revealing label may be any suitable label which allows the monomer to be detected. Suitable labels are described below.

[0353] The monomer derived from CsgG may also be produced using D-amino acids. For instance, the monomer derived from CsgG may comprise a mixture of L-amino acids and D-amino acids. This is conventional in the art for producing such proteins or peptides.

[0354] The monomer derived from CsgG contains one or more specific modifications to facilitate nucleotide discrimination. The monomer derived from CsgG may also contain other non-specific modifications as long as they do not interfere with pore formation. A number of non-specific side chain modifications are known in the art and may be made to the side chains of the monomer derived from CsgG. Such modifications include, for example, reductive alkylation of amino acids by reaction with an aldehyde followed by reduction with NaBH₄, amidination with methylacetimidate or acylation with acetic anhydride.

[0355] The monomer derived from CsgG can be produced using standard methods known in the art. The monomer derived from CsgG may be made synthetically or by recombinant means. For example, the monomer may be synthesised by in vitro translation and transcription (IVTT). Suitable methods for producing pores and monomers are discussed in International Application Nos. PCT/GB09/001690 (published as WO 2010/004273), PCT/GB09/001679 (published as WO 2010/004265) or PCT/GB10/

000133 (published as WO 2010/086603). Methods for inserting pores into membranes are discussed.

[0356] In some embodiments, the mutant monomer is chemically modified. The mutant monomer can be chemically modified in any way and at any site. The mutant monomer is preferably chemically modified by attachment of a molecule to one or more cysteines (cysteine linkage), attachment of a molecule to one or more lysines, attachment of a molecule to one or more non-natural amino acids, enzyme modification of an epitope or modification of a terminus. Suitable methods for carrying out such modifications are well-known in the art. The mutant monomer may be chemically modified by the attachment of any molecule. For instance, the mutant monomer may be chemically modified by attachment of a dye or a fluorophore.

[0357] In some embodiments, the mutant monomer is chemically modified with a molecular adaptor that facilitates the interaction between a pore comprising the monomer and a target nucleotide or target polynucleotide sequence. The presence of the adaptor improves the host-guest chemistry of the pore and the nucleotide or polynucleotide sequence and thereby improves the sequencing ability of pores formed from the mutant monomer. The principles of host-guest chemistry are well-known in the art. The adaptor has an effect on the physical or chemical properties of the pore that improves its interaction with the nucleotide or polynucleotide sequence. The adaptor may alter the charge of the barrel or channel of the pore or specifically interact with or bind to the nucleotide or polynucleotide sequence thereby facilitating its interaction with the pore.

[0358] The molecular adaptor is preferably a cyclic molecule, a cyclodextrin, a species that is capable of hybridization, a DNA binder or interchelator, a peptide or peptide analogue, a synthetic polymer, an aromatic planar molecule, a small positively-charged molecule or a small molecule capable of hydrogen-bonding.

[0359] The adaptor may be cyclic. A cyclic adaptor preferably has the same symmetry as the pore. The adaptor preferably has eight-fold or nine-fold symmetry since CsgG typically has eight or nine subunits around a central axis. This is discussed in more detail below.

[0360] The adaptor typically interacts with the nucleotide or polynucleotide sequence via host-guest chemistry. The adaptor is typically capable of interacting with the nucleotide or polynucleotide sequence. The adaptor comprises one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence. The one or more chemical groups preferably interact with the nucleotide or polynucleotide sequence by non-covalent interactions, such as hydrophobic interactions, hydrogen bonding, Van der Waal's forces, π -cation interactions and/or electrostatic forces. The one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence are preferably positively charged. The one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence more preferably comprise amino groups. The amino groups can be attached to primary, secondary or tertiary carbon atoms. The adaptor even more preferably comprises a ring of amino groups, such as a ring of 6, 7 or 8 amino groups. The adaptor most preferably comprises a ring of eight amino groups. A ring of protonated amino groups may interact with negatively charged phosphate groups in the nucleotide or polynucleotide sequence.

[0361] The correct positioning of the adaptor within the pore can be facilitated by host-guest chemistry between the adaptor and the pore comprising the mutant monomer. The adaptor preferably comprises one or more chemical groups that are capable of interacting with one or more amino acids in the pore. The adaptor more preferably comprises one or more chemical groups that are capable of interacting with one or more amino acids in the pore via non-covalent interactions, such as hydrophobic interactions, hydrogen bonding, Van der Waal's forces, π -cation interactions and/or electrostatic forces. The chemical groups that are capable of interacting with one or more amino acids in the pore are typically hydroxyls or amines. The hydroxyl groups can be attached to primary, secondary or tertiary carbon atoms. The hydroxyl groups may form hydrogen bonds with uncharged amino acids in the pore. Any adaptor that facilitates the interaction between the pore and the nucleotide or polynucleotide sequence can be used.

[0362] Suitable adaptors include, but are not limited to, cyclodextrins, cyclic peptides and cucurbiturils. The adaptor is preferably a cyclodextrin or a derivative thereof. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H.-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The adaptor is more preferably heptakis-6-amino- β -cyclodextrin (am_7 - β CD), 6-monodeoxy-6-monoamino- β -cyclodextrin (am_1 - β CD) or heptakis-(6-deoxy-6-guanidino)-cyclodextrin (gu_7 - β CD). The guanidino group in gu_7 - β CD has a much higher pKa than the primary amines in am_7 - β CD and so it is more positively charged. This gu_7 - β CD adaptor may be used to increase the dwell time of the nucleotide in the pore, to increase the accuracy of the residual current measured, as well as to increase the base detection rate at high temperatures or low data acquisition rates.

[0363] If a succinimidyl 3-(2-pyridyldithio) propionate (SPDP) crosslinker is used as discussed in more detail below, the adaptor is preferably heptakis (6-deoxy-6-amino)-6-N-mono(2-pyridyl)dithiopropionyl- β -cyclodextrin ($\text{am}_7\text{amPDP}_1$ - β CD).

[0364] More suitable adaptors include γ -cyclodextrins, which comprise 9 sugar units (and therefore have nine-fold symmetry). The γ -cyclodextrin may contain a linker molecule or may be modified to comprise all or more of the modified sugar units used in the β -cyclodextrin examples discussed above.

[0365] The molecular adaptor is preferably covalently attached to the mutant monomer. The adaptor can be covalently attached to the pore using any method known in the art. The adaptor is typically attached via chemical linkage. If the molecular adaptor is attached via cysteine linkage, the one or more cysteines have preferably been introduced to the mutant, for instance in the barrel, by substitution. The mutant monomer may be chemically modified by attachment of a molecular adaptor to one or more cysteines in the mutant monomer. The one or more cysteines may be naturally-occurring, i.e. at positions 1 and/or 215 in SEQ ID NO: 2. Alternatively, the mutant monomer may be chemically modified by attachment of a molecule to one or more cysteines introduced at other positions. The cysteine at position 215 may be removed, for instance by substitution, to ensure that the molecular adaptor does not attach to that position rather than the cysteine at position 1 or a cysteine introduced at another position. The reactivity of cysteine residues may be enhanced by modification of the adjacent

residues. For instance, the basic groups of flanking arginine, histidine or lysine residues will change the pKa of the cysteines thiol group to that of the more reactive S group. The reactivity of cysteine residues may be protected by thiol protective groups such as dTNB. These may be reacted with one or more cysteine residues of the mutant monomer before a linker is attached. The molecule may be attached directly to the mutant monomer. The molecule is preferably attached to the mutant monomer using a linker, such as a chemical crosslinker or a peptide linker.

[0366] Suitable chemical crosslinkers are well-known in the art. Preferred crosslinkers include 2,5-dioxopyrrolidin-1-yl 3-(pyridin-2-ylsulfany)propanoate, 2,5-dioxopyrrolidin-1-yl 4-(pyridin-2-ylsulfany)butanoate and 2,5-dioxopyrrolidin-1-yl 8-(pyridin-2-ylsulfany)octanoate. The most preferred crosslinker is succinimidyl 3-(2-pyridyldithio)propionate (SPDP). Typically, the molecule is covalently attached to the bifunctional crosslinker before the molecule/crosslinker complex is covalently attached to the mutant monomer but it is also possible to covalently attach the bifunctional crosslinker to the monomer before the bifunctional crosslinker/monomer complex is attached to the molecule.

[0367] The linker is preferably resistant to dithiothreitol (DTT). Suitable linkers include, but are not limited to, iodoacetamide-based and Maleimide-based linkers.

[0368] In other embodiment, the monomer may be attached to a polynucleotide binding protein. This forms a modular sequencing system that may be used in the methods of sequencing of the invention. Polynucleotide binding proteins are discussed below.

[0369] The polynucleotide binding protein is preferably covalently attached to the mutant monomer. The protein can be covalently attached to the monomer using any method known in the art. The monomer and protein may be chemically fused or genetically fused. The monomer and protein are genetically fused if the whole construct is expressed from a single polynucleotide sequence. Genetic fusion of a monomer to a polynucleotide binding protein is discussed in International Application No. PCT/GB09/001679 (published as WO 2010/004265).

[0370] If the polynucleotide binding protein is attached via cysteine linkage, the one or more cysteines have preferably been introduced to the mutant by substitution. The one or more cysteines are preferably introduced into loop regions which have low conservation amongst homologues indicating that mutations or insertions may be tolerated. They are therefore suitable for attaching a polynucleotide binding protein. In such embodiments, the naturally-occurring cysteine at position 251 may be removed. The reactivity of cysteine residues may be enhanced by modification as described above.

[0371] The polynucleotide binding protein may be attached directly to the mutant monomer or via one or more linkers. The molecule may be attached to the mutant monomer using the hybridization linkers described in International Application No. PCT/GB10/000132 (published as WO 2010/086602). Alternatively, peptide linkers may be used. Peptide linkers are amino acid sequences. The length, flexibility and hydrophilicity of the peptide linker are typically designed such that it does not disturb the functions of the monomer and molecule. Preferred flexible peptide linkers are stretches of 2 to 20, such as 4, 6, 8, 10 or 16, serine and/or glycine amino acids. More preferred flexible

linkers include (SG)₁, (SG)₂, (SG)₃, (SG)₄, (SG)₅ and (SG)₈ wherein S is serine and G is glycine. Preferred rigid linkers are stretches of 2 to 30, such as 4, 6, 8, 16 or 24, proline amino acids. More preferred rigid linkers include (P)₁₂ wherein P is proline.

[0372] The mutant monomer may be chemically modified with a molecular adaptor and a polynucleotide binding protein.

[0373] The molecule (with which the monomer is chemically modified) may be attached directly to the monomer or attached via a linker as disclosed in International Application Nos. PCT/GB09/001690 (published as WO 2010/004273), PCT/GB09/001679 (published as WO 2010/004265) or PCT/GB10/000133 (published as WO 2010/086603).

[0374] Any of the proteins described herein, such as the mutant monomers and pores of the invention, may be modified to assist their identification or purification, for example by the addition of histidine residues (a his tag), aspartic acid residues (an asp tag), a streptavidin tag, a flag tag, a SUMO tag, a GST tag or a MBP tag, or by the addition of a signal sequence to promote their secretion from a cell where the polypeptide does not naturally contain such a sequence. An alternative to introducing a genetic tag is to chemically react a tag onto a native or engineered position on the protein. An example of this would be to react a gel-shift reagent to a cysteine engineered on the outside of the protein. This has been demonstrated as a method for separating hemolysin hetero-oligomers (Chem Biol. 1997 July; 4 (7): 497-505).

[0375] Any of the proteins described herein, such as the mutant monomers and pores of the invention, may be labelled with a revealing label. The revealing label may be any suitable label which allows the protein to be detected. Suitable labels include, but are not limited to, fluorescent molecules, radioisotopes, e.g. ¹²⁵I, ³⁵S, enzymes, antibodies, antigens, polynucleotides and ligands such as biotin.

[0376] Any of the proteins described herein, such as the monomers or pores of the invention, may be made synthetically or by recombinant means. For example, the protein may be synthesised by in vitro translation and transcription (IVTT). The amino acid sequence of the protein may be modified to include non-naturally occurring amino acids or to increase the stability of the protein. When a protein is produced by synthetic means, such amino acids may be introduced during production. The protein may also be altered following either synthetic or recombinant production.

[0377] Proteins may also be produced using D-amino acids. For instance, the protein may comprise a mixture of L-amino acids and D-amino acids. This is conventional in the art for producing such proteins or peptides.

[0378] The protein may also contain other non-specific modifications as long as they do not interfere with the function of the protein. A number of non-specific side chain modifications are known in the art and may be made to the side chains of the protein(s). Such modifications include, for example, reductive alkylation of amino acids by reaction with an aldehyde followed by reduction with NaBH₄, amidation with methylacetimidate or acylation with acetic anhydride.

[0379] Any of the proteins described herein, including the monomers and pores of the invention, can be produced using standard methods known in the art. Polynucleotide

sequences encoding a protein may be derived and replicated using standard methods in the art. Polynucleotide sequences encoding a protein may be expressed in a bacterial host cell using standard techniques in the art. The protein may be produced in a cell by in situ expression of the polypeptide from a recombinant expression vector. The expression vector optionally carries an inducible promoter to control the expression of the polypeptide. These methods are described in Sambrook, J. and Russell, D. (2001). Molecular Cloning: A Laboratory Manual, 3rd Edition. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY.

[0380] Proteins may be produced in large scale following purification by any protein liquid chromatography system from protein producing organisms or after recombinant expression. Typical protein liquid chromatography systems include FPLC, AKTA systems, the Bio-Cad system, the Bio-Rad BioLogic system and the Gilson HPLC system.

Constructs

[0381] The invention also provides a construct comprising two or more covalently attached CsgG monomers, wherein at least one of the monomers is a mutant monomer of the invention. The construct of the invention retains its ability to form a pore. This may be determined as discussed above. One or more constructs of the invention may be used to form pores for characterising, such as sequencing, polynucleotides. The construct may comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9 or at least 10 monomers. The construct preferably comprises two monomers. The two or more monomers may be the same or different.

[0382] At least one monomer in the construct is a mutant monomer of the invention. 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more or 10 or more monomers in the construct may be mutant monomers of the invention. All of the monomers in the construct are preferably mutant monomers of the invention. The mutant monomers may be the same or different. In a preferred embodiment, the construct comprises two mutant monomers of the invention.

[0383] The mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0384] The construct may comprise one or more monomers which are not mutant monomers of the invention. CsgG mutant monomers which are non mutant monomers of the invention include monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 in which none of the amino acids/positions discussed above have been mutated. At least one monomer in the construct may comprise SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of the sequence shown in SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41. A comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 is at least 50% homologous to SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over its entire sequence based on

amino acid identity. More preferably, the comparative variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over the entire sequence.

[0385] The monomers in the construct are preferably genetically fused. Monomers are genetically fused if the whole construct is expressed from a single polynucleotide sequence. The coding sequences of the monomers may be combined in any way to form a single polynucleotide sequence encoding the construct.

[0386] The monomers may be genetically fused in any configuration. The monomers may be fused via their terminal amino acids. For instance, the amino terminus of the one monomer may be fused to the carboxy terminus of another monomer. The second and subsequent monomers in the construct (in the amino to carboxy direction) may comprise a methionine at their amino terminal ends (each of which is fused to the carboxy terminus of the previous monomer). For instance, if M is a monomer (without an amino terminal methionine) and mM is a monomer with an amino terminal methionine, the construct may comprise the sequence M-mM, M-mM-mM or M-mM-mM-mM. The presences of these methionines typically results from the expression of the start codons (i.e. ATGs) at the 5' end of the polynucleotides encoding the second or subsequent monomers within the polynucleotide encoding entire construct. The first monomer in the construct (in the amino to carboxy direction) may also comprise a methionine (e.g. mM-mM, mM-mM-mM or mM-mM-mM-mM).

[0387] The two or more monomers may be genetically fused directly together. The monomers are preferably genetically fused using a linker. The linker may be designed to constrain the mobility of the monomers. Preferred linkers are amino acid sequences (i.e. peptide linkers). Any of the peptide linkers discussed above may be used.

[0388] In another preferred embodiment, the monomers are chemically fused. Two monomers are chemically fused if the two parts are chemically attached, for instance via a chemical crosslinker. Any of the chemical crosslinkers discussed above may be used. The linker may be attached to one or more cysteine residues introduced into a mutant monomer of the invention. Alternatively, the linker may be attached to a terminus of one of the monomers in the construct.

[0389] If a construct contains different monomers, cross-linkage of monomers to themselves may be prevented by keeping the concentration of linker in a vast excess of the monomers. Alternatively, a “lock and key” arrangement may be used in which two linkers are used. Only one end of each linker may react together to form a longer linker and the other ends of the linker each react with a different monomers. Such linkers are described in International Application No. PCT/GB10/000132 (published as WO 2010/086602).

Polynucleotides

[0390] The present invention also provides polynucleotide sequences which encode a mutant monomer of the invention. The mutant monomer may be any of those discussed above. The polynucleotide sequence preferably comprises a sequence at least 50%, 60%, 70%, 80%, 90% or 95% homologous based on nucleotide identity to the sequence of

SEQ ID NO: 1 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95% nucleotide identity over a stretch of 300 or more, for example 375, 450, 525 or 600 or more, contiguous nucleotides (“hard homology”). Homology may be calculated as described above. The polynucleotide sequence may comprise a sequence that differs from SEQ ID NO: 1 on the basis of the degeneracy of the genetic code.

[0391] The present invention also provides polynucleotide sequences which encode any of the genetically fused constructs of the invention. The polynucleotide preferably comprises two or more variants of the sequence shown in SEQ ID NO: 1. The polynucleotide sequence preferably comprises two or more sequences having at least 50%, 60%, 70%, 80%, 90% or 95% homology to SEQ ID NO: 1 based on nucleotide identity over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95% nucleotide identity over a stretch of 600 or more, for example 750, 900, 1050 or 1200 or more, contiguous nucleotides (“hard homology”). Homology may be calculated as described above.

[0392] Polynucleotide sequences may be derived and replicated using standard methods in the art. Chromosomal DNA encoding wild-type CsgG may be extracted from a pore producing organism, such as *Escherichia coli*. The gene encoding the pore subunit may be amplified using PCR involving specific primers. The amplified sequence may then undergo site-directed mutagenesis. Suitable methods of site-directed mutagenesis are known in the art and include, for example, combine chain reaction. Polynucleotides encoding a construct of the invention can be made using well-known techniques, such as those described in Sambrook, J. and Russell, D. (2001). *Molecular Cloning: A Laboratory Manual*, 3rd Edition. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY.

[0393] The resulting polynucleotide sequence may then be incorporated into a recombinant replicable vector such as a cloning vector. The vector may be used to replicate the polynucleotide in a compatible host cell. Thus polynucleotide sequences may be made by introducing a polynucleotide into a replicable vector, introducing the vector into a compatible host cell, and growing the host cell under conditions which bring about replication of the vector. The vector may be recovered from the host cell. Suitable host cells for cloning of polynucleotides are known in the art and described in more detail below.

[0394] The polynucleotide sequence may be cloned into suitable expression vector. In an expression vector, the polynucleotide sequence is typically operably linked to a control sequence which is capable of providing for the expression of the coding sequence by the host cell. Such expression vectors can be used to express a pore subunit.

[0395] The term “operably linked” refers to a juxtaposition wherein the components described are in a relationship permitting them to function in their intended manner. A control sequence “operably linked” to a coding sequence is ligated in such a way that expression of the coding sequence is achieved under conditions compatible with the control sequences. Multiple copies of the same or different polynucleotide sequences may be introduced into the vector.

[0396] The expression vector may then be introduced into a suitable host cell. Thus, a mutant monomer or construct of the invention can be produced by inserting a polynucleotide sequence into an expression vector, introducing the vector

into a compatible bacterial host cell, and growing the host cell under conditions which bring about expression of the polynucleotide sequence. The recombinantly-expressed monomer or construct may self-assemble into a pore in the host cell membrane. Alternatively, the recombinant pore produced in this manner may be removed from the host cell and inserted into another membrane. When producing pores comprising at least two different monomers or constructs, the different monomers or constructs may be expressed separately in different host cells as described above, removed from the host cells and assembled into a pore in a separate membrane, such as a rabbit cell membrane or a synthetic membrane.

[0397] The vectors may be for example, plasmid, virus or phage vectors provided with an origin of replication, optionally a promoter for the expression of the said polynucleotide sequence and optionally a regulator of the promoter. The vectors may contain one or more selectable marker genes, for example a tetracycline resistance gene. Promoters and other expression regulation signals may be selected to be compatible with the host cell for which the expression vector is designed. A T7, trc, lac, ara or λ_L promoter is typically used.

[0398] The host cell typically expresses the monomer or construct at a high level. Host cells transformed with a polynucleotide sequence will be chosen to be compatible with the expression vector used to transform the cell. The host cell is typically bacterial and preferably *Escherichia coli*. Any cell with a λ DE3 lysogen, for example C41 (DE3), BL21 (DE3), JM109 (DE3), B834 (DE3), TUNER, Origami and Origami B, can express a vector comprising the T7 promoter. In addition to the conditions listed above any of the methods cited in Cao et al, 2014, PNAS, Structure of the nonameric bacterial amyloid secretion channel, doi-1411942111 and Goyal et al, 2014, Nature, 516, 250-253 structural and mechanistic insights into the bacterial amyloid secretion channel CsgG may be used to express the CsgG proteins.

[0399] The invention also comprises a method of producing a mutant monomer of the invention or a construct of the invention. The method comprises expressing a polynucleotide of the invention in a suitable host cell. The polynucleotide is preferably part of a vector and is preferably operably linked to a promoter.

Pores

[0400] The invention also provides various pores. The pores of the invention are ideal for characterising, such as sequencing, polynucleotide sequences because they can discriminate between different nucleotides with a high degree of sensitivity. The pores can surprisingly distinguish between the four nucleotides in DNA and RNA. The pores of the invention can even distinguish between methylated and unmethylated nucleotides. The base resolution of pores of the invention is surprisingly high. The pores show almost complete separation of all four DNA nucleotides. The pores further discriminate between deoxycytidine monophosphate (dCMP) and methyl-dCMP based on the dwell time in the pore and the current flowing through the pore.

[0401] The pores of the invention can also discriminate between different nucleotides under a range of conditions. In particular, the pores will discriminate between nucleotides under conditions that are favourable to the characterising, such as sequencing, of nucleic acids. The extent to which the

pores of the invention can discriminate between different nucleotides can be controlled by altering the applied potential, the salt concentration, the buffer, the temperature and the presence of additives, such as urea, betaine and DTT. This allows the function of the pores to be fine-tuned, particularly when sequencing. This is discussed in more detail below. The pores of the invention may also be used to identify polynucleotide polymers from the interaction with one or more monomers rather than on a nucleotide by nucleotide basis.

[0402] A pore of the invention may be isolated, substantially isolated, purified or substantially purified. A pore of the invention is isolated or purified if it is completely free of any other components, such as lipids or other pores. A pore is substantially isolated if it is mixed with carriers or diluents which will not interfere with its intended use. For instance, a pore is substantially isolated or substantially purified if it is present in a form that comprises less than 10%, less than 5%, less than 2% or less than 1% of other components, such as triblock copolymers, lipids or other pores. Alternatively, a pore of the invention may be present in a membrane. Suitable membranes are discussed below.

[0403] A pore of the invention may be present as an individual or single pore. Alternatively, a pore of the invention may be present in a homologous or heterologous population of two or more pores.

Homo-Oligomeric Pores

[0404] The invention also provides a homo-oligomeric pore derived from CsgG comprising identical mutant monomers of the invention. The homo-oligomeric pore may comprise any of the mutants of the invention. The homo-oligomeric pore of the invention is ideal for characterising, such as sequencing, polynucleotides. The homo-oligomeric pore of the invention may have any of the advantages discussed above.

[0405] The homo-oligomeric pore may contain any number of mutant monomers. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 identical mutant monomers, such as 7, 8, 9 or 10 mutant monomers. The pore preferably comprises eight or nine identical mutant monomers. One or more, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10, of the mutant monomers is preferably chemically modified as discussed above.

[0406] Methods for making pores are discussed in more detail below.

Hetero-Oligomeric Pores

[0407] The invention also provides a hetero-oligomeric pore derived from CsgG comprising at least one mutant monomer of the invention. The hetero-oligomeric pore of the invention is ideal for characterising, such as sequencing, polynucleotides. Hetero-oligomeric pores can be made using methods known in the art (e.g. Protein Sci. 2002 July; 11 (7): 1813-24).

[0408] The hetero-oligomeric pore contains sufficient monomers to form the pore. The monomers may be of any type. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 monomers, such as 7, 8, 9 or 10 monomers. The pore preferably comprises eight or nine monomers.

[0409] In a preferred embodiment, all of the monomers (such as 10, 9, 8 or 7 of the monomers) are mutant

monomers of the invention and at least one of them differs from the others. In a more preferred embodiment, the pore comprises eight or nine mutant monomers of the invention and at least one of them differs from the others. They may all differ from one another.

[0410] The mutant monomers of the invention in the pore are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the pore are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0411] In another preferred embodiment, at least one of the mutant monomers is not a mutant monomer of the invention. In this embodiment, the remaining monomers are preferably mutant monomers of the invention. Hence, the pore may comprise 9, 8, 7, 6, 5, 4, 3, 2 or 1 mutant monomers of the invention. Any number of the monomers in the pore may not be a mutant monomer of the invention. The pore preferably comprises seven or eight mutant monomers of the invention and a monomer which is not a monomer of the invention. The mutant monomers of the invention may be the same or different.

[0412] The mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0413] The pore may comprise one or more monomers which are not mutant monomers of the invention. CsgG monomers which are not mutant monomers of the invention include monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 in which none of the amino acids/positions discussed above in relation to the invention have been mutated/substituted. A comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 is typically at least 50% homologous to SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over its entire sequence based on amino acid identity. More preferably, the comparative variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over the entire sequence.

[0414] In all the embodiments discussed above, one or more, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10, of the mutant monomers is preferably chemically modified as discussed above.

[0415] Methods for making pores are discussed in more detail below.

Construct-Containing Pores

[0416] The invention also provides a pore comprising at least one construct of the invention. A construct of the invention comprises two or more covalently attached monomers derived from CsgG wherein at least one of the monomers is a mutant monomer of the invention. In other words, a construct must contain more than one monomer. The pore contains sufficient constructs and, if necessary, monomers to

form the pore. For instance, an octameric pore may comprise (a) four constructs each comprising two constructs, (b) two constructs each comprising four monomers or (b) one construct comprising two monomers and six monomers that do not form part of a construct. For instance, a nonameric pore may comprise (a) four constructs each comprising two constructs and one monomer that does not form part of a construct, (b) two constructs each comprising four monomers and a monomer that does not form part of a construct or (b) one construct comprising two monomers and seven monomers that do not form part of a construct. Other combinations of constructs and monomers can be envisaged by the skilled person.

[0417] At least two of the monomers in the pore are in the form of a construct of the invention. The construct, and hence the pore, comprises at least one mutant monomer of the invention. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 monomers, such as 7, 8, 9 or 10 monomers, in total (at least two of which must be in a construct). The pore preferably comprises eight or nine monomers (at least two of which must be in a construct).

[0418] The construct containing pore may be a homooligomer (i.e. include identical constructs) or be a heterooligomer (i.e. where at least one construct differs from the others).

[0419] A pore typically contains (a) one construct comprising two monomers and (b) 5, 6, 7 or 8 monomers. The construct may be any of those discussed above. The monomers may be any of those discussed above, including mutant monomers of the invention, monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 and mutant monomers comprising a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above.

[0420] Another typical pore comprises more than one construct of the invention, such as two, three or four constructs of the invention. If necessary, such pores further comprise sufficient additional monomers or constructs to form the pore. The additional monomer(s) may be any of those discussed above, including mutant monomers of the invention, monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 and mutant monomers comprising a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above. The additional construct(s) may be any of those discussed above or may be a construct comprising two or more covalently attached CsgG monomers each comprising a monomer comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above.

[0421] A further pore of the invention comprises only constructs comprising 2 monomers, for example a pore may comprise 4, 5, 6, 7 or 8 constructs comprising 2 monomers. At least one construct is a construct of the invention, i.e. at least one monomer in the at least one construct, and preferably each monomer in the at least one construct, is a mutant monomer of the invention. All of the constructs comprising 2 monomers may be constructs of the invention.

[0422] A specific pore according to the invention comprises four constructs of the invention each comprising two

monomers, wherein at least one monomer in each construct, and preferably each monomer in each construct, is a mutant monomer of the invention. The constructs may oligomerise into a pore with a structure such that only one monomer of each construct contributes to the channel of the pore. Typically the other monomers of the construct will be on the outside of the channel of the pore. For example, pores of the invention may comprise 7, 8, 9 or 10 constructs comprising 2 monomers where the channel comprises 7, 8, 9 or 10 monomers.

[0423] Mutations can be introduced into the construct as described above. The mutations may be alternating, i.e. the mutations are different for each monomer within a two monomer construct and the constructs are assembled as a homo-oligomer resulting in alternating modifications. In other words, monomers comprising MutA and MutB are fused and assembled to form an A-B: A-B: A-B: A-B pore. Alternatively, the mutations may be neighbouring, i.e. identical mutations are introduced into two monomers in a construct and this is then oligomerised with different mutant monomers or constructs. In other words, monomers comprising MutA are fused follow by oligomerisation with MutB-containing monomers to form A-A: B: B: B: B: B: B.

[0424] One or more of the monomers of the invention in a construct-containing pore may be chemically-modified as discussed above.

Analyte Characterisation

[0425] The invention provides a method of determining the presence, absence or one or more characteristics of a target analyte. The method involves contacting the target analyte with a pore of the invention such that the target analyte moves with respect to, such as through, the pore and taking one or more measurements as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte. The target analyte may also be called the template analyte or the analyte of interest.

[0426] Steps (a) and (b) are preferably carried out with a potential applied across the pore. As discussed in more detail below, the applied potential typically results in the formation of a complex between the pore and a polynucleotide binding protein. The applied potential may be a voltage potential. Alternatively, the applied potential may be a chemical potential. An example of this is using a salt gradient across an amphiphilic layer. A salt gradient is disclosed in Holden et al., J Am Chem Soc. 2007 Jul. 11; 129 (27): 8650-5.

[0427] The method is for determining the presence, absence or one or more characteristics of a target analyte. The method may be for determining the presence, absence or one or more characteristics of at least one analyte. The method may concern determining the presence, absence or one or more characteristics of two or more analytes. The method may comprise determining the presence, absence or one or more characteristics of any number of analytes, such as 2, 5, 10, 15, 20, 30, 40, 50, 100 or more analytes. Any number of characteristics of the one or more analytes may be determined, such as 1, 2, 3, 4, 5, 10 or more characteristics.

[0428] The target analyte is preferably a metal ion, an inorganic salt, a polymer, an amino acid, a peptide, a polypeptide, a protein, a nucleotide, an oligonucleotide, a polynucleotide, a dye, a bleach, a pharmaceutical, a diagnostic agent, a recreational drug, an explosive or an environmental pollutant. The method may concern determining

the presence, absence or one or more characteristics of two or more analytes of the same type, such as two or more proteins, two or more nucleotides or two or more pharmaceuticals. Alternatively, the method may concern determining the presence, absence or one or more characteristics of two or more analytes of different types, such as one or more proteins, one or more nucleotides and one or more pharmaceuticals.

[0429] The target analyte can be secreted from cells. Alternatively, the target analyte can be an analyte that is present inside cells such that the analyte must be extracted from the cells before the invention can be carried out.

[0430] The analyte is preferably an amino acid, a peptide, a polypeptides and/or a protein. The amino acid, peptide, polypeptide or protein can be naturally-occurring or non-naturally-occurring. The polypeptide or protein can include within them synthetic or modified amino acids. A number of different types of modification to amino acids are known in the art. Suitable amino acids and modifications thereof are above. For the purposes of the invention, it is to be understood that the target analyte can be modified by any method available in the art.

[0431] The protein can be an enzyme, an antibody, a hormone, a growth factor or a growth regulatory protein, such as a cytokine. The cytokine may be selected from interleukins, preferably IFN-1, IL-1, IL-2, IL-4, IL-5, IL-6, IL-10, IL-12 and IL-13, interferons, preferably IL- γ , and other cytokines such as TNF- α . The protein may be a bacterial protein, a fungal protein, a virus protein or a parasite-derived protein.

[0432] The target analyte is preferably a nucleotide, an oligonucleotide or a polynucleotide. Nucleotides and polynucleotides are discussed below. Oligonucleotides are short nucleotide polymers which typically have 50 or fewer nucleotides, such as 40 or fewer, 30 or fewer, 20 or fewer, 10 or fewer or 5 or fewer nucleotides. The oligonucleotides may comprise any of the nucleotides discussed below, including the abasic and modified nucleotides.

[0433] The target analyte, such as a target polynucleotide, may be present in any of the suitable samples discussed below.

[0434] The pore is typically present in a membrane as discussed below. The target analyte may be coupled or delivered to the membrane using of the methods discussed below.

[0435] Any of the measurements discussed below can be used to determine the presence, absence or one or more characteristics of the target analyte. The method preferably comprises contacting the target analyte with the pore such that the analyte moves with respect to, such as moves through, the pore and measuring the current passing through the pore as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte.

[0436] The target analyte is present if the current flows through the pore in a manner specific for the analyte (i.e. if a distinctive current associated with the analyte is detected flowing through the pore). The analyte is absent if the current does not flow through the pore in a manner specific for the nucleotide. Control experiments can be carried out in the presence of the analyte to determine the way in which it affects the current flowing through the pore.

[0437] The invention can be used to differentiate analytes of similar structure on the basis of the different effects they

have on the current passing through a pore. Individual analytes can be identified at the single molecule level from their current amplitude when they interact with the pore. The invention can also be used to determine whether or not a particular analyte is present in a sample. The invention can also be used to measure the concentration of a particular analyte in a sample. Analyte characterisation using pores other than CsgG is known in the art.

Polynucleotide Characterisation

[0438] The invention provides a method of characterising a target polynucleotide, such as sequencing a polynucleotide. There are two main strategies for characterising or sequencing polynucleotides using nanopores, namely strand characterisation/sequencing and exonuclease characterisation/sequencing. The method of the invention may concern either method.

[0439] In strand sequencing, the DNA is translocated through the nanopore either with or against an applied potential. Exonucleases that act progressively or processively on double stranded DNA can be used on the cis side of the pore to feed the remaining single strand through under an applied potential or the trans side under a reverse potential. Likewise, a helicase that unwinds the double stranded DNA can also be used in a similar manner. A polymerase may also be used. There are also possibilities for sequencing applications that require strand translocation against an applied potential, but the DNA must be first “caught” by the enzyme under a reverse or no potential. With the potential then switched back following binding the strand will pass cis to trans through the pore and be held in an extended conformation by the current flow. The single strand DNA exonucleases or single strand DNA dependent polymerases can act as molecular motors to pull the recently translocated single strand back through the pore in a controlled stepwise manner, trans to cis, against the applied potential.

[0440] In one embodiment, the method of characterising a target polynucleotide involves contacting the target sequence with a pore of the invention and a helicase enzyme. Any helicase may be used in the method. Suitable helicases are discussed below. Helicases may work in two modes with respect to the pore. First, the method is preferably carried out using a helicase such that it controls movement of the target sequence through the pore with the field resulting from the applied voltage. In this mode 5' end of the DNA is first captured in the pore, and the enzyme controls movement of the DNA into the pore such that the target sequence is passed through the pore with the field until it finally translocates through to the trans side of the bilayer. Alternatively, the method is preferably carried out such that a helicase enzyme controls movement of the target sequence through the pore against the field resulting from the applied voltage. In this mode 3' end of the DNA is first captured in the pore, and the enzyme controls movement of the DNA through the pore such that the target sequence is pulled out of the pore against the applied field until finally ejected back to the cis side of the bilayer.

[0441] In exonuclease sequencing, an exonuclease releases individual nucleotides from one end of the target polynucleotide and these individual nucleotides are identified as discussed below. In another embodiment, the method of characterising a target polynucleotide involves contacting the target sequence with a pore and an exonuclease enzyme. Any of the exonuclease enzymes discussed below may be

used in the method. The enzyme may be covalently attached to the pore as discussed below.

[0442] Exonucleases are enzymes that typically latch onto one end of a polynucleotide and digest the sequence one nucleotide at a time from that end. The exonuclease can digest the polynucleotide in 5' to 3' direction or 3' to 5' direction. The end of the polynucleotide to which the exonuclease binds is typically determined through the choice of enzyme used and/or using methods known in the art. Hydroxyl groups or cap structures at either end of the polynucleotide may typically be used to prevent or facilitate the binding of the exonuclease to a particular end of the polynucleotide.

[0443] The method involves contacting the polynucleotide with the exonuclease so that the nucleotides are digested from the end of the polynucleotide at a rate that allows characterisation or identification of a proportion of nucleotides as discussed above. Methods for doing this are well known in the art. For example, Edman degradation is used to successively digest single amino acids from the end of polypeptide such that they may be identified using High Performance Liquid Chromatography (HPLC). A homologous method may be used in the present invention.

[0444] The rate at which the exonuclease functions is typically slower than the optimal rate of a wild-type exonuclease. A suitable rate of activity of the exonuclease in the method of the invention involves digestion of from 0.5 to 1000 nucleotides per second, from 0.6 to 500 nucleotides per second, 0.7 to 200 nucleotides per second, from 0.8 to 100 nucleotides per second, from 0.9 to 50 nucleotides per second or 1 to 20 or 10 nucleotides per second. The rate is preferably 1, 10, 100, 500 or 1000 nucleotides per second. A suitable rate of exonuclease activity can be achieved in various ways. For example, variant exonucleases with a reduced optimal rate of activity may be used in accordance with the invention.

[0445] In the strand characterisation embodiment, the method comprises contacting the polynucleotide with a pore of the invention such that the polynucleotide moves with respect to, such as through, the pore and taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or more characteristics of the polynucleotide, and thereby characterising the target polynucleotide.

[0446] In the exonucleotide characterisation embodiment, the method comprises contacting the polynucleotide with a pore of the invention and an exonuclease such that the exonuclease digests individual nucleotides from one end of the target polynucleotide and the individual nucleotides move with respect to, such as through, the pore and taking one or more measurements as the individual nucleotides move with respect to the pore, wherein the measurements are indicative of one or more characteristics of the individual nucleotides, and thereby characterising the target polynucleotide.

[0447] An individual nucleotide is a single nucleotide. An individual nucleotide is one which is not bound to another nucleotide or polynucleotide by a nucleotide bond. A nucleotide bond involves one of the phosphate groups of a nucleotide being bound to the sugar group of another nucleotide. An individual nucleotide is typically one which is not bound by a nucleotide bond to another polynucleotide of at least 5, at least 10, at least 20, at least 50, at least 100, at least 200, at least 500, at least 1000 or at least 5000 nucleotides. For

example, the individual nucleotide has been digested from a target polynucleotide sequence, such as a DNA or RNA strand. The nucleotide can be any of those discussed below.

[0448] The individual nucleotides may interact with the pore in any manner and at any site. The nucleotides preferably reversibly bind to the pore via or in conjunction with an adaptor as discussed above. The nucleotides most preferably reversibly bind to the pore via or in conjunction with the adaptor as they pass through the pore across the membrane. The nucleotides can also reversibly bind to the barrel or channel of the pore via or in conjunction with the adaptor as they pass through the pore across the membrane.

[0449] During the interaction between the individual nucleotide and the pore, the nucleotide typically affects the current flowing through the pore in a manner specific for that nucleotide. For example, a particular nucleotide will reduce the current flowing through the pore for a particular mean time period and to a particular extent. In other words, the current flowing through the pore is distinctive for a particular nucleotide. Control experiments may be carried out to determine the effect a particular nucleotide has on the current flowing through the pore. Results from carrying out the method of the invention on a test sample can then be compared with those derived from such a control experiment in order to identify a particular nucleotide in the sample or determine whether a particular nucleotide is present in the sample. The frequency at which the current flowing through the pore is affected in a manner indicative of a particular nucleotide can be used to determine the concentration of that nucleotide in the sample. The ratio of different nucleotides within a sample can also be calculated. For instance, the ratio of dCMP to methyl-dCMP can be calculated.

[0450] The method involves measuring one or more characteristics of the target polynucleotide. The target polynucleotide may also be called the template polynucleotide or the polynucleotide of interest.

[0451] This embodiment also uses a pore of the invention. Any of the pores and embodiments discussed above with reference to the target analyte may be used.

Polynucleotide

[0452] A polynucleotide, such as a nucleic acid, is a macromolecule comprising two or more nucleotides. The polynucleotide or nucleic acid may comprise any combination of any nucleotides. The nucleotides can be naturally occurring or artificial. One or more nucleotides in the polynucleotide can be oxidized or methylated. One or more nucleotides in the polynucleotide may be damaged. For instance, the polynucleotide may comprise a pyrimidine dimer. Such dimers are typically associated with damage by ultraviolet light and are the primary cause of skin melanomas. One or more nucleotides in the polynucleotide may be modified, for instance with a label or a tag. Suitable labels are described below. The polynucleotide may comprise one or more spacers.

[0453] A nucleotide typically contains a nucleobase, a sugar and at least one phosphate group. The nucleobase and sugar form a nucleoside.

[0454] The nucleobase is typically heterocyclic. Nucleobases include, but are not limited to, purines and pyrimidines and more specifically adenine (A), guanine (G), thymine (T), uracil (U) and cytosine (C).

[0455] The sugar is typically a pentose sugar. Nucleotide sugars include, but are not limited to, ribose and deoxyribose. The sugar is preferably a deoxyribose.

[0456] The polynucleotide preferably comprises the following nucleosides: deoxyadenosine (dA), deoxyuridine (dU) and/or thymidine (dT), deoxyguanosine (dG) and deoxycytidine (dC).

[0457] The nucleotide is typically a ribonucleotide or deoxyribonucleotide. The nucleotide typically contains a monophosphate, diphosphate or triphosphate. The nucleotide may comprise more than three phosphates, such as 4 or 5 phosphates. Phosphates may be attached on the 5' of 3' side of a nucleotide. Nucleotides include, but are not limited to, adenosine monophosphate (AMP), guanosine monophosphate (GMP), thymidine monophosphate (TMP), uridine monophosphate (UMP), 5-methylcytidine monophosphate, 5-hydroxymethylcytidine monophosphate, cytidine monophosphate (CMP), cyclic adenosine monophosphate (cAMP), cyclic guanosine monophosphate (cGMP), deoxyadenosine monophosphate (dAMP), deoxyguanosine monophosphate (dGMP), deoxythymidine monophosphate (dTMP), deoxyuridine monophosphate (dUMP), deoxycytidine monophosphate (dCMP) and deoxymethylcytidine monophosphate. The nucleotides are preferably selected from AMP, TMP, GMP, CMP, UMP, dAMP, dTMP, dGMP, dCMP and dUMP.

[0458] A nucleotide may be abasic (i.e. lack a nucleobase). A nucleotide may also lack a nucleobase and a sugar (i.e. is a C3 spacer).

[0459] The nucleotides in the polynucleotide may be attached to each other in any manner. The nucleotides are typically attached by their sugar and phosphate groups as in nucleic acids. The nucleotides may be connected via their nucleobases as in pyrimidine dimers.

[0460] The polynucleotide may be single stranded or double stranded. The polynucleotide is preferably single stranded. Single stranded polynucleotide characterization is referred to as ID in the Examples. At least a portion of the polynucleotide may be double stranded.

[0461] The polynucleotide can be a nucleic acid, such as deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). The polynucleotide can comprise one strand of RNA hybridised to one strand of DNA. The polynucleotide may be any synthetic nucleic acid known in the art, such as peptide nucleic acid (PNA), glycerol nucleic acid (GNA), threose nucleic acid (TNA), locked nucleic acid (LNA) or other synthetic polymers with nucleotide side chains. The PNA backbone is composed of repeating N-(2-aminoethyl)-glycine units linked by peptide bonds. The GNA backbone is composed of repeating glycol units linked by phosphodiester bonds. The TNA backbone is composed of repeating threose sugars linked together by phosphodiester bonds. LNA is formed from ribonucleotides as discussed above having an extra bridge connecting the 2' oxygen and 4' carbon in the ribose moiety. Bridged nucleic acids (BNAs) are modified RNA nucleotides. They may also be called constrained or inaccessible RNA. BNA monomers can contain a five-membered, six-membered or even a seven-membered bridged structure with a "fixed" C3'-endo sugar puckering. The bridge is synthetically incorporated at the 2', 4'-position of the ribose to produce a 2', 4'-BNA monomer.

[0462] The polynucleotide is most preferably ribonucleic acid (RNA) or deoxyribonucleic acid (DNA).

[0463] The polynucleotide can be any length. For example, the polynucleotide can be at least 10, at least 50, at least 100, at least 150, at least 200, at least 250, at least 300, at least 400 or at least 500 nucleotides or nucleotide pairs in length. The polynucleotide can be 1000 or more nucleotides or nucleotide pairs, 5000 or more nucleotides or nucleotide pairs in length or 100000 or more nucleotides or nucleotide pairs in length.

[0464] Any number of polynucleotides can be investigated. For instance, the method of the invention may concern characterising 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 50, 100 or more polynucleotides. If two or more polynucleotides are characterised, they may be different polynucleotides or two instances of the same polynucleotide.

[0465] The polynucleotide can be naturally occurring or artificial. For instance, the method may be used to verify the sequence of a manufactured oligonucleotide. The method is typically carried out in vitro.

Sample

[0466] The polynucleotide is typically present in any suitable sample. The invention is typically carried out on a sample that is known to contain or suspected to contain the polynucleotide. Alternatively, the invention may be carried out on a sample to confirm the identity of a polynucleotide whose presence in the sample is known or expected.

[0467] The sample may be a biological sample. The invention may be carried out in vitro using a sample obtained from or extracted from any organism or microorganism. The organism or microorganism is typically archaeal, prokaryotic or eukaryotic and typically belongs to one of the five kingdoms: plantae, animalia, fungi, monera and protista. The invention may be carried out in vitro on a sample obtained from or extracted from any virus. The sample is preferably a fluid sample. The sample typically comprises a body fluid of the patient. The sample may be urine, lymph, saliva, mucus or amniotic fluid but is preferably blood, plasma or serum.

[0468] Typically, the sample is human in origin, but alternatively it may be from another mammal animal such as from commercially farmed animals such as horses, cattle, sheep, fish, chickens or pigs or may alternatively be pets such as cats or dogs. Alternatively, the sample may be of plant origin, such as a sample obtained from a commercial crop, such as a cereal, legume, fruit or vegetable, for example wheat, barley, oats, canola, maize, soya, rice, rhubarb, bananas, apples, tomatoes, potatoes, grapes, tobacco, beans, lentils, sugar cane, cocoa, cotton.

[0469] The sample may be a non-biological sample. The non-biological sample is preferably a fluid sample. Examples of non-biological samples include surgical fluids, water such as drinking water, sea water or river water, and reagents for laboratory tests.

[0470] The sample is typically processed prior to being used in the invention, for example by centrifugation or by passage through a membrane that filters out unwanted molecules or cells, such as red blood cells. The sample may be measured immediately upon being taken. The sample may also be typically stored prior to assay, preferably below -70° C.

Characterisation

[0471] The method may involve measuring two, three, four or five or more characteristics of the polynucleotide.

The one or more characteristics are preferably selected from (i) the length of the polynucleotide, (ii) the identity of the polynucleotide, (iii) the sequence of the polynucleotide, (iv) the secondary structure of the polynucleotide and (v) whether or not the polynucleotide is modified. Any combination of (i) to (v) may be measured in accordance with the invention, such as {i}, {ii}, {iii}, {iv}, {v}, {i,ii}, {i,iv}, {i,v}, {ii,iii}, {ii,iv}, {ii,v}, {iii,iv}, {iii,v}, {iv,v}, {i,ii,iii}, {i,ii,iv}, {i,ii,v}, {i,iii,iv}, {i,iii,v}, {i,iv,v}, {ii,iii,iv}, {ii,iii,v}, {ii,iv,v}, {iii,iv,v}, {i,ii,iii,iv}, {i,ii,iii,v}, {i,ii,iv,v}, {i,iii,iv,v}, {ii,iii,iv,v} or {i,ii,iii,iv,v}. Different combinations of (i) to (v) may be measured for the first polynucleotide compared with the second polynucleotide, including any of those combinations listed above.

[0472] For (i), the length of the polynucleotide may be measured for example by determining the number of interactions between the polynucleotide and the pore or the duration of interaction between the polynucleotide and the pore.

[0473] For (ii), the identity of the polynucleotide may be measured in a number of ways. The identity of the polynucleotide may be measured in conjunction with measurement of the sequence of the polynucleotide or without measurement of the sequence of the polynucleotide. The former is straightforward; the polynucleotide is sequenced and thereby identified. The latter may be done in several ways. For instance, the presence of a particular motif in the polynucleotide may be measured (without measuring the remaining sequence of the polynucleotide). Alternatively, the measurement of a particular electrical and/or optical signal in the method may identify the polynucleotide as coming from a particular source.

[0474] For (iii), the sequence of the polynucleotide can be determined as described previously. Suitable sequencing methods, particularly those using electrical measurements, are described in Stoddart D et al., *Proc Natl Acad Sci*, 12; 106 (19): 7702-7, Lieberman K R et al, *J Am Chem Soc*. 2010,132 (50): 17961-72, and International Application WO 2000/28312.

[0475] For (iv), the secondary structure may be measured in a variety of ways. For instance, if the method involves an electrical measurement, the secondary structure may be measured using a change in dwell time or a change in current flowing through the pore. This allows regions of single-stranded and double-stranded polynucleotide to be distinguished.

[0476] For (v), the presence or absence of any modification may be measured. The method preferably comprises determining whether or not the polynucleotide is modified by methylation, by oxidation, by damage, with one or more proteins or with one or more labels, tags or spacers. Specific modifications will result in specific interactions with the pore which can be measured using the methods described below. For instance, methylcytosine may be distinguished from cytosine on the basis of the current flowing through the pore during its interaction with each nucleotide.

[0477] The target polynucleotide is contacted with a pore of the invention. The pore is typically present in a membrane. Suitable membranes are discussed below. The method may be carried out using any apparatus that is suitable for investigating a membrane/pore system in which a pore is present in a membrane. The method may be carried out using any apparatus that is suitable for transmembrane pore sensing. For example, the apparatus comprises a chamber com-

prising an aqueous solution and a barrier that separates the chamber into two sections. The barrier typically has an aperture in which the membrane containing the pore is formed. Alternatively the barrier forms the membrane in which the pore is present.

[0478] The method may be carried out using the apparatus described in International Application No. PCT/GB08/000562 (WO 2008/102120).

[0479] A variety of different types of measurements may be made. This includes without limitation: electrical measurements and optical measurements. Possible electrical measurements include: current measurements, impedance measurements, tunnelling measurements (Ivanov A P et al., *Nano Lett.* 2011 Jan. 12; 11 (1): 279-85), and FET measurements (International Application WO 2005/124888). Optical measurements may be combined with electrical measurements (Soni G V et al., *Rev Sci Instrum.* 2010 January; 81 (1): 014301). The measurement may be a transmembrane current measurement such as measurement of ionic current flowing through the pore.

[0480] Electrical measurements may be made using standard single channel recording equipment as describe in Stoddart D et al., *Proc Natl Acad Sci*, 12; 106 (19): 7702-7, Lieberman K R et al, *J Am Chem Soc.* 2010; 132 (50): 17961-72, and International Application WO 2000/28312. Alternatively, electrical measurements may be made using a multi-channel system, for example as described in International Application WO 2009/077734 and International Application WO 2011/067559.

[0481] The method is preferably carried out with a potential applied across the membrane. The applied potential may be a voltage potential. Alternatively, the applied potential may be a chemical potential. An example of this is using a salt gradient across a membrane, such as an amphiphilic layer. A salt gradient is disclosed in Holden et al., *J Am Chem Soc.* 2007 Jul. 11; 129 (27): 8650-5. In some instances, the current passing through the pore as a polynucleotide moves with respect to the pore is used to estimate or determine the sequence of the polynucleotide. This is strand sequencing.

[0482] The method may involve measuring the current passing through the pore as the polynucleotide moves with respect to the pore. Therefore the apparatus used in the method may also comprise an electrical circuit capable of applying a potential and measuring an electrical signal across the membrane and pore. The methods may be carried out using a patch clamp or a voltage clamp. The methods preferably involve the use of a voltage clamp.

[0483] The method of the invention may involve the measuring of a current passing through the pore as the polynucleotide moves with respect to the pore. Suitable conditions for measuring ionic currents through transmembrane protein pores are known in the art and disclosed in the Example. The method is typically carried out with a voltage applied across the membrane and pore. The voltage used is typically from +5 V to -5 V, such as from +4 V to -4 V, +3 V to -3 V or +2 V to -2 V. The voltage used is typically from -600 mV to +600 mV or -400 mV to +400 mV. The voltage used is preferably in a range having a lower limit selected from -400 mV, -300 mV, -200 mV, -150 mV, -100 mV, -50 mV, -20 mV and 0 mV and an upper limit independently selected from +10 mV, +20 mV, +50 mV, +100 mV, +150 mV, +200 mV, +300 mV and +400 mV. The voltage used is more preferably in the range 100 mV to 240 mV and

most preferably in the range of 120 mV to 220 mV. It is possible to increase discrimination between different nucleotides by a pore by using an increased applied potential.

[0484] The method is typically carried out in the presence of any charge carriers, such as metal salts, for example alkali metal salt, halide salts, for example chloride salts, such as alkali metal chloride salt. Charge carriers may include ionic liquids or organic salts, for example tetramethyl ammonium chloride, trimethylphenyl ammonium chloride, phenyltrimethyl ammonium chloride, or 1-ethyl-3-methyl imidazolium chloride. In the exemplary apparatus discussed above, the salt is present in the aqueous solution in the chamber. Potassium chloride (KCl), sodium chloride (NaCl), caesium chloride (CsCl) or a mixture of potassium ferrocyanide and potassium ferricyanide is typically used. KCl, NaCl and a mixture of potassium ferrocyanide and potassium ferricyanide are preferred. The charge carriers may be asymmetric across the membrane. For instance, the type and/or concentration of the charge carriers may be different on each side of the membrane.

[0485] The salt concentration may be at saturation. The salt concentration may be 3 M or lower and is typically from 0.1 to 2.5 M, from 0.3 to 1.9 M, from 0.5 to 1.8 M, from 0.7 to 1.7 M, from 0.9 to 1.6 M or from 1 M to 1.4 M. The salt concentration is preferably from 150 mM to 1 M. The method is preferably carried out using a salt concentration of at least 0.3 M, such as at least 0.4 M, at least 0.5 M, at least 0.6 M, at least 0.8 M, at least 1.0 M, at least 1.5 M, at least 2.0 M, at least 2.5 M or at least 3.0 M. High salt concentrations provide a high signal to noise ratio and allow for currents indicative of the presence of a nucleotide to be identified against the background of normal current fluctuations.

[0486] The method is typically carried out in the presence of a buffer. In the exemplary apparatus discussed above, the buffer is present in the aqueous solution in the chamber. Any buffer may be used in the method of the invention. Typically, the buffer is phosphate buffer. Other suitable buffers are HEPES and Tris-HCl buffer. The methods are typically carried out at a pH of from 4.0 to 12.0, from 4.5 to 10.0, from 5.0 to 9.0, from 5.5 to 8.8, from 6.0 to 8.7 or from 7.0 to 8.8 or 7.5 to 8.5. The pH used is preferably about 7.5.

[0487] The method may be carried out at from 0° C. to 100° C., from 15° C. to 95° C., from 16° C. to 90° C., from 17° C. to 85° C., from 18° C. to 80° C., 19° C. to 70° C., or from 20° C. to 60° C. The methods are typically carried out at room temperature. The methods are optionally carried out at a temperature that supports enzyme function, such as about 37° C.

Polynucleotide Binding Protein

[0488] The strand characterisation method preferably comprises contacting the polynucleotide with a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore.

[0489] More preferably, the method comprises (a) contacting the polynucleotide with a pore of the invention and a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore and (b) taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or

more characteristics of the polynucleotide, and thereby characterising the polynucleotide.

[0490] More preferably, the method comprises (a) contacting the polynucleotide with a pore of the invention and a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore and (b) measuring the current through the pore as the polynucleotide moves with respect to the pore, wherein the current is indicative of one or more characteristics of the polynucleotide, and thereby characterising the polynucleotide.

[0491] The polynucleotide binding protein may be any protein that is capable of binding to the polynucleotide and controlling its movement through the pore. It is straightforward in the art to determine whether or not a protein binds to a polynucleotide. The protein typically interacts with and modifies at least one property of the polynucleotide. The protein may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The protein may modify the polynucleotide by orienting it or moving it to a specific position, i.e. controlling its movement.

[0492] The polynucleotide binding protein is preferably derived from a polynucleotide handling enzyme. A polynucleotide handling enzyme is a polypeptide that is capable of interacting with and modifying at least one property of a polynucleotide. The enzyme may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The enzyme may modify the polynucleotide by orienting it or moving it to a specific position. The polynucleotide handling enzyme does not need to display enzymatic activity as long as it is capable of binding the polynucleotide and controlling its movement through the pore. For instance, the enzyme may be modified to remove its enzymatic activity or may be used under conditions which prevent it from acting as an enzyme. Such conditions are discussed in more detail below.

[0493] The polynucleotide handling enzyme is preferably derived from a nucleolytic enzyme. The polynucleotide handling enzyme used in the construct of the enzyme is more preferably derived from a member of any of the Enzyme Classification (EC) groups 3.1.11, 3.1.13, 3.1.14, 3.1.15, 3.1.16, 3.1.21, 3.1.22, 3.1.25, 3.1.26, 3.1.27, 3.1.30 and 3.1.31. The enzyme may be any of those disclosed in International Application No. PCT/GB10/000133 (published as WO 2010/086603).

[0494] Preferred enzymes are polymerases, exonucleases, helicases and topoisomerases, such as gyrases. Suitable enzymes include, but are not limited to, exonuclease I from *E. coli* (SEQ ID NO: 11), exonuclease III enzyme from *E. coli* (SEQ ID NO: 13), RecJ from *T. thermophilus* (SEQ ID NO: 15) and bacteriophage lambda exonuclease (SEQ ID NO: 17), TatD exonuclease and variants thereof. Three subunits comprising the sequence shown in SEQ ID NO: 15 or a variant thereof interact to form a trimer exonuclease. These exonucleases can also be used in the exonuclease method of the invention. The polymerase may be Pyro-Phage® 3173 DNA Polymerase (which is commercially available from Lucigen® Corporation), SD Polymerase (commercially available from Bioron®) or variants thereof. The enzyme is preferably Phi29 DNA polymerase (SEQ ID NO: 9) or a variant thereof. The topoisomerase is preferably a member of any of the Moiety Classification (EC) groups 5.99.1.2 and 5.99.1.3.

[0495] The enzyme is most preferably derived from a helicase, such as Hel308 Mbu (SEQ ID NO: 18), Hel308 Csy (SEQ ID NO: 19), Hel308 Tga (SEQ ID NO: 20), Hel308 Mhu (SEQ ID NO: 21), TraI Eco (SEQ ID NO: 22), XPD Mbu (SEQ ID NO: 23) or a variant thereof. Any helicase may be used in the invention. The helicase may be or be derived from a Hel308 helicase, a RecD helicase, such as TraI helicase or a TrwC helicase, a XPD helicase or a Dda helicase. The helicase may be any of the helicases, modified helicases or helicase constructs disclosed in International Application Nos. PCT/GB2012/052579 (published as WO 2013/057495); PCT/GB2012/053274 (published as WO 2013/098562); PCT/GB2012/053273 (published as WO2013098561); PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

[0496] The helicase preferably comprises the sequence shown in SEQ ID NO: 25 (TrwC Cba) or as variant thereof, the sequence shown in SEQ ID NO: 18 (Hel308 Mbu) or a variant thereof or the sequence shown in SEQ ID NO: 24 (Dda) or a variant thereof. Variants may differ from the native sequences in any of the ways discussed below for transmembrane pores. A preferred variant of SEQ ID NO: 24 comprises (a) E94C and A360C or (b) E94C, A360C, C109A and C136A and then optionally (Δ M1) GIG2 (i.e. deletion of M1 and then addition G1 and G2).

[0497] Any number of helicases may be used in accordance with the invention. For instance, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more helicases may be used. In some embodiments, different numbers of helicases may be used.

[0498] The method of the invention preferably comprises contacting the polynucleotide with two or more helicases. The two or more helicases are typically the same helicase. The two or more helicases may be different helicases.

[0499] The two or more helicases may be any combination of the helicases mentioned above. The two or more helicases may be two or more Dda helicases. The two or more helicases may be one or more Dda helicases and one or more TrwC helicases. The two or more helicases may be different variants of the same helicase.

[0500] The two or more helicases are preferably attached to one another. The two or more helicases are more preferably covalently attached to one another. The helicases may be attached in any order and using any method. Preferred helicase constructs for use in the invention are described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

[0501] A variant of SEQ ID NOs: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 is an enzyme that has an amino acid sequence which varies from that of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 and which retains polynucleotide binding ability. This can be measured using any method known in the art. For instance, the variant can be contacted with a polynucleotide and its ability to bind to and move along the polynucleotide can be measured. The variant may include modifications that facilitate binding of the polynucleotide and/or facilitate its activity at high salt concentrations and/or room temperature. Variants may be modified such that they bind polynucleotides (i.e. retain polynucleotide binding ability) but do not function as a helicase (i.e. do not move along polynucleotides when

provided with all the necessary components to facilitate movement, e.g. ATP and Mg^{2+}). Such modifications are known in the art. For instance, modification of the Mg^{2+} binding domain in helicases typically results in variants which do not function as helicases. These types of variants may act as molecular brakes (see below).

[0502] Over the entire length of the amino acid sequence of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25, a variant will preferably be at least 50% homologous to that sequence based on amino acid identity. More preferably, the variant polypeptide may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95%, amino acid identity over a stretch of 200 or more, for example 230, 250, 270, 280, 300, 400, 500, 600, 700, 800, 900 or 1000 or more, contiguous amino acids ("hard homology"). Homology is determined as described above. The variant may differ from the wild-type sequence in any of the ways discussed above with reference to SEQ ID NO: 2 above. The enzyme may be covalently attached to the pore. Any method may be used to covalently attach the enzyme to the pore.

[0503] A preferred molecular brake is TrwC Cba-Q594A (SEQ ID NO: 25 with the mutation Q594A). This variant does not function as a helicase (i.e. binds polynucleotides but does not move along them when provided with all the necessary components to facilitate movement, e.g. ATP and Mg^{2+}).

[0504] In strand sequencing, the polynucleotide is translocated through the pore either with or against an applied potential. Exonucleases that act progressively or processively on double stranded polynucleotides can be used on the cis side of the pore to feed the remaining single strand through under an applied potential or the trans side under a reverse potential. Likewise, a helicase that unwinds the double stranded DNA can also be used in a similar manner. A polymerase may also be used. There are also possibilities for sequencing applications that require strand translocation against an applied potential, but the DNA must be first "caught" by the enzyme under a reverse or no potential. With the potential then switched back following binding the strand will pass cis to trans through the pore and be held in an extended conformation by the current flow. The single strand DNA exonucleases or single strand DNA dependent polymerases can act as molecular motors to pull the recently translocated single strand back through the pore in a controlled stepwise manner, trans to cis, against the applied potential.

[0505] Any helicase may be used in the method. Helicases may work in two modes with respect to the pore. First, the method is preferably carried out using a helicase such that it moves the polynucleotide through the pore with the field resulting from the applied voltage. In this mode the 5' end of the polynucleotide is first captured in the pore, and the helicase moves the polynucleotide into the pore such that it is passed through the pore with the field until it finally translocates through to the trans side of the membrane. Alternatively, the method is preferably carried out such that a helicase moves the polynucleotide through the pore against the field resulting from the applied voltage. In this mode 3'

end of the polynucleotide is first captured in the pore, and the helicase moves the polynucleotide through the pore such that it is pulled out of the pore against the applied field until finally ejected back to the cis side of the membrane.

[0506] The method may also be carried out in the opposite direction. The 3' end of the polynucleotide may be first captured in the pore and the helicase may move the polynucleotide into the pore such that it is passed through the pore with the field until it finally translocates through to the trans side of the membrane.

[0507] When the helicase is not provided with the necessary components to facilitate movement or is modified to hinder or prevent its movement, it can bind to the polynucleotide and act as a brake slowing the movement of the polynucleotide when it is pulled into the pore by the applied field. In the inactive mode, it does not matter whether the polynucleotide is captured either 3' or 5' down, it is the applied field which pulls the polynucleotide into the pore towards the trans side with the enzyme acting as a brake. When in the inactive mode, the movement control of the polynucleotide by the helicase can be described in a number of ways including ratcheting, sliding and braking. Helicase variants which lack helicase activity can also be used in this way.

[0508] The polynucleotide may be contacted with the polynucleotide binding protein and the pore in any order. It is preferred that, when the polynucleotide is contacted with the polynucleotide binding protein, such as a helicase, and the pore, the polynucleotide firstly forms a complex with the protein. When the voltage is applied across the pore, the polynucleotide/protein complex then forms a complex with the pore and controls the movement of the polynucleotide through the pore.

[0509] Any steps in the method using a polynucleotide binding protein are typically carried out in the presence of free nucleotides or free nucleotide analogues and an enzyme cofactor that facilitates the action of the polynucleotide binding protein. The free nucleotides may be one or more of any of the individual nucleotides discussed above. The free nucleotides include, but are not limited to, adenosine monophosphate (AMP), adenosine diphosphate (ADP), adenosine triphosphate (ATP), guanosine monophosphate (GMP), guanosine diphosphate (GDP), guanosine triphosphate (GTP), thymidine monophosphate (TMP), thymidine diphosphate (TDP), thymidine triphosphate (TTP), uridine monophosphate (UMP), uridine diphosphate (UDP), uridine triphosphate (UTP), cytidine monophosphate (CMP), cytidine diphosphate (CDP), cytidine triphosphate (CTP), cyclic adenosine monophosphate (cAMP), cyclic guanosine monophosphate (cGMP), deoxyadenosine monophosphate (dAMP), deoxyadenosine diphosphate (dADP), deoxyadenosine triphosphate (dATP), deoxyguanosine monophosphate (dGMP), deoxyguanosine diphosphate (dGDP), deoxyguanosine triphosphate (dGTP), deoxythymidine monophosphate (dTMP), deoxythymidine diphosphate (dTDP), deoxythymidine triphosphate (dTTP), deoxyuridine monophosphate (dUMP), deoxyuridine diphosphate (dUDP), deoxyuridine triphosphate (dUTP), deoxycytidine monophosphate (dCMP), deoxycytidine diphosphate (dCDP) and deoxycytidine triphosphate (dCTP). The free nucleotides are preferably selected from AMP, TMP, GMP, CMP, UMP, dAMP, dTMP, dGMP or dCMP. The free nucleotides are preferably adenosine triphosphate (ATP). The enzyme cofactor is a factor that allows the construct to

function. The enzyme cofactor is preferably a divalent metal cation. The divalent metal cation is preferably Mg^{2+} , Mn^{2+} , Ca^{2+} or Co^{2+} . The enzyme cofactor is most preferably Mg^{2+} .

Helicase(s) and Molecular Brake(s)

[0510] In a preferred embodiment, the method comprises:

[0511] (a) providing the polynucleotide with one or more helicases and one or more molecular brakes attached to the polynucleotide;

[0512] (b) contacting the polynucleotide with a pore of the invention and applying a potential across the pore such that the one or more helicases and the one or more molecular brakes are brought together and both control the movement of the polynucleotide with respect to, such as through, the pore;

[0513] (c) taking one or more measurements as the polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the polynucleotide and thereby characterising the polynucleotide.

[0514] This type of method is discussed in detail in the International Application PCT/GB2014/052737

[0515] The one or more helicases may be any of those discussed above. The one or more molecular brakes may be any compound or molecule which binds to the polynucleotide and slows the movement of the polynucleotide through the pore. The one or more molecular brakes preferably comprise one or more compounds which bind to the polynucleotide. The one or more compounds are preferably one or more macrocycles. Suitable macrocycles include, but are not limited to, cyclodextrins, calixarenes, cyclic peptides, crown ethers, cucurbiturils, pillararenes, derivatives thereof or a combination thereof. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H.-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The agent is more preferably heptakis-6-amino- β -cyclodextrin (am_7 - β CD), 6-monodeoxy-6-monoamino- β -cyclodextrin (am_1 - β CD) or heptakis-(6-deoxy-6-guanidino)-cyclodextrin (gu_7 - β CD).

[0516] The one or more molecular brakes are preferably one or more single stranded binding proteins (SSB). The one or more molecular brakes are more preferably a single-stranded binding protein (SSB) comprising a carboxy-terminal (C-terminal) region which does not have a net negative charge or (ii) a modified SSB comprising one or more modifications in its C-terminal region which decreases the net negative charge of the C-terminal region. The one or more molecular brakes are most preferably one of the SSBs disclosed in International Application No. PCT/GB2013/051924 (published as WO 2014/013259).

[0517] The one or more molecular brakes are preferably one or more polynucleotide binding proteins. The polynucleotide binding protein may be any protein that is capable of binding to the polynucleotide and controlling its movement through the pore. It is straightforward in the art to determine whether or not a protein binds to a polynucleotide. The protein typically interacts with and modifies at least one property of the polynucleotide. The protein may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The moiety may modify the polynucleotide by orienting it or moving it to a specific position, i.e. controlling its movement.

[0518] The polynucleotide binding protein is preferably derived from a polynucleotide handling enzyme. The one or more molecular brakes may be derived from any of the polynucleotide handling enzymes discussed above. Modified versions of Phi29 polymerase (SEQ ID NO: 8) which act as molecular brakes are disclosed in U.S. Pat. No. 5,576,204. The one or more molecular brakes are preferably derived from a helicase.

[0519] Any number of molecular brakes derived from a helicase may be used. For instance, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more helicases may be used as molecular brakes. If two or more helicases are used as molecular brakes, the two or more helicases are typically the same helicase. The two or more helicases may be different helicases.

[0520] The two or more helicases may be any combination of the helicases mentioned above. The two or more helicases may be two or more Dda helicases. The two or more helicases may be one or more Dda helicases and one or more TrwC helicases. The two or more helicases may be different variants of the same helicase.

[0521] The two or more helicases are preferably attached to one another. The two or more helicases are more preferably covalently attached to one another. The helicases may be attached in any order and using any method. The one or more molecular brakes derived from helicases are preferably modified to reduce the size of an opening in the polynucleotide binding domain through which in at least one conformational state the polynucleotide can unbind from the helicase. This is disclosed in WO 2014/013260.

[0522] Preferred helicase constructs for use in the invention are described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

[0523] If the one or more helicases are used in the active mode (i.e. when the one or more helicases are provided with all the necessary components to facilitate movement, e.g. ATP and Mg^{2+}), the one or more molecular brakes are preferably (a) used in an inactive mode (i.e. are used in the absence of the necessary components to facilitate movement or are incapable of active movement), (b) used in an active mode where the one or more molecular brakes move in the opposite direction to the one or more helicases or (c) used in an active mode where the one or more molecular brakes move in the same direction as the one or more helicases and move slowly than the one or more helicases.

[0524] If the one or more helicases are used in the inactive mode (i.e. when the one or more helicases are not provided with all the necessary components to facilitate movement, e.g. ATP and Mg^{2+} or are incapable of active movement), the one or more molecular brakes are preferably (a) used in an inactive mode (i.e. are used in the absence of the necessary components to facilitate movement or are incapable of active movement) or (b) used in an active mode where the one or more molecular brakes move along the polynucleotide in the same direction as the polynucleotide through the pore.

[0525] The one or more helicases and one or more molecular brakes may be attached to the polynucleotide at any positions so that they are brought together and both control the movement of the polynucleotide through the pore. The one or more helicases and one or more molecular brakes are at least one nucleotide apart, such as at least 5, at

least 10, at least 50, at least 100, at least 500, at least 1000, at least 5000, at least 10,000, at least 50,000 nucleotides or more apart. If the method concerns characterising a double stranded polynucleotide provided with a Y adaptor at one end and a hairpin loop adaptor at the other end, the one or more helicases are preferably attached to the Y adaptor and the one or more molecular brakes are preferably attached to the hairpin loop adaptor. In this embodiment, the one or more molecular brakes are preferably one or more helicases that are modified such that they bind the polynucleotide but do not function as a helicase. The one or more helicases attached to the Y adaptor are preferably stalled at a spacer as discussed in more detail below. The one or more molecular brakes attach to the hairpin loop adaptor are preferably not stalled at a spacer. The one or more helicases and the one or more molecular brakes are preferably brought together when the one or more helicases reach the hairpin loop. The one or more helicases may be attached to the Y adaptor before the Y adaptor is attached to the polynucleotide or after the Y adaptor is attached to the polynucleotide. The one or more molecular brakes may be attached to the hairpin loop adaptor before the hairpin loop adaptor is attached to the polynucleotide or after the hairpin loop adaptor is attached to the polynucleotide.

[0526] The one or more helicases and the one or more molecular brakes are preferably not attached to one another. The one or more helicases and the one or more molecular brakes are more preferably not covalently attached to one another. The one or more helicases and the one or more molecular brakes are preferably not attached as described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

Spacers

[0527] The one or more helicases may be stalled at one or more spacers as discussed in International Application No. PCT/GB2014/050175. Any configuration of one or more helicases and one or more spacers disclosed in the International Application may be used in this invention.

[0528] When a part of the polynucleotide enters the pore and moves through the pore along the field resulting from the applied potential, the one or more helicases are moved past the spacer by the pore as the polynucleotide moves through the pore. This is because the polynucleotide (including the one or more spacers) moves through the pore and the one or more helicases remain on top of the pore.

[0529] The one or more spacers are preferably part of the polynucleotide, for instance they interrupt(s) the polynucleotide sequence. The one or more spacers are preferably not part of one or more blocking molecules, such as speed bumps, hybridised to the polynucleotide.

[0530] There may be any number of spacers in the polynucleotide, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more spacers. There are preferably two, four or six spacers in the polynucleotide. There may be one or more spacers in different regions of the polynucleotide, such as one or more spacers in the Y adaptor and/or hairpin loop adaptor.

[0531] The one or more spacers each provides an energy barrier which the one or more helicases cannot overcome even in the active mode. The one or more spacers may stall the one or more helicases by reducing the traction of the helicase (for instance by removing the bases from the

nucleotides in the polynucleotide) or physically blocking movement of the one or more helicases (for instance using a bulky chemical group).

[0532] The one or more spacers may comprise any molecule or combination of molecules that stalls the one or more helicases. The one or more spacers may comprise any molecule or combination of molecules that prevents the one or more helicases from moving along the polynucleotide. It is straightforward to determine whether or not the one or more helicases are stalled at one or more spacers in the absence of a transmembrane pore and an applied potential. For instance, the ability of a helicase to move past a spacer and displace a complementary strand of DNA can be measured by PAGE.

[0533] The one or more spacers typically comprise a linear molecule, such as a polymer. The one or more spacers typically have a different structure from the polynucleotide. For instance, if the polynucleotide is DNA, the one or more spacers are typically not DNA. In particular, if the polynucleotide is deoxyribonucleic acid (DNA) or ribonucleic acid (RNA), the one or more spacers preferably comprise peptide nucleic acid (PNA), glycerol nucleic acid (GNA), threose nucleic acid (TNA), locked nucleic acid (LNA) or a synthetic polymer with nucleotide side chains. The one or more spacers may comprise one or more nucleotides in the opposite direction from the polynucleotide. For instance, the one or more spacers may comprise one or more nucleotides in 3' to 5' direction when the polynucleotide is in 5' to 3' direction. The nucleotides may be any of those discussed above.

[0534] The one or more spacers preferably comprises one or more nitroindoles, such as one or more 5-nitroindoles, one or more inosines, one or more acridines, one or more 2-aminopurines, one or more 2-6-diaminopurines, one or more 5-bromo-deoxyuridines, one or more inverted thymidines (inverted dTs), one or more inverted dideoxy-thymidines (ddTs), one or more dideoxy-cytidines (ddCs), one or more 5-methylcytidines, one or more 5-hydroxymethylcytidines, one or more 2'-O-Methyl RNA bases, one or more Iso-deoxycytidines (Iso-dCs), one or more Iso-deoxyguanosines (Iso-dGs), one or more iSpC3 groups (i.e. nucleotides which lack sugar and a base), one or more photo-cleavable (PC) groups, one or more hexandiol groups, one or more spacer 9 (iSp9) groups, one or more spacer 18 (iSp18) groups, a polymer or one or more thiol connections. The one or more spacers may comprise any combination of these groups. Many of these groups are commercially available from IDT® (Integrated DNA Technologies®).

[0535] The one or more spacers may contain any number of these groups. For instance, for 2-aminopurines, 2-6-diaminopurines, 5-bromo-deoxyuridines, inverted dTs, ddTs, ddCs, 5-methylcytidines, 5-hydroxymethylcytidines, 2'-O-Methyl RNA bases, Iso-dCs, Iso-dGs, iSpC3 groups, PC groups, hexandiol groups and thiol connections, the one or more spacers preferably comprise 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more. The one or more spacers preferably comprise 2, 3, 4, 5, 6, 7, 8 or more iSp9 groups. The one or more spacers preferably comprise 2, 3, 4, 5 or 6 or more iSp18 groups. The most preferred spacer is four iSp18 groups.

[0536] The polymer is preferably a polypeptide or a polyethylene glycol (PEG). The polypeptide preferably comprises 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more amino acids. The PEG preferably comprises 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more monomer units.

[0537] The one or more spacers preferably comprise one or more abasic nucleotides (i.e. nucleotides lacking a nucleobase), such as 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more abasic nucleotides. The nucleobase can be replaced by —H (idSp) or —OH in the abasic nucleotide. Abasic spacers can be inserted into polynucleotides by removing the nucleobases from one or more adjacent nucleotides. For instance, polynucleotides may be modified to include 3-methyladenine, 7-methylguanine, 1,N6-ethenoadenine inosine or hypoxanthine and the nucleobases may be removed from these nucleotides using Human Alkyladenine DNA Glycosylase (hAAG). Alternatively, polynucleotides may be modified to include uracil and the nucleobases removed with Uracil-DNA Glycosylase (UDG). In one embodiment, the one or more spacers do not comprise any abasic nucleotides.

[0538] The one or more helicases may be stalled by (i.e. before) or on each linear molecule spacers. If linear molecule spacers are used, the polynucleotide is preferably provided with a double stranded region of polynucleotide adjacent to the end of each spacer past which the one or more helicases are to be moved. The double stranded region typically helps to stall the one or more helicases on the adjacent spacer. The presence of the double stranded region (s) is particularly preferred if the method is carried out at a salt concentration of about 100 mM or lower. Each double stranded region is typically at least 10, such as at least 12, nucleotides in length. If the polynucleotide used in the invention is single stranded, a double stranded region may be formed by hybridising a shorter polynucleotide to a region adjacent to a spacer. The shorter polynucleotide is typically formed from the same nucleotides as the polynucleotide, but may be formed from different nucleotides. For instance, the shorter polynucleotide may be formed from LNA.

[0539] If linear molecule spacers are used, the polynucleotide is preferably provided with a blocking molecule at the end of each spacer opposite to the end past which the one or more helicases are to be moved. This can help to ensure that the one or more helicases remain stalled on each spacer. It may also help retain the one or more helicases on the polynucleotide in the case that it/they diffuse(s) off in solution. The blocking molecule may be any of the chemical groups discussed below which physically cause the one or more helicases to stall. The blocking molecule may be a double stranded region of polynucleotide.

[0540] The one or more spacers preferably comprise one or more chemical groups which physically cause the one or more helicases to stall. The one or more chemical groups are preferably one or more pendant chemical groups. The one or

cal groups may be attached to the polynucleotide backbone. Any number of these chemical groups may be present, such as 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more. Suitable groups include, but are not limited to, fluorophores, streptavidin and/or biotin, cholesterol, methylene blue, dinitrophenols (DNPs), digoxigenin and/or anti-digoxigenin and dibenzylcyclooctyne groups.

[0541] Different spacers in the polynucleotide may comprise different stalling molecules. For instance, one spacer may comprise one of the linear molecules discussed above and another spacer may comprise one or more chemical groups which physically cause the one or more helicases to stall. A spacer may comprise any of the linear molecules discussed above and one or more chemical groups which physically cause the one or more helicases to stall, such as one or more abasics and a fluorophore.

[0542] Suitable spacers can be designed depending on the type of polynucleotide and the conditions under which the method of the invention is carried out. Most helicases bind and move along DNA and so may be stalled using anything that is not DNA. Suitable molecules are discussed above.

[0543] The method of the invention is preferably carried out in the presence of free nucleotides and/or the presence of a helicase cofactor. This is discussed in more detail below. In the absence of the transmembrane pore and an applied potential, the one or more spacers are preferably capable of stalling the one or more helicases in the presence of free nucleotides and/or the presence of a helicase cofactor.

[0544] If the method of the invention is carried out in the presence of free nucleotides and a helicase cofactor as discussed below (such that the one or more helicases are in the active mode), one or more longer spacers are typically used to ensure that the one or more helicases are stalled on the polynucleotide before they are contacted with the transmembrane pore and a potential is applied. One or more shorter spacers may be used in the absence of free nucleotides and a helicase cofactor (such that the one or more helicases are in the inactive mode).

[0545] The salt concentration also affects the ability of the one or more spacers to stall the one or more helicases. In the absence of the transmembrane pore and an applied potential, the one or more spacers are preferably capable of stalling the one or more helicases at a salt concentration of about 100 mM or lower. The higher the salt concentration used in the method of the invention, the shorter the one or more spacers that are typically used and vice versa.

[0546] Preferred combinations of features are shown in Table 4 below.

TABLE 4

Polynucleotide	Spacer composition*	Spacer length (i.e. number of *)	Salt []	Free nucleotides?	Helicase cofactor?
DNA	iSpC3	4	1M	Yes	Yes
DNA	iSp18	4	100-1000 mM	Yes	Yes
DNA	iSp18	6	<100-1000 mM	Yes	Yes
DNA	iSp18	2	1M	Yes	Yes
DNA	iSpC3	12	<100-1000 mM	Yes	Yes
DNA	iSpC3	20	<100-1000 mM	Yes	Yes
DNA	iSp9	6	100-1000 mM	Yes	Yes
DNA	idSp	4	1M	Yes	Yes

more chemical groups may be attached to one or more nucleobases in the polynucleotide. The one or more chemi-

[0547] The method may concern moving two or more helicases past a spacer. In such instances, the length of the

spacer is typically increased to prevent the trailing helicase from pushing the leading helicase past the spacer in the absence of the pore and applied potential. If the method concerns moving two or more helicases past one or more spacers, the spacer lengths discussed above may be increased at least 1.5 fold, such 2 fold, 2.5 fold or 3 fold. For instance, if the method concerns moving two or more helicases past one or more spacers, the spacer lengths in the third column of Table 4 above may be increased 1.5 fold, 2 fold, 2.5 fold or 3 fold.

Membrane

[0548] The pore of the invention may be present in a membrane. In the methods of the invention, the polynucleotide is typically contacted with the pore of the invention in a membrane. Any membrane may be used in accordance with the invention. Suitable membranes are well-known in the art. The membrane is preferably an amphiphilic layer. An amphiphilic layer is a layer formed from amphiphilic molecules, such as phospholipids, which have both hydrophilic and lipophilic properties. The amphiphilic molecules may be synthetic or naturally occurring. Non-naturally occurring amphiphiles and amphiphiles which form a monolayer are known in the art and include, for example, block copolymers (Gonzalez-Perez et al., *Langmuir*, 2009, 25, 10447-10450). Block copolymers are polymeric materials in which two or more monomer sub-units that are polymerized together to create a single polymer chain. Block copolymers typically have properties that are contributed by each monomer sub-unit. However, a block copolymer may have unique properties that polymers formed from the individual sub-units do not possess. Block copolymers can be engineered such that one of the monomer sub-units is hydrophobic (i.e. lipophilic), whilst the other sub-unit(s) are hydrophilic whilst in aqueous media. In this case, the block copolymer may possess amphiphilic properties and may form a structure that mimics a biological membrane. The block copolymer may be a diblock (consisting of two monomer sub-units), but may also be constructed from more than two monomer sub-units to form more complex arrangements that behave as amphiphiles. The copolymer may be a triblock, tetrablock or pentablock copolymer. The membrane is preferably a triblock copolymer membrane.

[0549] Archaeobacterial bipolar tetraether lipids are naturally occurring lipids that are constructed such that the lipid forms a monolayer membrane. These lipids are generally found in extremophiles that survive in harsh biological environments, thermophiles, halophiles and acidophiles. Their stability is believed to derive from the fused nature of the final bilayer. It is straightforward to construct block copolymer materials that mimic these biological entities by creating a triblock polymer that has the general motif hydrophilic-hydrophobic-hydrophilic. This material may form monomeric membranes that behave similarly to lipid bilayers and encompass a range of phase behaviours from vesicles through to laminar membranes. Membranes formed from these triblock copolymers hold several advantages over biological lipid membranes. Because the triblock copolymer is synthesised, the exact construction can be carefully controlled to provide the correct chain lengths and properties required to form membranes and to interact with pores and other proteins.

[0550] Block copolymers may also be constructed from sub-units that are not classed as lipid sub-materials; for

example a hydrophobic polymer may be made from siloxane or other non-hydrocarbon based monomers. The hydrophilic sub-section of block copolymer can also possess low protein binding properties, which allows the creation of a membrane that is highly resistant when exposed to raw biological samples. This head group unit may also be derived from non-classical lipid head-groups.

[0551] Triblock copolymer membranes also have increased mechanical and environmental stability compared with biological lipid membranes, for example a much higher operational temperature or pH range. The synthetic nature of the block copolymers provides a platform to customise polymer based membranes for a wide range of applications.

[0552] The membrane is most preferably one of the membranes disclosed in International Application No. PCT/GB2013/052766 or PCT/GB2013/052767.

[0553] The amphiphilic molecules may be chemically-modified or functionalised to facilitate coupling of the polynucleotide.

[0554] The amphiphilic layer may be a monolayer or a bilayer. The amphiphilic layer is typically planar. The amphiphilic layer may be curved. The amphiphilic layer may be supported.

[0555] Amphiphilic membranes are typically naturally mobile, essentially acting as two dimensional fluids with lipid diffusion rates of approximately 10^{-8} cm s⁻¹. This means that the pore and coupled polynucleotide can typically move within an amphiphilic membrane.

[0556] The membrane may be a lipid bilayer. Lipid bilayers are models of cell membranes and serve as excellent platforms for a range of experimental studies. For example, lipid bilayers can be used for in vitro investigation of membrane proteins by single-channel recording. Alternatively, lipid bilayers can be used as biosensors to detect the presence of a range of substances. The lipid bilayer may be any lipid bilayer. Suitable lipid bilayers include, but are not limited to, a planar lipid bilayer, a supported bilayer or a liposome. The lipid bilayer is preferably a planar lipid bilayer. Suitable lipid bilayers are disclosed in International Application No. PCT/GB08/000563 (published as WO 2008/102121), International Application No. PCT/GB08/004127 (published as WO 2009/077734) and International Application No. PCT/GB2006/001057 (published as WO 2006/100484).

[0557] Methods for forming lipid bilayers are known in the art. Lipid bilayers are commonly formed by the method of Montal and Mueller (*Proc. Natl. Acad. Sci. USA.*, 1972; 69:3561-3566), in which a lipid monolayer is carried on aqueous solution/air interface past either side of an aperture which is perpendicular to that interface. The lipid is normally added to the surface of an aqueous electrolyte solution by first dissolving it in an organic solvent and then allowing a drop of the solvent to evaporate on the surface of the aqueous solution on either side of the aperture. Once the organic solvent has evaporated, the solution/air interfaces on either side of the aperture are physically moved up and down past the aperture until a bilayer is formed. Planar lipid bilayers may be formed across an aperture in a membrane or across an opening into a recess.

[0558] The method of Montal & Mueller is popular because it is a cost-effective and relatively straightforward method of forming good quality lipid bilayers that are suitable for protein pore insertion. Other common methods

of bilayer formation include tip-dipping, painting bilayers and patch-clamping of liposome bilayers.

[0559] Tip-dipping bilayer formation entails touching the aperture surface (for example, a pipette tip) onto the surface of a test solution that is carrying a monolayer of lipid. Again, the lipid monolayer is first generated at the solution/air interface by allowing a drop of lipid dissolved in organic solvent to evaporate at the solution surface. The bilayer is then formed by the *Langmuir*-Schaefer process and requires mechanical automation to move the aperture relative to the solution surface.

[0560] For painted bilayers, a drop of lipid dissolved in organic solvent is applied directly to the aperture, which is submerged in an aqueous test solution. The lipid solution is spread thinly over the aperture using a paintbrush or an equivalent. Thinning of the solvent results in formation of a lipid bilayer. However, complete removal of the solvent from the bilayer is difficult and consequently the bilayer formed by this method is less stable and more prone to noise during electrochemical measurement.

[0561] Patch-clamping is commonly used in the study of biological cell membranes. The cell membrane is clamped to the end of a pipette by suction and a patch of the membrane becomes attached over the aperture. The method has been adapted for producing lipid bilayers by clamping liposomes which then burst to leave a lipid bilayer sealing over the aperture of the pipette. The method requires stable, giant and unilamellar liposomes and the fabrication of small apertures in materials having a glass surface.

[0562] Liposomes can be formed by sonication, extrusion or the Mozafari method (Colas et al. (2007) *Micron* 38:841-847).

[0563] In a preferred embodiment, the lipid bilayer is formed as described in International Application No. PCT/GB08/004127 (published as WO 2009/077734). Advantageously in this method, the lipid bilayer is formed from dried lipids. In a most preferred embodiment, the lipid bilayer is formed across an opening as described in WO2009/077734 (PCT/GB08/004127).

[0564] A lipid bilayer is formed from two opposing layers of lipids. The two layers of lipids are arranged such that their hydrophobic tail groups face towards each other to form a hydrophobic interior. The hydrophilic head groups of the lipids face outwards towards the aqueous environment on each side of the bilayer. The bilayer may be present in a number of lipid phases including, but not limited to, the liquid disordered phase (fluid lamellar), liquid ordered phase, solid ordered phase (lamellar gel phase, interdigitated gel phase) and planar bilayer crystals (lamellar sub-gel phase, lamellar crystalline phase).

[0565] Any lipid composition that forms a lipid bilayer may be used. The lipid composition is chosen such that a lipid bilayer having the required properties, such surface charge, ability to support membrane proteins, packing density or mechanical properties, is formed. The lipid composition can comprise one or more different lipids. For instance, the lipid composition can contain up to 100 lipids. The lipid composition preferably contains 1 to 10 lipids. The lipid composition may comprise naturally-occurring lipids and/or artificial lipids.

[0566] The lipids typically comprise a head group, an interfacial moiety and two hydrophobic tail groups which may be the same or different. Suitable head groups include, but are not limited to, neutral head groups, such as dia-

cylglycerides (DG) and ceramides (CM); zwitterionic head groups, such as phosphatidylcholine (PC), phosphatidylethanolamine (PE) and sphingomyelin (SM); negatively charged head groups, such as phosphatidylglycerol (PG); phosphatidylserine (PS), phosphatidylinositol (PI), phosphatic acid (PA) and cardiolipin (CA); and positively charged headgroups, such as trimethylammonium-Propane (TAP). Suitable interfacial moieties include, but are not limited to, naturally-occurring interfacial moieties, such as glycerol-based or ceramide-based moieties. Suitable hydrophobic tail groups include, but are not limited to, saturated hydrocarbon chains, such as lauric acid (n-Dodecanoic acid), myristic acid (n-Tetradecanoic acid), palmitic acid (n-Hexadecanoic acid), stearic acid (n-Octadecanoic) and arachidic (n-Eicosanoic); unsaturated hydrocarbon chains, such as oleic acid (cis-9-Octadecanoic); and branched hydrocarbon chains, such as phytanoyl. The length of the chain and the position and number of the double bonds in the unsaturated hydrocarbon chains can vary. The length of the chains and the position and number of the branches, such as methyl groups, in the branched hydrocarbon chains can vary. The hydrophobic tail groups can be linked to the interfacial moiety as an ether or an ester. The lipids may be mycolic acid.

[0567] The lipids can also be chemically-modified. The head group or the tail group of the lipids may be chemically-modified. Suitable lipids whose head groups have been chemically-modified include, but are not limited to, PEG-modified lipids, such as 1,2-Diacyl-sn-Glycero-3-Phosphoethanolamine-N-[Methoxy (Polyethylene glycol)-2000]; functionalised PEG Lipids, such as 1,2-Distearoyl-sn-Glycero-3 Phosphoethanolamine-N-[Biotinyl(Polyethylene Glycol) 2000]; and lipids modified for conjugation, such as 1,2-Dioleoyl-sn-Glycero-3-Phosphoethanolamine-N-(succinyl) and 1,2-Dipalmitoyl-sn-Glycero-3-Phosphoethanolamine-N-(Biotinyl). Suitable lipids whose tail groups have been chemically-modified include, but are not limited to, polymerisable lipids, such as 1,2-bis(10,12-tricosadiynoyl)-sn-Glycero-3-Phosphocholine; fluorinated lipids, such as 1-Palmitoyl-2-(16-Fluoropalmitoyl)-sn-Glycero-3-Phosphocholine; deuterated lipids, such as 1,2-Dipalmitoyl-D62-sn-Glycero-3-Phosphocholine; and ether linked lipids, such as 1,2-Di-O-phytanoyl-sn-Glycero-3-Phosphocholine. The lipids may be chemically-modified or functionalised to facilitate coupling of the polynucleotide.

[0568] The amphiphilic layer, for example the lipid composition, typically comprises one or more additives that will affect the properties of the layer. Suitable additives include, but are not limited to, fatty acids, such as palmitic acid, myristic acid and oleic acid; fatty alcohols, such as palmitic alcohol, myristic alcohol and oleic alcohol; sterols, such as cholesterol, ergosterol, lanosterol, sitosterol and stigmasterol; lysophospholipids, such as 1-Acyl-2-Hydroxy-sn-Glycero-3-Phosphocholine; and ceramides.

[0569] In another preferred embodiment, the membrane comprises a solid state layer. Solid state layers can be formed from both organic and inorganic materials including, but not limited to, microelectronic materials, insulating materials such as Si₃N₄, Al₂O₃, and SiO₂, organic and inorganic polymers such as polyamide, plastics such as Teflon® or elastomers such as two-component addition-cure silicone rubber, and glasses. The solid state layer may be formed from graphene. Suitable graphene layers are disclosed in International Application No. PCT/US2008/

010637 (published as WO 2009/035647). If the membrane comprises a solid state layer, the pore is typically present in an amphiphilic membrane or layer contained within the solid state layer, for instance within a hole, well, gap, channel, trench or slit within the solid state layer. The skilled person can prepare suitable solid state/amphiphilic hybrid systems. Suitable systems are disclosed in WO 2009/020682 and WO 2012/005857. Any of the amphiphilic membranes or layers discussed above may be used.

[0570] The method is typically carried out using (i) an artificial amphiphilic layer comprising a pore, (ii) an isolated, naturally-occurring lipid bilayer comprising a pore, or (iii) a cell having a pore inserted therein. The method is typically carried out using an artificial amphiphilic layer, such as an artificial triblock copolymer layer. The layer may comprise other transmembrane and/or intramembrane proteins as well as other molecules in addition to the pore. Suitable apparatus and conditions are discussed below. The method of the invention is typically carried out in vitro.

Coupling

[0571] The polynucleotide is preferably coupled to the membrane comprising the pore. The method may comprise coupling the polynucleotide to the membrane comprising the pore. The polynucleotide is preferably coupled to the membrane using one or more anchors. The polynucleotide may be coupled to the membrane using any known method.

[0572] Each anchor comprises a group which couples (or binds) to the polynucleotide and a group which couples (or binds) to the membrane. Each anchor may covalently couple (or bind) to the polynucleotide and/or the membrane. If a Y adaptor and/or a hairpin loop adaptors are used, the polynucleotide is preferably coupled to the membrane using the adaptor(s).

[0573] The polynucleotide may be coupled to the membrane using any number of anchors, such as 2, 3, 4 or more anchors. For instance, a polynucleotide may be coupled to the membrane using two anchors each of which separately couples (or binds) to both the polynucleotide and membrane.

[0574] The one or more anchors may comprise the one or more helicases and/or the one or more molecular brakes.

[0575] If the membrane is an amphiphilic layer, such as a copolymer membrane or a lipid bilayer, the one or more anchors preferably comprise a polypeptide anchor present in the membrane and/or a hydrophobic anchor present in the membrane. The hydrophobic anchor is preferably a lipid, fatty acid, sterol, carbon nanotube, polypeptide, protein or amino acid, for example cholesterol, palmitate or tocopherol. In preferred embodiments, the one or more anchors are not the pore.

[0576] The components of the membrane, such as the amphiphilic molecules, copolymer or lipids, may be chemically-modified or functionalised to form the one or more anchors. Examples of suitable chemical modifications and suitable ways of functionalising the components of the membrane are discussed in more detail below. Any proportion of the membrane components may be functionalised, for example at least 0.01%, at least 0.1%, at least 1%, at least 10%, at least 25%, at least 50% or 100%.

[0577] The polynucleotide may be coupled directly to the membrane. The one or more anchors used to couple the polynucleotide to the membrane preferably comprise a linker. The one or more anchors may comprise one or more,

such as 2, 3, 4 or more, linkers. One linker may be used couple more than one, such as 2, 3, 4 or more, polynucleotides to the membrane.

[0578] Preferred linkers include, but are not limited to, polymers, such as polynucleotides, polyethylene glycols (PEGs), polysaccharides and polypeptides. These linkers may be linear, branched or circular. For instance, the linker may be a circular polynucleotide. The polynucleotide may hybridise to a complementary sequence on the circular polynucleotide linker.

[0579] The one or more anchors or one or more linkers may comprise a component that can be cut to broken down, such as a restriction site or a photolabile group.

[0580] Functionalised linkers and the ways in which they can couple molecules are known in the art. For instance, linkers functionalised with maleimide groups will react with and attach to cysteine residues in proteins. In the context of this invention, the protein may be present in the membrane or may be used to couple (or bind) to the polynucleotide. This is discussed in more detail below.

[0581] Crosslinkage of polynucleotides can be avoided using a "lock and key" arrangement. Only one end of each linker may react together to form a longer linker and the other ends of the linker each react with the polynucleotide or membrane respectively. Such linkers are described in International Application No. PCT/GB10/000132 (published as WO 2010/086602).

[0582] The use of a linker is preferred in the sequencing embodiments discussed below. If a polynucleotide is permanently coupled directly to the membrane in the sense that it does not uncouple when interacting with the pore (i.e. does not uncouple in step (b) or (e)), then some sequence data will be lost as the sequencing run cannot continue to the end of the polynucleotide due to the distance between the membrane and the pore. If a linker is used, then the polynucleotide can be processed to completion.

[0583] The coupling may be permanent or stable. In other words, the coupling may be such that the polynucleotide remains coupled to the membrane when interacting with the pore.

[0584] The coupling may be transient. In other words, the coupling may be such that the polynucleotide may decouple from the membrane when interacting with the pore.

[0585] For certain applications, such as aptamer detection, the transient nature of the coupling is preferred. If a permanent or stable linker is attached directly to either 5' or 3' end of a polynucleotide and the linker is shorter than the distance between the membrane and the transmembrane pore's channel, then some sequence data will be lost as the sequencing run cannot continue to the end of the polynucleotide. If the coupling is transient, then when the coupled end randomly becomes free of the membrane, then the polynucleotide can be processed to completion. Chemical groups that form permanent/stable or transient links are discussed in more detail below. The polynucleotide may be transiently coupled to an amphiphilic layer or triblock copolymer membrane using cholesterol or a fatty acyl chain. Any fatty acyl chain having a length of from 6 to 30 carbon atom, such as hexadecanoic acid, may be used.

[0586] In preferred embodiments, a polynucleotide, such as a nucleic acid, is coupled to an amphiphilic layer such as a triblock copolymer membrane or lipid bilayer. Coupling of nucleic acids to synthetic lipid bilayers has been carried out

previously with various different tethering strategies. These are summarised in Table 5 below.

TABLE 5

Anchor comprising	Type of coupling	Reference
Thiol	Stable	Yoshina-Ishii, C. and S. G. Boxer (2003). "Arrays of mobile tethered vesicles on supported lipid bilayers." <i>J Am Chem Soc</i> 125(13): 3696-7.
Biotin	Stable	Nikolov, V., R. Lipowsky, et al. (2007). "Behavior of giant vesicles with anchored DNA molecules." <i>Biophys J</i> 92(12): 4356-68
Cholesterol	Transient	Pfeiffer, I. and F. Hook (2004). "Bivalent cholesterol-based coupling of oligonucleotides to lipid membrane assemblies." <i>J Am Chem Soc</i> 126(33): 10224-5
Surfactant (e.g. Lipid, Palmitate, etc)	Stable	van Lengerich, B., R. J. Rawle, et al. "Covalent attachment of lipid vesicles to a fluid-supported bilayer allows observation of DNA-mediated vesicle interactions." <i>Langmuir</i> 26(11): 8666-72

[0587] Synthetic polynucleotides and/or linkers may be functionalised using a modified phosphoramidite in the synthesis reaction, which is easily compatible for the direct addition of suitable anchoring groups, such as cholesterol, tocopherol, palmitate, thiol, lipid and biotin groups. These different attachment chemistries give a suite of options for attachment to polynucleotides. Each different modification group couples the polynucleotide in a slightly different way and coupling is not always permanent so giving different dwell times for the polynucleotide to the membrane. The advantages of transient coupling are discussed above.

[0588] Coupling of polynucleotides to a linker or to a functionalised membrane can also be achieved by a number of other means provided that a complementary reactive group or an anchoring group can be added to the polynucleotide. The addition of reactive groups to either end of a polynucleotide has been reported previously. A thiol group can be added to the 5' of ssDNA or dsDNA using T4 polynucleotide kinase and ATP γ S (Grant, G. P. and P. Z. Qin (2007). "A facile method for attaching nitroxide spin labels at the 5' terminus of nucleic acids." *Nucleic Acids Res* 35 (10): e77). An azide group can be added to 5'-phosphate of ssDNA or dsDNA using T4 polynucleotide kinase and γ -[2-Azidoethyl]-ATP or γ -[6-Azidoethyl]-ATP. Using thiol or Click chemistry a tether, containing either a thiol, iodoacetamide OPSS or maleimide group (reactive to thiols) or a DIBO (dibenzocyclooctyne) or alkyne group (reactive to azides), can be covalently attached to the polynucleotide. A more diverse selection of chemical groups, such as biotin, thiols and fluorophores, can be added using terminal transferase to incorporate modified oligonucleotides to 3' of ssDNA (Kumar, A., P. Tchen, et al. (1988). "Nonradioactive labeling of synthetic oligonucleotide probes with terminal deoxynucleotidyl transferase." *Anal Biochem* 169 (2): 376-82). Streptavidin/biotin and/or streptavidin/desthiobiotin coupling may be used for any other polynucleotide. The Examples below describes how a polynucleotide can be coupled to a membrane using streptavidin/biotin and streptavidin/desthiobiotin. It may also be possible that anchors may be directly added to polynucleotides using terminal transferase with suitably modified nucleotides (e.g. cholesterol or palmitate).

[0589] The one or more anchors preferably couple the polynucleotide to the membrane via hybridisation. Hybridisation in the one or more anchors allows coupling in a transient manner as discussed above. The hybridisation may be present in any part of the one or more anchors, such as between the one or more anchors and the polynucleotide, within the one or more anchors or between the one or more anchors and the membrane. For instance, a linker may comprise two or more polynucleotides, such as 3, 4 or 5 polynucleotides, hybridised together. The one or more anchors may hybridise to the polynucleotide. The one or more anchors may hybridise directly to the polynucleotide or directly to a Y adaptor and/or leader sequence attached to the polynucleotide or directly to a hairpin loop adaptor attached to the polynucleotide (as discussed below). Alternatively, the one or more anchors may be hybridised to one or more, such as 2 or 3, intermediate polynucleotides (or "splints") which are hybridised to the polynucleotide, to a Y adaptor and/or leader sequence attached to the polynucleotide or to a hairpin loop adaptor attached to the polynucleotide (as discussed below).

[0590] The one or more anchors may comprise a single stranded or double stranded polynucleotide. One part of the anchor may be ligated to a single stranded or double stranded polynucleotide. Ligation of short pieces of ssDNA have been reported using T4 RNA ligase I (Troutt, A. B., M. G. McHeyzer-Williams, et al. (1992). "Ligation-anchored PCR: a simple amplification technique with single-sided specificity." *Proc Natl Acad Sci USA* 89 (20): 9823-5). Alternatively, either a single stranded or double stranded polynucleotide can be ligated to a double stranded polynucleotide and then the two strands separated by thermal or chemical denaturation. To a double stranded polynucleotide, it is possible to add either a piece of single stranded polynucleotide to one or both of the ends of the duplex, or a double stranded polynucleotide to one or both ends. For addition of single stranded polynucleotides to the a double stranded polynucleotide, this can be achieved using T4 RNA ligase I as for ligation to other regions of single stranded polynucleotides. For addition of double stranded polynucleotides to a double stranded polynucleotide then ligation can be "blunt-ended", with complementary 3' dA/dT tails on the polynucleotide and added polynucleotide respectively (as is routinely done for many sample prep applications to prevent concatemer or dimer formation) or using "sticky-ends" generated by restriction digestion of the polynucleotide and ligation of compatible adapters. Then, when the duplex is melted, each single strand will have either a 5' or 3' modification if a single stranded polynucleotide was used for ligation or a modification at the 5' end, 3' end or both if a double stranded polynucleotide was used for ligation.

[0591] If the polynucleotide is a synthetic strand, the one or more anchors can be incorporated during the chemical synthesis of the polynucleotide. For instance, the polynucleotide can be synthesised using a primer having a reactive group attached to it.

[0592] Adenylated polynucleotides are intermediates in ligation reactions, where an adenosine-monophosphate is attached to 5'-phosphate of the polynucleotide. Various kits are available for generation of this intermediate, such as 5' DNA Adenylation Kit from NEB. By substituting ATP in the reaction for a modified nucleotide triphosphate, then addition of reactive groups (such as thiols, amines, biotin, azides, etc) to 5' of a polynucleotide can be possible. It may also be

possible that anchors could be directly added to polynucleotides using a 5' DNA adenylation kit with suitably modified nucleotides (e.g. cholesterol or palmitate).

[0593] A common technique for the amplification of sections of genomic DNA is using polymerase chain reaction (PCR). Here, using two synthetic oligonucleotide primers, a number of copies of the same section of DNA can be generated, where for each copy the 5' of each strand in the duplex will be a synthetic polynucleotide. Single or multiple nucleotides can be added to 3' end of single or double stranded DNA by employing a polymerase. Examples of polymerases which could be used include, but are not limited to, Terminal Transferase, Klenow and *E. coli* Poly (A) polymerase. By substituting ATP in the reaction for a modified nucleotide triphosphate then anchors, such as a cholesterol, thiol, amine, azide, biotin or lipid, can be incorporated into double stranded polynucleotides. Therefore, each copy of the amplified polynucleotide will contain an anchor.

[0594] Ideally, the polynucleotide is coupled to the membrane without having to functionalise the polynucleotide. This can be achieved by coupling the one or more anchors, such as a polynucleotide binding protein or a chemical group, to the membrane and allowing the one or more anchors to interact with the polynucleotide or by functionalising the membrane. The one or more anchors may be coupled to the membrane by any of the methods described herein. In particular, the one or more anchors may comprise one or more linkers, such as maleimide functionalised linkers.

[0595] In this embodiment, the polynucleotide is typically RNA, DNA, PNA, TNA or LNA and may be double or single stranded. This embodiment is particularly suited to genomic DNA polynucleotides.

[0596] The one or more anchors can comprise any group that couples to, binds to or interacts with single or double stranded polynucleotides, specific nucleotide sequences within the polynucleotide or patterns of modified nucleotides within the polynucleotide, or any other ligand that is present on the polynucleotide.

[0597] Suitable binding proteins for use in anchors include, but are not limited to, *E. coli* single stranded binding protein, P5 single stranded binding protein, T4 gp32 single stranded binding protein, the TOPO V dsDNA binding region, human histone proteins, *E. coli* HU DNA binding protein and other archaeal, prokaryotic or eukaryotic single stranded or double stranded polynucleotide (or nucleic acid) binding proteins, including those listed below.

[0598] The specific nucleotide sequences could be sequences recognised by transcription factors, ribosomes, endonucleases, topoisomerases or replication initiation factors. The patterns of modified nucleotides could be patterns of methylation or damage.

[0599] The one or more anchors can comprise any group which couples to, binds to, intercalates with or interacts with a polynucleotide. The group may intercalate or interact with the polynucleotide via electrostatic, hydrogen bonding or Van der Waals interactions. Such groups include a lysine monomer, poly-lysine (which will interact with ssDNA or dsDNA), ethidium bromide (which will intercalate with dsDNA), universal bases or universal nucleotides (which can hybridise with any polynucleotide) and osmium complexes (which can react to methylated bases). A polynucleotide may therefore be coupled to the membrane using one

or more universal nucleotides attached to the membrane. Each universal nucleotide may be coupled to the membrane using one or more linkers. The universal nucleotide preferably comprises one of the following nucleobases: hypoxanthine, 4-nitroindole, 5-nitroindole, 6-nitroindole, formylindole, 3-nitropyrrole, nitroimidazole, 4-nitropyrrole, 4-nitrobenzimidazole, 5-nitroindazole, 4-aminobenzimidazole or phenyl (C6-aromatic ring). The universal nucleotide more preferably comprises one of the following nucleosides: 2'-deoxyinosine, inosine, 7-deaza-2'-deoxyinosine, 7-deaza-inosine, 2-aza-deoxyinosine, 2-aza-inosine, 2-O'-methyl-inosine, 4-nitroindole 2'-deoxyribonucleoside, 4-nitroindole ribonucleoside, 5-nitroindole 2'-deoxyribonucleoside, 5-nitroindole ribonucleoside, 6-nitroindole 2'-deoxyribonucleoside, 6-nitroindole ribonucleoside, 3-nitropyrrole 2'-deoxyribonucleoside, 3-nitropyrrole ribonucleoside, an acyclic sugar analogue of hypoxanthine, nitroimidazole 2'-deoxyribonucleoside, nitroimidazole ribonucleoside, 4-nitropyrrole 2'-deoxyribonucleoside, 4-nitropyrrole ribonucleoside, 4-nitrobenzimidazole 2'-deoxyribonucleoside, 4-nitrobenzimidazole ribonucleoside, 5-nitroindazole 2'-deoxyribonucleoside, 5-nitroindazole ribonucleoside, 4-aminobenzimidazole 2'-deoxyribonucleoside, 4-aminobenzimidazole ribonucleoside, phenyl C-ribonucleoside, phenyl C-2'-deoxyribosyl nucleoside, 2'-deoxynebularine, 2'-deoxysoguanosine, K-2'-deoxyribose, P-2'-deoxyribose and pyrrolidine. The universal nucleotide more preferably comprises 2'-deoxyinosine. The universal nucleotide is more preferably IMP or dIMP. The universal nucleotide is most preferably dPMP (2'-Deoxy-P-nucleoside monophosphate) or dKMP (N6-methoxy-2, 6-diaminopurine monophosphate).

[0600] The one or more anchors may couple to (or bind to) the polynucleotide via Hoogsteen hydrogen bonds (where two nucleobases are held together by hydrogen bonds) or reversed Hoogsteen hydrogen bonds (where one nucleobase is rotated through 180° with respect to the other nucleobase). For instance, the one or more anchors may comprise one or more nucleotides, one or more oligonucleotides or one or more polynucleotides which form Hoogsteen hydrogen bonds or reversed Hoogsteen hydrogen bonds with the polynucleotide. These types of hydrogen bonds allow a third polynucleotide strand to wind around a double stranded helix and form a triplex. The one or more anchors may couple to (or bind to) a double stranded polynucleotide by forming a triplex with the double stranded duplex.

[0601] In this embodiment at least 1%, at least 10%, at least 25%, at least 50% or 100% of the membrane components may be functionalised.

[0602] Where the one or more anchors comprise a protein, they may be able to anchor directly into the membrane without further functionalisation, for example if it already has an external hydrophobic region which is compatible with the membrane. Examples of such proteins include, but are not limited to, transmembrane proteins, intramembrane proteins and membrane proteins. Alternatively the protein may be expressed with a genetically fused hydrophobic region which is compatible with the membrane. Such hydrophobic protein regions are known in the art.

[0603] The one or more anchors are preferably mixed with the polynucleotide before contacting with the membrane, but the one or more anchors may be contacted with the membrane and subsequently contacted with the polynucleotide.

[0604] In another aspect the polynucleotide may be functionalised, using methods described above, so that it can be recognised by a specific binding group. Specifically the polynucleotide may be functionalised with a ligand such as biotin (for binding to streptavidin), amylose (for binding to maltose binding protein or a fusion protein), Ni-NTA (for binding to poly-histidine or poly-histidine tagged proteins) or a peptides (such as an antigen).

[0605] According to a preferred embodiment, the one or more anchors may be used to couple a polynucleotide to the membrane when the polynucleotide is attached to a leader sequence which preferentially threads into the pore. Leader sequences are discussed in more detail below. Preferably, the polynucleotide is attached (such as ligated) to a leader sequence which preferentially threads into the pore. Such a leader sequence may comprise a homopolymeric polynucleotide or an abasic region. The leader sequence is typically designed to hybridise to the one or more anchors either directly or via one or more intermediate polynucleotides (or splints). In such instances, the one or more anchors typically comprise a polynucleotide sequence which is complementary to a sequence in the leader sequence or a sequence in the one or more intermediate polynucleotides (or splints). In such instances, the one or more splints typically comprise a polynucleotide sequence which is complementary to a sequence in the leader sequence.

[0606] An example of a molecule used in chemical attachment is EDC (1-ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride). Reactive groups can also be added to the 5' of polynucleotides using commercially available kits (Thermo Pierce, Part No. 22980). Suitable methods include, but are not limited to, transient affinity attachment using histidine residues and Ni-NTA, as well as more robust covalent attachment by reactive cysteines, lysines or non natural amino acids.

Double Stranded Polynucleotide

[0607] The polynucleotide may be double stranded. If the polynucleotide is double stranded, the method preferably further comprises before the contacting step ligating a bridging moiety, such as a hairpin loop, to one end of the polynucleotide. The two strands of the polynucleotide may then be separated as or before the polynucleotide is contacted with the pore in accordance with the invention. The two strands may be separated as the polynucleotide movement through the pore is controlled by a polynucleotide binding protein, such as a helicase, or molecular brake.

[0608] Linking and interrogating both strands on a double stranded construct in this way increases the efficiency and accuracy of characterisation.

[0609] The bridging moiety is capable of linking the two strands of the target polynucleotide. The bridging moiety typically covalently links the two strands of the target polynucleotide. The bridging moiety can be anything that is capable of linking the two strands of the target polynucleotide, provided that the bridging moiety does not interfere with movement of the single stranded polynucleotide through the transmembrane pore.

[0610] The bridging moiety may be linked to the target polynucleotide by any suitable means known in the art. The bridging moiety may be synthesised separately and chemically attached or enzymatically ligated to the target polynucleotide. Alternatively, the bridging moiety may be generated in the processing of the target polynucleotide.

[0611] The bridging moiety is linked to the target polynucleotide at or near one end of the target polynucleotide. The bridging moiety is preferably linked to the target polynucleotide within 10 nucleotides of the end of the target polynucleotide.

[0612] Suitable bridging moieties include, but are not limited to a polymeric linker, a chemical linker, a polynucleotide or a polypeptide. Preferably, the bridging moiety comprises DNA, RNA, modified DNA (such as abasic DNA), RNA, PNA, LNA or PEG. The bridging moiety is more preferably DNA or RNA.

[0613] The bridging moiety is most preferably a hairpin loop or a hairpin loop adaptor. Suitable hairpin adaptors can be designed using methods known in the art. The hairpin loop may be any length. The hairpin loop is typically 110 or fewer nucleotides, such as 100 or fewer nucleotides, 90 or fewer nucleotides, 80 or fewer nucleotides, 70 or fewer nucleotides, 60 or fewer nucleotides, 50 or fewer nucleotides, 40 or fewer nucleotides, 30 or fewer nucleotides, 20 or fewer nucleotides or 10 or fewer nucleotides, in length. The hairpin loop is preferably from about 1 to 110, from 2 to 100, from 5 to 80 or from 6 to 50 nucleotides in length. Longer lengths of the hairpin loop, such as from 50 to 110 nucleotides, are preferred if the loop is involved in the differential selectability of the adaptor. Similarly, shorter lengths of the hairpin loop, such as from 1 to 5 nucleotides, are preferred if the loop is not involved in the selectable binding as discussed below.

[0614] The hairpin adaptor may be ligated to either end of the first and/or second polynucleotide, i.e. the 5' or 3' end. The hairpin adaptor may be ligated to the first and/or second polynucleotide using any method known in the art. The hairpin adaptor may be ligated using a ligase, such as T4 DNA ligase, *E. coli* DNA ligase, Taq DNA ligase, Tma DNA ligase and 9°N DNA ligase.

[0615] The two strands of the polynucleotide may be separated using any method known in the art. For instance, they may be separated by a polynucleotide binding protein or using conditions which favour dehybridisation (examples of conditions which favour dehybridisation include, but are not limited to, high temperature, high pH and the addition of agents that can disrupt hydrogen bonding or base pairing, such as formamide and urea).

[0616] The hairpin adaptor preferably comprises a selectable binding moiety. This allows the first and/or second polynucleotide to be purified or isolated. A selectable binding moiety is a moiety that can be selected on the basis of its binding properties. Hence, a selectable binding moiety is preferably a moiety that specifically binds to a surface. A selectable binding moiety specifically binds to a surface if it binds to the surface to a much greater degree than any other moiety used in the invention. In preferred embodiments, the moiety binds to a surface to which no other moiety used in the invention binds.

[0617] Suitable selective binding moieties are known in the art. Preferred selective binding moieties include, but are not limited to, biotin, a polynucleotide sequence, antibodies, antibody fragments, such as Fab and ScFv, antigens, polynucleotide binding proteins, poly histidine tails and GST tags. The most preferred selective binding moieties are biotin and a selectable polynucleotide sequence. Biotin specifically binds to a surface coated with avidins. Selectable polynucleotide sequences specifically bind (i.e. hybridise) to a surface coated with homologous sequences. Alter-

natively, selectable polynucleotide sequences specifically bind to a surface coated with polynucleotide binding proteins.

[0618] The hairpin adaptor and/or the selectable binding moiety may comprise a region that can be cut, nicked, cleaved or hydrolysed. Such a region can be designed to allow the first and/or second polynucleotide to be removed from the surface to which it is bound following purification or isolation. Suitable regions are known in the art. Suitable regions include, but are not limited to, an RNA region, a region comprising desthiobiotin and streptavidin, a disulphide bond and a photocleavable region.

[0619] The double stranded target polynucleotide preferably comprises a leader sequence at the opposite end of the bridging moiety, such as a hairpin loop or hairpin loop adaptor. Leader sequences are discussed in more detail below.

Round the Corner Sequencing

[0620] In a preferred embodiment, a target double stranded polynucleotide is provided with a bridging moiety, such as a hairpin loop or hairpin loop adaptor, at one end and the method comprises contacting the polynucleotide with the pore such that both strands of the polynucleotide move through the pore and taking one or more measurements as the both strands of the polynucleotide move with respect to the pore wherein the measurements are indicative of one or more characteristics of the strands of the polynucleotide and thereby characterising the target double stranded polynucleotide. In another preferred embodiment, a target double stranded polynucleotide is provided with a bridging moiety, such as a hairpin loop or hairpin loop adaptor, at one end and the method comprises contacting the polynucleotide with the pore and exonuclease such that both strands of the polynucleotide are digested to form individual nucleotides. Any of the embodiments discussed above equally apply to this embodiment.

Leader Sequence

[0621] Before the contacting step in the strand characterisation/sequencing method, the method preferably comprises attaching to the polynucleotide a leader sequence which preferentially threads into the pore. The leader sequence facilitates the method of the invention. The leader sequence is designed to preferentially thread into the pore and thereby facilitate the movement of polynucleotide through the pore. The leader sequence can also be used to link the polynucleotide to the one or more anchors as discussed above.

[0622] The leader sequence typically comprises a polymer. The polymer is preferably negatively charged. The polymer is preferably a polynucleotide, such as DNA or RNA, a modified polynucleotide (such as abasic DNA), PNA, LNA, polyethylene glycol (PEG) or a polypeptide. The leader preferably comprises a polynucleotide and more preferably comprises a single stranded polynucleotide. The leader sequence can comprise any of the polynucleotides discussed above. The single stranded leader sequence most preferably comprises a single strand of DNA, such as a poly dT section. The leader sequence preferably comprises the one or more spacers.

[0623] The leader sequence can be any length, but is typically 10 to 150 nucleotides in length, such as from 20 to

150 nucleotides in length. The length of the leader typically depends on the transmembrane pore used in the method.

[0624] The leader sequence is preferably part of a Y adaptor as defined below.

Double Coupling

[0625] The method of the invention may involve double coupling of a double stranded polynucleotide. In a preferred embodiment, the method of the invention comprises:

[0626] (a) providing the double stranded polynucleotide with a Y adaptor at one end and a bridging moiety adaptor, such as a hairpin loop adaptor, at the other end, wherein the Y adaptor comprises one or more first anchors for coupling the polynucleotide to the membrane, wherein the bridging moiety adaptor comprises one or more second anchors for coupling the polynucleotide to the membrane and wherein the strength of coupling of the bridging moiety adaptor to the membrane is greater than the strength of coupling of the Y adaptor to the membrane;

[0627] (b) contacting the polynucleotide provided in step (a) with a pore the invention such that the polynucleotide moves with respect to, such as through, the pore; and

[0628] (c) taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or more characteristics of the polynucleotide, and thereby characterising the target polynucleotide.

[0629] This type of method is discussed in detail in the UK Application No. 1406147.7.

[0630] The double stranded polynucleotide is provided with a Y adaptor at one end and a bridging moiety adaptor at the other end. The Y adaptor and/or the bridging moiety adaptor are typically polynucleotide adaptors. They may be formed from any of the polynucleotides discussed above.

[0631] The Y adaptor typically comprises (a) a double stranded region and (b) a single stranded region or a region that is not complementary at the other end. The Y adaptor may be described as having an overhang if it comprises a single stranded region. The presence of a non-complementary region in the Y adaptor gives the adaptor its Y shape since the two strands typically do not hybridise to each other unlike the double stranded portion. The Y adaptor comprises the one or more first anchors. Anchors are discussed in more detail above.

[0632] The Y adaptor preferably comprises a leader sequence which preferentially threads into the pore. This is discussed above.

[0633] The bridging moiety adaptor preferably comprises a selectable binding moiety as discussed above. The bridging moiety adaptor and/or the selectable binding moiety may comprise a region that can be cut, nicked, cleaved or hydrolysed as discussed above.

[0634] If one or more helicases and one or more molecular brakes are used as discussed above, the Y adaptor preferably comprises the one or more helicases and the bridging moiety adaptor preferably comprises the one or more molecular brakes.

[0635] The Y adaptor and/or the bridging moiety adaptor may be ligated to the polynucleotide using any method known in the art. One or both of the adaptors may be ligated using a ligase, such as T4 DNA ligase, *E. coli* DNA ligase, Taq DNA ligase, Tma DNA ligase and 9ⁿN DNA ligase.

Alternatively, the adaptors may be added to the polynucleotide using the methods of the invention discussed below.

[0636] In a preferred embodiment, step a) of the method comprises modifying the double stranded polynucleotide so that it comprises the Y adaptor at one end and the bridging moiety adaptor at the other end. Any manner of modification can be used. The method preferably comprises modifying the double stranded polynucleotide in accordance with the invention. This is discussed in more detail below. The methods of modification and characterisation may be combined in any way.

[0637] The strength of coupling (or binding) of the bridging moiety adaptor to the membrane is greater than the strength of coupling (or binding) of the Y adaptor to the membrane. This can be measured in any way. A suitable method for measuring the strength of coupling (or binding) is disclosed in the Examples of the UK Application No. 1406147.7.

[0638] The strength of coupling (or binding) of the bridging moiety adaptor is preferably at least 1.5 times the strength of coupling (or binding) of the Y adaptor, such as at least twice, at least three times, at least four times, at least five or at least ten times the strength of coupling (or binding) of the anchor adaptor. The affinity constant (Kd) of the bridging moiety adaptor for the membrane is preferably at least 1.5 times the affinity constant of the Y adaptor, such as at least twice, at least three times, at least four times, at least five or at least ten times the strength of coupling of the Y adaptor.

[0639] There are several ways in which the bridging moiety adaptor couples (or binds) more strongly to the membrane than the Y adaptor. For instance, the bridging moiety adaptor may comprise more anchors than the Y adaptor. For instance, the bridging moiety adaptor may comprise 2, 3 or more second anchors whereas the Y adaptor may comprise one first anchor.

[0640] The strength of coupling (or binding) of the one or more second anchors to the membrane may be greater than the strength of coupling (or binding) of the one or more first anchors to the membrane. The strength of coupling (or binding) of the one or more second anchors to the bridging moiety adaptor may be greater than the strength of coupling (or binding) of the one or more first anchors to the Y adaptor. The one or more first anchors and the one or more second anchors may be attached to their respective adaptors via hybridisation and the strength of hybridisation is greater in the one or more second anchors than in the one or more first anchors. Any combination of these embodiments may also be used in the invention. Strength of coupling (or binding) may be measured using known techniques in the art.

[0641] The one or more second anchors preferably comprise one or more groups which couple(s) (or bind(s)) to the membrane with a greater strength than the one or more groups in the one or more first anchors which couple(s) (or bind(s)) to the membrane. In preferred embodiments, the bridging moiety adaptor/one or more second anchors couple (or bind) to the membrane using cholesterol and the Y adaptor/one or more first anchors couple (or bind) to the membrane using palmitate. Cholesterol binds to triblock copolymer membranes and lipid membranes more strongly than palmitate. In an alternative embodiment, the bridging moiety adaptor/one or more second anchors couple (or bind) to the membrane using a mono-acyl species, such as palmitate, and the Y adaptor/one or more first anchors couple (or

bind) to the membrane using a diacyl species, such as dipalmitoylphosphatidylcholine.

Adding Hairpin Loops and Leader Sequences

[0642] Before provision, a double stranded polynucleotide may be contacted with a MuA transposase and a population of double stranded MuA substrates, wherein a proportion of the substrates in the population are Y adaptors comprising the leader sequence and wherein a proportion of the substrates in the population are hairpin loop adaptors. The transposase fragments the double stranded polynucleotide analyte and ligates MuA substrates to one or both ends of the fragments. This produces a plurality of modified double stranded polynucleotides comprising the leader sequence at one end and the hairpin loop at the other. The modified double stranded polynucleotides may then be investigated using the method of the invention.

[0643] Each substrate in the population preferably comprises at least one overhang of universal nucleotides such that the transposase fragments the template polynucleotide and ligates a substrate to one or both ends of the double stranded fragments and thereby produces a plurality of fragment/substrate constructs and wherein the method further comprises ligating the overhangs to the fragments in the constructs and thereby producing a plurality of modified double stranded polynucleotides. Suitable universal nucleotides are discussed above. The overhang is preferably five nucleotides in length.

[0644] Alternatively, each substrate in population preferably comprises (i) at least one overhang and (ii) at least one nucleotide in the same strand as the at least one overhang which comprises a nucleoside that is not present in the template polynucleotide such that the transposase fragments the template polynucleotide and ligates a substrate to one or both ends of the double stranded fragments and thereby produces a plurality of fragment/substrate constructs, and wherein the method further comprises (a) removing the overhangs from the constructs by selectively removing the at least one nucleotide and thereby producing a plurality of double stranded constructs comprising single stranded gaps and (b) repairing the single stranded gaps in the constructs and thereby producing a plurality of modified double stranded polynucleotides. The polynucleotide typically comprises the nucleosides deoxyadenosine (dA), deoxyuridine (dU) and/or thymidine (dT), deoxyguanosine (dG) and deoxycytidine (dC). The nucleoside that is not present in the polynucleotide is preferably abasic, adenosine (A), uridine (U), 5-methyluridine (m5U), cytidine (C) or guanosine (G) or comprises urea, 5, 6 dihydroxythymine, thymine glycol, 5-hydroxy-5 methylhydantoin, uracil glycol, 6-hydroxy-5, 6-dihydrothymine, methyltartronylurea, 7, 8-dihydro-8-oxoguanine (8-oxoguanine), 8-oxoadenine, fapy-guanine, methylfapy-guanine, fapy-adenine, aflatoxin B1-fapy-guanine, 5-hydroxy-cytosine, 5-hydroxy-uracil, 3-methyladenine, 7-methylguanine, 1,N6-ethenoadenine, hypoxanthine, 5-hydroxyuracil, 5-hydroxymethyluracil, 5-formyluracil or a cis-syn-cyclobutane pyrimidine dimer. The at least one nucleotide preferably is 10 nucleotides or fewer from the overhang. The at least one nucleotide is the first nucleotide in the overhang. All of the nucleotides in the overhang preferably comprise a nucleoside that is not present in the template polynucleotide.

[0645] These MuA based methods are disclosed in International Application No. PCT/GB2014/052505. They are also discussed in detail in the UK Application No. 1406147.7.

[0646] One or more helicases may be attached to the MuA substrate Y adaptors before they are contacted with the double stranded polynucleotide and MuA transposase. Alternatively, one or more helicases may be attached to the MuA substrate Y adaptors before they are contacted with the double stranded polynucleotide and MuA transposase.

[0647] One or more molecular brakes may be attached to the MuA substrate hairpin loop adaptors before they are contacted with the double stranded polynucleotide and MuA transposase. Alternatively, one or more molecular brakes may be attached to the MuA substrate hairpin loop adaptors before they are contacted with the double stranded polynucleotide and MuA transposase,

Uncoupling

[0648] The method of the invention may involve characterising multiple target polynucleotides and uncoupling of the at least the first target polynucleotide.

[0649] In a preferred embodiment, the invention involves characterising two or more target polynucleotides. The method comprises:

[0650] (a) providing a first polynucleotide in a first sample;

[0651] (b) providing a second polynucleotide in a second sample;

[0652] (c) coupling the first polynucleotide in the first sample to a membrane using one or more anchors;

[0653] (d) contacting the first polynucleotide with a pore of the invention such that the polynucleotide moves with respect to, such as through, the pore;

[0654] (e) taking one or more measurements as the first polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the first polynucleotide and thereby characterising the first polynucleotide;

[0655] (f) uncoupling the first polynucleotide from the membrane;

[0656] (g) coupling the second polynucleotide in the second sample to the membrane using one or more anchors;

[0657] (h) contacting the second polynucleotide with the pore of the invention such that the second polynucleotide moves with respect to, such as through, the pore; and

[0658] (i) taking one or more measurements as the second polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the second polynucleotide and thereby characterising the second polynucleotide.

[0659] This type of method is discussed in detail in the UK Application No. 1406155.0.

[0660] Step (f) (i.e. uncoupling of the first polynucleotide) may be performed before step (g) (i.e. before coupling the second polynucleotide to the membrane). Step (g) may be performed before step (f). If the second polynucleotide is coupled to the membrane before the first polynucleotide is uncoupled, step (f) preferably comprises selectively uncoupling the first polynucleotide from the membrane (i.e. uncoupling the first polynucleotide but not the second polynucleotide from the membrane). A skilled person can design

a system in which selective uncoupling is achieved. Steps (f) and (g) may be performed at the same time. This is discussed in more detail below.

[0661] In step (f), at least 10% of the first polynucleotide is preferably uncoupled from the membrane. For instance, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 95% of the first polynucleotide may be uncoupled from the membrane. Preferably, all of the first polynucleotide is uncoupled from the membrane. The amount of the first polynucleotide uncoupled from the membrane can be determined using the pore. This is disclosed in the Examples.

[0662] The first polynucleotide and second polynucleotide may be different from one another. Alternatively, the first and second polynucleotides may be different polynucleotides. In such instances, there may be no need to remove at least part of the first sample before adding the second polynucleotide. This is discussed in more detail below. If the method concerns investigating three or more polynucleotides, they may all be different from one another or some of them may be different from one another.

[0663] The first polynucleotide and the second polynucleotide may be two instances of the same polynucleotide. The first polynucleotide may be identical to the second polynucleotide. This allows proof reading. If the method concerns investigating three or more polynucleotides, they may all be three or more instances of the same polynucleotide or some of them may be separate instances of the same polynucleotide.

[0664] The first sample and second sample may be different from one another. For instance, the first sample may be derived from a human and the second sample may be derived from a virus. If the first and second samples are different from one another, they may contain or be suspected of containing the same first and second polynucleotides. If the method concerns investigating three or more samples, they may all be different from one another or some of them may be different from one another.

[0665] The first sample and the second sample are preferably two instances of the same sample. The first sample is preferably identical to the second sample. This allows proof reading. If the method concerns investigating three or more samples, they may all be three or more instances of the same sample or some of them may be separate instances of the same sample.

[0666] Any number of polynucleotides can be investigated. For instance, the method of the invention may concern characterising 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 50, 100 or more polynucleotides. If three or more polynucleotides are investigated using the method of the invention, the second polynucleotide is also uncoupled from the membrane and the requisite number of steps are added for the third polynucleotide. The same is true for four or more polynucleotides.

[0667] The method of the invention involves uncoupling the first polynucleotide from the membrane. The method of the invention may involve uncoupling the second polynucleotide from the membrane if three or more polynucleotides are being investigated.

[0668] The first polynucleotide can be uncoupled from the membrane using any known method. The first polynucleotide is preferably not uncoupled from the membrane in step

(f) using the transmembrane pore. The first polynucleotide is preferably not uncoupled from the membrane using a voltage or an applied potential.

[0669] Step (f) preferably comprises uncoupling the first polynucleotide from the membrane by removing the one or more anchors from the membrane. If the anchors are removed, the second polynucleotide is coupled to the membrane using other (or separate) anchors. The anchors used to couple the second polynucleotide may be the same type of anchors used to couple the first polynucleotide or different type of anchors.

[0670] Step (f) more preferably comprises contacting the one or more anchors with an agent which has a higher affinity for the one or more anchors than the anchors have for the membrane. A variety of protocols for competitive binding or immunoradiometric assays to determine the specific binding capability of molecules are well known in the art (see for example Maddox et al, *J. Exp. Med.* 158, 1211-1226, 1993). The agent removes the anchor(s) from the membrane and thereby uncouples the first polynucleotide. The agent is preferably a sugar. Any sugar which binds to the one or more anchors with a higher affinity than the one or more anchors have for the membrane may be used. The sugar may be a cyclodextrin or derivative thereof as discussed below.

[0671] If one or more anchors comprise a hydrophobic anchor, such as cholesterol, the agent is preferably a cyclodextrin or a derivative thereof or a lipid. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The agent is more preferably heptakis-6-amino- β -cyclodextrin ($\text{am}_7\text{-}\beta\text{CD}$), 6-monodeoxy-6-monamino- β -cyclodextrin ($\text{am}_1\text{-}\beta\text{CD}$) or heptakis-(6-deoxy-6-guainidino)-cyclodextrin ($\text{gu}_7\text{-}\beta\text{CD}$). Any of the lipids disclosed herein may be used.

[0672] If an anchor comprise(s) streptavidin, biotin or desthiobiotin, the agent is preferably biotin, desthiobiotin or streptavidin. Both biotin and desthiobiotin bind to streptavidin with a higher affinity than streptavidin binds to the membrane and vice versa. Biotin has a stronger affinity for streptavidin than desthiobiotin. An anchor comprising streptavidin may therefore be removed from the membrane using biotin or streptavidin and vice versa.

[0673] If an anchor comprises a protein, the agent is preferably an antibody or fragment thereof which specifically binds to the protein. An antibody specifically binds to a protein if it binds to the protein with preferential or high affinity, but does not bind or binds with only low affinity to other or different proteins. An antibody binds with preferential or high affinity if it binds with a K_d of 1×10^{-6} M or less, more preferably 1×10^{-7} M or less, 5×10^{-8} M or less, more preferably 1×10^{-8} M or less or more preferably 5×10^{-9} M or less. An antibody binds with low affinity if it binds with a K_d of 1×10^{-6} M or more, more preferably 1×10^{-5} M or more, more preferably 1×10^{-4} M or more, more preferably 1×10^{-3} M or more, even more preferably 1×10^{-2} M or more. Any method may be used to detect binding or specific binding. Methods of quantitatively measuring the binding of an antibody to a protein are well known in the art. The antibody may be a monoclonal antibody or a polyclonal antibody. Suitable fragments of antibodies include, but are not limited to, Fv, F(ab') and F(ab')₂ fragments, as well as single chain antibodies. Furthermore, the antibody or fragment thereof may be a chimeric antibody or fragment

thereof, a CDR-grafted antibody or fragment thereof or a humanised antibody or fragment thereof.

[0674] Step (f) preferably comprises contacting the one or more anchors with an agent which reduces ability of the one or more anchors to couple to the membrane. For instance, the agent could interfere with the structure and/or hydrophobicity of the one or more anchors and thereby reduce their ability to couple to the membrane. If an anchor comprises cholesterol, the agent is preferably cholesterol dehydrogenase. If an anchor comprises a lipid, the agent is preferably a phospholipase. If an anchor comprises a protein, the agent is preferably a proteinase or urea. Other combination of suitable anchors and agents will be clear to a person skilled in the art.

[0675] Step (f) preferably comprises uncoupling the first polynucleotide from the membrane by separating the first polynucleotide from the one or more anchors. This can be done in any manner. For instance, the linker could be cut in an anchor comprising a linker. This embodiment is particularly applicable to anchors which involve linkage via hybridisation. Such anchors are discussed above.

[0676] Step (f) more preferably comprises uncoupling the first polynucleotide from the membrane by contacting the first polynucleotide and the one or more anchors with an agent which competes with the first polynucleotide for binding to one or more anchors. Methods for determining and measuring competitive binding are known in the art. The agent is preferably a polynucleotide which competes with the first polynucleotide for hybridisation to the one or more anchors. For instance, if the first polynucleotide is coupled to the membrane using one or more anchors which involve hybridisation, the polynucleotide can be uncoupled by contacting the one or more anchors with a polynucleotide which also hybridises to the site of hybridisation. The polynucleotide agent is typically added at a concentration that is higher than the concentration of the first polynucleotide and one or more anchors. Alternatively, the polynucleotide agent may hybridise more strongly to the one or more anchors than the first polynucleotide.

[0677] Step (f) more preferably comprises (i) contacting the first polynucleotide and the one or more anchors with urea, tris(2-carboxyethyl) phosphine (TCEP), dithiothreitol (DTT), streptavidin or biotin, UV light, an enzyme or a binding agent; (ii) beating the first polynucleotide and the one or more anchors; or (iii) altering the pH. Urea, tris(2-carboxyethyl) phosphine (TCEP) or dithiothreitol (DTT) are capable of disrupting anchors and separating the first polynucleotide from the membrane. If an anchor comprises a streptavidin-biotin link, then a streptavidin agent will compete for binding to the biotin. If an anchor comprises a streptavidin-desthiobiotin link, then a biotin agent will compete for binding to the streptavidin. UV light can be used to breakdown photolabile groups. Enzymes and binding agents can be used to cut, breakdown or unravel the anchor. Preferred enzymes include, but are not limited to, an exonuclease, an endonuclease or a helicase. Preferred binding agents include, but are not limited to, an enzyme, an antibody or a fragment thereof or a single-stranded binding protein (SSB). Any of the enzymes discussed below or antibodies discussed above may be used. Heat and pH can be used to disrupt hybridisation and other linkages.

[0678] If the first polynucleotide is uncoupled from the membrane by separating the first polynucleotide from the one or more anchors, the one or more anchors will remain

in the membrane. Step (g) preferably comprises coupling the second polynucleotide to the membrane using the one or more anchors that was separated from the first polynucleotide. For instance, the second polynucleotide may also be provided with one or more polynucleotides which hybridise (s) to the one or more anchors that remain in the membrane. Alternatively, step (g) preferably comprises coupling the second polynucleotide to the membrane using one or more separate anchors from the ones separated from the first polynucleotide (i.e. one or more other anchors). The one or more separate anchors may be the same type of anchors used to couple the first polynucleotide to the membrane or may be different types of anchors. Step (g) preferably comprises coupling the second polynucleotide to the membrane using one or more different anchors from the one or more anchors separated from the first polynucleotide.

[0679] In a preferred embodiment, steps (f) and (g) comprise uncoupling the first polynucleotide from the membrane by contacting the membrane with the second polynucleotide such that the second polynucleotide competes with the first polynucleotide for binding to the one or more anchors and replaces the first polynucleotide. For instance, if the first polynucleotide is coupled to the membrane using one or more anchors which involve hybridisation, the first polynucleotide can be uncoupled by contacting the anchors with the second polynucleotide attached to polynucleotides which also hybridise to the sites of hybridisation in the one or more anchors. The second polynucleotide is typically added at a concentration that is higher than the concentration of the first polynucleotide and the one or more anchors. Alternatively, the second polynucleotide may hybridise more strongly to the one or more anchors than the first polynucleotide.

Removal or Washing

[0680] Although the first polynucleotide is uncoupled from the membrane in step (f), it is not necessarily removed or washed away. If the second polynucleotide can be easily distinguished from the first polynucleotide, there is no need to remove the first polynucleotide.

[0681] Between steps (f) and (g), the method preferably further comprises removing at least some of the first sample from the membrane. At least 10% of the first sample may be removed, such as at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80% or at least 90% of the first sample may be removed.

[0682] The method more preferably further comprises removing all of the first sample from the membrane. This can be done in any way. For instance, the membrane can be washed with a buffer after the first polynucleotide has been uncoupled. Suitable buffers are discussed below.

Modified Polynucleotides

[0683] Before characterisation, a target polynucleotide may be modified by contacting the polynucleotide with a polymerase and a population of free nucleotides under conditions in which the polymerase forms a modified polynucleotide using the target polynucleotide as a template, wherein the polymerase replaces one or more of the nucleotide species in the target polynucleotide with a different nucleotide species when forming the modified polynucleotide. The modified polynucleotide may then be provided with one or more helicases attached to the polynucleotide

and one or more molecular brakes attached to the polynucleotide. This type of modification is described in UK Application No. 1403096.9. Any of the polymerases discussed above may be used. The polymerase is preferably Klenow or 90 North.

[0684] The template polynucleotide is contacted with the polymerase under conditions in which the polymerase forms a modified polynucleotide using the template polynucleotide as a template. Such conditions are known in the art. For instance, the polynucleotide is typically contacted with the polymerase in commercially available polymerase buffer, such as buffer from New England Biolabs®. The temperature is preferably from 20 to 37° C. for Klenow or from 60 to 75° C. for 90 North. A primer or a 3' hairpin is typically used as the nucleation point for polymerase extension.

[0685] Characterisation, such as sequencing, of a polynucleotide using a transmembrane pore typically involves analyzing polymer units made up of k nucleotides where k is a positive integer (i.e. 'k-mers'). This is discussed in International Application No. PCT/GB2012/052343 (published as WO 2013/041878). While it is desirable to have clear separation between current measurements for different k-mers, it is common for some of these measurements to overlap. Especially with high numbers of polymer units in the k-mer, i.e. high values of k, it can become difficult to resolve the measurements produced by different k-mers, to the detriment of deriving information about the polynucleotide, for example an estimate of the underlying sequence of the polynucleotide.

[0686] By replacing one or more nucleotide species in the target polynucleotide with different nucleotide species in the modified polynucleotide, the modified polynucleotide contains k-mers which differ from those in the target polynucleotide. The different k-mers in the modified polynucleotide are capable of producing different current measurements from the k-mers in the target polynucleotide and so the modified polynucleotide provides different information from the target polynucleotide. The additional information from the modified polynucleotide can make it easier to characterise the target polynucleotide. In some instances, the modified polynucleotide itself may be easier to characterise. For instance, the modified polynucleotide may be designed to include k-mers with an increased separation or a clear separation between their current measurements or k-mers which have a decreased noise.

[0687] The polymerase preferably replaces two or more of the nucleotide species in the target polynucleotide with different nucleotide species when forming the modified polynucleotide. The polymerase may replace each of the two or more nucleotide species in the target polynucleotide with a distinct nucleotide species. The polymerase may replace each of the two or more nucleotide species in the target polynucleotide with the same nucleotide species.

[0688] If the target polynucleotide is DNA, the different nucleotide species in the modified typically comprises a nucleobase which differs from adenine, guanine, thymine, cytosine or methylcytosine and/or comprises a nucleoside which differs from deoxyadenosine, deoxyguanosine, thymidine, deoxycytidine or deoxymethylcytidine. If the target polynucleotide is RNA, the different nucleotide species in the modified polynucleotide typically comprises a nucleobase which differs from adenine, guanine, uracil, cytosine or methylcytosine and/or comprises a nucleoside which differs from adenosine, guanosine, uridine, cytidine or meth-

ylcytidine. The different nucleotide species may be any of the universal nucleotides discussed above.

[0689] The polymerase may replace the one or more nucleotide species with a different nucleotide species which comprises a chemical group or atom absent from the one or more nucleotide species. The chemical group may be a propynyl group, a thio group, an oxo group, a methyl group, a hydroxymethyl group, a formyl group, a carboxy group, a carbonyl group, a benzyl group, a propargyl group or a propargylamine group.

[0690] The polymerase may replace the one or more nucleotide species with a different nucleotide species which lacks a chemical group or atom present in the one or more nucleotide species. The polymerase may replace the one or more of the nucleotide species with a different nucleotide species having an altered electronegativity. The different nucleotide species having an altered electronegativity preferably comprises a halogen atom.

[0691] The method preferably further comprises selectively removing the nucleobases from the one or more different nucleotides species in the modified polynucleotide.

Analyte Delivery

[0692] The target analyte is preferably attached to a microparticle which delivers the analyte towards the membrane. This type of delivery is disclosed in UK Application No. 1418469.1. Any type of microparticle and attachment method may be used.

Other Characterisation Method

[0693] In another embodiment, a polynucleotide is characterised by detecting labelled species that are added to the target polynucleotide by a polymerase and then released. The polymerase uses the polynucleotide as a template. Each labelled species is specific for each nucleotide. The polynucleotide is contacted with a pore of the invention, such as a pore of the invention, a polymerase and labelled nucleotides such that phosphate labelled species are sequentially added to the polynucleotide by the polymerase, wherein the phosphate species contain a label specific for each nucleotide. The labelled species may be detected using the pore before they are released from the nucleotides (i.e. as they are added to the target polynucleotide) or after they are released from the nucleotides.

[0694] The polymerase may be any of those discussed above. The phosphate labelled species are detected using the pore and thereby characterising the polynucleotide. This type of method is disclosed in European Application No. 13187149.3 (published as EP 2682460). Any of the embodiments discussed above equally apply to this method.

[0695] Examples of labelled species include, but are not limited to, polymers, polyethylene glycols, sugars, cyclodextrins, fluorophores, drugs, metabolites, peptides. A non-limiting example of such tags can be found in the work of Kumar et al. *Sci Rep.* 2012; 2:684. Epub 2012 Sep. 21.

Methods of Forming Sensors

[0696] The invention also provides a method of forming a sensor for characterising a target polynucleotide. The method comprises forming a complex between a pore of the invention and a polynucleotide binding protein, such as a helicase or an exonuclease. The complex may be formed by contacting the pore and the protein in the presence of the

target polynucleotide and then applying a potential across the pore. The applied potential may be a chemical potential or a voltage potential as described above. Alternatively, the complex may be formed by covalently attaching the pore to the protein. Methods for covalent attachment are known in the art and disclosed, for example, in International Application Nos. PCT/GB09/001679 (published as WO 2010/004265) and PCT/GB 10/000133 (published as WO 2010/086603). The complex is a sensor for characterising the target polynucleotide. The method preferably comprises forming a complex between a pore of the invention and a helicase. Any of the embodiments discussed above equally apply to this method.

[0697] The invention also provides a sensor for characterising a target polynucleotide. The sensor comprises a complex between a pore of the invention and a polynucleotide binding protein. Any of the embodiments discussed above equally apply to the sensor of the invention.

Kits

[0698] The present invention also provides a kit for characterising a target polynucleotide. The kit comprises a pore of the invention and the components of a membrane. The membrane is preferably formed from the components. The pore is preferably present in the membrane. The kit may comprise components of any of the membranes disclosed above, such as an amphiphilic layer or a triblock copolymer membrane.

[0699] The kit may further comprise a polynucleotide binding protein. Any of the polynucleotide binding proteins discussed above may be used.

[0700] The kit may further comprise one or more anchors for coupling the polynucleotide to the membrane.

[0701] The kit is preferably for characterising a double stranded polynucleotide and preferably comprises a Y adaptor and a hairpin loop adaptor. The Y adaptor preferably has one or more helicases attached and the hairpin loop adaptor preferably has one or more molecular brakes attached. The Y adaptor preferably comprises one or more first anchors for coupling the polynucleotide to the membrane, the hairpin loop adaptor preferably comprises one or more second anchors for coupling the polynucleotide to the membrane and the strength of coupling of the hairpin loop adaptor to the membrane is preferably greater than the strength of coupling of the Y adaptor to the membrane.

[0702] The kit of the invention may additionally comprise one or more other reagents or instruments which enable any of the embodiments mentioned above to be carried out. Such reagents or instruments include one or more of the following: suitable buffer(s) (aqueous solutions), means to obtain a sample from a subject (such as a vessel or an instrument comprising a needle), means to amplify and/or express polynucleotides or voltage or patch clamp apparatus. Reagents may be present in the kit in a dry state such that a fluid sample resuspends the reagents. The kit may also, optionally, comprise instructions to enable the kit to be used in the method of the invention or details regarding for which organism the method may be used.

Apparatus

[0703] The invention also provides an apparatus for characterising a target analyte, such as a target polynucleotide. The apparatus comprises a plurality of pores of the invention

and a plurality of membranes. The plurality of pores are preferably present in the plurality of membranes. The number of pores and membranes is preferably equal. Preferably, a single pore is present in each membrane.

[0704] The apparatus preferably further comprises instructions for carrying out the method of the invention. The apparatus may be any conventional apparatus for analyte analysis, such as an array or a chip. Any of the embodiments discussed above with reference to the methods of the invention are equally applicable to the apparatus of the invention. The apparatus may further comprise any of the features present in the kit of the invention.

[0705] The apparatus is preferably set up to carry out the method of the invention.

[0706] The apparatus preferably comprises:

[0707] a sensor device that is capable of supporting the plurality of pores and membranes and being operable to perform analyte characterisation using the pores and membranes; and

[0708] at least one port for delivery of the material for performing the characterisation.

[0709] Alternatively, the apparatus preferably comprises:

[0710] a sensor device that is capable of supporting the plurality of pores and membranes being operable to perform analyte characterisation using the pores and membranes; and

[0711] at least one reservoir for holding material for performing the characterisation.

[0712] The apparatus more preferably comprises:

[0713] a sensor device that is capable of supporting the membrane and plurality of pores and membranes and being operable to perform analyte characterising using the pores and membranes;

[0714] at least one reservoir for holding material for performing the characterising;

[0715] a fluidics system configured to controllably supply material from the at least one reservoir to the sensor device; and

[0716] one or more containers for receiving respective samples, the fluidics system being configured to supply the samples selectively from one or more containers to the sensor device.

[0717] The apparatus may be any of those described in International Application No. No. PCT/GB08/004127 (published as WO 2009/077734), PCT/GB10/000789 (published as WO 2010/122293), International Application No. PCT/GB10/002206 (published as WO 2011/067559) or International Application No. PCT/US99/25679 (published as WO 00/28312).

[0718] The following Example illustrates the invention.

Example 1

[0719] This Example describes the simulations which were run to investigate DNA behaviour within CsgG.

Materials and Methods

[0720] Steered molecular dynamics simulations were performed to investigate the magnitude of the energetic barrier of CsgG-Eco and various mutants to DNA translocation. Simulations were performed using the GROMACS package version 4.0.5, with the GROMOS 53a6 forcefield and the SPC water model. The structure of CsgG-Eco (SEQ ID NO: 2) was taken from the protein data bank, accession code

4UV3. In order to make models of the CsgG-Eco mutants, the wild-type protein structure was mutated using PyMOL. The mutants studied were CsgG-Eco-(F56A) (SEQ ID NO: 2 with mutation F56A), CsgG-Eco-(F56A-N55S) (SEQ ID NO: 2 with mutations F56A/N55S) and CsgG-Eco-(F56A-N55S-Y51A) (SEQ ID NO: 2 with mutations F56A/N55S/Y51A).

[0721] DNA was then placed into the pores. Two different systems were set up:

[0722] i. A single guanine nucleotide was placed into the pore, just above the constriction region (approximately 5-10 Angstroms above the residue 56 ring)

[0723] ii. A single strand of DNA (ssDNA) was placed along the pore axis, with 5' end towards the beta-barrel side of the pore. In this set up, the ssDNA was pre-threaded through the entire length of the pore.

[0724] The simulation box was then solvated and then energy minimised using the steepest descents algorithm.

[0725] Each system was simulated in the NPT ensemble, using the Berendsen thermostat and Berendsen barostat to 300 K. Throughout the simulation, restraints were applied to the backbone of the pore.

[0726] In order to pull the DNA through the pore, a pulling force was applied to the phosphorus atom in the single guanine simulations. In the ssDNA simulations the pulling force was applied to the phosphorus atom at the 5' end of the strand. The pulling force was applied at a constant velocity by connecting a spring between the DNA phosphorus atom mentioned above and an imaginary point travelling at a constant velocity parallel to the pore axis. Note that the spring does not have any shape nor does it undergo any hydrodynamic drag. The spring constant was equal to 5 $\text{KJmol}^{-1} \text{\AA}^{-2}$.

Results

Single G Translocation

[0727] As shown in FIG. 3, a plot of the pulling force versus time shows that there is a large barrier for nucleotide entry into the ring of phenylalanine residues F56 in the wild type CsgG-Eco pore. There was no significant barrier to guanine translocation observed for the CsgG-Eco mutants studied.

ssDNA Translocation

[0728] For ssDNA translocation, two simulations were run per pore with each run having a different applied pulling velocity (100 $\text{\AA}/\text{ns}$ and 10 $\text{\AA}/\text{ns}$). As shown in FIG. 4, which illustrates the faster pulling velocity simulations, the CagG wild-type pore required the largest pulling force to enable ssDNA translocation. As shown in FIG. 5, which illustrates the slower pulling velocity simulations, both the CsgG-Eco (wild-type, SEQ ID NO: 2) and CsgG-Eco-(F56A) pores required the largest applied force to enable ssDNA translocation. Comparisons between the pulling force required for ssDNA translocation through CsgG and MspA baseline pore, suggest that mutation of the CsgG pore is required to allow a similar level of ssDNA translocation.

Example 2

[0729] This Example describes the characterisation of several CsgG mutants.

Materials and Methods

[0730] Prior to setting up the experiment, DNA construct X (final concentration 0.1 nM, see FIG. 13 for cartoon representation of construct X and description) was pre-incubated at room temperature for five minutes with T4 Dda-E94C/C109A/C136A/A360C (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C, final concentration added to the nanopore system 10 nM, which was provided in buffer (151.5 mM KCl, 25 mM potassium phosphate, 5% glycerol, pH 7.0, 1 mM EDTA)). After five minutes, TMAD (100 μ M) was added to the pre-mix and the mixture incubated for a further 5 minutes. Finally, MgCl₂ (1.5 mM final concentration added to the nanopore system), ATP (1.5 mM final concentration added to the nanopore system), KCl (500 mM final concentration added to the nanopore system) and potassium phosphate buffer (25 mM final concentration added to the nanopore system) were added to the pre-mix.

[0731] Electrical measurements were acquired from a variety of single CsgG nanopores inserted in block co-polymer in buffer (25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0). After achieving a single pore inserted in the block co-polymer, then buffer (2 mL, 25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0) was flowed through the system to remove any excess CsgG nanopores. 150 μ L of 500 mM KCl, 25 mM K Phosphate, 1.5 mM MgCl₂, 1.5 mM ATP, pH8.0 was then flowed through the system. After 10 minutes a 150 μ L of 500 mM KCl, 25 mM potassium phosphate, 1.5 mM MgCl₂, 1.5 mM ATP, pH8.0 was flowed through the system and then the enzyme (T4 Dda-E94C/C109A/C136A/A360C, 10 nM final concentration), DNA construct X (0.1 nM final concentration), fuel (MgCl₂ 1.5 mM final concentration, ATP 1.5 mM final concentration) pre-mix (150 μ L total) was then flowed into the single nanopore experimental system. The experiment was run at $\sim$ 120 mV and helicase-controlled DNA movement monitored.

Results

Pores Showing Increased Range (FIGS. 6 to 8, and 18 to 30)

CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) has a range of $\sim$ 10 pA (see FIG. 6 (a)) whereas the CsgG-Eco pore mutants below exhibited an increased current range-

[0732] 1-CsgG-Eco-(Y51N-F56A-D149N-E185R-E201N-E203N-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51N/F56A/D149N/E185R/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (See FIG. 6 (b)).

[0733] 2-CsgG-Eco-(N55A-StrepII(C))₉ (SEQ ID NO: 2 with mutation N55A where StrepII(C) is has SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 6 (c)).

[0734] 3-CsgG-Eco-(N55S-StrepII(C))₉ (SEQ ID NO: 2 with mutations N55S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 40 pA (see FIG. 7 (a)).

[0735] 4-CsgG-Eco-(Y51N-StrepII(C))₉ (SEQ ID NO: 2 with mutation Y51N where StrepII(C) is SEQ ID NO:

47 and is attached at the C-terminus) exhibited a range of $\sim$ 40 pA (see FIG. 7 (b)).

[0736] 5-CsgG-Eco-(Y51A-F56A-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51A/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 7 (c)).

[0737] 6-CsgG-Eco-(Y51A-F56N-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51A/F56N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 20 pA (see FIG. 8 (a)).

[0738] 7-CsgG-Eco-(Y51A-N55S-F56A-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51A/N55S/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 8 (b)).

[0739] 8-CsgG-Eco-(Y51A-N55S-F56N-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51A/N55S/F56N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 8 (c)).

[0740] 13-CsgG-Eco-(F56H-StrepII(C))₉ (SEQ ID NO: 2 with mutation F56H where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 18).

[0741] 14-CsgG-Eco-(F56Q-StrepII(C))₉ (SEQ ID NO: 2 with mutation F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 40 pA (see FIG. 19).

[0742] 15-CsgG-Eco-(F56T-StrepII(C))₉ (SEQ ID NO: 2 with mutation F56T where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 20).

[0743] 16-CsgG-Eco-(S54P/F56A-StrepII(C))₉ (SEQ ID NO: 2 with mutation S54P/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 21).

[0744] 17-CsgG-Eco-(Y51T/F56A-StrepII(C))₉ (SEQ ID NO: 2 with mutation Y51T/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 22).

[0745] 18-CsgG-Eco-(F56P-StrepII(C))₉ (SEQ ID NO: 2 with mutation F56P where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 23).

[0746] 19-CsgG-Eco-(F56A-StrepII(C))₉ (SEQ ID NO: 2 with mutation F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 40 pA (see FIG. 24).

[0747] 20-CsgG-Eco-(Y51T/F56Q-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 30 pA (see FIG. 25).

[0748] 21-CsgG-Eco-(N55S/F56Q-StrepII(C))₉ (SEQ ID NO: 2 with mutations N55S/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 26).

[0749] 22-CsgG-Eco-(Y51T/N55S/F56Q-StrepII(C))₉ (SEQ ID NO: 2 with mutations Y51T/N55S/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of $\sim$ 35 pA (see FIG. 27).

[0750] 23-CsgG-Eco-(F56Q/N102R-StrepII(C))₉ (SEQ ID NO: 2 with mutations F56Q/N102R where StrepII

(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 28).

[0751] 24-CsgG-Eco-(Y51Q/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51Q/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 29).

[0752] 25-CsgG-Eco-(Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 30).

Pores Showing Increased Throughput (FIGS. 9 and 10)

[0753] As can be seen from FIGS. 9 and 10, the following mutant pores (9-12 below) exhibited multiple helicase controlled DNA movements (Labelled as X in FIGS. 9 and 10) per channel in 4 hours, whereas CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) shown in FIG. 9 (a) frequently exhibited only 1 or 2 helicase controlled DNA movements (labelled as X in FIG. 9 (a)) per channel in 4 hours and instead exhibited prolonged block regions (labelled as Y in FIG. 9 (a)).

[0754] 9-CsgG-Eco-(D149N-E185N-E203N-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 9 (b))

[0755] 10-CsgG-Eco-(D149N-E185N-E201N-E203N-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185N/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 9 (c))

[0756] 11-CsgG-Eco-(D149N-E185R-D195N-E201N-E203N)-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185R/D195N/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 10 (a))

[0757] 12-CsgG-Eco-(D149N-E185R-D195N-E201R-E203N)-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185R/D195N/E201R/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 10 (b))

Pore Showing Increased Insertion (FIGS. 11 and 12)

[0758] As can be seen by comparing FIGS. 11 and 12, the mutant pore CagG-Eco-(T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) shown in FIG. 12, was present in the membrane in increased pore numbers (~4-5 fold) compared with the CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) pore (shown in FIG. 11). Arrows in FIGS. 11 and 12 illustrated the number of CsgG-Eco nanopores which inserted into the block co-polymer in a 4 hour experiment (130-140 in FIGS. 11 and 1-11 in FIG. 12 each corresponded to a separate nanopore experiment). For CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) three experiments showed insertion of one nanopore, whereas for the mutant pore (CsgG-Eco-(T150I-StrepII(C))9) each experiment showed insertion of at least one nanopore and several experiments showed multiple pore insertions.

Example 3

[0759] This example described an *E. Coli* purification method developed to purify the CsgG pore.

Materials and Methods

[0760] DNA encoding the polypeptide Pro-CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and where Pro is SEQ ID NO: 48 and is attached at the N-terminus) was synthesised in GenScript USA Inc. and cloned into a pT7 vector containing ampicillin resistance gene. Protein expression of the pT7 vector was induced by Isopropyl β -D-1-thiogalactopyranoside (IPTG). The concentration of the DNA solution was adjusted to 400 ng/ μ L. DNA (1 μ L) was used to transform Lemo21 (DE3) competent *E. coli* cells (50 μ L, NEB, catalogue number C2528H). Prior to transformation, the CsgG gene was knocked out from Lemo21 (DE3) cells (Gene Bridges GmbH, Germany). The cells were then plated out on LB agar containing ampicillin (0.1 mg/mL) and incubated for approx 16 hours at 37° C.

[0761] Bacterial colonies grown on LB plates, containing ampicillin, incorporated the CsgG plasmid. One such colony was used to inoculate a starter culture of LB media (100 mL) containing carbenicillin (0.1 mg/mL). The starter culture was grown at 37° C. with agitation until OD600 was reached to 1.0-1.2. The starter culture was used to inoculate a fresh 500 mL of LB media containing carbenicillin (0.1 mg/mL) and Rhamnose (500 μ M) to an O.D. 600 of 0.1. The culture was grown at 37° C. with agitation until OD600 reached 0.6. The temperature of the culture was then adjusted to 18° C. and induction was initiated by the addition of IPTG (0.2 mM final concentration). Induction was carried out for approximately 18 hours with agitation at 18° C.

[0762] Following induction, the culture was pelleted by centrifugation at 6,000 g for 30 minutes. The pellet was resuspended in 50 mM Tris, 300 mM NaCl, containing protease inhibitors (Merck Millipore 539138), benzonase nuclease (Sigma E1014) and 1 $\times$ bugbuster (Merck Millipore 70921) pH8.0 (approximately 10 mL of buffer per gram of pellet). Suspension was mixed well until it was fully homogeneous, the sample was then transferred to roller mixer at 4° C. for approx 5 hours. Lysate was pelleted by centrifugation at 20,000 g for 45 minutes and the supernatant was filtered through 0.22 μ m PES syringe filter. Supernatant which contained CsgG (known as sample 1) was taken forward for purification by column chromatography.

[0763] Sample 1 was applied to a 5 mL Strep Trap column (GE Healthcare). The column was washed with 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM pH8 until a stable baseline of 10 column volumes was maintained. The column was then washed with 25 mM Tris, 2M NaCl, 2 mM EDTA, 0.01% DDM pH8 before being returned to the 150 mM buffer. Elution was carried out with 10 mM desthiobiotin. An example of a chromatography trace of Strep trap (GE Healthcare) purification of a CsgG protein is shown in FIG. 14. The elution peak is labelled E1. FIG. 15 shows an example of a typical SDS-PAGE visualization of CsgG-Eco protein after the initial Strep purification. Lanes 1-3 shows the main elution peak (labelled E1 in FIG. 14) which contained CsgG protein as indicated by the arrow. Lanes 4-6 corresponded to elution fractions of the tail of the main elution peak (labelled E1 in FIG. 14) which contained contaminants.

[0764] The elution peak was pooled and heated to 65° C. for 15 minutes to remove heat unstable contaminated proteins. The heated solution was subjected to centrifugation at 20,000 g for 10 minutes and the pellet was discarded. The supernatant was subjected to gel filtration on a 120 mL Sephadex S200 column (GE Healthcare) in 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS pH8. Monitoring was carried out at 220 nM due to low Tryptophan component of protein. The sample was eluted at approximately 55 mL volume (FIG. 16 shows the size exclusion column trace with the 55 mL sample peak labelled with a star). The elution peak was run on a 4-20% TGX (see FIG. 17, Bio Rad) to confirm the presence of the pore of interest CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus). Identified fractions were pooled and concentrated by 50 kD Amicon spin column.

Example 4

[0765] This example describes the simulations which were run to investigate the interaction between CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20) with T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (ΔM1) G1G2).

Simulation Methods

[0766] Simulations were performed using the GROMACS package version 4.0.5, with the GROMOS 53a6 forcefield and the SPC water model.

[0767] The CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20) model was based on the crystal structures of CsgG found in the protein data bank, accession codes 4UV3 and 4Q79. The relevant mutations were made using PyMOL. The resultant pore model was then energy minimised using the steepest descents algorithm. The T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (ΔM1) G1G2) model was based on the Dda1993 structure found in the protein data bank, accession code 3UPU. Again, relevant mutations were made using PyMOL, and the model was energy minimised using the steepest descents algorithm.

[0768] The T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (ΔM1) G1G2) model was then placed above CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20). Three simulations were performed with a different initial enzyme conformation (Runs 1 to 3 (0 ns), see FIG. 31):

[0769] In all enzyme conformations, the enzyme was oriented such that the 5' end of the DNA was pointing towards the pore, and the enzyme was unrestrained throughout the simulation. The pore backbone was restrained and the simulation box was solvated. The system was simulated in the NPT ensemble for 40 ns, using the Berendsen thermostat and Berendsen barostat to 300 K.

[0770] The contacts between the enzyme and pore were analysed using both GROMACS analysis software and also locally written code. The tables below show the number of

contacts observed for both pore and enzyme amino acids. Tables 6-8 shows the amino acid contact points on pore which interact with the amino acid contact points on the enzyme. In two out of the three simulations the enzyme tilts on top of the pore (see run 2 and 3 (20, 30 and 40 ns), FIGS. 31 and 32). Run 1 shows that the enzyme has not tilted and so points that are shown to have high interaction in table 6 can be optimised in order to increase enzyme stability on the pore cap.

TABLE 6

=run 1 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
ASN 102	ASP 198	8200
ASN 102	TYR 438	8130
GLN 100	ASP 212	7369
GLU 101	TRP 195	5979
ARG 97	TYR 350	4873
GLU 101	LEU 215	4851
ASN 102	TRP 195	3988
ARG 97	TYR 415	3798
GLU 101	TYR 350	3759
LEU 113	ASP 212	3718
ASN 102	LYS 358	3124
ARG 97	GLY 211	2765
GLU 101	CYS 412	2715
ARG 97	GLY 193	2708
ASN 102	ILE 196	2342
GLU 101	TYR 415	2268
GLU 101	ARG 216	2158
ARG 110	THR 213	2094
ARG 110	ASP 212	2066
GLY 103	ARG 216	1456
GLU 101	TYR 318	1333
ASN 102	GLU 347	1316
GLU 101	LYS 194	1310
ARG 97	PRO 411	1203
GLU 101	LYS 358	1161
ASN 102	ARG 216	1132
ARG 97	TRP 195	888
LYS 94	TYR 415	793
ASN 102	PRO 315	696
ASN 102	LYS 247	541
GLU 101	ALA 214	449
ASN 102	ASP 346	440
ARG 97	ALA 214	366
ARG 97	LYS 194	336
GLU 101	ASP 212	302
ARG 97	VAL 439	267
ARG 110	THR 210	263
ARG 97	THR 210	259
ARG 97	GLN 422	257
GLU 101	TYR 409	228
ALA 98	TRP 195	207
GLU 101	LYS 247	201
ASN 102	GLU 317	179
ARG 110	ARG 216	147
ARG 97	ASP 212	108
ASN 102	VAL 314	87
GLU 101	THR 213	72
ASN 102	LYS 255	70
VAL 105	ARG 216	69
ASN 102	LEU 215	59
ASN 102	THR 210	55
ILE 111	ASP 212	48
ARG 97	HIS 414	48
THR 104	ARG 216	36
ASN 102	TYR 197	32
GLN 100	THR 213	30
ASN 102	GLU 361	28
ARG 97	VAL 418	28
ALA 98	TYR 415	27
GLU 101	LEU 354	17
GLU 101	TYR 197	16

TABLE 6-continued

=run 1 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
ASN 102	GLY 316	16
ARG 97	GLU 361	16
ARG 97	GLU 347	14
ILE 107	ARG 216	12
ASN 102	GLY 208	12
ARG 97	TYR 409	11
ARG 97	LYS 247	11
GLU 101	LYS 364	8
ARG 97	PHE 209	7
LYS 94	GLU 419	6
GLU 101	PRO 411	5
GLU 101	GLU 317	5
ASN 102	ILE 251	5
ARG 97	LEU 354	5
LYS 94	VAL 418	3
ASN 102	ARG 321	3
ARG 97	LYS 243	3
LYS 94	CYS 412	2
LEU 113	THR 210	2
GLY 103	GLU 317	2
GLU 101	LYS 351	2
ASN 102	TYR 318	2
ASN 102	MET 219	2
ASN 102	LYS 194	2
ARG 97	VAL 314	2
ARG 97	LYS 364	2
THR 104	PRO 315	1
GLY 103	THR 213	1
GLU 101	PRO 315	1

TABLE 7

=run 2 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
GLU 101	THR 210	14155
SER 115	ASP 202	9477
ARG 97	THR 210	9064
ASN 102	VAL 200	5323
THR 104	ASP 202	4476
ASN 102	ASN 221	3422
GLU 101	PHE 437	3171
ARG 97	ASP 217	2698
GLU 101	ARG 216	2198
ARG 97	GLY 208	1730
GLU 101	LYS 199	1710
SER 115	SER 224	1440
ASN 102	LYS 199	1351
ASN 102	ASP 212	1298
ASN 102	ARG 405	1219
GLU 101	ARG 207	1180
ASN 102	SER 224	1150
ASN 102	LYS 255	1114
ARG 97	ASP 198	946
GLU 101	PHE 209	931
ARG 97	THR 213	791
ARG 97	ARG 216	599
ASN 102	THR 210	589
GLN 114	ASP 202	530
ASN 102	ASP 202	492
ARG 97	ASP 212	490
GLY 103	ARG 405	474
THR 104	SER 224	451
GLU 101	LYS 255	429
ASN 102	ASP 198	405
ASN 102	PHE 209	400
ASN 102	ARG 178	316
ARG 110	GLU 258	309
ASN 102	ASN 180	257

TABLE 7-continued

=run 2 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
GLN 100	PHE 223	256
GLU 101	TYR 197	220
GLN 114	SER 228	212
LEU 113	PHE 223	210
ASN 102	ILE 225	204
GLN 114	LYS 227	194
GLU 101	GLY 211	189
GLU 101	ASP 212	174
LEU 113	SER 224	159
LEU 113	GLY 203	145
ARG 97	VAL 220	134
GLU 101	THR 213	133
THR 104	SER 228	125
ARG 97	TYR 197	123
LYS 94	ASP 212	118
ASN 102	ARG 216	110
ASN 102	ASN 235	108
ASN 102	GLY 211	104
GLU 101	ARG 405	79
GLN 114	SER 224	69
ASN 102	VAL 220	63
LEU 113	LYS 227	49
ASN 102	VAL 201	42
ARG 97	PHE 209	42
GLU 101	ASN 180	40
ARG 97	TYR 438	38
ARG 97	ARG 207	32
ASN 102	PHE 407	28
SER 115	ASN 221	23
ARG 110	HIS 204	22
GLU 101	PHE 223	21
ARG 97	ASP 189	19
ARG 110	PHE 223	16
THR 104	ILE 225	13
GLY 103	ASN 180	11
ARG 97	LYS 194	11
GLU 101	PHE 407	10
ARG 97	MET 219	9
THR 104	ASN 235	8
ARG 110	ARG 405	8
ARG 97	TRP 195	7
ILE 111	PHE 223	6
GLU 101	GLY 208	6
LEU 113	ASP 202	5
GLU 101	ARG 178	5
ASN 102	THR 213	5
ALA 98	ARG 216	5
ASN 102	ASP 217	4
ARG 97	LYS 199	4
THR 104	LEU 229	3
THR 104	ARG 405	3
GLU 101	VAL 201	3
GLU 101	MET 219	3
ARG 110	ASP 202	3
ARG 110	ARG 207	2
THR 104	VAL 201	1
GLY 103	SER 224	1
GLY 103	LYS 255	1
GLY 103	GLU 258	1
GLY 103	ASN 235	1
GLU 101	ASP 198	1
ASN 102	PHE 437	1
ARG 97	PHE 437	1
ARG 110	LYS 227	1

TABLE 8

=run 3 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
ARG 97	THR 174	15557
GLN 100	ASP 5	10353
GLU 101	LYS 177	9238
ARG 97	SER 179	6630
LEU 116	ASP 202	6545
GLU 101	TYR 434	6524
SER 115	ASP 202	5693
GLU 101	HIS 204	5457
ARG 97	GLN 10	5106
ARG 93	ASP 202	4646
ARG 93	GLU 8	4446
SER 115	LYS 11	4342
LEU 113	ASP 5	3871
ASN 102	SER 224	3605
GLU 101	ASN 12	3344
GLU 101	GLN 10	3327
ARG 97	GLU 175	3096
GLU 101	SER 224	3028
LEU 116	GLU 8	2936
LYS 94	ASP 185	2708
ARG 97	ASN 180	2700
GLU 101	PHE 3	2500
THR 104	LYS 11	2352
SER 115	GLU 8	2323
ARG 93	ASN 180	1912
ASN 102	LYS 177	1838
LYS 94	ASP 198	1828
ARG 110	ASP 5	1714
ALA 98	GLY 203	1701
ASN 102	ASN 12	1695
GLU 101	TYR 169	1691
ARG 97	THR 7	1593
ARG 110	ASP 4	1404
ARG 97	ASP 212	1381
ASN 102	HIS 204	1226
ASN 102	ASN 15	1173
ARG 97	VAL 176	1096
ALA 98	HIS 204	998
ARG 97	ASP 202	875
ASN 102	TYR 434	850
ALA 98	ASN 12	716
GLU 101	THR 213	702
GLU 101	ARG 178	642
GLU 101	ASN 221	600
ASN 102	LYS 11	588
ARG 97	ASP 217	585
ARG 97	ARG 207	537
GLU 101	ARG 207	525
ARG 97	PHE 437	511
GLU 101	ARG 216	510
ASN 102	LYS 19	482
ARG 97	HIS 204	473
LEU 113	LYS 11	409
ARG 97	THR 213	358
ARG 93	ASP 212	354
ARG 97	TYR 169	316
ARG 97	GLY 203	308
ARG 97	ASP 435	300
GLN 87	LYS 199	249
THR 104	ASN 15	221
ARG 97	ALA 181	220
ASN 102	LYS 227	198
LYS 94	ARG 178	184
ASN 102	GLU 8	183
LEU 113	LEU 6	182
ARG 93	SER 179	179
LEU 90	ASN 180	172
LEU 90	ASP 202	144
ARG 97	ILE 225	138
GLU 101	ASN 15	135
GLU 101	LYS 19	113
LYS 94	ASN 180	109
LYS 94	GLU 175	105

TABLE 8-continued

=run 3 enzyme and pore contact interactions		
Pore	Enzyme	# contacts
ARG 93	THR 7	81
LYS 94	ARG 207	77
GLN 100	PHE 3	72
ASN 102	ARG 216	66
ARG 97	LYS 177	62
GLU 101	THR 210	59
ARG 97	ARG 178	56
LYS 94	ASP 212	55
ARG 97	GLU 172	53
GLU 101	VAL 176	51
ALA 98	ARG 207	49
ARG 110	PHE 3	48
ALA 98	ASP 202	47
ARG 97	VAL 200	40
ALA 98	VAL 201	36
LYS 94	THR 210	35
ILE 111	ASP 5	32
ARG 97	ARG 405	27
LEU 90	VAL 200	26
ARG 97	THR 210	26
GLY 103	PHE 3	25
GLU 101	PHE 209	25
ARG 97	ARG 216	22
ASN 102	VAL 220	21
LYS 94	GLY 211	19
ARG 97	PHE 209	17
GLU 101	LYS 227	15
GLN 114	LYS 11	15
GLY 103	LYS 19	13
ARG 97	PHE 3	13
GLU 101	THR 2	12
GLU 101	ILE 225	12
ARG 97	ILE 184	12
ALA 98	GLU 8	12
ALA 98	ARG 178	12
ASN 102	ILE 225	11
LYS 94	LYS 199	10
GLU 101	ARG 433	8
ARG 97	ASN 221	8
LYS 94	VAL 200	7
ASN 102	ASP 202	7
ASN 102	ASN 221	7
ARG 97	LEU 173	7
SER 115	HIS 204	6
ASN 102	GLY 203	6
GLU 101	CYS 171	5
ARG 97	ASN 12	5
ASN 102	PHE 223	4
ASN 102	LYS 166	4
ARG 97	GLY 211	4
ARG 97	GLN 170	4
GLU 101	ARG 405	3
ASN 102	PHE 3	3
GLU 101	GLU 175	2
ARG 97	VAL 220	2
ARG 93	GLY 203	2
LYS 94	THR 174	1
LEU 90	LYS 199	1
LEU 116	ASN 180	1
LEU 113	ASP 212	1
LEU 113	ASP 202	1
GLY 103	ASN 15	1
GLU 101	THR 7	1
GLU 101	PHE 437	1
GLN 114	ASP 202	1
ASN 102	ARG 405	1
ARG 97	TYR 434	1
ARG 97	PRO 182	1
ARG 97	GLY 9	1
ARG 97	GLU 8	1
ALA 99	ASP 202	1

Example 5

[0771] This example describes the simulations which were run to investigate the interaction between a) CsgG-Eco-(Y51A/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 25) with T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2) and b) CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) with T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2).

Simulation Methods

[0772] Simulations were performed as described in Example 4.

[0773] The CsgG-Eco-(Y51A/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 25) and CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) models were based on the crystal structures of CsgG found in the protein data bank, accession codes 4UV3 and 4Q79. The relevant mutations were made using PyMOL. The resultant pore model was then energy minimised using the steepest descents algorithm.

[0774] The T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2) model was based on the Dda1993 structure found in the protein data bank, accession code 3UPU. Again, relevant mutations were made using PyMOL, and the model was energy minimised using the steepest descents algorithm.

[0775] The T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2) model was then placed above mutant pores 25 and 26.

[0776] In the simulations the enzyme was oriented such that the 5' end of the DNA was pointing towards the pore, and the enzyme was unrestrained throughout the simulation. For each of the mutant pores investigated two simulations were run-in the first the pore backbone was restrained and the simulation box was solvated and in the second the pore backbone was restrained except for the cap region and the simulation was box solvated. The system was simulated in the NPT ensemble for 40 ns, using the Berendsen thermostat and Berendsen barostat to 300 K.

[0777] The contacts between the enzyme and pore were analysed using both GROMACS analysis software and also locally written code. The tables below show the number of contacts observed for both pore and enzyme amino acids (for mutants 25 and 26 with T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 9 (pore backbone restrained) and 10 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 25 and the number of contacts that they make with the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 11 (pore backbone restrained) and 12 (pore

backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 25 (CsgG-Eco-(Y51A/F56Q)-StrepII(C)) 9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) which interact with the amino acid contact points on the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 13 (pore backbone restrained) and 14 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 26 and the number of contacts that they make with the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 15 (pore backbone restrained) and 16 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 26 (CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)) which interact with the amino acid contact points on the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). FIG. 33 shows an initial snapshot of pore mutant 26 and T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C).

TABLE 9

Amino Acid	Position	Number of Contacts
GLU	101	26081
ARG	97	12985
GLN	100	6485
ASN	102	4941
LEU	113	3923
LYS	94	3159
ARG	110	2348
GLN	114	617
ILE	111	195
ALA	98	74
ILE	107	67
THR	104	24
PRO	112	21
SER	115	17
GLY	103	17
GLN	87	13
LEU	90	6
ASN	108	3

TABLE 10

Amino Acid	Position	Number of Contacts
ARG	97	17462
GLU	101	12262
ASN	102	9008
LYS	94	5982
GLN	100	3471
GLY	103	3366
LEU	113	1511
GLN	87	734
THR	104	500
ARG	110	420
ASN	91	273
GLN	114	178
ALA	98	94
ARG	93	44
ILE	111	21
PRO	112	13
LEU	90	13
SER	115	4
VAL	105	2
ALA	99	2
ILE	95	1

TABLE 11

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
GLU	101	TYR	318	7116
GLU	101	THR	210	6306
GLN	100	ASN	365	4693
GLU	101	TRP	195	4608
LEU	113	ASN	367	3910
LYS	94	ASP	212	2369
ASN	102	LYS	371	2088
GLU	101	LYS	364	1903
ARG	97	SER	224	1883
ARG	97	TYR	438	1764
ARG	110	LYS	368	1727
GLU	101	LYS	358	1443
GLN	100	ASN	367	1390
GLU	101	GLN	422	1356
ARG	97	TRP	195	1318
ARG	97	PHE	209	1095
ARG	97	TYR	318	965
ARG	97	TYR	350	895
ASN	102	LYS	358	833
ARG	97	GLU	361	818
ARG	97	GLY	208	706
GLU	101	GLY	316	697
GLU	101	GLU	317	678
ARG	97	GLU	317	647
GLN	114	ASN	367	617
GLU	101	LYS	371	506
ASN	102	TRP	366	466
ARG	97	THR	210	463
ASN	102	ARG	207	457
LYS	94	ASP	217	423
GLU	101	ARG	207	409
ASN	102	GLU	419	405
GLN	100	LYS	368	402
GLU	101	ARG	321	354
ARG	110	GLY	369	332
ARG	97	GLN	422	324
ASN	102	GLY	313	313
ARG	97	ASN	221	306
ARG	97	VAL	439	301
ARG	110	ASN	365	274
SER	257	GLU	258	267
GLU	101	LYS	310	217
ILE	111	ASN	367	195
ARG	97	ASP	198	154
GLU	101	SER	224	154
ARG	97	VAL	418	145
ARG	97	ALA	157	137
GLU	101	GLY	208	136
ARG	97	GLU	419	135
LYS	94	TYR	415	111
ARG	97	VAL	220	105
ASN	102	PRO	315	104
ASN	102	GLU	93	103
ARG	97	LYS	364	97
LYS	94	TYR	350	94
ARG	97	GLU	154	92
ARG	97	ALA	416	82
GLU	101	PHE	223	76
ILE	107	LYS	368	67
ARG	97	GLU	347	66
ARG	97	TYR	415	60
ASN	102	TYR	197	57
LYS	94	GLY	211	49
LYS	94	THR	213	47
GLU	101	ASN	365	46
ARG	97	LEU	354	41
LYS	94	ARG	216	41
ALA	98	TRP	195	37
ASN	102	TRP	195	34
ALA	98	LYS	358	32
ASN	102	LEU	194	27
THR	104	TRP	366	24
ARG	97	GLY	211	23

TABLE 11-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
PRO	112	ASN	367	21
GLU	101	PHE	209	21
ASN	102	LYS	364	20
LYS	94	ALA	416	19
SER	115	TRP	366	17
GLY	103	LEU	194	17
ARG	97	ASP	212	16
GLU	101	TYR	415	15
ARG	110	GLY	370	15
ARG	97	LYS	351	14
ASN	102	LYS	310	14
ASN	102	GLU	154	14
ARG	97	CYS	360	13
ARG	97	THR	156	12
GLU	101	LEU	194	11
GLN	87	ASP	212	9
ARG	97	PHE	308	8
GLU	101	LEU	354	8
LEU	113	TRP	366	7
ARG	97	ARG	321	7
GLU	101	ALA	157	7
LEU	113	ASN	365	6
GLU	101	GLY	193	6
ARG	97	ASP	217	5
ALA	98	THR	210	5
GLN	87	ARG	216	4
ARG	97	LEU	319	4
ARG	97	ASN	155	4
LYS	94	VAL	220	4
LEU	90	TYR	415	3
LEU	90	ASP	212	3
ARG	97	PHE	223	3
ARG	97	GLY	193	3
ARG	97	ALA	421	3
ASN	108	LYS	368	3
ASN	102	GLY	316	3
GLU	101	VAL	418	3
ARG	97	ILE	159	2
GLU	101	PRO	315	2
SER	257	ASP	260	1
ARG	97	GLY	357	1
ASN	102	THR	210	1
ASN	102	ILE	196	1
ASN	102	GLU	317	1
GLU	101	LYS	227	1
GLU	101	GLU	361	1
GLU	101	ASN	221	1
LYS	94	VAL	418	1
LYS	94	GLU	154	1

TABLE 12

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
ARG	97	ASN	365	3386
GLU	101	GLY	316	2967
GLY	103	ASP	189	2773
LYS	94	ASP	212	2524
GLU	101	GLU	317	2401
ASN	102	TYR	197	2377
GLN	100	ASN	365	2241
GLU	101	LYS	358	2110
ARG	97	GLU	317	2031
GLU	101	THR	210	2025
ASN	102	ASP	189	1985
LYS	94	THR	213	1689
LEU	113	ASN	367	1509
ARG	97	GLU	361	1421

TABLE 12-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	
ARG	97	TRP	195	1361
ARG	97	TYR	438	1315
ARG	97	VAL	439	1145
LYS	94	ASP	217	957
GLU	101	TYR	318	852
GLU	101	ASN	365	797
SER	257	ASP	260	784
ARG	97	GLY	193	744
ARG	97	GLU	419	738
ARG	97	PHE	209	731
ASN	102	ARG	321	719
ASN	102	TRP	195	708
ASN	102	LYS	371	686
ARG	97	GLN	422	637
GLN	87	ASP	212	603
ARG	97	TYR	318	603
ARG	97	ASN	367	575
ARG	97	ASP	198	543
GLN	100	ASN	367	517
LYS	94	TYR	350	470
ARG	97	ALA	157	446
ARG	97	GLY	208	444
SER	257	GLU	258	425
GLU	101	LYS	371	407
GLU	101	LYS	364	400
ASN	102	GLU	93	389
GLN	100	LYS	368	383
ARG	110	LYS	368	357
ARG	97	THR	210	303
ASN	91	ARG	216	273
THR	104	ASN	367	214
GLY	103	TYR	197	204
GLN	114	ASN	367	178
GLU	101	ARG	207	173
ASN	102	TRP	366	171
ASN	102	GLY	369	153
ASN	102	LEU	194	123
ASN	102	GLY	313	120
ARG	97	THR	362	100
THR	104	TRP	366	99
GLY	103	THR	164	97
LYS	94	TRP	195	94
GLN	100	GLY	193	84
GLY	103	LYS	177	81
ASN	102	LYS	166	80
ARG	97	ILE	159	78
LYS	94	ALA	416	76
GLN	87	ARG	216	75
ASN	102	SER	306	73
ASN	102	ARG	207	73
ALA	98	TRP	195	68
ARG	97	TYR	158	65
ASN	102	GLU	347	65
ASN	102	GLN	170	65
SER	257	LYS	261	62
ARG	97	ASN	192	62
GLN	100	GLY	369	61
THR	104	LYS	199	60
ARG	97	LEU	194	57
ASN	102	PHE	163	54
ASN	102	GLU	348	53
GLY	103	GLY	313	53
GLN	100	GLU	419	53
ASN	102	LYS	351	52
GLN	87	THR	213	51
LYS	94	GLY	211	49
GLN	100	GLN	423	49
THR	104	LYS	166	48
GLU	101	ARG	321	45
ARG	97	PHE	308	44
ARG	93	GLU	419	44
ASN	102	GLY	370	41
ARG	110	GLY	369	41

TABLE 12-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	
SER	257	THR	259	40
ARG	97	GLY	211	40
ARG	97	THR	156	39
ASN	102	THR	164	39
LYS	94	ARG	216	39
ARG	97	ARG	321	38
GLY	103	LYS	166	35
ASN	102	GLY	208	33
GLU	101	GLY	370	30
LYS	94	GLU	361	28
ASN	102	SER	224	27
GLU	101	ILE	159	25
THR	104	PHE	308	23
THR	104	ASN	365	21
ASN	102	PRO	315	21
LYS	94	GLU	154	21
ILE	111	ASN	367	21
ARG	110	ILE	159	20
GLY	103	ARG	321	20
GLN	100	LEU	194	20
ARG	97	GLU	347	18
ARG	97	ARG	122	16
GLY	103	TRP	366	16
GLY	103	GLY	369	15
PRO	112	ASN	367	13
ALA	98	ARG	216	13
ARG	97	SER	345	12
ASN	102	LYS	364	12
GLN	100	GLY	370	12
GLU	101	PRO	315	11
GLN	100	ILE	159	11
GLN	100	GLU	317	11
LEU	90	VAL	418	10
ASN	102	VAL	220	10
ASN	102	LYS	255	10
ALA	98	LYS	358	10
THR	104	ASN	180	9
GLY	103	LEU	194	9
THR	104	LYS	368	8
LYS	94	TYR	415	8
GLN	100	THR	164	8
ARG	97	VAL	418	7
ARG	97	VAL	220	7
ARG	97	TRP	366	7
ASN	102	LEU	319	7
GLY	103	LYS	255	7
LYS	94	ILE	413	7
GLN	100	PHE	163	7
THR	104	LYS	255	6
THR	104	LYS	177	6
LYS	94	GLU	347	6
GLN	100	ASN	192	6
GLN	87	ALA	416	5
ARG	97	LYS	368	5
ARG	97	ASP	217	5
ASN	102	ARG	216	5
LYS	94	VAL	439	5
ASN	102	VAL	200	4
ASN	102	ASN	365	4
GLU	101	LYS	310	4
GLU	101	GLY	313	4
GLY	103	PHE	223	4
LYS	94	ASN	155	4
GLN	100	LEU	420	4
ARG	97	SER	224	3
ARG	97	ASP	417	3
ASN	102	THR	362	3
SER	115	GLU	419	3
GLU	101	LEU	194	3
GLU	101	GLY	369	3
GLY	103	ASN	192	3
LYS	94	ASP	417	3
VAL	105	LYS	166	2

TABLE 12-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
LEU	113	ASN	365	2
LEU	90	ASP	212	2
ARG	97	TYR	92	2
ARG	97	GLU	154	2
ARG	97	ARG	207	2
THR	104	TYR	197	2
THR	104	ASP	185	2
ASN	102	PHE	209	2
ASN	102	ILE	159	2
GLY	103	PHE	308	2
GLY	103	LEU	319	2
GLY	103	ILE	159	2
ALA	98	THR	210	2
ALA	99	ILE	159	2
GLN	100	THR	210	2
SER	257	LYS	254	1
ILE	95	ARG	216	1
LEU	90	THR	213	1
ARG	97	TYR	304	1
ARG	97	ASP	212	1
THR	104	GLY	369	1
THR	104	ASN	192	1
ASN	102	VAL	314	1
ASN	102	THR	210	1
ASN	102	PHE	223	1
ASN	102	LYS	199	1
SER	115	ASP	212	1
ARG	110	GLU	317	1
ARG	110	ASN	365	1
GLY	103	PRO	315	1
GLY	103	ASN	367	1
ALA	98	ALA	157	1
LYS	94	VAL	418	1

TABLE 13

Amino Acid	Position	Number of Contacts
GLU	101	18418
TRP	97	12195
ASN	102	5232
GLY	103	212
ALA	98	113
SER	115	89
THR	104	57
LYS	94	52
LEU	113	11
GLN	100	5
ARG	110	5
GLN	114	4
ARG	93	1

TABLE 14

Amino Acid	Position	Number of Contacts
TRP	97	16770
ASN	102	11609
GLU	101	4947
GLY	103	2211
THR	104	2187
GLN	100	1589
LYS	94	686
ALA	98	289
SER	115	274
ARG	110	251
ARG	93	44
ILE	95	14

TABLE 14-continued

Amino Acid	Position	Number of Contacts
LEU	113	5
ASN	91	5
LEU	116	4
VAL	105	1
LEU	90	1

TABLE 15

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
GLU	101	TRP	195	5230
TRP	97	GLY	211	4360
GLU	101	THR	210	3265
GLU	101	LYS	358	3046
GLU	101	GLN	422	2476
ASN	102	ASP	212	1980
TRP	97	ARG	216	1707
TRP	97	ASN	365	1445
TRP	97	GLU	361	944
GLU	101	LYS	368	937
TRP	97	GLU	419	909
GLU	101	ALA	157	906
ASN	102	LYS	368	842
GLU	101	TYR	318	764
TRP	97	THR	210	720
ASN	102	TRP	366	626
GLU	101	ARG	216	518
TRP	97	ASN	155	487
GLU	101	LYS	364	482
ASN	102	GLU	361	409
ASN	102	THR	210	292
GLU	101	ASN	365	284
GLU	101	GLY	211	261
TRP	97	TRP	366	253
ASN	102	TRP	195	243
ASN	102	ASN	365	239
TRP	97	GLN	422	230
TRP	97	VAL	418	205
ASN	102	LYS	358	204
TRP	97	THR	156	195
GLY	103	GLY	193	152
TRP	97	GLU	154	148
ASN	102	THR	213	129
TRP	97	ASP	212	128
ALA	98	ASP	212	112
TRP	97	THR	213	112
TRP	97	ALA	157	101
SER	115	ASP	212	76
ASN	102	LEU	194	75
GLU	101	PRO	315	61
ASN	102	VAL	314	59
THR	104	TRP	366	56
TRP	97	TYR	350	48
GLU	101	GLY	193	46
TRP	97	LYS	364	45
ASN	102	PRO	315	45
GLU	101	THR	156	38
ASN	102	GLU	317	37
GLU	101	LEU	194	36
TRP	97	PHE	209	32
GLY	103	LEU	194	31
TRP	97	TYR	318	30
LYS	94	GLU	154	30
TRP	97	GLY	193	28
TRP	97	ALA	214	27
GLU	101	GLU	317	25
ASN	102	GLU	154	22
TRP	97	LEU	354	18
GLY	103	TRP	366	14
ALA	115	TRP	366	13

TABLE 15-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	
TRP	97	ASP	417	12
LEU	113	TRP	366	11
GLU	101	VAL	418	11
ASN	102	GLY	211	10
GLU	101	ARG	207	10
GLY	103	LYS	368	9
GLU	101	VAL	314	8
LYS	94	VAL	418	8
ASN	102	LYS	166	7
GLU	101	LYS	166	7
GLY	103	ASN	365	6
TRP	97	LYS	358	5
ASN	102	LYS	364	5
ARG	110	ASN	365	5
LYS	94	TYR	350	5
GLN	114	TRP	366	4
LYS	94	ASN	155	4
GLN	100	ASN	365	4
ASN	102	GLY	313	3
GLU	101	ASP	212	3
LYS	94	ALA	416	3
ASN	102	GLY	316	2
ASN	102	ALA	157	2
GLU	101	LYS	126	2
ALA	98	GLY	211	1
TRP	97	VAL	439	1
TRP	97	VAL	314	1
TRP	97	TRP	195	1
TRP	97	GLU	347	1
TRP	97	GLU	317	1
TRP	97	ARG	122	1
THR	104	LYS	368	1
ARG	93	TRP	366	1
ASN	102	THR	362	1
GLU	101	CYS	360	1
GLU	101	ARG	321	1
LYS	94	HIS	414	1
LYS	94	ASP	417	1
GLN	100	LYS	368	1

TABLE 16

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	
TRP	97	GLU	361	3862
TRP	97	GLU	317	3269
TRP	97	GLU	154	2166
ASN	102	GLY	313	2086
ASN	102	THR	210	2009
GLU	101	TRP	195	1896
ASN	102	ASN	365	1656
THR	104	GLU	419	1510
TRP	97	ASN	365	1488
TRP	97	ARG	216	1435
GLN	100	ALA	157	1204
GLY	103	TRP	195	1191
GLU	101	ASN	365	1106
GLU	101	LYS	368	1100
ASN	102	TRP	195	822
SER	257	ASP	260	779
ASN	102	GLU	361	738
TRP	97	TRP	195	719
TRP	97	ASN	155	591
ASN	102	ARG	207	535
ASN	102	TRP	366	524
GLY	103	LYS	364	514
GLU	101	THR	210	402
TRP	97	THR	210	377

TABLE 16-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	
ASN	102	GLY	208	371
TRP	97	TYR	318	365
TRP	97	GLN	422	333
TRP	97	PRO	315	316
TRP	97	VAL	220	309
THR	104	LYS	368	307
TRP	97	THR	156	304
ASN	102	ALA	157	274
SER	115	ASP	212	274
TRP	97	GLY	316	266
GLU	101	GLN	422	264
GLY	103	TYR	197	214
ASN	102	ARG	148	203
ASN	102	LYS	368	199
ASN	102	ASP	198	199
ASN	102	LYS	255	192
ASN	102	PHE	308	180
TRP	97	GLU	419	174
LYS	94	GLY	211	174
GLN	100	ASN	155	168
ASN	102	PHE	223	153
TRP	97	LYS	358	152
TRP	97	ASP	198	151
ARG	110	GLY	193	148
GLN	100	GLU	419	144
ASN	102	LYS	364	139
ALA	98	ALA	157	135
ASN	102	ILE	159	122
ALA	98	TRP	195	113
THR	104	TYR	197	113
TRP	97	THR	362	103
GLY	103	GLY	369	103
LYS	94	THR	210	103
LYS	94	ASP	212	89
ASN	102	GLU	154	87
ASN	102	ILE	196	84
LYS	94	TYR	415	84
THR	104	GLU	154	77
TRP	97	GLY	193	75
ASN	102	ARG	321	71
TRP	97	VAL	418	69
GLU	101	TRP	366	69
THR	104	ASP	198	67
THR	104	ALA	157	64
GLY	103	GLU	154	60
LYS	94	GLY	153	60
LYS	94	GLU	154	59
TRP	97	TRP	366	58
GLU	101	ARG	207	56
ARG	110	LEU	194	52
ASN	102	GLY	369	46
LYS	94	GLU	361	46
GLY	103	LYS	368	42
ARG	93	ASP	212	40
ASN	102	GLU	317	39
GLN	100	THR	156	34
TRP	97	VAL	314	32
ALA	98	ASP	212	31
GLY	103	ARG	207	31
LYS	94	PHE	209	29
TRP	97	GLY	211	28
ASN	102	ARG	312	28
ARG	110	GLY	313	25
TRP	97	ARG	321	22
ASN	102	VAL	220	22
TRP	97	ASP	417	18
THR	104	GLY	369	18
ASN	102	THR	164	17
LYS	94	ARG	216	17
TRP	97	GLY	153	16
ASN	102	TYR	158	16
GLY	103	LEU	194	16
GLU	101	ILE	159	15

TABLE 16-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
GLN	100	PRO	315	15
ILE	95	ASP	212	14
GLY	103	TRP	366	14
TRP	97	LEU	194	13
GLU	101	ARG	321	13
GLN	100	LYS	368	13
ASN	102	GLY	153	11
ARG	110	ASN	192	11
GLN	100	TYR	158	10
ASN	102	LYS	145	9
GLU	101	ARG	216	9
ARG	110	PRO	315	8
ALA	98	THR	210	7
ASN	102	GLY	211	7
ASN	102	ARG	216	7
ARG	110	LYS	368	7
LYS	94	HIS	414	7
LYS	94	ALA	214	7
SER	257	GLU	258	6
TRP	97	LYS	255	6
THR	104	ASP	212	6
LYS	94	ASP	417	6
ASN	91	ASP	212	5
TRP	97	THR	213	5
TRP	97	LYS	364	5
THR	104	TRP	195	5
THR	104	PHE	308	5
GLY	103	LYS	255	5
GLY	103	GLU	419	5
GLY	103	ARG	321	5
LEU	116	ASP	212	4
THR	104	TYR	438	4
THR	104	THR	210	4
THR	104	ILE	159	4
ARG	93	ARG	216	4
GLU	101	ALA	157	4
GLY	103	ILE	159	4
GLY	103	ASP	198	4
LYS	94	ASN	155	4
LEU	113	ARG	216	3
ASN	102	LEU	319	3
GLU	101	VAL	314	3
GLU	101	GLY	193	3
LEU	113	TRP	366	2
ASN	102	VAL	314	2
ASN	102	TYR	197	2
ASN	102	PHE	209	2
ASN	102	LEU	194	2
GLU	101	THR	362	2
GLU	101	GLY	313	2
GLY	103	ILE	196	2
ALA	98	PRO	315	1
ALA	98	GLY	211	1
ALA	98	ASN	155	1
VAL	105	LYS	368	1
LEU	90	ASP	212	1
TRP	97	PHE	223	1
TRP	97	PHE	209	1
TRP	97	ASP	217	1
TRP	97	ASP	212	1
THR	104	VAL	200	1
THR	104	GLY	370	1
THR	104	ARG	207	1
ASN	102	PRO	152	1
ASN	102	LYS	310	1
ASN	102	LYS	227	1
ASN	102	GLU	419	1
ASN	102	ARG	122	1
ASN	102	ALA	311	1
GLU	101	VAL	418	1
GLU	101	PHE	308	1
GLU	101	LEU	194	1
GLY	103	PHE	308	1

TABLE 16-continued

Pore		Enzyme		Number of
Amino Acid	Position	Amino Acid	Position	Contacts
LYS	94	VAL	439	1
GLN	100	ARG	122	1

Example 6

[0778] This Example describes the characterisation of several CsgG mutants which show improved characterisation accuracy.

Materials and Methods

[0779] The materials and methods that were used in this example are the same as those described above for example 2. The enzyme used to control movement was either Enzyme 1=T4 Dda-E94C/C109A/C136A/A360C (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C) or Enzyme 2=T4 Dda-E94C/F98W/C109A/C136A/K194L/A360C (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C).

[0780] The ID accuracy characterisation measurements were calculated using methods as disclosed in the International Application PCT/GB2012/052343 (published as WO/2013/041878)

Results

[0781] The 1D basecall characterisation accuracy for MspA mutant x=MspA-((Del-L74/G75/D118/L119) D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K) 8 (SEQ ID NO: 50 with mutations D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K and deletion of the amino acids L74/G75/D118/L119) with DNA translocation controlled by T4 Dda-E94C/F98W/C109A/C136A/K194L/A360C was 68.7%. All of the mutants tested (see Table 17 below) showed improved 1D basecall characterisation accuracy in comparison to MspA mutant X.

[0782] 27-CsgG-Eco-(Y51A/F56Q/R97W/R192Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0783] 28-CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0784] 29-CsgG-Eco-(Y51A/F56Q/K135L/T150I/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L/T150I/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0785] 30-CsgG-Eco-(Y51A/F56Q/T150I/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/T150I/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0786] 31-CsgG-Eco-(Y51A/F56Q/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0787] 32-CsgG-Eco-(Y51A/F56Q/T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

- [0788] 33-CsgG-Eco-(Y51A/F56Q/K135V/T150Y-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135V/T150Y where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0789] 34-CsgG-Eco-(Y51A/F56Q/K135L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0790] 35-CsgG-Eco-(Y51A/F56Q/R97F/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97F/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0791] 36-CsgG-Eco-(Y51A/F56Q/K135L/T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L/T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0792] 37-CsgG-Eco-((Del-D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids D195/Y196/Q197/R198/L199)
- [0793] 38-CsgG-Eco-((Del-R192/F193/I194/D195/Y196/Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids R192/F193/I194/D195/Y196/Q197/R198/L199/L200)
- [0794] 39-CsgG-Eco-((Del-Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Q197/R198/L199/L200)
- [0795] 40-CsgG-Eco-((Del-I194/D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids I194/D195/Y196/Q197/R198/L199)
- [0796] 41-CsgG-Eco-((Del-V139/G140/D149/T150/V186/Q187/V204/G205)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids V139/G140/D149/T150/V186/Q187/V204/G205)
- [0797] 42-CsgG-Eco-((Del-D195/Y196/Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids D195/Y196/Q197/R198/L199/L200)
- [0798] 43-CsgG-Eco-((Del-Y196/Q197/R198/L199/L200/E201)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Y196/Q197/R198/L199/L200/E201)
- [0799] 44-CsgG-Eco-((Del-Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Q197/R198/L199)
- [0800] 45-CsgG-Eco-((Del-F193/I194/D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids F193/I194/D195/Y196/Q197/R198/L199)
- [0801] 46-CsgG-Eco-(Y51A/F56Q/R192T-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192T where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0802] 47-CsgG-Eco-(Y51A/F56Q/N102S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/N102S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0803] 48-CsgG-Eco-(Y51A/F56Q/Q42R-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/Q42R where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0804] 49-CsgG-Eco-(Y51A/F56Q/R192S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0805] 50-CsgG-Eco-(Y51A/F56Q/G103N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/G103N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0806] 51-CsgG-Eco-(Y51A/F56Q/R97N/N102R-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N/N102R where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0807] 52-CsgG-Eco-(Y51A/F56Q/R97L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0808] 53-CsgG-Eco-(Y51A/F56Q/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0809] 54-CsgG-Eco-(Y51A/F56Q/R97N/N102G-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N/N102G where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0810] 55-CsgG-Eco-(Y51A/F56Q/F48S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/F48S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0811] 56-CsgG-Eco-(Y51A/F56Q/G103S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/G103S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0812] 57-CsgG-Eco-(Y51A/F56Q/E101L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/E101L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0813] 58-CsgG-Eco-(Y51A/F56Q/R192Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0814] 59-CsgG-Eco-(Y51A/F56Q/K135N/R142N/R192N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135N/R142N/R192N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0815] 60-CsgG-Eco-(Y51A/F56Q/R97N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

- [0816] 61-CsgG-Eco-(Y51A/F56Q/R192N-StrepII (C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0817] 62-CsgG-Eco-(Y51A/F56Q/Y130W-StrepII (C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/Y130W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
- [0818] 63-CsgG-Eco-(Y51A/F56Q/E101G-StrepII (C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/E101G where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

TABLE 17

CsgC Mutant No.	Enzyme	1D Basecall Characterisation Accuracy (%)
25	2	72.6
26	2	79.2
27	2	81.3
28	2	80.2
29	2	72.2
30	2	74.7
31	2	74.5
32	2	73.3
33	2	73.8
34	2	74.3
35	2	75.5
36	2	73.5
37	2	73.4
38	2	75.3
39	2	72.0
40	2	74.5
41	2	74.8
42	2	71.0
43	2	70.9
44	2	71.9
45	2	73.0
46	1	75.0
47	1	72.1
48	1	75.0
49	1	74.8
50	1	75.2
51	1	73.1
52	1	75.4
53	1	77.1
54	1	75.9
55	1	73.7
56	1	73.1
57	1	73.2
58	1	76.8
59	1	72.7
60	1	76.1
61	1	75.0
62	1	74.7
63	1	73.1
25	1	74.8
26	1	77.4

Example 7

[0819] This example compares DNA capture of a number of different mutant nanopores.

Materials and Methods

[0820] Electrical measurements were acquired from a variety of single CsgG or MspA nanopores inserted in block co-polymer in buffer (25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0). After achieving a single pore inserted in the block co-polymer, then buffer (2 mL, 25 mM K Phos-

phate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0) was flowed through the system to remove any excess nanopores. 150 µL of 500 mM KCl, 25 mM K Phosphate, 1.5 mM MgCl2, 1.5 mM ATP, pH8.0 was then flowed through the system. After 10 minutes a 150 µL of DNA (SEQ ID NO: 51, 200 nM) was then flowed into the single nanopore experimental system. The experiment was run at -120 mV and helicase-controlled DNA movement monitored.

Results

- [0821] The CsgG mutant CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/E101S/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (see FIG. 36) shows a higher rate of capture (e.g. captures DNA polynucleotides more easily) than MspA-((Del-L74/G75/D118/L119) D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K) 8 (see FIG. 34). Each spike in the current traces corresponds to translocation of the DNA polynucleotide (SEQ ID NO: 51) through the nanopore without being controlled by an enzyme. The 10 second current traces in FIG. 34 show fewer DNA translocations than the 10 second current traces for the CsgG nanopore-CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9.
- [0822] The mutation of position E101 to E101S resulted in an increase in the capture rate when compared to a CsgG mutant without the E101S mutation. FIG. 35 shows that the 10 second current traces for CsgG mutant CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII (C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited fewer translocations than CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9. The average number of translocations for CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 was 7.25 per second (n=12) and the average number of translocations for CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 was 18 per second (n=14).

Example 8

[0823] This example compares the level of expression of two different CsgG mutant pores.

Materials and Methods

[0824] The materials and methods that were used to make the nanopores (A=CsgG-Eco-(Y51A/F56Q/R97W)-StrepII (C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B=CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)) in this example are the same as those described above for example 3.

Results

[0825] The two nanopores A=CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B=CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) were expressed

and purified using exactly the same protocol and the same volumes of each nanopore were analysed using gel filtration chromatograms (120 mL S200 Column, see FIG. 37) and SDS-PAGE analysis (see FIG. 38). The absorbance value for B (470.3 mAu) was much higher than A (11.4 mAu) which indicated that CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StreptII(C))9 expressed at a much higher level than CsgG-Eco-(Y51A/F56Q/R97W)-StreptII(C))9. The intensity of the bands in FIG. 38 also indicate the expression level of the two pores. Bands A-C (containing CsgG-Eco-(Y51A/F56Q/R97W)-StreptII(C))9) were less intense than bands D-E (containing CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StreptII(C))9). The two methods of analysis both indicated that the addition of R192D mutation greatly increased the observed expression of the CsgG mutant.

Example 9

[0826] This Example describes the characterisation of several CsgG mutants which show improved characterisation accuracy.

Materials and Methods

CsG Pores

[0827] The following 8 CsgG mutant pores were tested. Mutant 28 described in Example X above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StreptII has the sequence shown in SEQ ID NO: 47.

[0828] Baseline pore (mutant 28): CsgG-(WT-Y51A/F56Q/R97W/R192D-StreptII)9

[0829] Mutant A: CsgG-(WT-Y51A/F56Q/R97W/R192D-StreptII)9-del (D195-L199)

[0830] Mutant B: CsgG-(WT-Y51A/F56Q/R97W/R192D-StreptII)9-del (F193-L199)

[0831] Mutant C: CsgG-(WT-Y51A/F56Q/R97W/R192D-StreptII)9-F191T

[0832] Mutant D: CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StreptII)9

[0833] Mutant E: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)

[0834] Mutant F: CsgG-(WT-Y51A/F56Q/R192D-StreptII)9-R93W

[0835] Mutant G: CsgG-(WT-Y51A/F56Q/R192D-StreptII)9-R93W-del (D195-L199)

[0836] Mutant H: CsgG-(WT-Y51A/F56Q/R192D-StreptII)9-R93Y/R97Y

Nanopore Preparation

[0837] To prepare a nanopore array chip that contain multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted the following method was used. CsgG mutants expressed in *E. coli* were purified and stored in buffer containing 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, 0.1% Brij 58 at pH8. These mutant CsgG pores were diluted to 1 in 1 million using buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 and added to chips to obtain single pores in each of the wells. After pore insertion, the array chip was washed with 1 mL buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 to remove excess

pores. After a few minutes, each chip was flushed twice with 500 mL of sequencing mix containing 470 mM KCl, 25 mM HEPES, 11 mM ATP and 10 mM MgCl₂.

DNA Sample Preparation

[0838] DNA sample was prepared for sequencing using the following method. 1 µg of DNA analyte was incubated with the 40 nM of adapter mix containing a T4 Dda helicase enzyme prebound to the adapter and blunt TA ligase for 10 minutes (available from <https://store.nanoporetech.com/>). The structure of the adapter is shown in FIG. 43 and the sequences contained in the adapter are set out in SEQ ID NOs: 52 to 55. The ligation mixture was then purified to remove unligated free adapter using Spri purification. The final ligated mixture was eluted in 25 µL elution buffer containing 40 mM CAPS at pH10, 40 mM KCl and 400 nM cholesterol tether. For each chip, 12 µL of DNA-adapter ligated mix was mixed with the sequencing mix (final volume of 150 µL) and added to chip for sequencing. The experiment was then run for 6 hours at 160 mV.

[0839] The 1D accuracy characterisation measurements were calculated using methods as disclosed in WO2013/041878.

Measuring Template Speed

[0840] The template speed was measured by the following method. The basecall of each squiggle was aligned to the reference sequence. The number of bases that spanned the alignment (alignment end position subtracted from the alignment start position) was divided by the time between the event that corresponded to the end of the alignment and by the event that corresponded to the start of the alignment.

Results

Basecall Accuracy

[0841] As shown in FIG. 39, all 8 CsgG mutant pores were found to have an improved basecall accuracy compared to the baseline pore, mutant 28 (CsgG-(WT-Y51A/F56Q/R97W/R192D-StreptII)9) described in Example 6. As shown in Table 17, mutant 28 showed the highest basecall accuracy (80.2%) of all the CsgG mutants tested in Example 6, in which mutant 25 (CsgG-(WT-Y51A/F56Q-StreptII)9) described in Example 5 was the baseline mutant. Therefore the deletion of D195-L199, of F193-L199 or V105-I107, or the substitution of F191T results in a further improvement in accuracy in addition to the improvement in accuracy resulting from the R97W and R192D substitutions in mutant 28.

[0842] The deletion of D195-L199, of F193-L199 or V105-I107, or the substitution of F191T, would each also be expected to improve accuracy in the presence of other mutations in the CsgG sequence or in the absence of the R97W, R192D, Y51A and/or F56Q mutations. For example, mutant E, which contains the mutation K94Q in addition to the del (V105-I107) mutation has an improved accuracy compared to mutant 28, as does the del (V105-I107)-containing mutant G which does not contain the R97W substitution, but instead contains a R93W substitution.

[0843] Table 17 in Example 26 shows that mutant 26, which contains the R97W substitution has almost as good a basecall accuracy (79.2%) as mutant 28 (80.2%). This suggests that the R97W mutation underlies the increased basecall accuracy. In this Example, two mutants which do

not contain the R97W mutation (and also do not contain the R192D mutation), but instead contain a R93W substitution (mutant F) or both a R93Y substitution and a R97Y substitution (mutant H) were tested and found to have a higher basecall accuracy than mutant 28. This shows that R93W and R93Y/R97Y substitutions can be used to improve the basecall accuracy of CsgG nanopores.

Template Speed and Template Accuracy

[0844] As shown in FIG. 40A, mutant D (CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StrepII)9) has a tightened distribution of speed population as compared to the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9.

[0845] FIG. 40B shows that mutant D (CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StrepII)9) has tightened distribution of template accuracy as compared to the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9.

[0846] It is an advantage to have a tightened speed and accuracy distribution since there is less variation in the data. Also, the median accuracy of the data will increase by reducing the amount of lower accuracy data generated. Thus, deleting the amino acids from V105 to I107 of a CsgG nanopore can be used to produce a CsgG nanopore with improved properties for characterising polynucleotides.

Example 10

[0847] This Example describes the characterisation of CsgG mutants which show a reduction in noisy pore signal.

Materials and Methods

CsG Pores

[0848] The following CsgG mutant pores were tested. Mutant 28 described in Example X above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StrepII has the sequence shown in SEQ ID NO: 47.

[0849] Baseline pore (mutant 28): CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9

[0850] Mutant I: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-K94N.

[0851] Mutant J: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-K94Q.

Nanopore Preparation

[0852] Chips containing multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted were prepared as described in Example 9.

DNA Sample Preparation

[0853] DNA sample was prepared for sequencing using the method described in Example 9.

Determining Time Spent in Noisy Pore State

[0854] The percentage of time spent in a noisy pore state was calculated as follows. The event detected signal within each channel was split up into non-overlapping short windows. For each window the mean average of the current levels and the dispersion of the current levels was calculated. The values obtained were then passed to a classifier which

returned a label denoting whether the window contained a noisy signal. The classifier was trained to detect noisy signals by providing it with pre-labelled data.

Results

Noisy Pore State

[0855] FIG. 41 displays an example “squiggle” that shows the “noisy” pore error mode exhibited by baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9. The top panel of FIG. 41 shows the difference in flow of current through the pore during the “good” and “noisy” pore states. The bottom panel of FIG. 41 shows an expanded view of the transition from “good” state to “noisy” state.

[0856] FIG. 42 shows the reduction in noisy pore state of mutant pores I and J when compared to baseline mutant 28, averaged over at least 5 runs.

[0857] The percentage of time spent in noise pore state by both Mutant J and Mutant I is significantly reduced compared to the baseline. Mutants I and J differ from the mutant 28 used as a baseline at just one residue. Both mutant I and mutant J contain a substitution of K94. Mutant I contains a K94N mutation and mutant II contains a K94Q mutation. Therefore substitution of K94 in a CsgG nanopore, particularly substitution with N or Q, can be used to produce a CsgG nanopore with improved properties for characterizing polynucleotides.

Example 11

[0858] This Example describes the characterisation of several CsgG mutants which show increased capture activity.

Materials and Methods

CsG Pores

[0859] The following CsgG mutant pores were tested. Mutant E described in Example 9 above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StrepII has the sequence shown in SEQ ID NO: 47.

[0860] Baseline (mutant E): CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)).

[0861] Mutant K: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-Q42K

[0862] Mutant L: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E44N

[0863] Mutant M: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E44Q

[0864] Mutant N: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-L90R

[0865] Mutant O: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-N91R

[0866] Mutant P: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-195R

[0867] Mutant Q: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-A99R

[0868] Mutant R: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E101H

[0869] Mutant S: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E101K

[0870] Mutant T: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E101N

- [0871] Mutant U: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E101Q
 [0872] Mutant V: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-E101T
 [0873] Mutant W: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)-Q114K

Nanopore Preparation

[0874] To prepare nanopore array chips that contain multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted the following method was used. CsgG mutants expressed in *E. coli* were purified and stored in buffer containing 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, 0.1% Brij 58 at pH8. These mutant CsgG pores were diluted to 1 in 1 million using buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 and added to the chips to obtain single pores in each of the wells. After pore insertion, the chips were washed with 1 mL buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 to remove excess pores. 1 mL of solution containing 240 nM TBA analyte in 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 was flushed into the chip.

Determining Capture Ability

[0875] The ability of a mutant pore to capture DNA analyte is then assessed by its ability to capture Thrombin Binding Aptamer (TBA) (SEQ ID NO: 51). The experiment was run at 180 mV. In order to measure TBA capture by a pore, the median time between TBA events was calculated.

Results

[0876] As shown in FIG. 44, the median time between TBA events for 13 mutants was significantly reduced compared to the baseline, indicating that all 13 mutants display increased capture rates of template DNA.

[0877] Each of the 13 mutants had a single amino acid substitution compared to the baseline pore. The particular substitutions were: Q42K, E44N, E44Q, L90R, N91R, I95R, A99R, E101H, E101K, E101N, E101Q, E101T and Q114K. All of these mutations involve the substitution of a negatively charged amino acid with an uncharged amino acid or a positively charged amino acid, or of an uncharged amino acid with a positively charged amino acid. Therefore, it can be concluded that substitution of the amino acid at one or more of positions Q42, E44, E44, L90, N91, I95, A99, E101 and Q114 with amino acids that remove the negative charge and/or increase the positive charge at these positions results in increased capture of a polynucleotide.

Sequence Alignment of the Various CsgG Homologues

[0878] FIG. 45 shows the sequence alignment between the twenty-one CsgG homologues as detailed above. A multiple sequence alignment was performed on SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30, SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40 and SEQ ID NO: 41.

[0879] Praline software, a multiple sequence alignment toolbox that integrates homology-extended and secondary structure information was used to perform the alignment <http://www.ibi.vu.nl/programs/pralinewww/>, see also Simonsis V A1, Heringa J.; Nucleic Acids Res. 2005 Jul. 1; 33 (Web Server issue): W289-94. The alignment was scored using the BLOSUM62 residue exchange matrix. For details of this method, see for example Henikoff S, Henikoff J G; Proc. Natl. Acad. Sci. USA Vol. 89, pp. 10915-10919, November 1992. Gap opening and extension penalties of 12 and 1 were used respectively. Secondary structure prediction using PSIPRED was used to guide the alignment. For details of this method, see for example Jones D. T.; J Mol Biol. 1999 Sep. 17; 292 (2): 195-202. The above methods used to align the sequences are exemplary and other methods of sequence alignment known in the art may be used.

[0880] With reference to the sequence alignments of FIG. 45, each section of sequence alignment the conservation at each position is indicated by a histogram and a score. Numbers 0-9 on the scale indicate increasing conservation, columns with mutations which result in similar properties of that amino being conserved are marked with a plus ('+') and the star symbol (*) indicates 100% sequence identity at that position. It can be seen from the conservation values of the sequence alignment that many of the residues show very high or even perfect sequence identity indicating that these 21 homologues are closely related.

[0881] FIG. 46 shows the same relative sequence alignments as FIG. 45 with predicted alpha helical secondary structure regions additionally shaded in grey. FIG. 47 shows the same relative sequence alignments as FIG. 45 with predicted beta sheet secondary structure regions additionally shaded in grey. FIGS. 46 and 47 show that the regions of predicted alpha helices and beta sheets of these homologues, important secondary structures for CsgG nanopores are highly conserved.

[0882] The multiple sequence alignment strongly suggests the sequences are related; not only is there a high degree of conservation along the alignment, but the predicted secondary structural elements are also aligned.

[0883] The sequence alignments in FIGS. 45, 46 and 47 may be used as a reference to show the relative positions that align with each other. Thus amino-acid residues identified with respect to SEQ ID NO 2 and the corresponding amino-acid residues in other CsgG homologues can be identified. For ease of identification, residues R97 and R192 have been located with an asterisk. It can be seen from the table that for example R192 of SEQ ID NO: 2 corresponds to residue R191 of SEQ ID NO: 32 and residue K177 of SEQ ID NO: 37.

[0884] As will be readily appreciated with reference to FIGS. 45 to 47, the CsgG monomers are highly conserved. Furthermore, from knowledge of the mutations in relation to SEQ ID NO: 2 it is possible to determine the equivalent positions for mutations of CsgG monomers other than that of SEQ ID NO: 2.

[0885] Thus reference to a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 2 and specific amino-acid mutations thereof as set out in the claims and elsewhere in the specification also encompasses a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30, SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40 and SEQ ID NO: 41.

NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40 and SEQ ID NO: 41 and corresponding amino-acid mutations thereof. Likewise reference to a construct, pore or method involving the use of a pore relating to a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 2 and specific amino-acid mutations thereof as set out in the claims

and elsewhere in the specification also encompasses a construct, pore or method relating to a mutant CsgG monomer comprising a variant of the sequence according to the above disclosed SEQ ID NOS and corresponding amino-acid mutations thereof. It will further be appreciated that the invention extends to other variant CsgG monomers not expressly identified in the specification that show highly conserved regions.

SEQUENCE LISTING

Sequence total quantity: 56

SEQ ID NO: 1 moltype = DNA length = 786
 FEATURE Location/Qualifiers
 source 1..786
 mol_type = genomic DNA
 organism = Escherichia coli

SEQUENCE: 1
 tgtctgaccg caccgcccga agaagccgca cgtccgaccc tgaatgcccg tgcacagtct 60
 tataaagatc tgaccatctt gccggctccg acgggcaaaa tttttgttag cgtctataac 120
 atccaggacg aaaccgggtca atttaaacgg taccggcgca gtaattttct caccggccgtt 180
 ccgcagagtg caaccgctat gctgggtcac gcaactgaaag attcccggtg gttcattccg 240
 ctggaacgcc agggcctgca aaacctgctg aatgaacgta aaattatccg cgcagctcag 300
 gaaaaacggt cgtgggccat taacaatcgt attccgctgc aaagcctgac cgccgcaaac 360
 atcatgggtg aaggctctat catcgggttac gaatcaaacg tcaaatccgg cggtgtgggc 420
 gcacgttatt ttggcatttg tgctgatacc cagtaccaac tggaccagat cgcagttaac 480
 ctgcgctgtg ttaatgtcag caccggcgaa attctgagct ctgtgaatac cagcaaaacg 540
 atcctgtctt acgaagtga ggctgggtgt tttcgtttca ttgattatca acgcctgctg 600
 gaaggcgaag tcggttacac ctcaaacgaa ccggtgatgc tgtgtctgat gtcggcgatt 660
 gaaacgggtg ttattttcct gatcaatgat ggcatcgacc gtggctctgt ggatctgcag 720
 aacaaagccg aacgtcaaaa tgacattctg gtgaaatacc gccacatgag tgttccgcg 780
 gaatcc 786

SEQ ID NO: 2 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 2
 CLTAPPKEAA RPTLMPRAQS YKDLTHLPAP TGKIFVSVYN IQDETGQFKP YPASNFSTAV 60
 PQSATAMLVT ALKDSRWFIPL LERQGLQNLL NERKIIRAAQ ENGTVAINNR IPLQSLTAAN 120
 IMVEGSIIGY ESNVKSggvg ARYFGIGADT QYQLDQIAVN LRVVNSTGE ILSSVNTSKT 180
 ILSYEVQAGV FRFIDYQRL L EGEVGYTSNE PVMLCLMSAI ETGVIPLIND GIDRGLWDLQ 240
 NKAERQNDIL VKYRHMSVPP ES 262

SEQ ID NO: 3 moltype = AA length = 248
 FEATURE Location/Qualifiers
 source 1..248
 mol_type = protein
 organism = Citrobacter koseri

SEQUENCE: 3
 MPRAQSYKDL THLPMTGKI FVSVYNIQDE TGQFKPYPAS NFSTAVPQSA TAMLVTALKD 60
 SRWFIPLERQ GLQNLLNERK IIRAAQENGTV VAINNRIPLO SLTAANIMVE GSIIGYESNV 120
 KSGGVGARYF GIGADTQYQL DQIAVNLRVV NVSTGEILSS VNTSKTILSY EVQAGVFRFI 180
 DYQRLLEGEI GYTSNEPVML CLMSAIETGV IFLINDGIDR GLWDLQNKAE RQNDILVKYR 240
 HMSVPPES 248

SEQ ID NO: 4 moltype = AA length = 223
 FEATURE Location/Qualifiers
 source 1..223
 mol_type = protein
 organism = Salmonella enterica

SEQUENCE: 4
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 PQSATAMLVT ALKDSRWFIPL LERQGLQNLL NERKIIRAAQ ENGTVAMNRR IPLQSLTAAN 120
 IMVEGSIIGY ESNVKSggvg ARYFGIGADT QYQLDQIAVN LRVVNSTGE ILSSVNTSKT 180
 ILSYEVQAGV FRFIDYQRL L EGEVGYTSNE PVMLCLMSAI ETG 223

SEQ ID NO: 5 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Citrobacter amalonaticus

SEQUENCE: 5
 CLTAPPKEAA KPTLMPRAQS YKDLTHLPAP TGKIFVSVYN IQDETGQFKP YPASNFSTAV 60

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PQSATAMLVT	ALKDSRWFPV	LERQGLQNLL	NERKIIRAAQ	ENGTVAINNRR	IPLQSLTAAN	120
IMVEGSIIGY	ESNVKSGGVG	ARYFGIGADT	QYQLDQIAVN	LRVVNVSTGE	ILSSVNTSKT	180
ILSYEVQAGV	FRFIDYQRL	EGEIGYTSNE	PVMLCLMSAI	ETGVIIFLIND	GIDRGLWDLQ	240
NKADRQNDIL	VKYRHMVSVP	ES				262

SEQ ID NO: 6 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Citrobacter rodentium

SEQUENCE: 6

CLTTPPKKEAA	KPTLMPRAQS	YKDLTHLPVP	TGKIFVSVYN	IQDETGGQFKP	YPASNFTSTAV	60
PQSATAMLVT	ALKDSRWFPV	LERQGLQNLL	NERKIIRAAQ	ENGTVAINNRR	IPLPSLTAAN	120
IMVEGSIIGY	ESNVKSGGVG	ARYFGIGADT	QYQLDQIAVN	LRVVNVSTGE	ILSSVNTSKT	180
ILSYEVQAGV	FRFIDYQRL	EGEIGYTSNE	PVMLCLMSAI	ETGVIIFLIND	GIDRGLWDLQ	240
NKADRQNDIL	VKYRQMSVPP	ES				262

SEQ ID NO: 7 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Enterobacter asburiae

SEQUENCE: 7

CLTAPPKEAA	KPTLMPRAQS	YRDLTHLPAP	TGKIFVSVYN	IQDETGGQFKP	YPASNFTSTAV	60
PQSATAMLVT	ALKDSRWFPV	LERQGLQNLL	NERKIIRAAQ	ENGTVAINNRR	MPLQSLAAAN	120
VMIEGSIIGY	ESNVKSGGVG	ARYFGIGADT	QYQLDQIAVN	LRVVNVSTGE	VLSSVNTSKT	180
ILSYEVQAGV	FRFIDYQRL	EGEIGYTSNE	PVMMLMSAI	ETGVIIFLIND	GIDRGLWDLQ	240
NKADARQNPVL	VKYRDMVSVP	ES				262

SEQ ID NO: 8 moltype = DNA length = 1830
 FEATURE Location/Qualifiers
 source 1..1830
 mol_type = genomic DNA
 note = Bacteriophage phi-29
 organism = unidentified

SEQUENCE: 8

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gattgtcgcg	tgtgggccta	tggctacatg	aacatcgag	atcattctga	atacaaaatc	120
ggtaacagtc	tggatgaatt	tatggcatgg	gtgctgaaag	ttcaggcgga	tctgtacttc	180
cacaacctga	aatttgatgg	cgcattcatt	atcaactggc	tggaaacgta	tggctttaa	240
tggagcgcg	atggtctgcc	gaacacgtat	aataaccatta	tctctcgtat	gggcccagtg	300
tatatgattg	atatctgcct	gggctacaaa	ggtaaacgca	aaattcatac	cgtgatctat	360
gatagcctga	aaaaactgcc	gtttccggtg	aagaaaaattg	cgaagattt	caaaactgac	420
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gaagaatacg	catacatcaa	aaacgatata	cagatcatcg	cagaagcgct	gctgattcag	540
tttaaacagg	gcctggatcg	catgaccgcg	ggcagtgata	gcctgaaagg	tttcaaatg	600
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gaagtctggt	atgcctaccg	cggcggtttt	acctgggctg	acgatcggtt	caaagaaaaa	720
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ctggccaac	tgatgctgaa	cagcctgtat	ggcaaatctg	cctctaattc	ggatgtgacc	1200
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atccaggata	tctacatgaa	agaagtggat	ggcaaaactg	ttgaagggtc	tccggatgat	1560
tacaccgata	tcaaatcag	tgtgaaatgc	gccggcatga	cggataaaat	caaaaaagaa	1620
gtgaccttcg	aaaacttcaa	agttggtttc	agccgcaaaa	tgaacccgaa	accggtgcag	1680
gttccggcgc	gtctgggttc	ggtagtatga	acgtttacca	ttaaatctgg	cggtagtgcg	1740
tggagccatc	cgcagttcga	aaaaggcgg	ggctctgggt	gcggttctgg	cggtagtgcg	1800
tggagccacc	cgcagtttga	aaaataataa				1830

SEQ ID NO: 9 moltype = AA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = protein
 note = Bacteriophage phi-29
 organism = unidentified

SEQUENCE: 9

MKHMPRKMY	CAFETTTKVE	DCRVWAYGYM	NIEDHSEYKI	GNSLDEFMAW	VLVQADLYF	60
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HNLKPDGAFI	INWLERNGFK	WSADGLPNTY	NTIISRMGQW	YMIDICLGYK	GKRKIHTVIY	120
DSLKKLPFPV	KKIAKDFKLT	VLKGDIDYHK	ERPVGKITP	EEYAYIKNDI	QIIAEALLIQ	180
FKQGLDRMTA	GSDSLKGFKD	IITTKKFKKV	FPTLSLGLDK	EVRYAYRGGF	TWLNDRFKKEK	240
EIGEGMVDFV	NSLYPAQMYS	RLLPYGEPV	FEGKYVWDED	YPLHIQHIRC	EFEKKEGYIP	300
TIQIKRSRFY	KGNEYLKSSG	GEIADLWLSN	VDLELMKEHY	DLYNVEYISG	LKFKATTGLF	360
KDFIDKWTYI	KTTSEGAIKQ	LAKLMLNSLY	GKFASNPDTV	GKVPYLKENG	ALGFRLGEEE	420
TKDPVYTPMG	VFITAWARYT	TITAAQACYD	RIIYCDTDSI	HLTGTETPDV	IKDIVDPKKL	480
GYWAHESTFK	RAKYLRQKTY	IQDIYMKEVD	GKLVEGSPDD	YTDIKFSVKC	AGMTDKIKKE	540
VTFENFKVGF	SRKMKPKPVQ	VPGGVVLVDD	TFTIKSGGSA	WSHPQFEKGG	GSGGSGGSA	600
WSHPQFEK						608

SEQ ID NO: 10 moltype = DNA length = 1390
 FEATURE Location/Qualifiers
 source 1..1390
 mol_type = genomic DNA
 organism = Escherichia coli

SEQUENCE: 10

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acccatccgg	ccctggatcg	tccggcgag	tttgcggcca	ttcgcaccga	tagcgaattc	120
aatgtgattg	gcgaaccgga	agtgttttat	tgcaaacggg	ccgatgatta	tctgccgcag	180
ccgggtgcgg	tgctgattac	cggtattacc	ccgcagggaag	cgccgcgcaa	aggtgaaaaa	240
gaagcggcgt	ttgccgcgcg	cattcatagc	ctgtttaccg	tgccgaaaaa	ctgcattctg	300
ggctataaca	atgtgcgctt	cgatgatgaa	gttaccgcga	atatctttta	tctgaacttt	360
tatgatccgt	atgcgtggag	ctggcagcat	gataacagcc	gttgggatct	gctgggatgtg	420
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cagccgcgcc	tggttgatta	tctgtttacc	caccgtaaca	aacacaaact	gatggcgctg	660
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atggttgatc	tgcccggtga	tattagcccg	ctgctggaac	tggatagcga	taccctgcgt	840
gaacgcctgt	ataccgccaa	aaccgatctg	ggcgataatg	ccgcgcgtgc	ggtgaaactg	900
gttcacatta	acaaatgccc	ggtgctggcc	caggcgaaaca	ccctgcgccc	ggaagatgcg	960
gatcgtctgg	gtattaatcg	ccagcattgt	ctggataatc	tgaaaatcct	gcgtgaaaac	1020
ccgcaggtgc	gtgaaaaagt	ggtggcgatc	ttcgcgggaag	cggaaccggt	caccccgagc	1080
gataacgtgg	atgcgcagct	gtataacggc	ttcttttagcg	atgccgatcg	cgccgcatg	1140
aaaatcgttc	tggaaaccca	accgcgcaat	ctgccggcgc	tggatattac	ctttgttgat	1200
aaacgtattg	aaaaactgct	gtttaattat	cgtgcgcgca	attttccggg	taccctggat	1260
tatgccgaac	agcagcgttg	gctggaacat	cgctcgtcagg	ttttaccccc	ggaatttctg	1320
caagggttatg	cggatgaact	gcagatgctg	gttcagcagt	atgccgatga	taaagaaaaa	1380
gtggcgctgc						1390

SEQ ID NO: 11 moltype = AA length = 485
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 source 1..485
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 11

MMNDGKQST	FLPHDYETFG	THPALDRPAQ	FAAIRTDSEF	NVIGEPVEFY	CKPADDYLPQ	60
PGAVLITGIT	PQEARAKGEN	EAAFAARIHS	LFTVPKTCIL	GYNNVRFDE	VTRNIFYRNF	120
YDPYAWSQH	DNSRWDLLDV	MRACYALRPE	GINWPENDDG	LPSFRLEHLT	KANGIEHSNA	180
HDAMADVYAT	IAMAKLVKTR	QPRLFDYLF	HRNKHKLML	IDVPQMKPLV	HVSGMFGAWR	240
GNTSWVAPLA	WHPENRNAV	MVDLAGDISP	LLELSDTLR	ERLYTAKTDL	GDNAAVPVKL	300
VHINKCPVLA	QANTLRPEDA	DRLGINRQHC	LDNLKILREN	PQVREKVVAI	FAEAEPTPS	360
DNVDAQLYNG	FFSDADRAAM	KIVLETEPRN	LPALDITFVD	KRIEKLLEFNY	RARNFPPTLD	420
YAEQQRWLEH	RRQVFTPEFL	QGYADELQML	VQQYADDKEK	VALLKALWQY	ABEIVSGSGH	480
HHHHH						485

SEQ ID NO: 12 moltype = DNA length = 804
 FEATURE Location/Qualifiers
 source 1..804
 mol_type = genomic DNA
 organism = Escherichia coli

SEQUENCE: 12

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atgtttccgc	tccaagaggt	ggcgaagctc	ggctacaacg	tgttttatca	cgggcagaaa	180
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cccggtgacg	acgaagaggc	gcagcggcgg	attattatgg	cggaatcccc	ctcactgctg	300
ggtaaatgca	ccgtgatcaa	cggttacttc	ccgcagggtg	aaagccgcga	ccatccgata	360
aaattcccg	caaaagcgca	gatttatcag	aactcgcaaa	actacctgga	aaccgaaactc	420
aaacgtgata	atccggctact	gattatgggc	gatatgaata	tcagccctac	agatctggat	480
atcgccattg	gcgaagaaaa	ccgtaagcgc	tggtgcgcta	ccggtaaatg	ctctttcctg	540
ccggaagagc	gcgaatggat	ggacaggtcg	atgagctggg	gggttgctga	taccttcgcg	600
catgcgaatc	gcgaacacgc	agatcgtttc	tcatgggttg	attaccgctc	aaaaggtttt	660
gacgataacc	gtggtctgcg	catcgacctg	ctgctcgcca	gccaaaccgt	ggcagaatgt	720

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tgcgtagaaa cgcgcacga ctagaaatc cgcagcatgg aaaaaccgtc cgatcacgcc 780
ccgctctggg cgacottccg ccgc 804

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SEQ ID NO: 13      moltype = AA length = 268
FEATURE           Location/Qualifiers
source            1..268
                  mol_type = protein
                  organism = Escherichia coli

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SEQUENCE: 13
MKFVSFNING LRARPHQLEA IVEKHQPDVI GLQETKVHDD MFPLEEVAKL GYNVIFYHGQK 60
GHYGVALLTK ETPIAVRRGF PGDDEEAQRR IIMAEIPSLG GNVTVINGYF PQGESRDHPI 120
KFPKAAQFYQ NLQNYLETEL KRDNPVLIMG DMNISPTDLD IGIGEENRKR WLRTGKCSPL 180
PEEREWMDDL MSWGLVDLFR HANPQTADRF SWFDYRSKGF DDNRGLRIDL LLASQPLAEC 240
CVETGIDYEI RSMKPSDHA PVWATFR 268

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SEQ ID NO: 14      moltype = DNA length = 1275
FEATURE           Location/Qualifiers
source            1..1275
                  mol_type = genomic DNA
                  organism = Thermus thermophilus

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SEQUENCE: 14
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ggcgactatg atgcggatgg cctgaccggc accgcgatcc tggttcgtgg tctggccgcc 180
ctgggtgcgg atgttcattc gtttatccc caccgcctgg aagaaggcta tgggtgctcg 240
atggaacgcg tcccggaaca tctggaagcc tccgacctgt tctgacctg tgactgcggc 300
attaccaacc atgcggaact gcgcgaactg ctggaaaatg gcgtggaagt cattgttacc 360
gatcatcata cgcggggcaa aacgcgcgcg ccgggtctgg tctgcatcc ggcgctgacg 420
ccgcatctga aagaaaaaac gaccggcgca gccgtggcgt tctgctgct gtgggcactg 480
catgaacgcc tgggcctgcc gccgcgcgtg gaatacgcgg acctggcagc cgttggcacc 540
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cgcatcccg cttcatcttg ggtgggcctg cgtctgctgg ctgaagccgt gggctatacc 660
ggcaaacgcg tcgaagtgcg ttcccgcatc gcgcgcgcga tcaatgcggc ttccgcctg 720
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gaaccgctgt tctctg 1275

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SEQ ID NO: 15      moltype = AA length = 425
FEATURE           Location/Qualifiers
source            1..425
                  mol_type = protein
                  organism = Thermus thermophilus

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SEQUENCE: 15
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DHHTPGKTPP PGLVVHPALT PDLKEKPTGA GVAFLLWLAL HERLGLPPPL EYADLAAVGT 180
IADVAPLWGW NRALVKEGLA RIPASSWVGL RLLAEAVGYT GKAVEVAFRI APRINAASRL 240
GEAEKALRLR LTDDAAEAQA LVGELHRLNA RRQTLLEAML RKLLPQADPE AKAIIVLLDPE 300
GHPGVMGIVA SRILEATLRP VFLVAQKGT VRSLAPISAV EALRSAEDLL LRYGGHKEAA 360
GFAMDEALFP AFKARVEAYA ARFPDPVREV ALLDLLPEPG LLPQVFRELA LLEPYGEGNP 420
EPLFL 425

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SEQ ID NO: 16      moltype = DNA length = 738
FEATURE           Location/Qualifiers
source            1..738
                  mol_type = genomic DNA
                  note = Bacteriophage lambda
                  organism = unidentified

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SEQUENCE: 16
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gtcatcaccg cttcagaagt tcacaacgtg atagcaaaac cccgctccgg aaagaagtgg 180
cctgacatga aaatgtccta cttccacacc ctgcttctg aggtttgcac cgggtgtggc 240
ccggaagtta acgctaaaag actggcctgg ggaaacagt acgagaacga cgcagaaacc 300
ctgtttgaat tcacttcagg cgtgaatggt actgaatccc cgatcatcta tcgcgacgaa 360
agtatgcgta ccgcctgctc tcccgatggt ttatgcagt acggcaacgg ccttgaactg 420
aaatgcccg ttacctcccg ggatttcacg aagttccggc tcggtgggtt cgaggccata 480
aagtcagctt acatggccca ggtgcagtac agcatgtggg tgacgcgaaa aaatgcctgg 540
tactttgcca actatgacct gcgtatgaag cgtgaaggcc tgcattatgt cgtgattgag 600

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cgggatgaaa	agtagcatggc	gagttttgac	gagatcgtgc	cggagttcat	cgaaaaaatg	660
gacgaggcac	tggctgaaat	tggttttgta	tttggggagc	aatggcgatc	tggctctggt	720
tccgcgacgc	gttccgga					738

SEQ ID NO: 17 moltype = AA length = 226
 FEATURE Location/Qualifiers
 source 1..226
 mol_type = protein
 note = Bacteriophage lambda
 organism = unidentified

SEQUENCE: 17
 MTPDIILQRT GIDVRAVEQG DDAWHKLRLLG VITASEVHNV IAKPRSGKKW PDMKMSYFHT 60
 LLAEVCTGVA PEVNAKALAW GKQYENDART LFEFTSGVNV TESPIIYRDE SMRTACSPDG 120
 LCSDNGNLEL KCPFTSRDFM KFRLLGGFEAI KSAYMAQVQY SMWVTRKNW YFANYDPRMK 180
 REGLHYVVIE RDEKYMASF D EIVPEPIEKM DEALAEIGFV FGEQWR 226

SEQ ID NO: 18 moltype = AA length = 760
 FEATURE Location/Qualifiers
 source 1..760
 mol_type = protein
 organism = Methanococcoides burtonii

SEQUENCE: 18
 MMIRELDIPR DIIGFYEDSG IKELYPPQAE AIEMGLEKK NLLAAIPTAS GKTLLAELAM 60
 IKAIREGGKA LYIVPLRALA SEKFERFKEL APFGIKVGIS TGDLDLRADW LGVNDIIVAT 120
 SEKTDLSLLRN GTSMWDEITT VVVEIHLLD SKNRGPTLEV TITKLMRLNP DVQVVALSAT 180
 VGNAREMADW LGAAVLVLEW RPTDLHEGVL PGDAINFPGS QKKIDRLEKD DAVNLVLDTI 240
 KAEGQCLVFE SSRNCAGFA KTASSKVAKI LDNDIMIKLA GIAEEVESTG ETDTAIVLAN 300
 CIRKGVAFHH AGLNSNHRKL VENGFRQNLV KVISSTPTLA AGLNLPARRV IIRSYRRFDS 360
 NFGMQPIPV L EYKQ MAGRAG RPHLDPYGES VLLAKTYDEF AQLMENYVEA DAEDIWSKLG 420
 TENALRTHVL STIVNGFAST RQELDFDFGA TFFAYQQDKW MLEEVINDCL EPLIDKAMVS 480
 ETEDIEDASK LFLRGTRLGS LVSMLYIDPL SGSKIYDGFK DIGKSTGNGM GSLEDDKGDD 540
 ITVTDMTLLH LVCSTPDMRQ LYLRNTDYTI VNEYIVAHSD EFHEIPDKLK ETDYEWFMGE 600
 VKTAMLLEEW VTEVSAEDIT RHFNVGEGDI HALADTSEWL MHAATAKLAEL LGVEYSSHAY 660
 SLEKRIRYGS GLDLMELVGI RGVRVRARK LYNAGFVSA KLKADISVL SKLVGPKVAY 720
 NILSGIGVRV NDKHENSAPI SSNTLDTLDD KNQKTFNDFQ 760

SEQ ID NO: 19 moltype = AA length = 707
 FEATURE Location/Qualifiers
 source 1..707
 mol_type = protein
 note = Cenarchaeum symbiosum
 organism = unidentified

SEQUENCE: 19
 MRISELDIPR PAIEFLEGEG YKKLYPPQAA AAKAGLTDGK SVLVSAPTAS GKTLLIAIAM 60
 ISHLSRNRGK AVYLSPLRAL AAEKFAEFGK IGGIPLGRPV RVGVSTGDFE KAGRSLGNND 120
 ILVLTNERMD SLIRRRPDWM DEVGLVIAD EHLIGDRSRG PTLEMLVTKL RGLRSSPQVV 180
 ALSATISNAD EIAGWLDCTL VHSWTRPVPL SEGVIQDGEV AMGDGSRHEV AATGGGPAVD 240
 LAAESVAEGG QSLIFADTRA RSASLAAKAS AVIPEAKGAD AAKLAAAKK IISGGGETKL 300
 AKTLAELVKE GAAPHHAGLN QDCRSVVVEE FRSGRIRLLA STPTLAAGVN LPARRVVISS 360
 VMRYNSSSGM SEPISILEYK QLCGRAGRPO YDKSGEAIIV GGVNADEIFD RYIGGEPEPI 420
 RSAMVDDRAL RIHVLSTVTT SPGIKEDDVT EFFLGLTGGQ QSGESTVKFS VAVALRFLQE 480
 EGMLGRRGGR LAATKMGRLV SRLYMDPMTA VTLRDAVGEA SPGRMHTLGF LHLVSECSEF 540
 MPRFALRQKD HEVAEMMLEA GRGELLRPVY SYECGRGLLA LHRWIGESPE AKLAEDLKFE 600
 SGDVHRMVES SGWLLRCIWE ISKHQERPD L GELDVLSR VAYGIKAEV PLVSIKIGIR 660
 VRSRLRFRGG IKGPGDLAAV PVERLSRVEG IGATLANNIK SQLRKGG 707

SEQ ID NO: 20 moltype = AA length = 720
 FEATURE Location/Qualifiers
 source 1..720
 mol_type = protein
 organism = Thermococcus gammatolerans

SEQUENCE: 20
 MKVDELVD E RLKAVLKERG IEELYPPQAE ALKSGALEGR NLVLAIPTAS GKTLLVSEIVM 60
 VNKLIQEGGK AVYLVPLKAL ABEKYREFKE WEKLGKLVAA TTGDYDSTDD WLGRIYDIIV 120
 TAEKFDLLR HGARWINDVK LVVADEVHLI GSYDRGATLE MILTHMLGRA QILALSATVG 180
 NAEELAELWD ASLVVSWDRP VQLRRGVFHL GTLIWEDGKV ESYPENWYSL VVDAVKRGKG 240
 ALVFNTRRS AEKEALALSK LVSSHLTKPE KRALESASQ LEDNPTSEKL KRALRGGVAF 300
 HHAGLSRVER TLIEDAFREG LIKIVITATP LSAGVNLPSF RVIIIRDTKRY AGFGWTDIPV 360
 LEIQQMMGRA GRPRYDKYGE AIIIVARTDEP GKLMERYIRG KPEKLFMSLA NEQAFRSQVL 420
 ALITNFGIRS FPPELVRLER TFYAHQRKDL SSLEYKAKEV VYFLIENEFI DLDLEDRFIP 480
 LPFGKRTSGL YIDPLTAKKF KDAFPAIERN PNPFGIFQLI ASTPDMATLT ARREMEDYL 540
 DLALELEDKL YASIPYYEDS RFQGFGLQVK TAKVLLDWIN EVPEARIYET YSIDPGDLR 600
 LLELADWLMY SLIELYLKFE PKEEILNLYR DLHLRLRHGV REELLEVLRL PNIGRKRARA 660
 LYNAGFRSVE AIANAKPAEL LAVEGIGAKI LDGIYRHLGI EKRVTTEKPK RKGTLEDFLR 720

SEQ ID NO: 21 moltype = AA length = 799

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FEATURE	Location/Qualifiers	
source	1..799	
	mol_type = protein	
	organism = Methanospirillum hungatei	
SEQUENCE: 21		
MEIASLPLPD	SFIRACHAKG	IRSLYPPQAE CIEKGLLEGK NLLISIPTAS GKTLLAEMAM 60
WSRIAAGGKC	LYIIVPLRALA	SEKYDEFSKK GVIIRVGATG DLDRTDAYLG ENDIIVATSE 120
KTDSLRRNRT	PWLSQITCIV	LDEVHLIGSE NRGATLEMVI TKLRYTNPVM QIIGLSATIG 180
NPAQLAEWLD	ATLITSTWRP	VDLRQGVYYN GKIRFSDSER PIQGTKHDD LNLCLDTIEE 240
GGQCLVVFSS	RRNAEGFAKK	AAGALKAGSP DSKALAEQLR RLRDRDEGNV LADCVERGAA 300
FHHAGLIRQE	RTIIEEGFRN	GYIEVIAATP TLAAGLNLPV RRVIIIRDYNR FASGLGMVPI 360
PVGEYHQMAG	RAGRPPLDPY	GEAVLLAKDA PSVERLFETF IDAEAERVDS QCVDDASLCA 420
HILSLIATGF	AHDQEALSSF	MERTFYFFQH PKTRSLPRLV ADAIRFLTTA GMVEERENTL 480
SATRLGSLVS	RLYLNPCATP	LILDSLKSKC TPTLIGLLHV ICSVSPDMQRL YLKAADTQLL 540
RTFLPKHKDD	LILPLPFQEQ	EEELWLSGLK TALVLTWDAD EFSEGMIEER YGIGAGDLYN 600
IVDSGKWLHL	GTERLVSVEM	PMSQVVKTL SVRVHHGVKS ELLPLVALRN IGRVRARTLY 660
NAGYPDPEAV	ARAGLSTIAR	IIEGEGARQV IDEITGVKRS GIHSSDDDYQ QKTPELLTDI 720
PGIGKKMAEK	LQNAGIITVS	DLTDADEVLL SDVLGAARAR KVLAFLSNSE KENSSSDKTE 780
EIPDTQKIRG	QSSWEDEFC	
SEQ ID NO: 22		
FEATURE	moltype = AA length = 1756	
source	Location/Qualifiers	
	1..1756	
	mol_type = protein	
	organism = Escherichia coli	
SEQUENCE: 22		
MMSIAQVRS	GSAGNYITDK	DNYVVLGSMG ERWAGKGAEQ LGLQGSVDKD VFTRLLEGRL 60
PDGADLSRMQ	DGSNKHRRPGY	DLTFSAPKSV SMMAMLGDK RLIDAHNQAV DFAVRQVEAL 120
ASTRVMTDQG	SETVLTGNLV	MALFNHDTSR DQEPQLHTHA VVANVTQHNG EWKTLSSDKV 180
GKTGFIENVY	ANQIAFGRLY	REKLKEQVEA LGYETEVVGK HGMWEMPGVP VEAFFSGRSQA 240
IREAVGEDAS	LKSRDVAALD	TRKSKQHVDP EIRMAEWMQT LKETGFDIRA YRDAADORTE 300
IRTQAPGPAS	QDGPVQQAQ	TQAIAGLSEK KVQFTYTDVL ARTVGILPPE NGVIERARAG 360
IDEAISREQL	IPLDREKGLF	TSGIHVLDEL SVRALSRDIM KQNRVTVHPE KSPRTAGYS 420
DAVSVAQDR	PSLAIVSQGQ	GAAGQREVA ELVMMAREQG REVQIIAADR RSQMNKQDE 480
RLSGELITGR	RQLLEGMAFT	PGSTVIVDQG EKLSLKETLT LLDGAARHNV QVLITDSGQR 540
TGTGSALMAM	KDAGVNTYRW	QGGEQRPATI ISEPDNRVRY ARLAGDFAAS VKAGEESVAQ 600
VSGVREQAIL	TQAIRSELTK	QGVLGHEPVT MTALSPVWLD SRSRYLRDMY RPGMVMEQWN 660
PETRSHDRYV	IDRVTAQSHS	LTLRDAQGET QVVRISLDS SWSLFRPEKM PVADGERLRV 720
TGKIPGLRVS	GGDRLQVASV	SEDAMTVVVP GRAEPASLPV SDSPTALKL ENGWEVETPGH 780
SVSDSATVFA	SGQRAATFMI	LTNGLARSGR DVRLYSSLDE TRTAEKLARH PSFTTVSEKI 840
KARAGETLLE	TAISLQKAGL	HTPAQQAIHL ALPVLESKNL AFSMVDLLTE AKSFAAEGTG 900
FTELGGEINA	QIKRGDILLY	DVAKGYGTGL LVSRASYEAE KSILRHILEG KEAVTPLMER 960
VPGEMLMETLT	SGQRAATFMI	LETSRRETVV QGYAGVGKTT QFRVMSAVN MLPASERPRV 1020
VGLGPTHRAV	GEMRSAGVDA	QTLASFLHDT QLQQRSGETP DFSNTLFLLD ESSMVGNTM 1080
ARAYALIAAG	GGRVAVSGDT	DQLQAIAPGQ SFRLQQTRSA ADVVIMKEIV RQTPELREAV 1140
YSLINRDVER	ALSGLSEVYP	SQVPRLEGAW APEHSTVEFS HSQEAKLAEA QQKAMLKGEA 1200
FPDIPMTLYE	AIVRDYTGRT	PEAREQTLIV THLNEDRRVL NSMIHDAREK AGELGKEQVM 1260
VPVLNTANIR	DGELRRRLSTW	EKNPDALALV DNVYHRIAGI SKDDGLITLQ DAEGNTRLIS 1320
PREAVAEGVT	LYTPDKIRVG	TGDRMRFTKS DRERGVYANS VWTVTAVSGD SVTLSDGQQT 1380
RVIRPGQERA	EQHIDLAYAI	TAHGAQGASE TFAIALEGTE GNRKLMAGFE SAYVALSRMK 1440
QHVQVYTDNR	QGWDAINNA	VQKGTAMDVL EPKPDREVMN AQRLFSTARE LRDVAAGRAV 1500
LRQAGLAGGD	SPARFIAPGR	KYPQPVVALP AFDRNGKSAG IWLNLPLTDD GNGLRIGFSGE 1560
GRVKSGSDAQ	FVALQGSRNG	ESLLADNMQD GVRIARDNPD SGVVRIAGE GRPWNPGAIT 1620
GGRVWGDIPD	NSVQPGAGNG	EPVTAEVLAQ RQAEAEAIRRE TERRADEIVR KMAENKPDLP 1680
DGKTELAVRD	IAGQERDRSA	ISERETALPE SVLRESQRER EAVREVAREN LLQERLQQME 1740
RDMVRDLQKE	KTLGGD	
SEQ ID NO: 23		
FEATURE	moltype = AA length = 726	
source	Location/Qualifiers	
	1..726	
	mol_type = protein	
	organism = Methanococcoides burtonii	
SEQUENCE: 23		
MSDKPAPMKY	FTQSSCPYQ	QEAMDRHISA LMQQQLVLF E GACGTGKTLS ALVPALHVGK 60
MLGKTVIIAT	NVHQMQYQFI	NEARDIKKVQ DVKVAVIKGK TAMCPQEADE EECVSKRENT 120
FELMETEREI	YLRQELNSA	RDSYKKSHPD AFVTLRDELS KEIDAVEEKA RGLRDRACND 180
LYEVLRSDE	KFEWLYKEV	RSPEEINDHA IKDGMCGYEL VKRELKHADL LICNYHHVLN 240
PDIISTVLWG	IEKEPQETIV	IFDEAHNLES AARSHSSLSL TEHSIEKAIT ELEANLDLLA 300
DDNIHNLFIN	PLEVISDTYN	SRKFGGERER VRKNWYDIRI SDPYERNDIV RGKFLRQAKG 360
DFGEKDDIQI	LLSEASLGA	KLDETYRDQY KKGLSVVMKR SHIRYVADFM SAYIELSHNL 420
NYYPILNVR	DMNDEIVGRV	ELFTCIPKNV TEPLFNSLFS VILMSATLHP FEMVKKTLGI 480
TRDTCEMSYG	TSPPEEKRLS	IAVSIPPLFA KNRDDRHVTE LLEQVLLDSI ENSKGNVILF 540
FQSAFEAKRY	YSKIEPLVNV	PVFLDEVGIS SQDVREEFFS IGEENGKAVL LSYLWGTLS 600
GIDYRDGRGR	TVIIIGVGYP	ALNDRMNAVE SAYDHVFGY AGWEFAIQVP TIRKIRQAMG 660
RVRSPTDYG	ARILLDGRFL	TDSKKRFGKF SVFEVFPFAE RSEFVDVDPE KVKYSLMNF 720
MDNDEQ		

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SEQ ID NO: 24 moltype = AA length = 439
 FEATURE Location/Qualifiers
 source 1..439
 mol_type = protein
 organism = Dickeya dadantii

SEQUENCE: 24

MTFDDLTEGQ	KNAFNIVMKA	IKEKKHHVTI	NGPAGTGKTT	LTKFIIEALI	STGETGIILA	60
APTHAAKKIL	SKLSGKEAST	IHSILKINPV	TYBENVLFEQ	KEVPDLAKCR	VLICDEVSMY	120
DRKLFKILLS	TIPPWCTIIG	IGDNKQIRPV	DPGENTAYIS	PFFTHKDFYQ	CELTEVKRSN	180
APIIDVATDV	RNGKWIYDKV	VDGHGVRGFT	GDTALRDFMV	NYFSIVKSLD	DLFENRVMAF	240
TNKSVDKLNS	IIRKKIFETD	KDFIVGEIIV	MQEPLFKTYK	IDGKPVSEII	FNNGQLVRII	300
EAEYTSTFVK	ARGVPGEYLI	RHWDLTVETY	GDDEYYREKI	KIISSEELY	KFNLFLGKTA	360
ETYKNWNKGG	KAPWSDFWDA	KSQFSKVKAL	PASTFHKAQG	MSVDRAFIYT	PCIHYADVEL	420
AQQLLYVGVV	RGRYDVIFYV					439

SEQ ID NO: 25 moltype = AA length = 970
 FEATURE Location/Qualifiers
 REGION 1..970
 note = Synthetic polypeptide
 source 1..970
 mol_type = protein
 organism = unidentified

SEQUENCE: 25

MLSVANVRSP	SAAASYFASD	NYVASADADR	SGQWIGDGAK	RLGLEGKVEA	RAFDALLRGE	60
LPDGSSVGNP	GQAHPRPGTDL	TFSVPKSWSL	LALVGKDERI	IAAYREAVVE	ALHWAENKAA	120
ETRVVEKGMV	VTQATGNLAI	GLFQHDTRN	QEPNLHFHAV	IANVTQKGKG	KWRTLKNDRL	180
WQLNTTLNSI	AMARFVRVAVE	KLGYEPGPVL	KHGNFEARGI	SREQVMAFST	RRKEVLEARR	240
GPGLDAGRIA	ALDTRASKEG	IEDRATLSKQ	WSEAAQSIGL	DLKPLVDRAR	TKALGQGMEA	300
TRIGSLVERG	RAWLSRFAAH	VRGDPADPLV	PPSVLKQDRQ	TIAAAQAVAS	AVRHLSQREA	360
AFERTALYKA	ALDFGLPTTI	ADVEKRTRAL	VRSGDLIAGK	GEHKGWLASR	DAVVTEQRIL	420
SEVAAGKGDS	SPAITPQKAA	ASVQAAALTG	QGFRNLNEGQL	AAARLILISK	DRTIAVQGIA	480
GAGKSSVLKP	VAEVLRDEGH	PVIGLAIQNT	LVQMLERDTG	IGSQTARFL	GGWNKLLDDP	540
GNVALRAEAQ	ASLKDHLVLV	DEASMVSNE	KEKLVRLANL	AGVHRLVLIG	DRKQLGAVDA	600
GKPPFALLQRA	GIARAEMATN	LRARDPVVRE	AQAAAQAGDV	RKALRHLKSH	TVEARGDGAQ	660
VAAETWLALD	KETRARTSIY	ASGRAIRSAV	NAAVQQGLLA	SREIGPAKMK	LEVLDVRVNT	720
REELRHLPAY	RAGRVLEVSR	KQQALGLFIG	EYRVIGQDRK	GKLVEVEDKR	GKRFRFDPAR	780
IRAGKGDDNL	TLLEPRKLEI	HEGDRIWTR	NDHRRGLFNA	DQARVVEIAN	GKVTFTETSKG	840
DLVELKKDDP	MLKRIDLAYA	LNVMHAQGLT	SDRGIAVMD	RERNLSNQKT	FLVTVTRLRD	900
HLTLVVDSAD	KLGAAVARNK	GEKASAEIVT	GSVKPTATKG	SGVDQPKSVE	ANKAEKELTR	960
SKSKTLDFGI						970

SEQ ID NO: 26 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Yokenella regensburgei

SEQUENCE: 26

CLTAPPKEAA	KPTLMPRAQS	YRDLTHLPLP	SGKVFVSVYN	IQDETGQFKP	YPASNFSTAV	60
PQSATAMLV	ALKDSRWFPV	LERQGLQNLL	NERKIIRAAQ	ENGTVAENNR	IPLQSLTAAN	120
VMIEGSIIGY	ESNVKSGGVG	ARYFGIGADT	QYQLDQIAVN	LRVVNVSTGE	VLSSVNTSKT	180
ILSYEVQAGV	FRFVDYQRL	EGEIGYTSNE	PVMLCLMSAI	ETGVIYLIND	GIERGLWDLQ	240
QKADVDNPIL	ARYRMSAPP	ES				262

SEQ ID NO: 27 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 note = Cronobacter pulveris
 organism = unidentified

SEQUENCE: 27

CLTAPPKEAA	KPTLMPRAQS	YRDLTNLPDP	KGKLFVSVYN	IQDETGQFKP	YPASNFSTAV	60
PQSATAMLV	ALKDSRWFPV	LERQGLQNLL	NERKIIRAAQ	ENGTVAENNR	MPLQSLVAAN	120
VMIEGSIIGY	ESNVKSGGVG	ARYFGIGGDT	QYQLDQIAVN	LRVVNVSTGE	VLSSVNTSKT	180
ILSYEVQAGV	FRFVDYQRL	EGEIGYTANE	PVMLCLMSAI	ETGVIHLIND	GINRGLWELK	240
NKGDAKNTIL	AKYRSMVAPP	ES				262

SEQ ID NO: 28 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Rahnella aquatilis

SEQUENCE: 28

CLTAAPEKAA	RPTLLPRAPS	YTDLTHLPSP	QGRIFVSVYN	IQDETGQFKP	YPACNFSTAV	60
PQSATAMLV	ALKDSKWFIP	LERQGLQNLL	NERKIIRAAQ	ENGVAINNQ	RPLSSSLVAAN	120
ILIEGSIIGY	ESNVKSGGVG	ARYFGIGAST	QYQLDQIAVN	LRAVDVNTGE	VLSSVNTSKT	180

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ILSYEVQAGV	FRFIDYQRL	EGELGYTTNE	PVMLCLMSAI	ESGVIYLVND	GIERNLWQLQ	240
NPSEINSPIL	QRYKNNIVPA	ES				262

SEQ ID NO: 29	moltype = AA	length = 259
FEATURE	Location/Qualifiers	
source	1..259	
	mol_type = protein	
	organism = Kluyvera ascorbata	

SEQUENCE: 29						
CITSPPKQAA	KPTLLPRSQS	YQDLTHLPEP	QGRLFVSVYN	ISDETQGFKP	YPASNFSTSV	60
PQSATAMLV	ALKDSNWFI	LERQGLQNL	NERKIIRAAQ	ENGTVAVNNR	TQLPSLVAAN	120
ILIEGSIIGY	ESNVKSGGAG	ARYFGIGAST	QYQLDQIAVN	LRVNVSTGE	VLSSVNTSKT	180
ILSYEVQAGV	FRYIDYQRL	EGEVGYTVNE	PVMLCLMSAI	ETGVIYLVND	GISRNLWQLK	240
NASDINSPVL	EKYKSIIVP					259

SEQ ID NO: 30	moltype = AA	length = 259
FEATURE	Location/Qualifiers	
source	1..259	
	mol_type = protein	
	organism = Hafnia alvei	

SEQUENCE: 30						
CLTAPPKQAA	KPTLMPRAQS	YQDLTHLPEP	AGKLFVSVYN	IQDETQGFKP	YPASNFSTAV	60
PQSATAMLV	ALKDSGWFI	LERQGLQNL	NERKIIRAAQ	ENGTAAVNNQ	HQLSSLVAAN	120
VLVEGSIIGY	ESNVKSGGAG	ARFFGIGAST	QYQLDQIAVN	LRVVDVNTGQ	VLSSVNTSKT	180
ILSYEVQAGV	FRYIDYQRL	EGEIGYTTNE	PVMCLVMSAI	ETGVIYLVND	GINRNLWTLK	240
NPQDAKSSVL	ERYKSTIVP					259

SEQ ID NO: 31	moltype = AA	length = 255
FEATURE	Location/Qualifiers	
REGION	1..255	
	note = Enterobacteriaceae bacterium	
source	1..255	
	mol_type = protein	
	organism = unidentified	

SEQUENCE: 31						
CITTPPQEA	KPTLLPRDAT	YKDLVSLPQP	RGKIYVAVYN	IQDETQGFQP	YPASNFSTSV	60
PQSATAMLV	SLKDSRWFP	LERQGLQNL	NERKIIRAAQ	QNGTVGDNNA	SPLPSLYSAN	120
VIVEGSIIGY	ASNVTGGGFG	ARYFGIGGST	QYQLDQVAVN	LRIVNVHTGE	VLSSVNTSKT	180
ILSYEVQAGV	FRFIDYQRL	EGEAGFTTNE	PVMTCLMSAI	EEGVIHLIND	GINKKLWALS	240
NAADINSEVL	TRYRK					255

SEQ ID NO: 32	moltype = AA	length = 258
FEATURE	Location/Qualifiers	
source	1..258	
	mol_type = protein	
	organism = Plesiomonas shigelloides	

SEQUENCE: 32						
ITEVPKEAAK	PTLMPRASTY	KDLVALPKPN	GKIIIVSVYSV	QDETQGFQPL	PASNFSTAVP	60
QSGNAMLTS	LKDSGWFP	LERQGLQNL	ERKIIRAAQ	NGTVANNQ	PLPSLLSANV	120
VIEGAIIGYD	SDIKTGGAGA	RYFGIGADGK	YRVDQVAVNL	RAVDVRTGEV	LLSVNTSKTI	180
LSSELSAGVF	RFIEYQRLLE	LEAGYTTNEP	VMMCMSALE	AGVAHLIVEG	IRQNLWSLQN	240
PSDINNPIIQ	RYMKEDVP					258

SEQ ID NO: 33	moltype = AA	length = 248
FEATURE	Location/Qualifiers	
source	1..248	
	mol_type = protein	
	note = Vibrio fischeri	
	organism = unidentified	

SEQUENCE: 33						
PETSESPTLM	QRGANYIDLI	SLPKPQGGKIF	VSVYDFRDQT	GQYKQPNSN	FSTAVPQGGT	60
ALLTMALLDS	EWFYPLERQG	LQNLTERKI	IRAAQKKQES	ISNHGSTLPS	LLSANVMIEG	120
GIVAYDSNIK	TGGAGARYLG	IGGSGQYRAD	QVTNIRAVD	VRSGKILTSV	TTSKTILSYE	180
VSAGAFRFVD	YKELLEVLG	YTNNEPVNIA	LMSAIDSAVI	HLIVKGVQGG	LWRPANLDR	240
NNPIFKKY						248

SEQ ID NO: 34	moltype = AA	length = 248
FEATURE	Location/Qualifiers	
source	1..248	
	mol_type = protein	
	organism = Aliivibrio logei	

SEQUENCE: 34						
PDASESPTLM	QRGATYLDLI	SLPKPQGGKIY	VSVYDFRDQT	GQYKQPNSN	FSTAVPQGGT	60
ALLTMALLDS	EWFYPLERQG	LQNLTERKI	IRAAQKKQES	ISNHGSTLPS	LLSANVMIEG	120
GIVAYDSNIK	TGGAGARYLG	IGGSGQYRAD	QVTNIRAVD	VRSGKILTSV	TTSKTILSYE	180
LSAGAFRFVD	YKELLEVLG	YTNNEPVNIA	LMSAIDSAVI	HLIVKGIEEG	LWRPENQNGK	240

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ENPIFRKY 248

SEQ ID NO: 35 moltype = AA length = 254
 FEATURE Location/Qualifiers
 source 1..254
 mol_type = protein
 organism = Photobacterium sp.

SEQUENCE: 35

PETSKPTLM	ARGTAYQDLV	SLPLPKGKVY	VSVYDFRDQT	GQYKQPNSN	FSTAVPQGGA	60
ALLTTALLDS	RWFMPLEREG	LQNLLTERKI	IRAAQKKDEI	PTNHGVHLPS	LASANIMVEG	120
GIVAYDTNIQ	TGGAGARYLG	VGASGQYRTD	QVTVNIRAVD	VRTGRILLSV	TTSKTILSKE	180
LQTVGFKFVD	YKDLLEAELG	YTTNEPVNLA	VMSAIDAAV	HVIVDGIKTG	LWEPLRGEDL	240
QHPIIQEYMN	RSKP					254

SEQ ID NO: 36 moltype = AA length = 261
 FEATURE Location/Qualifiers
 source 1..261
 mol_type = protein
 organism = Aeromonas veronii

SEQUENCE: 36

CATHIGSPVA	DEKATLMPRS	VSYKELISLP	KPKGKIVAAV	YDFRDQTGQY	LPAPASNFST	60
AVTQGGVAML	STALWDSQWF	VPLEREGLQN	LLTERKIVRA	AQNKPNVPGN	NANQLPSLVA	120
ANILIEGGIV	AYDSNVRTGG	AGAKYFGIGA	SGEYRVDQVT	VNLRAVDIRS	GRILNSVTTS	180
KTVMSSQQVQA	GVFRFVEYKR	LLEAEAGFST	NEPVQMCVMS	AIESGVIRLI	ANGVRDNLWQ	240
LADQRDIDNP	ILQEYLQDNA	P				261

SEQ ID NO: 37 moltype = AA length = 239
 FEATURE Location/Qualifiers
 source 1..239
 mol_type = protein
 organism = Shewanella sp.

SEQUENCE: 37

ASSSLMPKGE	SYDYDLINLPA	PQGVMLAAVY	DFRDQTGQYK	PIPSSNFSTA	VPQSGTAFLA	60
QALNDSSWFI	PVEREGLQNL	LTERKIVRAG	LKGDANKLPQ	LNSAQILMEG	GIVAYDTNVR	120
TGGAGARYLG	IGAATQFRVD	TVTNNLRAVD	IRTGRLLSSV	TTTKSILSKE	ITAGVFKFID	180
AQELLESELG	YTSNEPVSLC	VASAIESA	VHMIADGIWKG	AWNLAQASG	LRSPVLQKY	239

SEQ ID NO: 38 moltype = AA length = 233
 FEATURE Location/Qualifiers
 source 1..233
 mol_type = protein
 organism = Pseudomonas putida

SEQUENCE: 38

QDSEPTLTLP	RASTYYDLIN	MPRPKGRMLA	VVYGFDRDQTG	QYKPTPASSF	STSVTQGAAS	60
MLMDALSASG	WFFVLEREGL	QNLLTERKII	RASQKPDVA	ENIMGELPPL	QAANLMLEGG	120
ITAYDTNVRS	GEGGARYLGI	DISREYRVDQ	VTNNLRAVDV	RTGQVLANVM	TSKTIYSVGR	180
SAGVFKFIEF	KKLLEAEVGY	TTNEPAQLCV	LSAIESAVGH	LLAQGIEQRL	WQV	233

SEQ ID NO: 39 moltype = AA length = 234
 FEATURE Location/Qualifiers
 source 1..234
 mol_type = protein
 organism = Shewanella violacea

SEQUENCE: 39

MPKSDTYIDL	IGLPHPPQGS	LAAYDFRDQ	TGQYKAIPSS	NFSTAVPQSG	TAFLAQALND	60
SSWFPVVERE	GLQNLLTERK	IVRAGLKGEA	NQLPQLSSAQ	ILMEGGIVAY	DTNIKTGGAG	120
ARYLGIGVNS	KFRVDTVTN	LRAVDIRTGR	LLSSVTTTKS	ILSKEVSAGV	FKFIDAQDLL	180
ESELGYTSNE	PVSLCVAQAI	ESAVVHMIAD	GIWKRAWNLA	DTASGLNNPV	LQKY	234

SEQ ID NO: 40 moltype = AA length = 245
 FEATURE Location/Qualifiers
 source 1..245
 mol_type = protein
 organism = Marinobacterium jannaschii

SEQUENCE: 40

LTRRMSTYQD	LIDMPAPRGK	IVTAVYSFRD	QSGQYKPAPS	SSFSTAVTQG	AAAMLVNVLN	60
DSGWFIPLER	EGLQNILTER	KIIRAALKKD	NVPVNSAGL	PSLLAANIML	EGGIVGYDSN	120
IHTGGAGARY	FGIGASEKYR	VDEVTVNLRA	IDIRTGRILH	SVLTSKKILS	REIRSDVYRF	180
IEFKHLLEME	AGITTNDPAQ	LCVLSAIESA	VAHLIVDGI	KKSWSLADPN	ELNSPVIQAY	240
QQQRI						245

SEQ ID NO: 41 moltype = AA length = 234
 FEATURE Location/Qualifiers
 source 1..234
 mol_type = protein
 organism = Chryseobacterium oranimense

-continued

SEQUENCE: 41
 PSDPERSTMG ELTPSTAE LR NLPLPNEKIV IGVYKFRDQT GQYKPSENGN NWSTAVPQGT 60
 TTILIKALED SRWFIPIERE NIANLLNERQ IIRSTRQEYM KDADKNSQSL PPLLYAGILL 120
 EGGVISYDSN TMTGGFGARY FGIGASTQYR QDRITIYLRA VSTLNGEILK TVYTSKTILS 180
 TSVNGSFFRY IDTERLLEAE VGLTQNEPVQ LAVTEAIEKA VRSLIEGTR DKIW 234

SEQ ID NO: 42 moltype = DNA length = 12
 FEATURE Location/Qualifiers
 misc_feature 1..12
 note = Synthetic polynucleotide
 source 1..12
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 42
 tttttttttt tt 12

SEQ ID NO: 43 moltype = DNA length = 27
 FEATURE Location/Qualifiers
 misc_feature 1..27
 note = Synthetic polynucleotide
 source 1..27
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 43
 ggttggttct gttggtgctg atattgc 27

SEQ ID NO: 44 moltype = DNA length = 3635
 FEATURE Location/Qualifiers
 misc_feature 1..3635
 note = Synthetic polynucleotide
 source 1..3635
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 44
 gccatcagat tgtgtttggt agtcgctgcc atcagattgt gtttgtagt cgcttttttt 60
 ttttggaatt ttttttttgg aatttttttt ttgcgctaac aacctcctgc cgcttttgccc 120
 gtgcataatcg gtcacgaaca aatctgatta ctaaacacag tagcctggat ttgttctatc 180
 agtaatcgac cttatttccta attaaataga gcaaatcccc ttattggggg taagacatga 240
 agatgccaga aaaacatgac ctgtttggccg ccattctcgc ggcaaaaggaa caaggcatcg 300
 gggcaatccct tcgcttttgc atggcggtacc ttccgcgagc atataatggc ggtgctgtta 360
 caaaaacagt aatcgacgca acgatgtgct ccattatcgc ctagtctcatt cgtgaccttc 420
 tcgacttcgc cggactaagt agcaatctcg cttatataac gagcgtgttt atcggctaca 480
 tcggtactga ctcgatttgt tcgcttatca aacgcttcgc tgctaaaaaa gccggagtag 540
 aagatggtag aaatcaataa tcaacgtaag gcgttcctcg atatgctggc gttgctggag 600
 ggaactgata acggacgtca gaaaaccaga aatcatggtt atgacgtcat ttaggcgga 660
 gagctattta ctgattactc cgatcacccct cgcaaaacttg tcacgctaaa cccaaaactc 720
 aaatcaacag gcgcgggacg ctaccagctt ctttcccggt ggtgggagtc ctaccgcaag 780
 cagcttgccc tgaagactt ctctccgaaa agtcaggacg ctgtggcatt gcagcagatt 840
 aaggagcgtc cgcgtttacc tatgattgat cgtggtgata tccgtcaggc aatcgaccgt 900
 tgcagcaata tctgggcttc actgcccggc gctggttatg gtcagttcga gcataaggct 960
 gacagcctga tgcgaaaatt caaagaagcg ggcggaacgg tcagagagat tgatgtatga 1020
 gcagagtcac ctcgactcgg ttatctgcat catcgtctgc ctgtcatggg 1080
 ctgttaatac ttaccgtgat aacgccatta cctacaaaag ccagcgcgac aaaaatgcca 1140
 gagaactgaa gctggcgaaac gcggcaatta ctgacatgca gatgcgtcag cgtgatgttg 1200
 ctgctgctga tgcgaaaatac acgaaggagt tagctgatgc taaagctgaa aatgatgctc 1260
 tgcgtgatga tgttgccgct ggtcgtcgtc ggttgacat caaagcagtc tgtcagtcag 1320
 tgcgtgaagc caccaccgctc tccggcgtgg ataatgcagc ctccccccga ctggcagaca 1380
 ccgctgaacg ggattatttc accctcagag agaggctgat cactatgcaa aaacaactgg 1440
 aaggaaacca gaagtatatt aatgagcagt gcagatagag ttgcccata cgtatgggcaa 1500
 ctcatgcaat tattgtgagc aatcacacag cgcttcacag ggagatataa tgcctaaagt 1560
 aataaaaccg agcaaatccat ttacgaatgt ttgctgggtt tctgttttaa caacattttc 1620
 tgcgcccgca caaatttttg ctgcatcgac agttttcttc tgcccatttc cagaaacgaa 1680
 gaaatgatgg gtgatgtgtt cctttggtgc tactgctgcc ggtttgtttt gaacagtaaa 1740
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 aaaccctaaa caatgagttg aaatttcata ttgttaatat ttattaatgt atgtcaggtg 1920
 cgatgaatcg tcattgtatt cccggattaa ctatgtccac agccctgacg gggaaactct 1980
 ctgcgggagt tcccggaat aattaaaacg atgcacacag ggtttagcgc gtacacgtat 2040
 tgcattatgc caacgccccg gtgctgacac ggaagaaacc ggacgttatg atttagcgtg 2100
 gaaagatttg tgtagtgttc tgaatgctct cagtaaatag taatgaatta tcaaaaggat 2160
 agtaatatct tttatgttca tggatatttg taaccatcg gaaaactcct gcttttagca 2220
 gattttccct gtattgtgta aatgtgattt ctcttgattt caacctatca taggacgttt 2280
 ctataagatg cgtgtttctt gagaatttaa catttacaac ctttttaagt ccttttatta 2340
 acacgggtgt atcgtttttt aacacgatgt gaattatttc tgtggctaga tagtaaatat 2400
 aatgtgagac gttgtgacgt ttagtttcag aataaaacaa ttcacagtct aaatcttttc 2460
 gcacttgatc gaatttttct ttaaaaatgg caacctgagc cattggtaaa accttccatg 2520

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tgatacgagg	gcgcgtagtt	tgcattatcg	tttttatcgt	ttcaatctgg	tctgacctcc	2580
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tcgctaacaa	agtgcggtcc	tgctggcatt	ctggaggagg	atacaaccga	cagatgtatg	2700
taaggccaac	gtgctcaaat	cttcatacag	aaagatttga	agtaatat	taaccgctag	2760
atgaagagca	agcgcatgga	gcgacaaaat	gaataaagaa	caatctgctg	atgatccctc	2820
cgtggatctg	attcgtgtaa	aaaatatgct	taatagcacc	atttctatga	gttaccctga	2880
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tgagaaattc	ccggaccctt	tttgcacaag	agcgatgtta	atttgttcaa	tcatttgggt	3300
aggaagacgg	actgttcggg	ttgtttgtct	gcgggttctg	ttcttcgttg	acatgagggt	3360
gccccgtatt	cagtgctcgt	gatttgtatt	gtctgaagtt	gtttttacgt	taagtgtatg	3420
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cacgcacgtt	gtgatattga	gatgataatc	attatcactt	tacgggtcct	ttccggtgaa	3540
aaaaaaggtg	ccaaaaaaat	catcgctcgt	agtagtgaac	cgtaagccgt	tctgtttatg	3600
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SEQ ID NO: 45 moltype = DNA length = 3636
FEATURE Location/Qualifiers
misc_feature 1..3636
 note = Synthetic polynucleotide
source 1..3636
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 45

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gataatgatt	tcatctctaca	tatcacaaacg	tgcggtggagg	ccatcaaaacc	acgtcaaaata	180
atcaattatg	acgcagggtat	cgtattaatt	gatctgcatc	aacttaacgt	aaaaacaact	240
tcagacaata	caaatcagcg	acactgaata	cggggcaacc	tcagtgtcaac	gaagaacaga	300
acccgcagaa	caacaaccocg	caacatccgc	tttcttaacc	aaatgattga	acaaattaac	360
atcgctcttg	agcaaaaagg	gtccgggaat	ttctcagcct	gggtcattga	agcctgccgt	420
cggagactaa	cgtcagaaaa	gagagcatat	acatcaatta	aaagtgtatg	agaatgaaca	480
ttcccgcttc	ttccctccga	acaggacgat	attgtaaatt	cacttaatta	cgagggcatt	540
gcagtaattg	agttgcagtt	ttaccacttt	cctgacagtg	acagactcgg	tggtggctct	600
gtccacagct	aaatagtttg	aatgattagc	agttatgggt	atcagtcac	caccagggaa	660
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ctattaagca	tattttttac	acgaatcaga	tccacggagg	gatcatcagc	agattgttct	840
ttattcattt	tgtcgctcca	tgcgcttgct	cttcatctag	cggttaaaat	attacttcaa	900
atctttctgt	atgaagattt	gagcacgttg	gccttacata	catctgtcgg	ttgtatttcc	960
tcocagaaatg	ccagcaggag	cgcactttgt	tacgcaacca	atactattaa	gtgaaaacat	1020
tcctaataat	tgacataaat	catcaacaaa	acacaaggag	gtcagaccag	attgaaacga	1080
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agggtgccat	ttttaaagaa	atatctgatc	aagtgcgaaa	agatttagac	tgatgaattgt	1200
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gcagtagcac	caaggaacac	catcacccat	catttcttcg	tttctggaat	tgggcagaag	1980
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gaagcgctgt	tgtattgctc	acaataattg	catgagttgc	ccatcgatat	gggcaactct	2160
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accagcgccc	ggcagtgaa	cccagatatt	gctgcaacgg	tcgatgtcct	gacggatatc	2760
accacgatca	atcataggta	aagcgccacg	ctccttaate	tgctgcaatg	ccacagcgtc	2820
ctgacttttc	ggagagaagt	ctttcaggcc	aagctgcttg	cggtaggcat	cccaccaacg	2880

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ggaaagaagc  tggtagcgtc  cggcgccctgt  tgatttgagt  tttgggttta  gcgtgacaag  2940
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gaacgcctta  cgttgattat  tgattttctac  catctttctac  tccggctttt  ttagcagcga  3120
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gaatggcgcg  caacaggta  tgtttttctg  gcatcttcat  gtcttacct  caataagggg  3420
atttgctcta  ttaattagg  aataaggctg  attactgata  gaacaaatcc  aggctactgt  3480
gtttagtaat  cagatttggt  cgtgaccgat  atgcacgggc  aaaacggcag  gaggttgta  3540
gcgcaaaaa  aaaattccaa  aaaaaaatt  ccaaaaaaa  aaagcgacta  acaaacacaa  3600
tctgatggca  gcgactaaca  aacacaatct  gatggc  3636

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SEQ ID NO: 46      moltype = DNA  length = 28
FEATURE
misc_feature      1..28
                  note = Synthetic polynucleotide
source            1..28
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 46
gcaatatcag  caccaacaga  aacaacct  28

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SEQ ID NO: 47      moltype = AA  length = 10
FEATURE
REGION            1..10
                  note = StrepII (C)
source            1..10
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 47
SAWSHPQFEK  10

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SEQ ID NO: 48      moltype = AA  length = 15
FEATURE
REGION            1..15
                  note = Pro
source            1..15
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 48
MQRLFLLVAV  MLLSG  15

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```

SEQ ID NO: 49      moltype = DNA  length = 558
FEATURE
source            1..558
                  mol_type = genomic DNA
                  organism = Mycobacterium smegmatis

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```

SEQUENCE: 49
atgggcctgg  ataacgaact  tagcctgggtg  gacggccaag  atcgcacgct  gacggtgcaa  60
caatgggata  ccttcttgaa  tgggtgtgttt  ccgctggatc  gtaaccgcct  gaccctgaa  120
tgggtttcatt  ccggtcgcg  aaaatatatc  gtcgcaggcc  cgggtgctga  cgaattcgaa  180
ggcacgctgg  aactgggtta  tcagattggc  tttccgtggt  cactgggcgt  tggatatcaac  240
ttctcgta  ccacgcgcaa  tattctgata  gatgacgggt  atattaccgc  accgcggttt  300
ggcctgaaca  gcgtgattac  gccgaacctg  tttccgggtg  ttagcatctc  tgcgcatctg  360
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ggcggtgtcg  cggtgtctaa  tgcccacgg  accgttacgg  gcgcggccgg  cggtgtctcg  480
ctgcgtccgt  tcgcgcgct  gattgcgagc  accggcgact  ctgttacgac  ctatggcgaa  540
ccgtggaata  tgaactaa  558

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```

SEQ ID NO: 50      moltype = AA  length = 184
FEATURE
source            1..184
                  mol_type = protein
                  organism = Mycobacterium smegmatis

```

```

SEQUENCE: 50
GLDNELSLVD  GQDRTLTVQQ  WDTFLNGVFP  LDRNRLTREW  FHSGRAKYIV  AGPGADEFEG  60
TLELG YQIGF  PWSLGVGINF  SYTTPNILID  DGDITAPPPG  LNSVITPNLF  PGVSISADLG  120
NGPGIQEVAT  FSDVDSGAE  GVAVSNAHGT  VTGAAGGVLL  RPFARLIAS  T  GDSVTTYGEP  180
WNMN  184

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SEQ ID NO: 51      moltype = DNA  length = 35
FEATURE
misc_feature      1..35
                  note = Thrombin Binding Aptamer

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source                1..35
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 51
tttttttttt tttttttttt ggttggtgtg gttgg                35

SEQ ID NO: 52          moltype = DNA length = 95
FEATURE               Location/Qualifiers
misc_feature          1..95
                      note = Y Adaptor Top Strand
misc_feature          1..30
                      note = C3 Spacer - C3 Spacer phosphoramidite (Integrated
                      DNA technologies: 5SpC3)
misc_feature          80..83
                      note = 18 Spacer - 18-atom hexa-ethyleneglycol spacer
                      (Integrated DNA technologies: iSp18)
source                1..95
                      mol_type = other DNA
                      organism = synthetic construct
misc_difference       1..30
                      note = n is a, c, g, or t
misc_difference       64..67
                      note = n is a, c, g, or t

SEQUENCE: 52
nnnnnnnnnn nnnnnnnnnn nnnnnnnnnn ggcgtctgct tgggtgttta accttttttt 60
tttnnnnaat gtacttcggt cagttacgta ttgct                95

SEQ ID NO: 53          moltype = DNA length = 25
FEATURE               Location/Qualifiers
misc_feature          1..25
                      note = Y Adaptor Blocker Strand
source                1..25
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 53
ttccgcagac gaaccacaaa attgg                25

SEQ ID NO: 54          moltype = DNA length = 15
FEATURE               Location/Qualifiers
misc_feature          1..15
                      note = Y Adaptor Cholesterol Tether
misc_feature          1
                      note = 5' Cholesterol tag
source                1..15
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 54
ttgaccgctc gcctc                15

SEQ ID NO: 55          moltype = DNA length = 45
FEATURE               Location/Qualifiers
misc_feature          1..45
                      note = Y Adaptor Bottom Strand
misc_feature          45
                      note = 5 Phos
source                1..45
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 55
aactggcgag cggagttttt acatgaagca agtcaatgca taacg                45

SEQ ID NO: 56          moltype = DNA length = 3561
FEATURE               Location/Qualifiers
misc_feature          1..3561
                      note = 3.6kb target sequence
source                1..3561
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 56
gccatcagat tgtgtttgtt agtcgctttt ttttttttga attttttttt tggaattttt 60
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1-46. (canceled)

47. A kit for characterising a target analyte, the kit comprising:

- (a) a CsgG pore comprising a mutant monomer which is a variant of the amino acid sequence set forth in SEQ ID NO: 2, and the variant comprises R192D, R192Q, R192F, R192S, or R192T substitutions; and
- (b) the components of a membrane.

48. The kit of claim **47** wherein the variant comprises R192D substitution.

49. The kit of claim **47**, wherein the variant comprises

- (a) one or more mutations at the following positions: I41, R93, A98, Q100, G103, T104, A106, I107, N108, L113, S115, T117, Y130, K135, E170, S208, D233, D238 and E244 and/or
- (b) one or more of D43S, E44S, F48S/N/Q/Y/W/I/V/H/R/K, Q87N/R/K, N91K/R, K94R/F/Y/W/L/S/N, R97F/Y/W/V/I/K/S/Q/H, E101I/

L/A/H, N102K/Q/L/I/V/S/H, R110F/G/N, Q114R/K, R142Q/S, T150Y/A/V/L/S/Q/N and D248S/N/Q/K/R;

- (A) deletion of one or more positions F193, I194, D195, Y196, Q197, R198, L199, L200 and E201 and/or (B) deletion of one or more of V139/G140/D149/T150/V186/Q187/V204/G205, G137/G138/Q151/Y152/Y184/E185/Y206/T207 and A141/R142/G147/A148/A188/G189/G202/E203; and/or

one or more of the following:

- (i) one or more mutations at the following positions N40, D43, E44, S54, S57, Q62, R97, E101, E124, E131, R142 and T150;
- (ii) mutations at Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56;
- (iii) Q42R or Q42K;
- (iv) K49R or K49Q;
- (v) N102R, N102F, N102Y or N102W;

- (vi) D149N, D149Q or D149R;
- (vii) E185N, E185Q or E185R;
- (viii) D195N, D195Q or D195R;
- (ix) E201N, E201Q or E201R;
- (x) E203N, E203Q or E203R;
- (xi) deletion of one or more of the following positions
F48, K49, P50, Y51, P52, A53, S54, N55, F56 and
S57;
- (xii) one or more mutations at the following positions
L90, N91, I95, A99, Q114; and
- (xiii) R97W; R93W; R93Y and R97Y; F191T; deletion
of V105, A106 and I107; and/or deletion of F193,
I194, D195, Y196, Q197, R198 and L199 or deletion
of D195, Y196, Q197, R198 and L199.

50. The kit of claim **47**, wherein the variant comprises a mutation at Y51 and/or Y56.

51. The kit of claim **50**, wherein the mutation at Y51 is Y51A and/or the mutation at Y56 is F56Q.

52. The kit of claim **47**, wherein

- (a) the variant comprises one or more of the following substitutions N40R, N40K, D43N, D43Q, D43R, D43K, E44N, E44Q, E44R, E44K, S54P, S57P, Q62R, Q62K, R97N, R97G, R97L, E101N, E101Q, E101R, E101K, E101F, E101Y, E101W, E101T, E124N, E124Q, E124R, E124K, E124F, E124Y, E124W, E131D, R142E, R142N and T150I;

- (b) the variant comprises F56N/N55Q, F56N/N55R, F56N/N55K, F56N/N55S, F56N/N55G, F56N/N55A, F56N/N55T, F56Q/N55Q, F56Q/N55R, F56Q/N55K, F56Q/N55S, F56Q/N55G, F56Q/N55A, F56Q/N55T, F56R/N55Q, F56R/N55R, F56R/N55K, F56R/N55S, F56R/N55G, F56R/N55A, F56R/N55T, F56S/N55Q, F56S/N55R, F56S/N55K, F56S/N55S, F56S/N55G, F56S/N55A, F56S/N55T, F56G/N55Q, F56G/N55R, F56G/N55K, F56G/N55S, F56G/N55G, F56G/N55A, F56G/N55T, F56A/N55Q, F56A/N55R, F56A/N55K, F56A/N55S, F56A/N55G, F56A/N55A, F56A/N55T, F56K/N55Q, F56K/N55R, F56K/N55K, F56K/N55S, F56K/N55G, F56K/N55A, F56K/N55T, F56N/Y51L, F56N/Y51V, F56N/Y51A, F56N/Y51N, F56N/Y51Q, F56N/Y51S, F56N/Y51G, F56Q/Y51L, F56Q/Y51V, F56Q/Y51A, F56Q/Y51N, F56Q/Y51Q, F56Q/Y51S, F56Q/Y51G, F56R/Y51L, F56R/Y51V, F56R/Y51A, F56R/Y51N, F56R/Y51Q, F56R/Y51S, F56R/Y51G, F56S/Y51L, F56S/Y51V, F56S/Y51A, F56S/Y51N, F56S/Y51Q, F56S/Y51S, F56S/Y51G, F56G/Y51L, F56G/Y51V, F56G/Y51A, F56G/Y51N, F56G/Y51Q, F56G/Y51S, F56G/Y51G, F56A/Y51L, F56A/Y51V, F56A/Y51A, F56A/Y51N, F56A/Y51Q, F56A/Y51S, F56A/Y51G, F56K/Y51L, F56K/Y51V, F56K/Y51A, F56K/Y51N, F56K/Y51Q, F56K/Y51S, F56K/Y51G, N55Q/Y51L, N55Q/Y51V, N55Q/Y51A, N55Q/Y51N, N55Q/Y51Q, N55Q/Y51S, N55Q/Y51G, N55R/Y51L, N55R/Y51V, N55R/Y51A, N55R/Y51N, N55R/Y51Q, N55R/Y51S, N55R/Y51G, N55K/Y51L, N55K/Y51V, N55K/Y51A, N55K/Y51N, N55K/Y51Q, N55K/Y51S, N55K/Y51G, N55S/Y51L, N55S/Y51V, N55S/Y51A, N55S/Y51N, N55S/Y51Q, N55S/Y51S, N55S/Y51G, N55G/Y51L, N55G/Y51V, N55G/Y51A, N55G/Y51N, N55G/Y51Q, N55G/Y51S, N55G/Y51G, N55A/Y51L, N55A/Y51V, N55A/Y51A, N55A/Y51N, N55A/Y51Q, N55A/Y51S, N55A/Y51G, N55T/Y51L, N55T/Y51V, N55T/Y51A, N55T/Y51N, N55T/Y51Q, N55T/Y51S, N55T/Y51G,

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F56A/N55A/Y51L, F56A/N55A/Y51V, F56A/N55A/
Y51A, F56A/N55A/Y51N, F56A/N55A/Y51Q, F56A/
N55A/Y51S, F56A/N55A/Y51G, F56A/N55T/Y51L,
F56A/N55T/Y51V, F56A/N55T/Y51A, F56A/N55T/
Y51N, F56A/N55T/Y51Q, F56A/N55T/Y51S, F56A/
N55T/Y51G, F56K/N55Q/Y51L, F56K/N55Q/Y51V,
F56K/N55Q/Y51A, F56K/N55Q/Y51N, F56K/N55Q/
Y51Q, F56K/N55Q/Y51S, F56K/N55Q/Y51G, F56K/
N55R/Y51L, F56K/N55R/Y51V, F56K/N55R/Y51A,
F56K/N55R/Y51N, F56K/N55R/Y51Q, F56K/N55R/
Y51S, F56K/N55R/Y51G, F56K/N55K/Y51L, F56K/
N55K/Y51V, F56K/N55K/Y51A, F56K/N55K/Y51N,
F56K/N55K/Y51Q, F56K/N55K/Y51S, F56K/N55K/
Y51G, F56K/N55S/Y51L, F56K/N55S/Y51V, F56K/
N55S/Y51A, F56K/N55S/Y51N, F56K/N55S/Y51Q,
F56K/N55S/Y51S, F56K/N55S/Y51G, F56K/N55G/
Y51L, F56K/N55G/Y51V, F56K/N55G/Y51A, F56K/
N55G/Y51N, F56K/N55G/Y51Q, F56K/N55G/Y51S,
F56K/N55G/Y51G, F56K/N55A/Y51L, F56K/N55A/
Y51V, F56K/N55A/Y51A, F56K/N55A/Y51N, F56K/
N55A/Y51Q, F56K/N55A/Y51S, F56K/N55A/Y51G,
F56K/N55T/Y51L, F56K/N55T/Y51V, F56K/N55T/
Y51A, F56K/N55T/Y51N, F56K/N55T/Y51Q, F56K/

N55T/Y51S, F56K/N55T/Y51G, F56E/N55R, F56E/N55K, F56D/N55R, F56D/N55K, F56R/N55E, F56R/N55D, F56K/N55E or F56K/N55D; and/or

(c) the variant comprises deletion of Y51/P52, Y51/P52/A53, P50 to P52, P50 to A53, K49 to Y51, K49 to A53 and replacement with a single proline (P), K49 to S54 and replacement with a single P, Y51 to A53, Y51 to S54, N55/F56, N55 to S57, N55/F56 and replacement with a single P, N55/F56 and replacement with a single glycine (G), N55/F56 and replacement with a single alanine (A), N55/F56 and replacement with a single P and Y51N, N55/F56 and replacement with a single P and Y51Q, N55/F56 and replacement with a single P and Y51S, N55/F56 and replacement with a single G and Y51N, N55/F56 and replacement with a single G and Y51Q, N55/F56 and replacement with a single G and Y51S, N55/F56 and replacement with a single A and Y51N, N55/F56 and replacement with a single A/Y51Q or N55/F56 and replacement with a single A and Y51S.

[illegible]

E185R/D195Q/E201Q/E203N, D149Q/E185R/D195Q/
E201N/E203Q, D149Q/E185R/D195N/E201Q/E203Q,
D149N/E185R/D195Q/E201Q/E203Q, D149Q/E185R/
D195Q/E201Q/E203Q, D149N/E185R/D195N/E201R/
E203N, D149Q/E185R/D195N/E201R/E203N, D149N/
E185R/D195Q/E201R/E203N, D149N/E185R/D195N/
E201R/E203Q, D149Q/E185R/D195Q/E201R/E203N,
D149Q/E185R/D195N/E201R/E203Q, D149N/E185R/
D195Q/E201R/E203Q, D149Q/E185R/D195Q/E201R/
E203Q, E131D/K49R, E101N/N102F, E101N/N102Y,
E101N/N102W, E101F/N102F, E101F/N102Y, E101F/
N102W, E101Y/N102F, E101Y/N102Y, E101Y/N102W,
E101W/N102F, E101W/N102Y, E101W/N102W, E101N/
N102R, E101F/N102R, E101Y/N102R or E101W/N102F.

55. The kit of claim **47**, wherein the CsgG pore comprises nine monomers.

56. The kit of claim **55**, wherein the CsgG pore comprises two or more mutant monomers.

58. The kit of claim **55**, wherein the two or more mutant monomers are attached via one or more linkers.

59. The kit of claim **55**, wherein the target analyte is a

metal ion, an inorganic salt, a polymer, an amino acid, a peptide, a polypeptide, a protein, a nucleotide, an oligonucleotide, a polynucleotide, a dye, a bleach, a pharmaceutical, a diagnostic agent, a recreational drug, an explosive or an environmental pollutant.

60. The kit of claim **59**, wherein the target analyte is a polynucleotide.

61. The kit of claim **59**, wherein the target analyte is a polypeptide.

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